

Thesis Project by Group CLS

THESIS 1-3

Thesis Project

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Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1–F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1–F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid, and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL - k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power <i>(Deprecated)</i>	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200

No.	Component Functions	Global Optimum
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1–F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11–F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21–F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
!pip install cec2017 --quiet  
  
ERROR: Could not find a version that satisfies the requirement cec2017  
(from versions: none)  
ERROR: No matching distribution found for cec2017
```

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')

cec2017 loaded!
```

Import `sys` control over warning messages for verification that `dpllde.py` exists

```
from google.colab import files
import sys

uploaded = files.upload()

if 'dpllde.py' in uploaded:
    if 'dpllde' in sys.modules:
        del sys.modules['dpllde']
    import dpllde
    from dpllde import DPLLDE
    print('dpllde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named dpllde.py')

<IPython.core.display.HTML object>

Saving dpllde.py to dpllde (1).py
ERROR: Please upload a file named dpllde.py

import dpllde
from dpllde import DPLLDE
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
D = 10 # Change this when testing different dimensions
```

```

bounds = np.array([-100.0, 100.0]) * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42

```

Store all results for summary table under `all_results` dict

```
all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```

def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,)
    input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper

```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```

def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run DP-LLDE on a function and plot convergence + diversity
    graphs."""
    print('=' * 80)
    print(f'DP-LLDE on CEC 2017 {fn_name} - Single Run')
    print('=' * 80)
    print(f'Function           : {fn_label} ({D}D)')
    print(f'Bounds              : [-100, 100]')
    print(f'Global optimum      : {f_global}')

    opt = DPLLDE(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history =

```

```

opt.optimize(verbose=True)

error = abs(best_fit - f_global)
print(f'\n{"="*80}')
print('Results:')
print(f'  Best Fitness           : {best_fit:.6e}')
print(f'  Global Optimum          : {f_global:.6e}')
print(f'  Error (|best-optimum|)   : {error:.6e}')
print(f'  Best Solution (5 dim)    : {best_sol[:5]}')
print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
print(f'{"="*80}')
```

Store for summary

```

all_results[fn_name] = {
    'label': fn_label,
    'best_fit': best_fit,
    'f_global': f_global,
    'error': error
}

fes_axis = np.linspace(0, MAX_FES, len(history))
gen_axis = np.arange(len(diversity_history)) # Generations (0, 1,
2, ...)

# — Two-panel figure

```

```

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

# Convergence curve (top) – FEs on x-axis
ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
ax1.set_ylabel('Best Fitness', fontsize=12)
ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
ax1.set_title(
    f'DP-LLDE – CEC 2017 {fn_name}: {fn_label} ({D}D)',
    fontsize=12
)
ax1.grid(True, which='both', linestyle='--', alpha=0.6)

# Diversity curve (bottom) – Generations on x-axis
ax2.plot(gen_axis, diversity_history, color='darkorange',
linewidth=1.5)
ax2.set_xlabel('Generation', fontsize=12)
ax2.set_ylabel('Population Diversity\n(Mean Std Dev)',
fontsize=12)
ax2.set_title(f'Diversity Across Generations – {fn_name}',
fontsize=12)
ax2.grid(True, which='both', linestyle='--', alpha=0.6)

plt.tight_layout()
fname = f'convergence_{fn_name}.png'

```



```
plt.savefig(fname, dpi=150)
plt.show()
print(f'Graph saved as {fname}\n')
```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```
# =====
# F1 – Shifted and Rotated Bent Cigar
# =====
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and
Rotated Bent Cigar')

=====
=====
DP-LLDE on CEC 2017 F1 – Single Run
=====
=====
Function      : Shifted and Rotated Bent Cigar (10D)
Bounds        : [-100, 100]
Global optimum : 100.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 1.871674e+11
Generation 10: Best = 1.145665e+10 FES = 1100  $\mu_F$  = 0.580  $\mu_{CR}$ 
= 0.349 Diversity = 1.9041e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 7.373016e+08 FES = 2100  $\mu_F$  = 0.635  $\mu_{CR}$ 
= 0.352 Diversity = 5.7959e+00  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 7.460979e+07 FES = 3100  $\mu_F$  = 0.661  $\mu_{CR}$ 
= 0.351 Diversity = 2.4609e+00  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 6.348839e+06 FES = 4100  $\mu_F$  = 0.685  $\mu_{CR}$ 
= 0.352 Diversity = 2.1100e+00  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 2.552607e+06 FES = 5100  $\mu_F$  = 0.699  $\mu_{CR}$ 
= 0.346 Diversity = 2.2196e+00  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 1.759662e+05 FES = 6100  $\mu_F$  = 0.709  $\mu_{CR}$ 
= 0.325 Diversity = 3.7061e+00  $k_t$  = [2, 2, 4, 4]
Generation 70: Best = 1.117800e+05 FES = 7100  $\mu_F$  = 0.727  $\mu_{CR}$ 
= 0.315 Diversity = 4.8627e+00  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 5.366448e+03 FES = 8100  $\mu_F$  = 0.745  $\mu_{CR}$ 
= 0.289 Diversity = 4.3980e+00  $k_t$  = [3, 3, 2, 4]
Generation 90: Best = 4.710831e+03 FES = 9100  $\mu_F$  = 0.756  $\mu_{CR}$ 
= 0.292 Diversity = 4.4793e+00  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 1.351412e+03 FES = 10100  $\mu_F$  = 0.755  $\mu_{CR}$ 
= 0.257 Diversity = 3.7372e+00  $k_t$  = [2, 3, 4, 4]
Generation 110: Best = 7.014613e+02 FES = 11100  $\mu_F$  = 0.757  $\mu_{CR}$ 
= 0.246 Diversity = 2.2772e+00  $k_t$  = [1, 1, 1, 1]
```

Generation 120: Best = 2.243539e+02	FES = 12100	$\mu_F = 0.762$	μ_{CR}
= 0.213 Diversity = 1.5262e+00	$k_t = [2, 2, 2, 2]$		
Generation 130: Best = 1.330577e+02	FES = 13100	$\mu_F = 0.770$	μ_{CR}
= 0.194 Diversity = 1.1273e+00	$k_t = [1, 1, 1, 1]$		
Generation 140: Best = 1.144315e+02	FES = 14100	$\mu_F = 0.755$	μ_{CR}
= 0.184 Diversity = 5.7805e-01	$k_t = [2, 3, 3, 3]$		
Generation 150: Best = 1.048549e+02	FES = 15100	$\mu_F = 0.761$	μ_{CR}
= 0.180 Diversity = 4.4483e-01	$k_t = [1, 1, 1, 1]$		
Generation 160: Best = 1.016541e+02	FES = 16100	$\mu_F = 0.765$	μ_{CR}
= 0.173 Diversity = 2.6086e-01	$k_t = [4, 4, 3, 3]$		
Generation 170: Best = 1.003603e+02	FES = 17100	$\mu_F = 0.758$	μ_{CR}
= 0.151 Diversity = 1.7188e-01	$k_t = [1, 1, 1, 1]$		
Generation 180: Best = 1.002049e+02	FES = 18100	$\mu_F = 0.755$	μ_{CR}
= 0.144 Diversity = 7.8975e-02	$k_t = [3, 2, 4, 4]$		
Generation 190: Best = 1.000823e+02	FES = 19100	$\mu_F = 0.765$	μ_{CR}
= 0.140 Diversity = 5.3918e-02	$k_t = [1, 1, 1, 1]$		
Generation 200: Best = 1.000560e+02	FES = 20100	$\mu_F = 0.756$	μ_{CR}
= 0.137 Diversity = 3.7550e-02	$k_t = [4, 4, 4, 2]$		
Generation 210: Best = 1.000251e+02	FES = 21100	$\mu_F = 0.743$	μ_{CR}
= 0.141 Diversity = 2.6362e-02	$k_t = [1, 1, 1, 1]$		
Generation 220: Best = 1.000078e+02	FES = 22100	$\mu_F = 0.740$	μ_{CR}
= 0.131 Diversity = 1.7829e-02	$k_t = [3, 2, 4, 2]$		
Generation 230: Best = 1.000021e+02	FES = 23100	$\mu_F = 0.745$	μ_{CR}
= 0.128 Diversity = 1.1895e-02	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.000021e+02	FES = 24100	$\mu_F = 0.755$	μ_{CR}
= 0.132 Diversity = 7.9780e-03	$k_t = [3, 4, 4, 2]$		
Generation 250: Best = 1.000010e+02	FES = 25100	$\mu_F = 0.756$	μ_{CR}
= 0.132 Diversity = 5.1965e-03	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.000005e+02	FES = 26100	$\mu_F = 0.764$	μ_{CR}
= 0.131 Diversity = 3.7784e-03	$k_t = [4, 4, 3, 4]$		
Generation 270: Best = 1.000002e+02	FES = 27100	$\mu_F = 0.756$	μ_{CR}
= 0.125 Diversity = 2.2956e-03	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.000001e+02	FES = 28100	$\mu_F = 0.758$	μ_{CR}
= 0.125 Diversity = 1.3281e-03	$k_t = [4, 2, 3, 2]$		
Generation 290: Best = 1.000000e+02	FES = 29100	$\mu_F = 0.761$	μ_{CR}
= 0.119 Diversity = 1.1412e-03	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.000000e+02	FES = 30100	$\mu_F = 0.778$	μ_{CR}
= 0.120 Diversity = 6.8490e-04	$k_t = [2, 3, 2, 3]$		
Generation 310: Best = 1.000000e+02	FES = 31100	$\mu_F = 0.790$	μ_{CR}
= 0.122 Diversity = 5.2385e-04	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.000000e+02	FES = 32100	$\mu_F = 0.790$	μ_{CR}
= 0.120 Diversity = 4.1043e-04	$k_t = [2, 3, 2, 3]$		
Generation 330: Best = 1.000000e+02	FES = 33100	$\mu_F = 0.794$	μ_{CR}
= 0.118 Diversity = 3.5173e-04	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.000000e+02	FES = 34100	$\mu_F = 0.792$	μ_{CR}
= 0.119 Diversity = 2.7503e-04	$k_t = [2, 3, 4, 4]$		
Generation 350: Best = 1.000000e+02	FES = 35100	$\mu_F = 0.797$	μ_{CR}
= 0.120 Diversity = 2.4291e-04	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.000000e+02	FES = 36100	$\mu_F = 0.817$	μ_{CR}

= 0.134	Diversity = 1.7026e-04	k_t = [4, 4, 3, 4]	μ_F = 0.816	μ_{CR}
Generation 370:	Best = 1.000000e+02	FES = 37100		
= 0.130	Diversity = 1.2551e-04	k_t = [1, 1, 1, 1]	μ_F = 0.815	μ_{CR}
Generation 380:	Best = 1.000000e+02	FES = 38100		
= 0.131	Diversity = 8.9442e-05	k_t = [2, 2, 2, 4]	μ_F = 0.798	μ_{CR}
Generation 390:	Best = 1.000000e+02	FES = 39100		
= 0.132	Diversity = 5.3184e-05	k_t = [1, 1, 1, 1]	μ_F = 0.788	μ_{CR}
Generation 400:	Best = 1.000000e+02	FES = 40100		
= 0.125	Diversity = 3.8400e-05	k_t = [4, 2, 4, 4]	μ_F = 0.792	μ_{CR}
Generation 410:	Best = 1.000000e+02	FES = 41100		
= 0.118	Diversity = 2.8386e-05	k_t = [1, 1, 1, 1]	μ_F = 0.804	μ_{CR}
Generation 420:	Best = 1.000000e+02	FES = 42100		
= 0.120	Diversity = 2.1947e-05	k_t = [3, 4, 4, 4]	μ_F = 0.790	μ_{CR}
Generation 430:	Best = 1.000000e+02	FES = 43100		
= 0.134	Diversity = 1.6389e-05	k_t = [1, 1, 1, 1]	μ_F = 0.784	μ_{CR}
Generation 440:	Best = 1.000000e+02	FES = 44100		
= 0.132	Diversity = 1.3488e-05	k_t = [4, 3, 2, 2]	μ_F = 0.785	μ_{CR}
Generation 450:	Best = 1.000000e+02	FES = 45100		
= 0.130	Diversity = 1.1964e-05	k_t = [1, 1, 1, 1]	μ_F = 0.781	μ_{CR}
Generation 460:	Best = 1.000000e+02	FES = 46100		
= 0.116	Diversity = 1.0825e-05	k_t = [4, 2, 3, 3]	μ_F = 0.780	μ_{CR}
Generation 470:	Best = 1.000000e+02	FES = 47100		
= 0.123	Diversity = 8.9263e-06	k_t = [1, 1, 1, 1]	μ_F = 0.792	μ_{CR}
Generation 480:	Best = 1.000000e+02	FES = 48100		
= 0.122	Diversity = 1.0839e-05	k_t = [3, 2, 2, 4]	μ_F = 0.805	μ_{CR}
Generation 490:	Best = 1.000000e+02	FES = 49100		
= 0.126	Diversity = 1.4559e-05	k_t = [1, 1, 1, 1]	μ_F = 0.810	μ_{CR}
Generation 500:	Best = 1.000000e+02	FES = 50100		
= 0.133	Diversity = 1.3347e-05	k_t = [2, 4, 3, 4]	μ_F = 0.802	μ_{CR}
Generation 510:	Best = 1.000000e+02	FES = 51100		
= 0.133	Diversity = 9.5664e-06	k_t = [1, 1, 1, 1]	μ_F = 0.811	μ_{CR}
Generation 520:	Best = 1.000000e+02	FES = 52100		
= 0.134	Diversity = 5.9909e-06	k_t = [4, 4, 2, 3]	μ_F = 0.792	μ_{CR}
Generation 530:	Best = 1.000000e+02	FES = 53100		
= 0.134	Diversity = 3.4120e-06	k_t = [1, 1, 1, 1]	μ_F = 0.794	μ_{CR}
Generation 540:	Best = 1.000000e+02	FES = 54100		
= 0.140	Diversity = 2.2005e-06	k_t = [4, 2, 4, 3]	μ_F = 0.795	μ_{CR}
Generation 550:	Best = 1.000000e+02	FES = 55100		
= 0.132	Diversity = 1.5589e-06	k_t = [1, 1, 1, 1]	μ_F = 0.783	μ_{CR}
Generation 560:	Best = 1.000000e+02	FES = 56100		
= 0.127	Diversity = 9.8783e-07	k_t = [2, 3, 3, 4]	μ_F = 0.800	μ_{CR}
Generation 570:	Best = 1.000000e+02	FES = 57100		
= 0.128	Diversity = 6.4491e-07	k_t = [1, 1, 1, 1]	μ_F = 0.796	μ_{CR}
Generation 580:	Best = 1.000000e+02	FES = 58100		
= 0.130	Diversity = 4.2709e-07	k_t = [2, 3, 3, 3]	μ_F = 0.795	μ_{CR}
Generation 590:	Best = 1.000000e+02	FES = 59100		
= 0.129	Diversity = 3.2168e-07	k_t = [1, 1, 1, 1]	μ_F = 0.790	μ_{CR}
Generation 600:	Best = 1.000000e+02	FES = 60100		
= 0.132	Diversity = 2.4781e-07	k_t = [4, 4, 3, 2]		

Generation 610: Best = 1.000000e+02	FES = 61100	$\mu_F = 0.804$	μ_{CR}
= 0.127 Diversity = 1.8542e-07 $k_t = [1, 1, 1, 1]$			
Generation 620: Best = 1.000000e+02	FES = 62100	$\mu_F = 0.800$	μ_{CR}
= 0.134 Diversity = 1.3220e-07 $k_t = [4, 2, 4, 3]$			
Generation 630: Best = 1.000000e+02	FES = 63100	$\mu_F = 0.794$	μ_{CR}
= 0.133 Diversity = 9.7668e-08 $k_t = [1, 1, 1, 1]$			
Generation 640: Best = 1.000000e+02	FES = 64100	$\mu_F = 0.794$	μ_{CR}
= 0.143 Diversity = 6.5430e-08 $k_t = [2, 4, 3, 4]$			
Generation 650: Best = 1.000000e+02	FES = 65100	$\mu_F = 0.801$	μ_{CR}
= 0.136 Diversity = 5.1534e-08 $k_t = [1, 1, 1, 1]$			
Generation 660: Best = 1.000000e+02	FES = 66100	$\mu_F = 0.782$	μ_{CR}
= 0.123 Diversity = 3.4919e-08 $k_t = [4, 3, 4, 3]$			
Generation 670: Best = 1.000000e+02	FES = 67100	$\mu_F = 0.793$	μ_{CR}
= 0.121 Diversity = 3.0356e-08 $k_t = [1, 1, 1, 1]$			
Generation 680: Best = 1.000000e+02	FES = 68100	$\mu_F = 0.802$	μ_{CR}
= 0.117 Diversity = 2.1542e-08 $k_t = [3, 4, 3, 2]$			
Generation 690: Best = 1.000000e+02	FES = 69100	$\mu_F = 0.802$	μ_{CR}
= 0.116 Diversity = 1.8221e-08 $k_t = [1, 1, 1, 1]$			
Generation 700: Best = 1.000000e+02	FES = 70100	$\mu_F = 0.810$	μ_{CR}
= 0.112 Diversity = 1.4071e-08 $k_t = [2, 4, 4, 3]$			
Generation 710: Best = 1.000000e+02	FES = 71100	$\mu_F = 0.813$	μ_{CR}
= 0.113 Diversity = 9.7875e-09 $k_t = [1, 1, 1, 1]$			
Generation 720: Best = 1.000000e+02	FES = 72100	$\mu_F = 0.823$	μ_{CR}
= 0.121 Diversity = 8.3053e-09 $k_t = [4, 2, 4, 3]$			
Generation 730: Best = 1.000000e+02	FES = 73100	$\mu_F = 0.824$	μ_{CR}
= 0.121 Diversity = 7.8898e-09 $k_t = [1, 1, 1, 1]$			
Generation 740: Best = 1.000000e+02	FES = 74100	$\mu_F = 0.833$	μ_{CR}
= 0.110 Diversity = 8.7464e-09 $k_t = [2, 3, 4, 2]$			
Generation 750: Best = 1.000000e+02	FES = 75100	$\mu_F = 0.835$	μ_{CR}
= 0.111 Diversity = 8.7305e-09 $k_t = [1, 1, 1, 1]$			
Generation 760: Best = 1.000000e+02	FES = 76100	$\mu_F = 0.839$	μ_{CR}
= 0.115 Diversity = 7.3295e-09 $k_t = [3, 2, 2, 4]$			
Generation 770: Best = 1.000000e+02	FES = 77100	$\mu_F = 0.842$	μ_{CR}
= 0.119 Diversity = 8.1964e-09 $k_t = [1, 1, 1, 1]$			
Generation 780: Best = 1.000000e+02	FES = 78100	$\mu_F = 0.835$	μ_{CR}
= 0.118 Diversity = 6.0661e-09 $k_t = [4, 4, 3, 2]$			
Generation 790: Best = 1.000000e+02	FES = 79100	$\mu_F = 0.836$	μ_{CR}
= 0.117 Diversity = 6.1182e-09 $k_t = [1, 1, 1, 1]$			
Generation 800: Best = 1.000000e+02	FES = 80100	$\mu_F = 0.830$	μ_{CR}
= 0.117 Diversity = 6.0521e-09 $k_t = [2, 4, 3, 3]$			
Generation 810: Best = 1.000000e+02	FES = 81100	$\mu_F = 0.833$	μ_{CR}
= 0.117 Diversity = 6.1372e-09 $k_t = [1, 1, 1, 1]$			
Generation 820: Best = 1.000000e+02	FES = 82100	$\mu_F = 0.823$	μ_{CR}
= 0.118 Diversity = 5.7034e-09 $k_t = [4, 3, 3, 4]$			
Generation 830: Best = 1.000000e+02	FES = 83100	$\mu_F = 0.837$	μ_{CR}
= 0.116 Diversity = 5.8457e-09 $k_t = [1, 1, 1, 1]$			
Generation 840: Best = 1.000000e+02	FES = 84100	$\mu_F = 0.841$	μ_{CR}
= 0.118 Diversity = 5.9564e-09 $k_t = [4, 4, 4, 3]$			
Generation 850: Best = 1.000000e+02	FES = 85100	$\mu_F = 0.838$	μ_{CR}

```

= 0.123 Diversity = 6.1222e-09 k_t = [1, 1, 1, 1]
Generation 860: Best = 1.000000e+02 FES = 86100  $\mu_F$  = 0.824  $\mu_{CR}$ 
= 0.124 Diversity = 5.9846e-09 k_t = [4, 3, 4, 2]
Generation 870: Best = 1.000000e+02 FES = 87100  $\mu_F$  = 0.823  $\mu_{CR}$ 
= 0.120 Diversity = 5.8552e-09 k_t = [1, 1, 1, 1]
Generation 880: Best = 1.000000e+02 FES = 88100  $\mu_F$  = 0.822  $\mu_{CR}$ 
= 0.115 Diversity = 5.6676e-09 k_t = [3, 3, 3, 3]
Generation 890: Best = 1.000000e+02 FES = 89100  $\mu_F$  = 0.819  $\mu_{CR}$ 
= 0.116 Diversity = 5.8610e-09 k_t = [1, 1, 1, 1]
Generation 900: Best = 1.000000e+02 FES = 90100  $\mu_F$  = 0.829  $\mu_{CR}$ 
= 0.115 Diversity = 6.0195e-09 k_t = [3, 4, 2, 4]
Generation 910: Best = 1.000000e+02 FES = 91100  $\mu_F$  = 0.828  $\mu_{CR}$ 
= 0.118 Diversity = 6.2157e-09 k_t = [1, 1, 1, 1]
Generation 920: Best = 1.000000e+02 FES = 92100  $\mu_F$  = 0.825  $\mu_{CR}$ 
= 0.125 Diversity = 5.5945e-09 k_t = [2, 4, 3, 3]
Generation 930: Best = 1.000000e+02 FES = 93100  $\mu_F$  = 0.821  $\mu_{CR}$ 
= 0.118 Diversity = 5.4298e-09 k_t = [1, 1, 1, 1]
Generation 940: Best = 1.000000e+02 FES = 94100  $\mu_F$  = 0.820  $\mu_{CR}$ 
= 0.122 Diversity = 5.6236e-09 k_t = [2, 3, 4, 4]
Generation 950: Best = 1.000000e+02 FES = 95100  $\mu_F$  = 0.822  $\mu_{CR}$ 
= 0.121 Diversity = 5.9935e-09 k_t = [1, 1, 1, 1]
Generation 960: Best = 1.000000e+02 FES = 96100  $\mu_F$  = 0.836  $\mu_{CR}$ 
= 0.122 Diversity = 6.1661e-09 k_t = [3, 4, 4, 3]
Generation 970: Best = 1.000000e+02 FES = 97100  $\mu_F$  = 0.829  $\mu_{CR}$ 
= 0.124 Diversity = 6.6019e-09 k_t = [1, 1, 1, 1]
Generation 980: Best = 1.000000e+02 FES = 98100  $\mu_F$  = 0.822  $\mu_{CR}$ 
= 0.120 Diversity = 6.2752e-09 k_t = [4, 3, 4, 3]
Generation 990: Best = 1.000000e+02 FES = 99100  $\mu_F$  = 0.813  $\mu_{CR}$ 
= 0.116 Diversity = 6.7148e-09 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

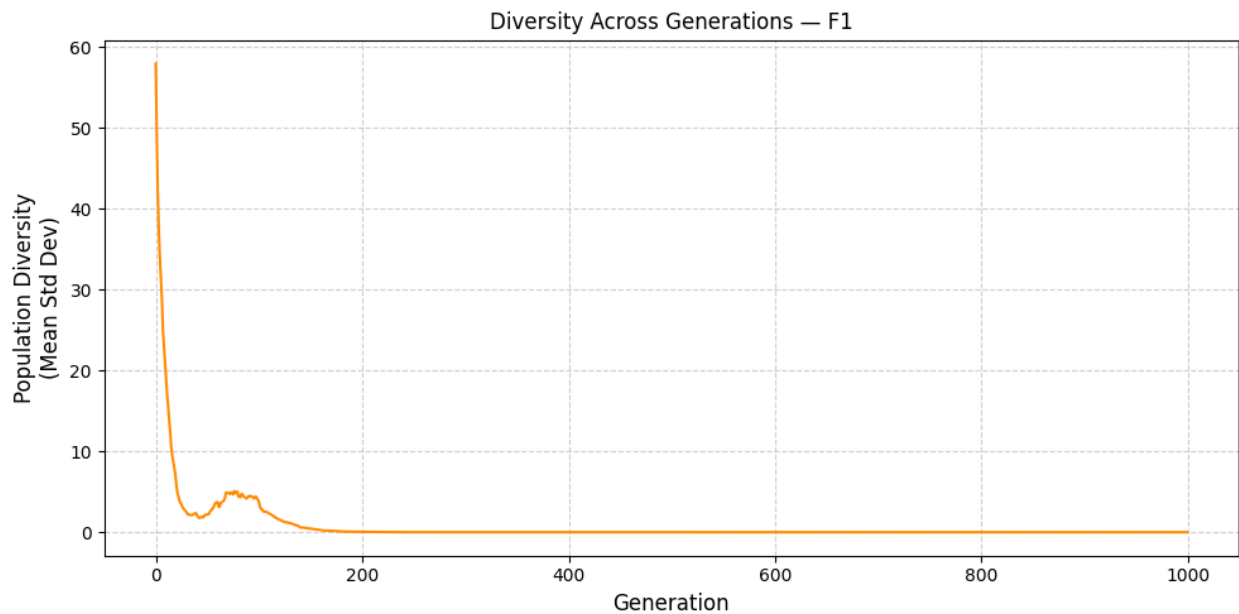
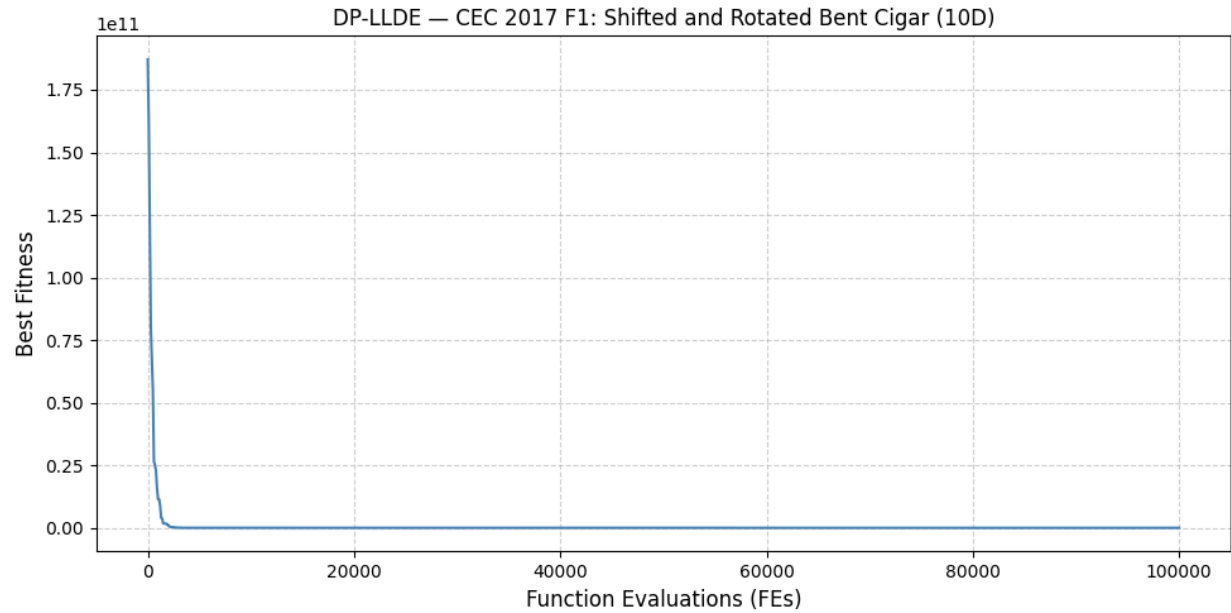
Best Fitness          = 1.000000e+02
Total Function Evals  = 100000
Total Successful Updates = 31823

```

```

=====
=====
Results:
  Best Fitness          : 1.000000e+02
  Global Optimum        : 1.000000e+02
  Error (|best-optimum|) : 1.421085e-14
  Best Solution (5 dim) : [-55.2763985 -70.42955972 -29.61018187 -
58.32676328 22.08960188]
  D^(0) (init. diversity): 5.7928e+01
=====
=====

```



Graph saved as convergence_F1.png

```
# =====
# F2 – Shifted and Rotated Sum of Different Power
# =====
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and
Rotated Sum of Different Power (Deprecated)')
```

```
=====
=====
DP-LLDE on CEC 2017 F2 – Single Run
=====
```

=====

Function : Shifted and Rotated Sum of Different Power
(Deprecated) (10D)

Bounds : [-100, 100]

Global optimum : 200.0

WARNING: f2 has been deprecated from the CEC 2017 benchmark suite

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.603565e+14

Generation 10: Best = 3.307877e+08 FES = 1100 $\mu_F = 0.578$ μ_{CR}
= 0.347 Diversity = 2.7434e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 9.373572e+05 FES = 2100 $\mu_F = 0.625$ μ_{CR}
= 0.347 Diversity = 1.7729e+01 $k_t = [3, 4, 2, 3]$

Generation 30: Best = 7.764745e+04 FES = 3100 $\mu_F = 0.650$ μ_{CR}
= 0.351 Diversity = 1.1410e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 1.655447e+04 FES = 4100 $\mu_F = 0.677$ μ_{CR}
= 0.352 Diversity = 8.0397e+00 $k_t = [3, 3, 2, 4]$

Generation 50: Best = 6.156778e+03 FES = 5100 $\mu_F = 0.693$ μ_{CR}
= 0.361 Diversity = 4.7416e+00 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 4.259731e+03 FES = 6100 $\mu_F = 0.709$ μ_{CR}
= 0.357 Diversity = 4.6995e+00 $k_t = [2, 2, 4, 4]$

Generation 70: Best = 1.209699e+03 FES = 7100 $\mu_F = 0.734$ μ_{CR}
= 0.360 Diversity = 4.2166e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 6.240057e+02 FES = 8100 $\mu_F = 0.749$ μ_{CR}
= 0.352 Diversity = 4.1154e+00 $k_t = [3, 3, 2, 4]$

Generation 90: Best = 3.253432e+02 FES = 9100 $\mu_F = 0.763$ μ_{CR}
= 0.349 Diversity = 3.9947e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.355936e+02 FES = 10100 $\mu_F = 0.759$ μ_{CR}
= 0.343 Diversity = 2.9505e+00 $k_t = [2, 3, 4, 4]$

Generation 110: Best = 2.093675e+02 FES = 11100 $\mu_F = 0.759$ μ_{CR}
= 0.351 Diversity = 1.1535e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.014198e+02 FES = 12100 $\mu_F = 0.774$ μ_{CR}
= 0.348 Diversity = 6.8336e-01 $k_t = [2, 2, 2, 2]$

Generation 130: Best = 2.012142e+02 FES = 13100 $\mu_F = 0.766$ μ_{CR}
= 0.343 Diversity = 5.2329e-01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.005741e+02 FES = 14100 $\mu_F = 0.752$ μ_{CR}
= 0.337 Diversity = 3.2279e-01 $k_t = [3, 2, 2, 4]$

Generation 150: Best = 2.000497e+02 FES = 15100 $\mu_F = 0.764$ μ_{CR}
= 0.331 Diversity = 2.5371e-01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.000078e+02 FES = 16100 $\mu_F = 0.738$ μ_{CR}
= 0.331 Diversity = 1.6469e-01 $k_t = [2, 2, 2, 3]$

Generation 170: Best = 2.000047e+02 FES = 17100 $\mu_F = 0.722$ μ_{CR}
= 0.321 Diversity = 1.2858e-01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 2.000013e+02 FES = 18100 $\mu_F = 0.714$ μ_{CR}
= 0.304 Diversity = 9.3526e-02 $k_t = [3, 2, 2, 3]$

Generation 190: Best = 2.000007e+02 FES = 19100 $\mu_F = 0.712$ μ_{CR}
= 0.293 Diversity = 8.1010e-02 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 2.000006e+02 FES = 20100 $\mu_F = 0.710$ μ_{CR}
= 0.283 Diversity = 6.2620e-02 $k_t = [3, 2, 3, 4]$

Generation 210: Best = 2.000004e+02 FES = 21100 $\mu_F = 0.732$ μ_{CR}

= 0.257	Diversity = 5.7799e-02	k_t = [1, 1, 1, 1]	μ_F = 0.707	μ_{CR}
Generation 220:	Best = 2.000003e+02	FES = 22100		
= 0.236	Diversity = 5.0460e-02	k_t = [2, 2, 3, 4]	μ_F = 0.699	μ_{CR}
Generation 230:	Best = 2.000002e+02	FES = 23100		
= 0.200	Diversity = 4.0779e-02	k_t = [1, 1, 1, 1]	μ_F = 0.714	μ_{CR}
Generation 240:	Best = 2.000002e+02	FES = 24100		
= 0.180	Diversity = 3.5583e-02	k_t = [4, 3, 4, 4]	μ_F = 0.714	μ_{CR}
Generation 250:	Best = 2.000001e+02	FES = 25100		
= 0.159	Diversity = 3.0700e-02	k_t = [1, 1, 1, 1]	μ_F = 0.716	μ_{CR}
Generation 260:	Best = 2.000001e+02	FES = 26100		
= 0.137	Diversity = 2.7625e-02	k_t = [4, 4, 2, 2]	μ_F = 0.733	μ_{CR}
Generation 270:	Best = 2.000001e+02	FES = 27100		
= 0.128	Diversity = 2.5734e-02	k_t = [1, 1, 1, 1]	μ_F = 0.731	μ_{CR}
Generation 280:	Best = 2.000001e+02	FES = 28100		
= 0.121	Diversity = 2.3294e-02	k_t = [2, 2, 4, 4]	μ_F = 0.745	μ_{CR}
Generation 290:	Best = 2.000001e+02	FES = 29100		
= 0.118	Diversity = 2.3584e-02	k_t = [1, 1, 1, 1]	μ_F = 0.766	μ_{CR}
Generation 300:	Best = 2.000001e+02	FES = 30100		
= 0.118	Diversity = 2.2249e-02	k_t = [2, 3, 4, 4]	μ_F = 0.775	μ_{CR}
Generation 310:	Best = 2.000001e+02	FES = 31100		
= 0.117	Diversity = 2.2610e-02	k_t = [1, 1, 1, 1]	μ_F = 0.773	μ_{CR}
Generation 320:	Best = 2.000001e+02	FES = 32100		
= 0.115	Diversity = 2.3208e-02	k_t = [4, 3, 3, 3]	μ_F = 0.782	μ_{CR}
Generation 330:	Best = 2.000001e+02	FES = 33100		
= 0.110	Diversity = 2.4627e-02	k_t = [1, 1, 1, 1]	μ_F = 0.783	μ_{CR}
Generation 340:	Best = 2.000001e+02	FES = 34100		
= 0.112	Diversity = 2.1451e-02	k_t = [4, 3, 3, 3]	μ_F = 0.786	μ_{CR}
Generation 350:	Best = 2.000001e+02	FES = 35100		
= 0.119	Diversity = 2.1089e-02	k_t = [1, 1, 1, 1]	μ_F = 0.795	μ_{CR}
Generation 360:	Best = 2.000001e+02	FES = 36100		
= 0.106	Diversity = 1.8210e-02	k_t = [3, 2, 3, 4]	μ_F = 0.783	μ_{CR}
Generation 370:	Best = 2.000001e+02	FES = 37100		
= 0.115	Diversity = 1.8402e-02	k_t = [1, 1, 1, 1]	μ_F = 0.785	μ_{CR}
Generation 380:	Best = 2.000001e+02	FES = 38100		
= 0.122	Diversity = 1.6346e-02	k_t = [2, 3, 4, 3]	μ_F = 0.795	μ_{CR}
Generation 390:	Best = 2.000001e+02	FES = 39100		
= 0.112	Diversity = 1.6566e-02	k_t = [1, 1, 1, 1]	μ_F = 0.779	μ_{CR}
Generation 400:	Best = 2.000001e+02	FES = 40100		
= 0.112	Diversity = 1.5785e-02	k_t = [4, 4, 2, 2]	μ_F = 0.797	μ_{CR}
Generation 410:	Best = 2.000001e+02	FES = 41100		
= 0.116	Diversity = 1.7270e-02	k_t = [1, 1, 1, 1]	μ_F = 0.804	μ_{CR}
Generation 420:	Best = 2.000001e+02	FES = 42100		
= 0.108	Diversity = 1.6590e-02	k_t = [4, 2, 2, 4]	μ_F = 0.805	μ_{CR}
Generation 430:	Best = 2.000001e+02	FES = 43100		
= 0.107	Diversity = 1.7531e-02	k_t = [1, 1, 1, 1]	μ_F = 0.802	μ_{CR}
Generation 440:	Best = 2.000001e+02	FES = 44100		
= 0.109	Diversity = 1.6501e-02	k_t = [3, 2, 4, 4]	μ_F = 0.808	μ_{CR}
Generation 450:	Best = 2.000001e+02	FES = 45100		
= 0.110	Diversity = 1.7341e-02	k_t = [1, 1, 1, 1]		

Generation 460: Best = 2.000001e+02	FES = 46100	$\mu_F = 0.799$	μ_{CR}
= 0.101 Diversity = 1.5729e-02	$k_t = [2, 4, 3, 4]$		
Generation 470: Best = 2.000001e+02	FES = 47100	$\mu_F = 0.794$	μ_{CR}
= 0.109 Diversity = 1.5998e-02	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.000001e+02	FES = 48100	$\mu_F = 0.804$	μ_{CR}
= 0.124 Diversity = 1.7315e-02	$k_t = [3, 3, 3, 3]$		
Generation 490: Best = 2.000001e+02	FES = 49100	$\mu_F = 0.801$	μ_{CR}
= 0.121 Diversity = 1.8024e-02	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 2.000001e+02	FES = 50100	$\mu_F = 0.792$	μ_{CR}
= 0.128 Diversity = 1.6278e-02	$k_t = [4, 4, 2, 2]$		
Generation 510: Best = 2.000001e+02	FES = 51100	$\mu_F = 0.793$	μ_{CR}
= 0.118 Diversity = 1.6696e-02	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.000001e+02	FES = 52100	$\mu_F = 0.797$	μ_{CR}
= 0.121 Diversity = 1.6349e-02	$k_t = [2, 4, 3, 3]$		
Generation 530: Best = 2.000001e+02	FES = 53100	$\mu_F = 0.794$	μ_{CR}
= 0.130 Diversity = 1.6813e-02	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.000001e+02	FES = 54100	$\mu_F = 0.794$	μ_{CR}
= 0.112 Diversity = 1.6032e-02	$k_t = [4, 4, 4, 2]$		
Generation 550: Best = 2.000001e+02	FES = 55100	$\mu_F = 0.794$	μ_{CR}
= 0.113 Diversity = 1.7029e-02	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.000001e+02	FES = 56100	$\mu_F = 0.781$	μ_{CR}
= 0.111 Diversity = 1.7482e-02	$k_t = [4, 2, 2, 3]$		
Generation 570: Best = 2.000001e+02	FES = 57100	$\mu_F = 0.757$	μ_{CR}
= 0.115 Diversity = 2.0161e-02	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.000001e+02	FES = 58100	$\mu_F = 0.764$	μ_{CR}
= 0.114 Diversity = 1.8204e-02	$k_t = [3, 4, 2, 2]$		
Generation 590: Best = 2.000001e+02	FES = 59100	$\mu_F = 0.756$	μ_{CR}
= 0.109 Diversity = 1.9342e-02	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.000001e+02	FES = 60100	$\mu_F = 0.767$	μ_{CR}
= 0.090 Diversity = 1.7659e-02	$k_t = [2, 3, 3, 4]$		
Generation 610: Best = 2.000001e+02	FES = 61100	$\mu_F = 0.763$	μ_{CR}
= 0.101 Diversity = 1.7268e-02	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.000001e+02	FES = 62100	$\mu_F = 0.773$	μ_{CR}
= 0.094 Diversity = 1.7129e-02	$k_t = [2, 3, 2, 2]$		
Generation 630: Best = 2.000001e+02	FES = 63100	$\mu_F = 0.761$	μ_{CR}
= 0.095 Diversity = 1.6809e-02	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.000001e+02	FES = 64100	$\mu_F = 0.757$	μ_{CR}
= 0.093 Diversity = 1.6374e-02	$k_t = [2, 2, 2, 4]$		
Generation 650: Best = 2.000001e+02	FES = 65100	$\mu_F = 0.761$	μ_{CR}
= 0.099 Diversity = 1.6930e-02	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.000001e+02	FES = 66100	$\mu_F = 0.773$	μ_{CR}
= 0.100 Diversity = 1.5840e-02	$k_t = [2, 2, 3, 3]$		
Generation 670: Best = 2.000001e+02	FES = 67100	$\mu_F = 0.756$	μ_{CR}
= 0.106 Diversity = 1.5947e-02	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.000001e+02	FES = 68100	$\mu_F = 0.747$	μ_{CR}
= 0.099 Diversity = 1.5889e-02	$k_t = [3, 4, 4, 4]$		
Generation 690: Best = 2.000001e+02	FES = 69100	$\mu_F = 0.778$	μ_{CR}
= 0.102 Diversity = 1.5348e-02	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.000001e+02	FES = 70100	$\mu_F = 0.778$	μ_{CR}

= 0.108	Diversity = 1.5198e-02	k_t = [3, 3, 3, 3]		
Generation 710:	Best = 2.000001e+02	FES = 71100	$\mu_F = 0.773$	μ_{CR}
= 0.095	Diversity = 1.5636e-02	k_t = [1, 1, 1, 1]		
Generation 720:	Best = 2.000001e+02	FES = 72100	$\mu_F = 0.771$	μ_{CR}
= 0.102	Diversity = 1.3059e-02	k_t = [3, 3, 4, 4]		
Generation 730:	Best = 2.000001e+02	FES = 73100	$\mu_F = 0.796$	μ_{CR}
= 0.101	Diversity = 1.2915e-02	k_t = [1, 1, 1, 1]		
Generation 740:	Best = 2.000001e+02	FES = 74100	$\mu_F = 0.799$	μ_{CR}
= 0.096	Diversity = 1.3988e-02	k_t = [3, 4, 3, 3]		
Generation 750:	Best = 2.000001e+02	FES = 75100	$\mu_F = 0.796$	μ_{CR}
= 0.099	Diversity = 1.4048e-02	k_t = [1, 1, 1, 1]		
Generation 760:	Best = 2.000001e+02	FES = 76100	$\mu_F = 0.777$	μ_{CR}
= 0.118	Diversity = 1.4509e-02	k_t = [4, 4, 2, 2]		
Generation 770:	Best = 2.000001e+02	FES = 77100	$\mu_F = 0.765$	μ_{CR}
= 0.121	Diversity = 1.4509e-02	k_t = [1, 1, 1, 1]		
Generation 780:	Best = 2.000001e+02	FES = 78100	$\mu_F = 0.773$	μ_{CR}
= 0.120	Diversity = 1.1740e-02	k_t = [2, 3, 4, 4]		
Generation 790:	Best = 2.000001e+02	FES = 79100	$\mu_F = 0.770$	μ_{CR}
= 0.127	Diversity = 1.2219e-02	k_t = [1, 1, 1, 1]		
Generation 800:	Best = 2.000001e+02	FES = 80100	$\mu_F = 0.768$	μ_{CR}
= 0.114	Diversity = 1.2296e-02	k_t = [3, 3, 3, 2]		
Generation 810:	Best = 2.000001e+02	FES = 81100	$\mu_F = 0.764$	μ_{CR}
= 0.116	Diversity = 1.1983e-02	k_t = [1, 1, 1, 1]		
Generation 820:	Best = 2.000001e+02	FES = 82100	$\mu_F = 0.772$	μ_{CR}
= 0.118	Diversity = 1.0598e-02	k_t = [2, 2, 4, 3]		
Generation 830:	Best = 2.000001e+02	FES = 83100	$\mu_F = 0.782$	μ_{CR}
= 0.092	Diversity = 9.8979e-03	k_t = [1, 1, 1, 1]		
Generation 840:	Best = 2.000001e+02	FES = 84100	$\mu_F = 0.787$	μ_{CR}
= 0.094	Diversity = 9.4674e-03	k_t = [4, 2, 4, 2]		
Generation 850:	Best = 2.000001e+02	FES = 85100	$\mu_F = 0.790$	μ_{CR}
= 0.100	Diversity = 1.0181e-02	k_t = [1, 1, 1, 1]		
Generation 860:	Best = 2.000001e+02	FES = 86100	$\mu_F = 0.784$	μ_{CR}
= 0.094	Diversity = 9.5190e-03	k_t = [3, 3, 4, 3]		
Generation 870:	Best = 2.000001e+02	FES = 87100	$\mu_F = 0.770$	μ_{CR}
= 0.102	Diversity = 9.8854e-03	k_t = [1, 1, 1, 1]		
Generation 880:	Best = 2.000001e+02	FES = 88100	$\mu_F = 0.772$	μ_{CR}
= 0.096	Diversity = 9.0430e-03	k_t = [2, 2, 2, 2]		
Generation 890:	Best = 2.000001e+02	FES = 89100	$\mu_F = 0.754$	μ_{CR}
= 0.087	Diversity = 9.1630e-03	k_t = [1, 1, 1, 1]		
Generation 900:	Best = 2.000001e+02	FES = 90100	$\mu_F = 0.767$	μ_{CR}
= 0.094	Diversity = 8.8956e-03	k_t = [2, 3, 3, 2]		
Generation 910:	Best = 2.000001e+02	FES = 91100	$\mu_F = 0.768$	μ_{CR}
= 0.098	Diversity = 9.2419e-03	k_t = [1, 1, 1, 1]		
Generation 920:	Best = 2.000001e+02	FES = 92100	$\mu_F = 0.785$	μ_{CR}
= 0.103	Diversity = 1.0715e-02	k_t = [3, 4, 2, 2]		
Generation 930:	Best = 2.000001e+02	FES = 93100	$\mu_F = 0.776$	μ_{CR}
= 0.106	Diversity = 1.0558e-02	k_t = [1, 1, 1, 1]		
Generation 940:	Best = 2.000001e+02	FES = 94100	$\mu_F = 0.776$	μ_{CR}
= 0.098	Diversity = 1.0680e-02	k_t = [2, 2, 4, 4]		

```

Generation 950: Best = 2.000001e+02  FES = 95100   $\mu_F$  = 0.797   $\mu_{CR}$ 
= 0.097  Diversity = 1.0625e-02   $k_t$  = [1, 1, 1, 1]
Generation 960: Best = 2.000001e+02  FES = 96100   $\mu_F$  = 0.800   $\mu_{CR}$ 
= 0.091  Diversity = 1.1464e-02   $k_t$  = [2, 3, 2, 4]
Generation 970: Best = 2.000001e+02  FES = 97100   $\mu_F$  = 0.795   $\mu_{CR}$ 
= 0.098  Diversity = 1.1022e-02   $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 2.000001e+02  FES = 98100   $\mu_F$  = 0.792   $\mu_{CR}$ 
= 0.118  Diversity = 1.1406e-02   $k_t$  = [4, 4, 2, 4]
Generation 990: Best = 2.000001e+02  FES = 99100   $\mu_F$  = 0.813   $\mu_{CR}$ 
= 0.116  Diversity = 1.2334e-02   $k_t$  = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 2.000001e+02

Total Function Evals = 100000

Total Successful Updates = 24046

```

=====
=====

```

Results:

Best Fitness : 2.000001e+02

Global Optimum : 2.000000e+02

Error (|best-optimum|) : 8.558419e-05

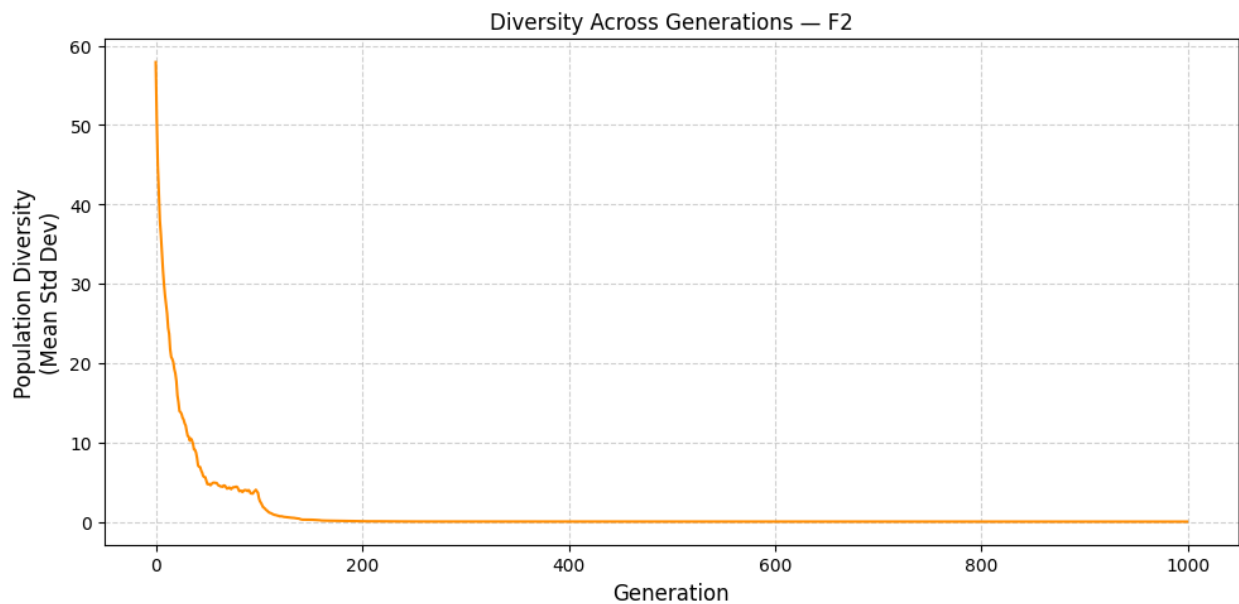
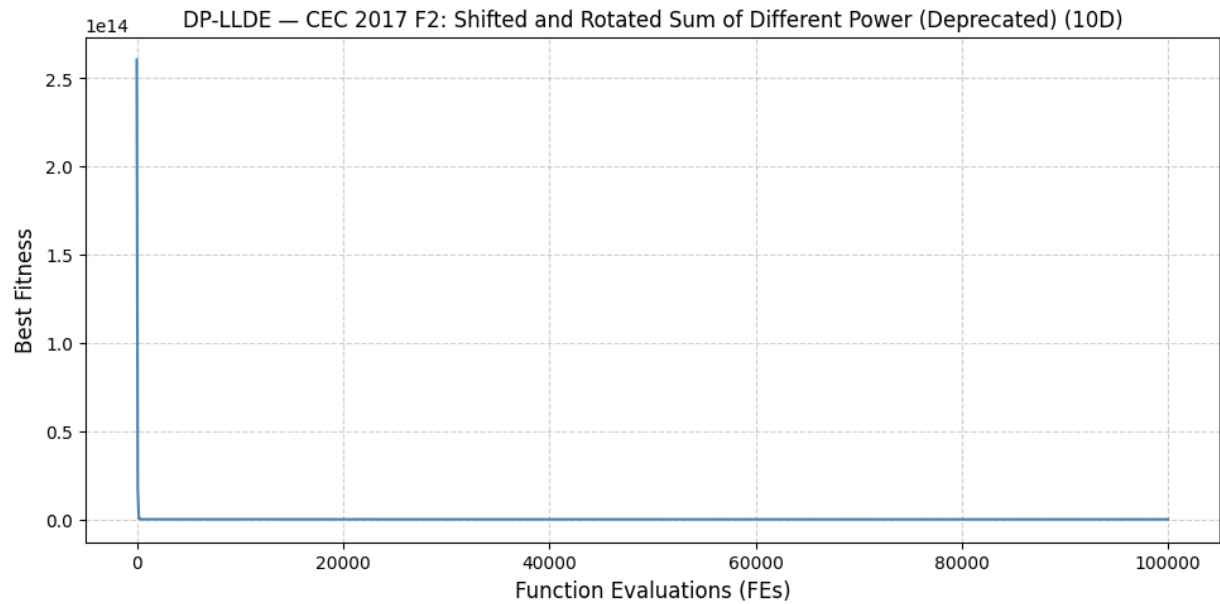
Best Solution (5 dim) : [-29.33202482 -16.80030815 -75.48366001 -
0.66389823 4.26160218]

D^0 (init. diversity): 5.7928e+01

```

=====
=====

```



Graph saved as convergence_F2.png

```
# =====
# F3 – Shifted and Rotated Zakharov
# =====
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and
Rotated Zakharov')
```

```
=====
=====
DP-LLDE on CEC 2017 F3 – Single Run
=====
```

=====

Function : Shifted and Rotated Zakharov (10D)

Bounds : [-100, 100]

Global optimum : 300.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.625937e+04

Generation 10: Best = 1.549451e+04 FES = 1100 $\mu_F = 0.572$ μ_{CR}

= 0.357 Diversity = 4.1542e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 6.173756e+03 FES = 2100 $\mu_F = 0.612$ μ_{CR}

= 0.360 Diversity = 2.4400e+01 $k_t = [3, 4, 2, 3]$

Generation 30: Best = 2.065660e+03 FES = 3100 $\mu_F = 0.643$ μ_{CR}

= 0.377 Diversity = 1.3908e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 9.772182e+02 FES = 4100 $\mu_F = 0.683$ μ_{CR}

= 0.394 Diversity = 8.0290e+00 $k_t = [3, 3, 2, 4]$

Generation 50: Best = 5.582284e+02 FES = 5100 $\mu_F = 0.706$ μ_{CR}

= 0.408 Diversity = 5.0305e+00 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.605158e+02 FES = 6100 $\mu_F = 0.706$ μ_{CR}

= 0.411 Diversity = 2.9960e+00 $k_t = [2, 2, 4, 4]$

Generation 70: Best = 3.416351e+02 FES = 7100 $\mu_F = 0.724$ μ_{CR}

= 0.430 Diversity = 1.9320e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 3.064396e+02 FES = 8100 $\mu_F = 0.729$ μ_{CR}

= 0.437 Diversity = 1.2782e+00 $k_t = [3, 3, 2, 4]$

Generation 90: Best = 3.040772e+02 FES = 9100 $\mu_F = 0.696$ μ_{CR}

= 0.441 Diversity = 9.0750e-01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 3.022924e+02 FES = 10100 $\mu_F = 0.674$ μ_{CR}

= 0.441 Diversity = 6.0314e-01 $k_t = [3, 3, 3, 4]$

Generation 110: Best = 3.006276e+02 FES = 11100 $\mu_F = 0.660$ μ_{CR}

= 0.456 Diversity = 3.9601e-01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 3.002822e+02 FES = 12100 $\mu_F = 0.646$ μ_{CR}

= 0.460 Diversity = 2.6560e-01 $k_t = [2, 2, 2, 2]$

Generation 130: Best = 3.000834e+02 FES = 13100 $\mu_F = 0.627$ μ_{CR}

= 0.456 Diversity = 1.7504e-01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 3.000326e+02 FES = 14100 $\mu_F = 0.615$ μ_{CR}

= 0.460 Diversity = 9.2391e-02 $k_t = [3, 2, 2, 4]$

Generation 150: Best = 3.000168e+02 FES = 15100 $\mu_F = 0.622$ μ_{CR}

= 0.467 Diversity = 5.5630e-02 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 3.000057e+02 FES = 16100 $\mu_F = 0.601$ μ_{CR}

= 0.472 Diversity = 3.1778e-02 $k_t = [2, 3, 2, 3]$

Generation 170: Best = 3.000034e+02 FES = 17100 $\mu_F = 0.597$ μ_{CR}

= 0.472 Diversity = 1.9879e-02 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 3.000026e+02 FES = 18100 $\mu_F = 0.584$ μ_{CR}

= 0.469 Diversity = 1.0663e-02 $k_t = [4, 4, 2, 2]$

Generation 190: Best = 3.000014e+02 FES = 19100 $\mu_F = 0.603$ μ_{CR}

= 0.474 Diversity = 6.7714e-03 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.000012e+02 FES = 20100 $\mu_F = 0.609$ μ_{CR}

= 0.477 Diversity = 4.6946e-03 $k_t = [3, 2, 3, 4]$

Generation 210: Best = 3.000010e+02 FES = 21100 $\mu_F = 0.617$ μ_{CR}

= 0.482 Diversity = 3.5743e-03 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 3.000009e+02 FES = 22100 $\mu_F = 0.629$ μ_{CR}

= 0.484	Diversity = 2.4131e-03	k_t = [4, 3, 2, 4]		
Generation 230:	Best = 3.000008e+02	FES = 23100	μ_F = 0.628	μ_{CR}
= 0.481	Diversity = 1.5750e-03	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 3.000008e+02	FES = 24100	μ_F = 0.622	μ_{CR}
= 0.483	Diversity = 1.1063e-03	k_t = [2, 3, 4, 2]		
Generation 250:	Best = 3.000008e+02	FES = 25100	μ_F = 0.621	μ_{CR}
= 0.497	Diversity = 7.6607e-04	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 3.000008e+02	FES = 26100	μ_F = 0.639	μ_{CR}
= 0.489	Diversity = 5.2120e-04	k_t = [4, 4, 2, 3]		
Generation 270:	Best = 3.000007e+02	FES = 27100	μ_F = 0.647	μ_{CR}
= 0.491	Diversity = 4.2960e-04	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 3.000007e+02	FES = 28100	μ_F = 0.661	μ_{CR}
= 0.488	Diversity = 4.1256e-04	k_t = [2, 2, 2, 2]		
Generation 290:	Best = 3.000007e+02	FES = 29100	μ_F = 0.693	μ_{CR}
= 0.494	Diversity = 4.4512e-04	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 3.000007e+02	FES = 30100	μ_F = 0.704	μ_{CR}
= 0.502	Diversity = 4.6641e-04	k_t = [4, 2, 3, 2]		
Generation 310:	Best = 3.000007e+02	FES = 31100	μ_F = 0.706	μ_{CR}
= 0.499	Diversity = 3.9745e-04	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 3.000007e+02	FES = 32100	μ_F = 0.721	μ_{CR}
= 0.509	Diversity = 3.0936e-04	k_t = [3, 3, 4, 4]		
Generation 330:	Best = 3.000007e+02	FES = 33100	μ_F = 0.718	μ_{CR}
= 0.513	Diversity = 2.6617e-04	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 3.000007e+02	FES = 34100	μ_F = 0.718	μ_{CR}
= 0.522	Diversity = 2.2297e-04	k_t = [4, 2, 3, 2]		
Generation 350:	Best = 3.000007e+02	FES = 35100	μ_F = 0.723	μ_{CR}
= 0.537	Diversity = 2.3412e-04	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 3.000007e+02	FES = 36100	μ_F = 0.731	μ_{CR}
= 0.562	Diversity = 2.0408e-04	k_t = [3, 2, 2, 2]		
Generation 370:	Best = 3.000007e+02	FES = 37100	μ_F = 0.716	μ_{CR}
= 0.570	Diversity = 1.9350e-04	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 3.000007e+02	FES = 38100	μ_F = 0.705	μ_{CR}
= 0.580	Diversity = 1.4405e-04	k_t = [4, 3, 2, 2]		
Generation 390:	Best = 3.000007e+02	FES = 39100	μ_F = 0.720	μ_{CR}
= 0.590	Diversity = 1.4613e-04	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 3.000007e+02	FES = 40100	μ_F = 0.716	μ_{CR}
= 0.593	Diversity = 1.2742e-04	k_t = [3, 4, 3, 4]		
Generation 410:	Best = 3.000007e+02	FES = 41100	μ_F = 0.729	μ_{CR}
= 0.585	Diversity = 1.1516e-04	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 3.000007e+02	FES = 42100	μ_F = 0.736	μ_{CR}
= 0.583	Diversity = 1.1306e-04	k_t = [3, 3, 3, 4]		
Generation 430:	Best = 3.000007e+02	FES = 43100	μ_F = 0.726	μ_{CR}
= 0.592	Diversity = 1.3590e-04	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 3.000007e+02	FES = 44100	μ_F = 0.720	μ_{CR}
= 0.584	Diversity = 1.3250e-04	k_t = [2, 4, 2, 4]		
Generation 450:	Best = 3.000007e+02	FES = 45100	μ_F = 0.719	μ_{CR}
= 0.590	Diversity = 1.2365e-04	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 3.000007e+02	FES = 46100	μ_F = 0.725	μ_{CR}
= 0.594	Diversity = 9.4853e-05	k_t = [2, 2, 2, 4]		

Generation 470: Best = 3.000007e+02	FES = 47100	$\mu_F = 0.711$	μ_{CR}
= 0.598 Diversity = 1.0578e-04 $k_t = [1, 1, 1, 1]$			
Generation 480: Best = 3.000007e+02	FES = 48100	$\mu_F = 0.708$	μ_{CR}
= 0.607 Diversity = 8.2781e-05 $k_t = [4, 3, 3, 2]$			
Generation 490: Best = 3.000007e+02	FES = 49100	$\mu_F = 0.712$	μ_{CR}
= 0.613 Diversity = 5.7902e-05 $k_t = [1, 1, 1, 1]$			
Generation 500: Best = 3.000007e+02	FES = 50100	$\mu_F = 0.710$	μ_{CR}
= 0.608 Diversity = 4.0657e-05 $k_t = [3, 2, 2, 3]$			
Generation 510: Best = 3.000007e+02	FES = 51100	$\mu_F = 0.718$	μ_{CR}
= 0.614 Diversity = 3.9794e-05 $k_t = [1, 1, 1, 1]$			
Generation 520: Best = 3.000007e+02	FES = 52100	$\mu_F = 0.706$	μ_{CR}
= 0.615 Diversity = 3.0052e-05 $k_t = [4, 2, 2, 3]$			
Generation 530: Best = 3.000007e+02	FES = 53100	$\mu_F = 0.721$	μ_{CR}
= 0.616 Diversity = 2.4347e-05 $k_t = [1, 1, 1, 1]$			
Generation 540: Best = 3.000007e+02	FES = 54100	$\mu_F = 0.722$	μ_{CR}
= 0.614 Diversity = 2.5205e-05 $k_t = [4, 2, 2, 3]$			
Generation 550: Best = 3.000007e+02	FES = 55100	$\mu_F = 0.729$	μ_{CR}
= 0.615 Diversity = 3.1112e-05 $k_t = [1, 1, 1, 1]$			
Generation 560: Best = 3.000007e+02	FES = 56100	$\mu_F = 0.713$	μ_{CR}
= 0.612 Diversity = 2.5927e-05 $k_t = [3, 4, 4, 3]$			
Generation 570: Best = 3.000007e+02	FES = 57100	$\mu_F = 0.728$	μ_{CR}
= 0.614 Diversity = 2.6067e-05 $k_t = [1, 1, 1, 1]$			
Generation 580: Best = 3.000007e+02	FES = 58100	$\mu_F = 0.715$	μ_{CR}
= 0.624 Diversity = 2.1894e-05 $k_t = [4, 2, 2, 3]$			
Generation 590: Best = 3.000007e+02	FES = 59100	$\mu_F = 0.712$	μ_{CR}
= 0.641 Diversity = 1.8189e-05 $k_t = [1, 1, 1, 1]$			
Generation 600: Best = 3.000007e+02	FES = 60100	$\mu_F = 0.713$	μ_{CR}
= 0.643 Diversity = 1.6792e-05 $k_t = [3, 3, 3, 4]$			
Generation 610: Best = 3.000007e+02	FES = 61100	$\mu_F = 0.718$	μ_{CR}
= 0.661 Diversity = 1.6644e-05 $k_t = [1, 1, 1, 1]$			
Generation 620: Best = 3.000007e+02	FES = 62100	$\mu_F = 0.715$	μ_{CR}
= 0.670 Diversity = 1.1858e-05 $k_t = [4, 4, 3, 3]$			
Generation 630: Best = 3.000007e+02	FES = 63100	$\mu_F = 0.703$	μ_{CR}
= 0.669 Diversity = 1.0495e-05 $k_t = [1, 1, 1, 1]$			
Generation 640: Best = 3.000007e+02	FES = 64100	$\mu_F = 0.703$	μ_{CR}
= 0.665 Diversity = 8.5315e-06 $k_t = [3, 2, 2, 4]$			
Generation 650: Best = 3.000007e+02	FES = 65100	$\mu_F = 0.699$	μ_{CR}
= 0.660 Diversity = 8.3654e-06 $k_t = [1, 1, 1, 1]$			
Generation 660: Best = 3.000007e+02	FES = 66100	$\mu_F = 0.696$	μ_{CR}
= 0.658 Diversity = 6.7680e-06 $k_t = [2, 2, 3, 4]$			
Generation 670: Best = 3.000007e+02	FES = 67100	$\mu_F = 0.692$	μ_{CR}
= 0.664 Diversity = 6.8306e-06 $k_t = [1, 1, 1, 1]$			
Generation 680: Best = 3.000007e+02	FES = 68100	$\mu_F = 0.675$	μ_{CR}
= 0.656 Diversity = 5.0356e-06 $k_t = [4, 2, 2, 4]$			
Generation 690: Best = 3.000007e+02	FES = 69100	$\mu_F = 0.682$	μ_{CR}
= 0.674 Diversity = 4.0628e-06 $k_t = [1, 1, 1, 1]$			
Generation 700: Best = 3.000007e+02	FES = 70100	$\mu_F = 0.701$	μ_{CR}
= 0.693 Diversity = 2.9422e-06 $k_t = [4, 2, 2, 3]$			
Generation 710: Best = 3.000007e+02	FES = 71100	$\mu_F = 0.719$	μ_{CR}

= 0.696	Diversity = 2.7604e-06	k_t = [1, 1, 1, 1]	μ_F = 0.738	μ_{CR}
Generation 720:	Best = 3.000007e+02	FES = 72100		
= 0.706	Diversity = 2.4485e-06	k_t = [3, 4, 4, 4]	μ_F = 0.736	μ_{CR}
Generation 730:	Best = 3.000007e+02	FES = 73100		
= 0.713	Diversity = 2.2600e-06	k_t = [1, 1, 1, 1]	μ_F = 0.716	μ_{CR}
Generation 740:	Best = 3.000007e+02	FES = 74100		
= 0.723	Diversity = 1.7295e-06	k_t = [4, 2, 3, 4]	μ_F = 0.718	μ_{CR}
Generation 750:	Best = 3.000007e+02	FES = 75100		
= 0.728	Diversity = 1.2724e-06	k_t = [1, 1, 1, 1]	μ_F = 0.727	μ_{CR}
Generation 760:	Best = 3.000007e+02	FES = 76100		
= 0.733	Diversity = 9.9107e-07	k_t = [2, 3, 3, 4]	μ_F = 0.722	μ_{CR}
Generation 770:	Best = 3.000007e+02	FES = 77100		
= 0.739	Diversity = 8.7155e-07	k_t = [1, 1, 1, 1]	μ_F = 0.737	μ_{CR}
Generation 780:	Best = 3.000007e+02	FES = 78100		
= 0.745	Diversity = 7.1735e-07	k_t = [3, 2, 3, 3]	μ_F = 0.738	μ_{CR}
Generation 790:	Best = 3.000007e+02	FES = 79100		
= 0.753	Diversity = 9.7699e-07	k_t = [1, 1, 1, 1]	μ_F = 0.717	μ_{CR}
Generation 800:	Best = 3.000007e+02	FES = 80100		
= 0.772	Diversity = 8.7389e-07	k_t = [2, 3, 2, 3]	μ_F = 0.719	μ_{CR}
Generation 810:	Best = 3.000007e+02	FES = 81100		
= 0.780	Diversity = 7.3184e-07	k_t = [1, 1, 1, 1]	μ_F = 0.719	μ_{CR}
Generation 820:	Best = 3.000007e+02	FES = 82100		
= 0.790	Diversity = 5.7847e-07	k_t = [2, 3, 2, 3]	μ_F = 0.715	μ_{CR}
Generation 830:	Best = 3.000007e+02	FES = 83100		
= 0.796	Diversity = 4.8685e-07	k_t = [1, 1, 1, 1]	μ_F = 0.719	μ_{CR}
Generation 840:	Best = 3.000007e+02	FES = 84100		
= 0.807	Diversity = 4.4214e-07	k_t = [2, 2, 2, 3]	μ_F = 0.728	μ_{CR}
Generation 850:	Best = 3.000007e+02	FES = 85100		
= 0.810	Diversity = 4.7618e-07	k_t = [1, 1, 1, 1]	μ_F = 0.718	μ_{CR}
Generation 860:	Best = 3.000007e+02	FES = 86100		
= 0.808	Diversity = 3.6259e-07	k_t = [3, 3, 4, 4]	μ_F = 0.744	μ_{CR}
Generation 870:	Best = 3.000007e+02	FES = 87100		
= 0.811	Diversity = 4.9944e-07	k_t = [1, 1, 1, 1]	μ_F = 0.727	μ_{CR}
Generation 880:	Best = 3.000007e+02	FES = 88100		
= 0.820	Diversity = 3.5610e-07	k_t = [2, 2, 3, 2]	μ_F = 0.711	μ_{CR}
Generation 890:	Best = 3.000007e+02	FES = 89100		
= 0.827	Diversity = 3.2980e-07	k_t = [1, 1, 1, 1]	μ_F = 0.697	μ_{CR}
Generation 900:	Best = 3.000007e+02	FES = 90100		
= 0.837	Diversity = 2.0607e-07	k_t = [2, 3, 4, 4]	μ_F = 0.708	μ_{CR}
Generation 910:	Best = 3.000007e+02	FES = 91100		
= 0.831	Diversity = 2.3584e-07	k_t = [1, 1, 1, 1]	μ_F = 0.713	μ_{CR}
Generation 920:	Best = 3.000007e+02	FES = 92100		
= 0.833	Diversity = 2.1929e-07	k_t = [3, 4, 3, 2]	μ_F = 0.696	μ_{CR}
Generation 930:	Best = 3.000007e+02	FES = 93100		
= 0.838	Diversity = 1.7956e-07	k_t = [1, 1, 1, 1]	μ_F = 0.708	μ_{CR}
Generation 940:	Best = 3.000007e+02	FES = 94100		
= 0.844	Diversity = 1.4709e-07	k_t = [2, 4, 4, 3]	μ_F = 0.719	μ_{CR}
Generation 950:	Best = 3.000007e+02	FES = 95100		
= 0.857	Diversity = 1.9200e-07	k_t = [1, 1, 1, 1]		


```

Generation 960: Best = 3.000007e+02  FES = 96100   $\mu_F$  = 0.715   $\mu_{CR}$ 
= 0.865  Diversity = 1.5561e-07  k_t = [3, 3, 4, 2]
Generation 970: Best = 3.000007e+02  FES = 97100   $\mu_F$  = 0.721   $\mu_{CR}$ 
= 0.867  Diversity = 1.6562e-07  k_t = [1, 1, 1, 1]
Generation 980: Best = 3.000007e+02  FES = 98100   $\mu_F$  = 0.717   $\mu_{CR}$ 
= 0.873  Diversity = 1.3810e-07  k_t = [4, 3, 2, 4]
Generation 990: Best = 3.000007e+02  FES = 99100   $\mu_F$  = 0.714   $\mu_{CR}$ 
= 0.881  Diversity = 1.1373e-07  k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 3.000007e+02

Total Function Evals = 100000

Total Successful Updates = 34584

```

=====
=====

```

Results:

Best Fitness : 3.000007e+02

Global Optimum : 3.000000e+02

Error (|best-optimum|) : 6.879505e-04

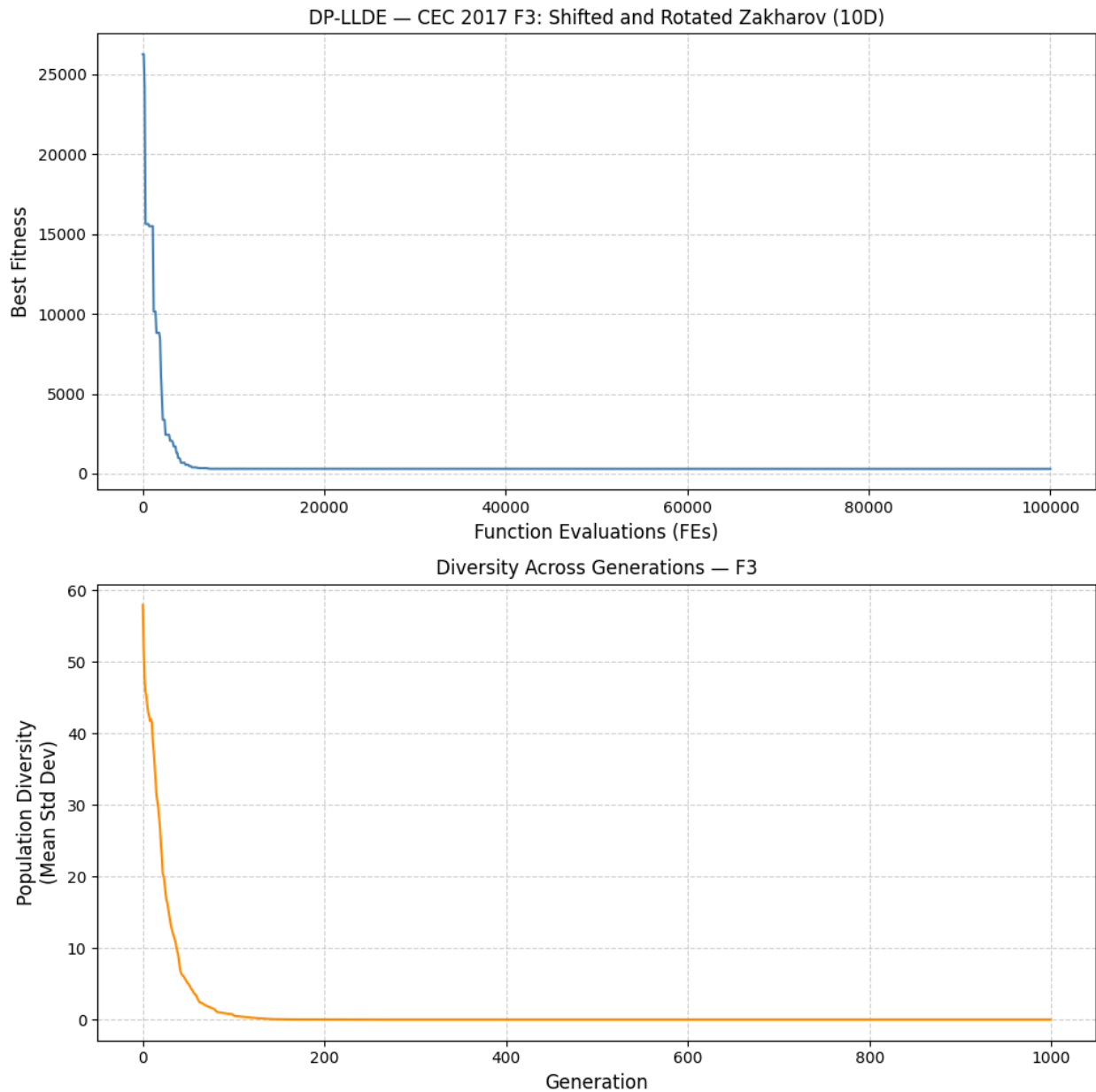
Best Solution (5 dim) : [-55.94071194 4.54717019 35.29913901
8.27893663 -47.43635776]

D^(0) (init. diversity): 5.7928e+01

```

=====
=====

```



Graph saved as convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
# =====  
# F4 – Shifted and Rotated Rosenbrock  
# =====  
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and  
Rotated Rosenbrock's")
```

DP-LLDE on CEC 2017 F4 – Single Run

```

Function      : Shifted and Rotated Rosenbrock's (10D)
Bounds       : [-100, 100]
Global optimum : 400.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 3.004346e+03
Generation 10: Best = 5.207037e+02 FES = 1100  $\mu_F$  = 0.570  $\mu_{CR}$ 
= 0.351 Diversity = 2.1609e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 4.241273e+02 FES = 2100  $\mu_F$  = 0.611  $\mu_{CR}$ 
= 0.358 Diversity = 1.0583e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 4.053987e+02 FES = 3100  $\mu_F$  = 0.641  $\mu_{CR}$ 
= 0.368 Diversity = 5.5865e+00  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 4.037930e+02 FES = 4100  $\mu_F$  = 0.661  $\mu_{CR}$ 
= 0.378 Diversity = 2.3553e+00  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 4.030347e+02 FES = 5100  $\mu_F$  = 0.657  $\mu_{CR}$ 
= 0.382 Diversity = 1.9636e+00  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 4.024577e+02 FES = 6100  $\mu_F$  = 0.668  $\mu_{CR}$ 
= 0.395 Diversity = 1.2296e+00  $k_t$  = [2, 2, 4, 4]
Generation 70: Best = 4.022593e+02 FES = 7100  $\mu_F$  = 0.676  $\mu_{CR}$ 
= 0.400 Diversity = 8.4703e-01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 4.021321e+02 FES = 8100  $\mu_F$  = 0.678  $\mu_{CR}$ 
= 0.401 Diversity = 5.5280e-01  $k_t$  = [3, 3, 2, 4]
Generation 90: Best = 4.020067e+02 FES = 9100  $\mu_F$  = 0.691  $\mu_{CR}$ 
= 0.395 Diversity = 4.8721e-01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 4.019698e+02 FES = 10100  $\mu_F$  = 0.676  $\mu_{CR}$ 
= 0.406 Diversity = 3.8346e-01  $k_t$  = [3, 3, 3, 4]
Generation 110: Best = 4.019095e+02 FES = 11100  $\mu_F$  = 0.690  $\mu_{CR}$ 
= 0.417 Diversity = 3.4749e-01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 4.018679e+02 FES = 12100  $\mu_F$  = 0.703  $\mu_{CR}$ 
= 0.424 Diversity = 3.2288e-01  $k_t$  = [2, 2, 2, 2]
Generation 130: Best = 4.018364e+02 FES = 13100  $\mu_F$  = 0.706  $\mu_{CR}$ 
= 0.418 Diversity = 2.8634e-01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 4.018175e+02 FES = 14100  $\mu_F$  = 0.691  $\mu_{CR}$ 
= 0.419 Diversity = 2.4281e-01  $k_t$  = [3, 2, 2, 4]
Generation 150: Best = 4.018044e+02 FES = 15100  $\mu_F$  = 0.700  $\mu_{CR}$ 
= 0.423 Diversity = 2.0867e-01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 4.017952e+02 FES = 16100  $\mu_F$  = 0.686  $\mu_{CR}$ 
= 0.432 Diversity = 2.0454e-01  $k_t$  = [2, 2, 2, 3]
Generation 170: Best = 4.017854e+02 FES = 17100  $\mu_F$  = 0.678  $\mu_{CR}$ 
= 0.430 Diversity = 1.9314e-01  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 4.017660e+02 FES = 18100  $\mu_F$  = 0.690  $\mu_{CR}$ 
= 0.418 Diversity = 1.7714e-01  $k_t$  = [2, 2, 4, 3]
Generation 190: Best = 4.017547e+02 FES = 19100  $\mu_F$  = 0.689  $\mu_{CR}$ 
= 0.410 Diversity = 1.8973e-01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 4.017422e+02 FES = 20100  $\mu_F$  = 0.667  $\mu_{CR}$ 
= 0.400 Diversity = 1.4254e-01  $k_t$  = [3, 2, 3, 4]

```

Generation 210: Best = 4.017240e+02	FES = 21100	$\mu_F = 0.668$	μ_{CR}
= 0.408 Diversity = 1.3547e-01 $k_t = [1, 1, 1, 1]$			
Generation 220: Best = 4.017103e+02	FES = 22100	$\mu_F = 0.690$	μ_{CR}
= 0.402 Diversity = 1.3893e-01 $k_t = [4, 3, 2, 3]$			
Generation 230: Best = 4.017037e+02	FES = 23100	$\mu_F = 0.704$	μ_{CR}
= 0.407 Diversity = 1.5542e-01 $k_t = [1, 1, 1, 1]$			
Generation 240: Best = 4.016950e+02	FES = 24100	$\mu_F = 0.702$	μ_{CR}
= 0.396 Diversity = 1.4084e-01 $k_t = [3, 4, 3, 4]$			
Generation 250: Best = 4.016913e+02	FES = 25100	$\mu_F = 0.717$	μ_{CR}
= 0.399 Diversity = 1.1703e-01 $k_t = [1, 1, 1, 1]$			
Generation 260: Best = 4.016819e+02	FES = 26100	$\mu_F = 0.722$	μ_{CR}
= 0.389 Diversity = 1.1658e-01 $k_t = [3, 4, 2, 4]$			
Generation 270: Best = 4.016787e+02	FES = 27100	$\mu_F = 0.720$	μ_{CR}
= 0.394 Diversity = 1.0113e-01 $k_t = [1, 1, 1, 1]$			
Generation 280: Best = 4.016783e+02	FES = 28100	$\mu_F = 0.704$	μ_{CR}
= 0.397 Diversity = 9.8598e-02 $k_t = [2, 2, 2, 2]$			
Generation 290: Best = 4.016753e+02	FES = 29100	$\mu_F = 0.711$	μ_{CR}
= 0.384 Diversity = 9.2145e-02 $k_t = [1, 1, 1, 1]$			
Generation 300: Best = 4.016678e+02	FES = 30100	$\mu_F = 0.707$	μ_{CR}
= 0.410 Diversity = 1.0501e-01 $k_t = [4, 2, 3, 2]$			
Generation 310: Best = 4.016629e+02	FES = 31100	$\mu_F = 0.714$	μ_{CR}
= 0.421 Diversity = 1.1718e-01 $k_t = [1, 1, 1, 1]$			
Generation 320: Best = 4.016616e+02	FES = 32100	$\mu_F = 0.712$	μ_{CR}
= 0.424 Diversity = 9.1169e-02 $k_t = [3, 3, 4, 4]$			
Generation 330: Best = 4.016596e+02	FES = 33100	$\mu_F = 0.719$	μ_{CR}
= 0.418 Diversity = 8.5266e-02 $k_t = [1, 1, 1, 1]$			
Generation 340: Best = 4.016584e+02	FES = 34100	$\mu_F = 0.696$	μ_{CR}
= 0.433 Diversity = 5.8196e-02 $k_t = [4, 2, 3, 2]$			
Generation 350: Best = 4.016565e+02	FES = 35100	$\mu_F = 0.685$	μ_{CR}
= 0.423 Diversity = 6.0260e-02 $k_t = [1, 1, 1, 1]$			
Generation 360: Best = 4.016550e+02	FES = 36100	$\mu_F = 0.687$	μ_{CR}
= 0.414 Diversity = 6.5626e-02 $k_t = [4, 2, 2, 2]$			
Generation 370: Best = 4.016538e+02	FES = 37100	$\mu_F = 0.698$	μ_{CR}
= 0.397 Diversity = 6.1973e-02 $k_t = [1, 1, 1, 1]$			
Generation 380: Best = 4.016521e+02	FES = 38100	$\mu_F = 0.717$	μ_{CR}
= 0.376 Diversity = 5.5510e-02 $k_t = [4, 4, 3, 3]$			
Generation 390: Best = 4.016513e+02	FES = 39100	$\mu_F = 0.716$	μ_{CR}
= 0.371 Diversity = 5.2031e-02 $k_t = [1, 1, 1, 1]$			
Generation 400: Best = 4.016504e+02	FES = 40100	$\mu_F = 0.728$	μ_{CR}
= 0.373 Diversity = 4.8337e-02 $k_t = [3, 3, 2, 3]$			
Generation 410: Best = 4.016492e+02	FES = 41100	$\mu_F = 0.725$	μ_{CR}
= 0.369 Diversity = 4.6002e-02 $k_t = [1, 1, 1, 1]$			
Generation 420: Best = 4.016484e+02	FES = 42100	$\mu_F = 0.720$	μ_{CR}
= 0.359 Diversity = 4.2967e-02 $k_t = [2, 2, 2, 2]$			
Generation 430: Best = 4.016479e+02	FES = 43100	$\mu_F = 0.688$	μ_{CR}
= 0.336 Diversity = 4.9701e-02 $k_t = [1, 1, 1, 1]$			
Generation 440: Best = 4.016476e+02	FES = 44100	$\mu_F = 0.691$	μ_{CR}
= 0.338 Diversity = 4.3447e-02 $k_t = [3, 4, 3, 3]$			
Generation 450: Best = 4.016467e+02	FES = 45100	$\mu_F = 0.707$	μ_{CR}
= 0.333 Diversity = 4.9665e-02 $k_t = [1, 1, 1, 1]$			

Generation 460: Best = 4.016453e+02	FES = 46100	$\mu_F = 0.712$	μ_{CR}
= 0.344 Diversity = 4.7617e-02	$k_t = [2, 4, 4, 4]$		
Generation 470: Best = 4.016448e+02	FES = 47100	$\mu_F = 0.705$	μ_{CR}
= 0.348 Diversity = 4.3648e-02	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 4.016436e+02	FES = 48100	$\mu_F = 0.703$	μ_{CR}
= 0.349 Diversity = 4.2321e-02	$k_t = [2, 3, 2, 4]$		
Generation 490: Best = 4.016432e+02	FES = 49100	$\mu_F = 0.720$	μ_{CR}
= 0.339 Diversity = 4.1099e-02	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 4.016423e+02	FES = 50100	$\mu_F = 0.705$	μ_{CR}
= 0.345 Diversity = 3.8149e-02	$k_t = [4, 3, 4, 2]$		
Generation 510: Best = 4.016415e+02	FES = 51100	$\mu_F = 0.725$	μ_{CR}
= 0.353 Diversity = 4.3304e-02	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 4.016412e+02	FES = 52100	$\mu_F = 0.713$	μ_{CR}
= 0.355 Diversity = 4.1520e-02	$k_t = [2, 2, 4, 4]$		
Generation 530: Best = 4.016407e+02	FES = 53100	$\mu_F = 0.750$	μ_{CR}
= 0.341 Diversity = 3.7932e-02	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 4.016395e+02	FES = 54100	$\mu_F = 0.745$	μ_{CR}
= 0.325 Diversity = 3.6849e-02	$k_t = [3, 2, 2, 3]$		
Generation 550: Best = 4.016384e+02	FES = 55100	$\mu_F = 0.742$	μ_{CR}
= 0.329 Diversity = 5.5778e-02	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 4.016343e+02	FES = 56100	$\mu_F = 0.743$	μ_{CR}
= 0.324 Diversity = 6.5449e-02	$k_t = [3, 4, 4, 3]$		
Generation 570: Best = 4.016293e+02	FES = 57100	$\mu_F = 0.757$	μ_{CR}
= 0.319 Diversity = 8.8319e-02	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 4.016235e+02	FES = 58100	$\mu_F = 0.750$	μ_{CR}
= 0.311 Diversity = 1.0371e-01	$k_t = [2, 2, 3, 3]$		
Generation 590: Best = 4.016214e+02	FES = 59100	$\mu_F = 0.752$	μ_{CR}
= 0.302 Diversity = 1.1843e-01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 4.016144e+02	FES = 60100	$\mu_F = 0.745$	μ_{CR}
= 0.310 Diversity = 9.6961e-02	$k_t = [4, 2, 2, 3]$		
Generation 610: Best = 4.016093e+02	FES = 61100	$\mu_F = 0.750$	μ_{CR}
= 0.318 Diversity = 1.1734e-01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 4.016063e+02	FES = 62100	$\mu_F = 0.736$	μ_{CR}
= 0.346 Diversity = 1.1440e-01	$k_t = [3, 2, 3, 3]$		
Generation 630: Best = 4.016010e+02	FES = 63100	$\mu_F = 0.728$	μ_{CR}
= 0.351 Diversity = 9.8832e-02	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 4.016004e+02	FES = 64100	$\mu_F = 0.716$	μ_{CR}
= 0.351 Diversity = 1.0402e-01	$k_t = [4, 3, 2, 2]$		
Generation 650: Best = 4.015983e+02	FES = 65100	$\mu_F = 0.710$	μ_{CR}
= 0.341 Diversity = 8.3777e-02	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 4.015969e+02	FES = 66100	$\mu_F = 0.713$	μ_{CR}
= 0.342 Diversity = 6.7874e-02	$k_t = [3, 4, 3, 4]$		
Generation 670: Best = 4.015946e+02	FES = 67100	$\mu_F = 0.713$	μ_{CR}
= 0.355 Diversity = 6.2819e-02	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 4.015932e+02	FES = 68100	$\mu_F = 0.706$	μ_{CR}
= 0.359 Diversity = 6.1314e-02	$k_t = [2, 2, 4, 4]$		
Generation 690: Best = 4.015927e+02	FES = 69100	$\mu_F = 0.705$	μ_{CR}
= 0.361 Diversity = 5.6973e-02	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 4.015883e+02	FES = 70100	$\mu_F = 0.715$	μ_{CR}
= 0.345 Diversity = 5.6793e-02	$k_t = [2, 2, 4, 4]$		

Generation 710: Best = 4.015841e+02	FES = 71100	$\mu_F = 0.735$	μ_{CR}
= 0.336 Diversity = 7.4262e-02 $k_t = [1, 1, 1, 1]$			
Generation 720: Best = 4.015802e+02	FES = 72100	$\mu_F = 0.732$	μ_{CR}
= 0.336 Diversity = 8.1407e-02 $k_t = [3, 4, 4, 4]$			
Generation 730: Best = 4.015796e+02	FES = 73100	$\mu_F = 0.740$	μ_{CR}
= 0.331 Diversity = 8.7959e-02 $k_t = [1, 1, 1, 1]$			
Generation 740: Best = 4.015784e+02	FES = 74100	$\mu_F = 0.724$	μ_{CR}
= 0.300 Diversity = 7.4002e-02 $k_t = [2, 3, 3, 3]$			
Generation 750: Best = 4.015770e+02	FES = 75100	$\mu_F = 0.724$	μ_{CR}
= 0.286 Diversity = 7.3402e-02 $k_t = [1, 1, 1, 1]$			
Generation 760: Best = 4.015747e+02	FES = 76100	$\mu_F = 0.721$	μ_{CR}
= 0.298 Diversity = 8.0661e-02 $k_t = [4, 4, 4, 2]$			
Generation 770: Best = 4.015735e+02	FES = 77100	$\mu_F = 0.711$	μ_{CR}
= 0.282 Diversity = 7.2104e-02 $k_t = [1, 1, 1, 1]$			
Generation 780: Best = 4.015698e+02	FES = 78100	$\mu_F = 0.710$	μ_{CR}
= 0.269 Diversity = 8.5462e-02 $k_t = [3, 3, 4, 3]$			
Generation 790: Best = 4.015669e+02	FES = 79100	$\mu_F = 0.727$	μ_{CR}
= 0.274 Diversity = 8.6535e-02 $k_t = [1, 1, 1, 1]$			
Generation 800: Best = 4.015647e+02	FES = 80100	$\mu_F = 0.718$	μ_{CR}
= 0.264 Diversity = 8.8754e-02 $k_t = [4, 2, 2, 4]$			
Generation 810: Best = 4.015584e+02	FES = 81100	$\mu_F = 0.718$	μ_{CR}
= 0.261 Diversity = 8.4421e-02 $k_t = [1, 1, 1, 1]$			
Generation 820: Best = 4.015581e+02	FES = 82100	$\mu_F = 0.737$	μ_{CR}
= 0.242 Diversity = 9.5861e-02 $k_t = [4, 4, 2, 4]$			
Generation 830: Best = 4.015573e+02	FES = 83100	$\mu_F = 0.729$	μ_{CR}
= 0.216 Diversity = 8.3215e-02 $k_t = [1, 1, 1, 1]$			
Generation 840: Best = 4.015570e+02	FES = 84100	$\mu_F = 0.733$	μ_{CR}
= 0.214 Diversity = 6.6552e-02 $k_t = [3, 4, 2, 4]$			
Generation 850: Best = 4.015564e+02	FES = 85100	$\mu_F = 0.733$	μ_{CR}
= 0.215 Diversity = 5.4507e-02 $k_t = [1, 1, 1, 1]$			
Generation 860: Best = 4.015554e+02	FES = 86100	$\mu_F = 0.737$	μ_{CR}
= 0.225 Diversity = 4.9763e-02 $k_t = [2, 4, 4, 3]$			
Generation 870: Best = 4.015543e+02	FES = 87100	$\mu_F = 0.750$	μ_{CR}
= 0.202 Diversity = 4.7739e-02 $k_t = [1, 1, 1, 1]$			
Generation 880: Best = 4.015541e+02	FES = 88100	$\mu_F = 0.763$	μ_{CR}
= 0.192 Diversity = 4.6832e-02 $k_t = [3, 2, 4, 2]$			
Generation 890: Best = 4.015537e+02	FES = 89100	$\mu_F = 0.750$	μ_{CR}
= 0.185 Diversity = 4.8022e-02 $k_t = [1, 1, 1, 1]$			
Generation 900: Best = 4.015536e+02	FES = 90100	$\mu_F = 0.740$	μ_{CR}
= 0.176 Diversity = 4.2993e-02 $k_t = [2, 3, 2, 2]$			
Generation 910: Best = 4.015530e+02	FES = 91100	$\mu_F = 0.750$	μ_{CR}
= 0.144 Diversity = 4.1195e-02 $k_t = [1, 1, 1, 1]$			
Generation 920: Best = 4.015527e+02	FES = 92100	$\mu_F = 0.769$	μ_{CR}
= 0.132 Diversity = 4.1591e-02 $k_t = [2, 3, 3, 3]$			
Generation 930: Best = 4.015524e+02	FES = 93100	$\mu_F = 0.758$	μ_{CR}
= 0.147 Diversity = 3.9083e-02 $k_t = [1, 1, 1, 1]$			
Generation 940: Best = 4.015524e+02	FES = 94100	$\mu_F = 0.750$	μ_{CR}
= 0.140 Diversity = 3.4211e-02 $k_t = [2, 4, 3, 2]$			
Generation 950: Best = 4.015522e+02	FES = 95100	$\mu_F = 0.726$	μ_{CR}
= 0.158 Diversity = 3.1810e-02 $k_t = [1, 1, 1, 1]$			

```
Generation 960: Best = 4.015521e+02 FES = 96100  $\mu_F$  = 0.706  $\mu_{CR}$ 
= 0.153 Diversity = 2.7577e-02  $k_t$  = [2, 2, 4, 2]
Generation 970: Best = 4.015520e+02 FES = 97100  $\mu_F$  = 0.701  $\mu_{CR}$ 
= 0.159 Diversity = 2.7316e-02  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 4.015520e+02 FES = 98100  $\mu_F$  = 0.712  $\mu_{CR}$ 
= 0.157 Diversity = 1.8359e-02  $k_t$  = [3, 3, 4, 4]
Generation 990: Best = 4.015519e+02 FES = 99100  $\mu_F$  = 0.707  $\mu_{CR}$ 
= 0.158 Diversity = 1.6508e-02  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 4.015519e+02

Total Function Evals = 100000

Total Successful Updates = 24661

```
=====
=====
```

Results:

Best Fitness : 4.015519e+02

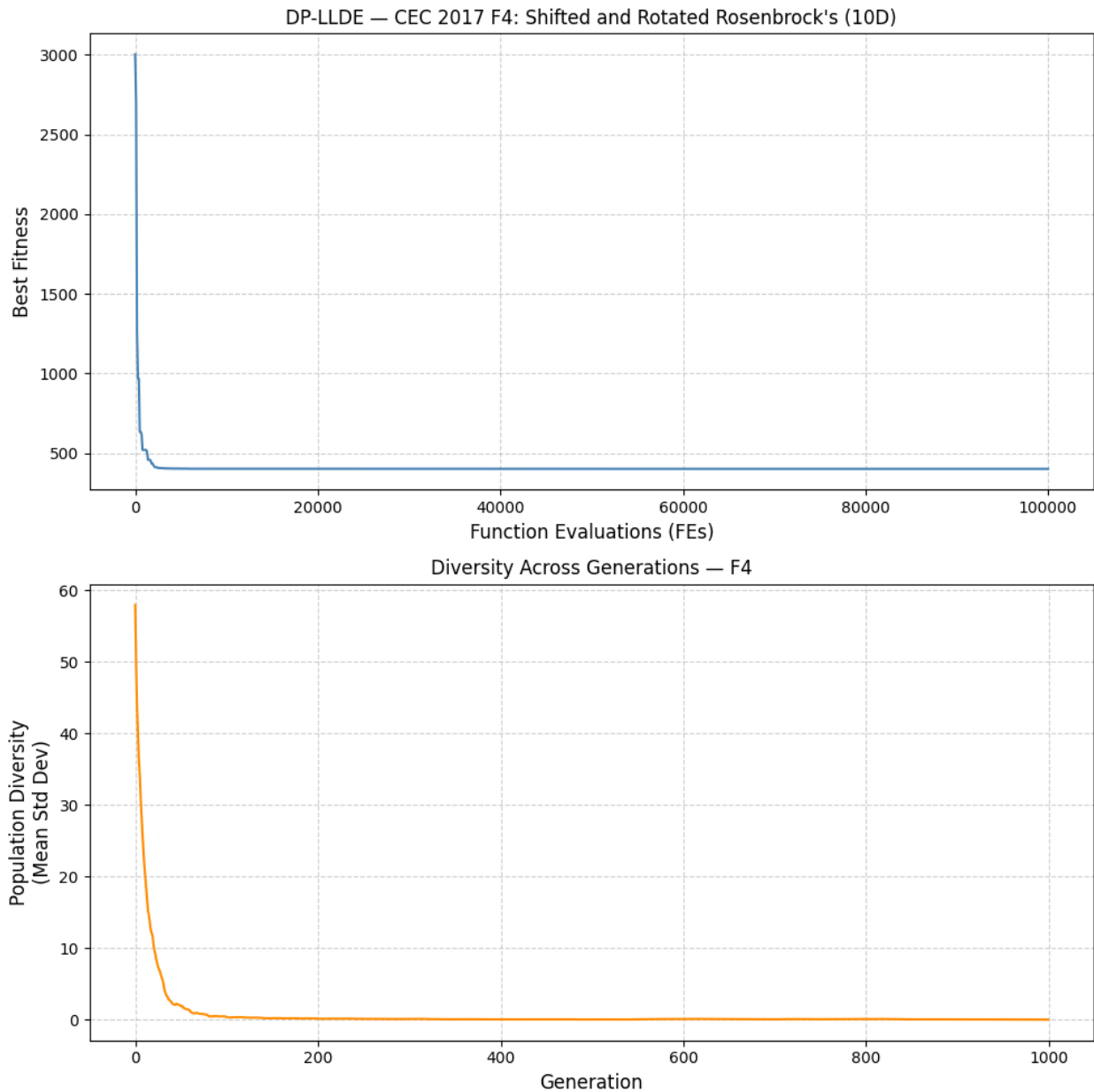
Global Optimum : 4.000000e+02

Error (|best-optimum|) : 1.551874e+00

Best Solution (5 dim) : [56.68145204 27.14252751 -68.38936064
54.58219164 30.12995621]

D^(0) (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F4.png

```
# =====
# F5 – Shifted and Rotated Rastrigin
# =====
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and
Rotated Rastrigin's")
```

```
=====
=====
DP-LLDE on CEC 2017 F5 – Single Run
=====
```


=====

Function : Shifted and Rotated Rastrigin's (10D)

Bounds : [-100, 100]

Global optimum : 500.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 6.377184e+02

Generation 10: Best = 5.667070e+02 FES = 1100 $\mu_F = 0.587$ μ_{CR}

= 0.342 Diversity = 2.7135e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 5.380114e+02 FES = 2100 $\mu_F = 0.657$ μ_{CR}

= 0.324 Diversity = 2.2104e+01 $k_t = [3, 4, 2, 3]$

Generation 30: Best = 5.380114e+02 FES = 3100 $\mu_F = 0.660$ μ_{CR}

= 0.303 Diversity = 1.9775e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 5.307100e+02 FES = 4100 $\mu_F = 0.690$ μ_{CR}

= 0.243 Diversity = 1.8886e+01 $k_t = [3, 3, 4, 4]$

Generation 50: Best = 5.307100e+02 FES = 5100 $\mu_F = 0.692$ μ_{CR}

= 0.216 Diversity = 1.7949e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 5.272589e+02 FES = 6100 $\mu_F = 0.699$ μ_{CR}

= 0.187 Diversity = 1.7021e+01 $k_t = [3, 3, 4, 3]$

Generation 70: Best = 5.272589e+02 FES = 7100 $\mu_F = 0.717$ μ_{CR}

= 0.189 Diversity = 1.6632e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 5.245801e+02 FES = 8100 $\mu_F = 0.729$ μ_{CR}

= 0.146 Diversity = 1.6262e+01 $k_t = [3, 2, 3, 2]$

Generation 90: Best = 5.233753e+02 FES = 9100 $\mu_F = 0.740$ μ_{CR}

= 0.140 Diversity = 1.6171e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 5.232637e+02 FES = 10100 $\mu_F = 0.737$ μ_{CR}

= 0.121 Diversity = 1.6052e+01 $k_t = [3, 4, 4, 2]$

Generation 110: Best = 5.232054e+02 FES = 11100 $\mu_F = 0.705$ μ_{CR}

= 0.129 Diversity = 1.6041e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 5.206306e+02 FES = 12100 $\mu_F = 0.701$ μ_{CR}

= 0.110 Diversity = 1.5861e+01 $k_t = [2, 4, 2, 4]$

Generation 130: Best = 5.191735e+02 FES = 13100 $\mu_F = 0.764$ μ_{CR}

= 0.109 Diversity = 1.5767e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 5.171031e+02 FES = 14100 $\mu_F = 0.719$ μ_{CR}

= 0.085 Diversity = 1.5722e+01 $k_t = [3, 3, 2, 2]$

Generation 150: Best = 5.154868e+02 FES = 15100 $\mu_F = 0.682$ μ_{CR}

= 0.077 Diversity = 1.5760e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 5.154868e+02 FES = 16100 $\mu_F = 0.698$ μ_{CR}

= 0.115 Diversity = 1.5716e+01 $k_t = [2, 4, 3, 4]$

Generation 170: Best = 5.114984e+02 FES = 17100 $\mu_F = 0.736$ μ_{CR}

= 0.086 Diversity = 1.5628e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 5.103976e+02 FES = 18100 $\mu_F = 0.696$ μ_{CR}

= 0.090 Diversity = 1.5559e+01 $k_t = [2, 4, 4, 3]$

Generation 190: Best = 5.103976e+02 FES = 19100 $\mu_F = 0.699$ μ_{CR}

= 0.093 Diversity = 1.5330e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 5.103976e+02 FES = 20100 $\mu_F = 0.667$ μ_{CR}

= 0.084 Diversity = 1.5339e+01 $k_t = [2, 4, 4, 2]$

Generation 210: Best = 5.103976e+02 FES = 21100 $\mu_F = 0.716$ μ_{CR}

= 0.081 Diversity = 1.5305e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 5.103976e+02 FES = 22100 $\mu_F = 0.655$ μ_{CR}

= 0.075	Diversity = 1.5136e+01	k_t = [4, 3, 2, 2]	μ_F = 0.631	μ_{CR}
Generation 230:	Best = 5.103976e+02	FES = 23100		
= 0.105	Diversity = 1.5135e+01	k_t = [1, 1, 1, 1]	μ_F = 0.559	μ_{CR}
Generation 240:	Best = 5.087442e+02	FES = 24100		
= 0.088	Diversity = 1.5133e+01	k_t = [2, 3, 2, 3]	μ_F = 0.582	μ_{CR}
Generation 250:	Best = 5.087442e+02	FES = 25100		
= 0.081	Diversity = 1.5105e+01	k_t = [1, 1, 1, 1]	μ_F = 0.622	μ_{CR}
Generation 260:	Best = 5.084847e+02	FES = 26100		
= 0.081	Diversity = 1.5158e+01	k_t = [2, 2, 3, 2]	μ_F = 0.561	μ_{CR}
Generation 270:	Best = 5.084847e+02	FES = 27100		
= 0.083	Diversity = 1.5222e+01	k_t = [1, 1, 1, 1]	μ_F = 0.611	μ_{CR}
Generation 280:	Best = 5.084847e+02	FES = 28100		
= 0.104	Diversity = 1.5217e+01	k_t = [2, 4, 3, 4]	μ_F = 0.590	μ_{CR}
Generation 290:	Best = 5.075798e+02	FES = 29100		
= 0.132	Diversity = 1.5190e+01	k_t = [1, 1, 1, 1]	μ_F = 0.485	μ_{CR}
Generation 300:	Best = 5.075798e+02	FES = 30100		
= 0.114	Diversity = 1.5174e+01	k_t = [2, 4, 4, 2]	μ_F = 0.527	μ_{CR}
Generation 310:	Best = 5.074115e+02	FES = 31100		
= 0.101	Diversity = 1.5088e+01	k_t = [1, 1, 1, 1]	μ_F = 0.462	μ_{CR}
Generation 320:	Best = 5.074115e+02	FES = 32100		
= 0.097	Diversity = 1.5069e+01	k_t = [4, 2, 3, 3]	μ_F = 0.387	μ_{CR}
Generation 330:	Best = 5.073091e+02	FES = 33100		
= 0.103	Diversity = 1.5047e+01	k_t = [1, 1, 1, 1]	μ_F = 0.362	μ_{CR}
Generation 340:	Best = 5.072050e+02	FES = 34100		
= 0.106	Diversity = 1.5000e+01	k_t = [3, 2, 2, 2]	μ_F = 0.416	μ_{CR}
Generation 350:	Best = 5.068876e+02	FES = 35100		
= 0.077	Diversity = 1.5077e+01	k_t = [1, 1, 1, 1]	μ_F = 0.360	μ_{CR}
Generation 360:	Best = 5.068876e+02	FES = 36100		
= 0.095	Diversity = 1.5051e+01	k_t = [3, 4, 2, 3]	μ_F = 0.357	μ_{CR}
Generation 370:	Best = 5.065296e+02	FES = 37100		
= 0.088	Diversity = 1.4920e+01	k_t = [1, 1, 1, 1]	μ_F = 0.337	μ_{CR}
Generation 380:	Best = 5.056739e+02	FES = 38100		
= 0.071	Diversity = 1.4914e+01	k_t = [2, 2, 2, 4]	μ_F = 0.299	μ_{CR}
Generation 390:	Best = 5.048505e+02	FES = 39100		
= 0.095	Diversity = 1.4883e+01	k_t = [1, 1, 1, 1]	μ_F = 0.323	μ_{CR}
Generation 400:	Best = 5.048505e+02	FES = 40100		
= 0.087	Diversity = 1.4874e+01	k_t = [4, 4, 3, 2]	μ_F = 0.315	μ_{CR}
Generation 410:	Best = 5.048505e+02	FES = 41100		
= 0.070	Diversity = 1.4851e+01	k_t = [1, 1, 1, 1]	μ_F = 0.386	μ_{CR}
Generation 420:	Best = 5.048488e+02	FES = 42100		
= 0.082	Diversity = 1.4846e+01	k_t = [2, 2, 2, 3]	μ_F = 0.319	μ_{CR}
Generation 430:	Best = 5.048488e+02	FES = 43100		
= 0.088	Diversity = 1.4788e+01	k_t = [1, 1, 1, 1]	μ_F = 0.303	μ_{CR}
Generation 440:	Best = 5.048488e+02	FES = 44100		
= 0.099	Diversity = 1.4767e+01	k_t = [4, 2, 3, 3]	μ_F = 0.316	μ_{CR}
Generation 450:	Best = 5.048488e+02	FES = 45100		
= 0.113	Diversity = 1.4837e+01	k_t = [1, 1, 1, 1]	μ_F = 0.252	μ_{CR}
Generation 460:	Best = 5.048488e+02	FES = 46100		
= 0.102	Diversity = 1.4813e+01	k_t = [2, 4, 3, 3]		

Generation 470: Best = 5.048488e+02	FES = 47100	$\mu_F = 0.278$	μ_{CR}
= 0.126 Diversity = 1.4707e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 5.048488e+02	FES = 48100	$\mu_F = 0.272$	μ_{CR}
= 0.118 Diversity = 1.4687e+01	$k_t = [2, 2, 3, 2]$		
Generation 490: Best = 5.043723e+02	FES = 49100	$\mu_F = 0.287$	μ_{CR}
= 0.105 Diversity = 1.4685e+01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 5.043723e+02	FES = 50100	$\mu_F = 0.382$	μ_{CR}
= 0.093 Diversity = 1.4680e+01	$k_t = [4, 4, 2, 3]$		
Generation 510: Best = 5.043723e+02	FES = 51100	$\mu_F = 0.411$	μ_{CR}
= 0.090 Diversity = 1.4633e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 5.043601e+02	FES = 52100	$\mu_F = 0.334$	μ_{CR}
= 0.089 Diversity = 1.4586e+01	$k_t = [4, 4, 2, 3]$		
Generation 530: Best = 5.043601e+02	FES = 53100	$\mu_F = 0.304$	μ_{CR}
= 0.090 Diversity = 1.4589e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 5.043601e+02	FES = 54100	$\mu_F = 0.283$	μ_{CR}
= 0.091 Diversity = 1.4590e+01	$k_t = [2, 4, 4, 2]$		
Generation 550: Best = 5.043601e+02	FES = 55100	$\mu_F = 0.314$	μ_{CR}
= 0.079 Diversity = 1.4497e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 5.043601e+02	FES = 56100	$\mu_F = 0.407$	μ_{CR}
= 0.081 Diversity = 1.4415e+01	$k_t = [4, 3, 4, 4]$		
Generation 570: Best = 5.043005e+02	FES = 57100	$\mu_F = 0.508$	μ_{CR}
= 0.088 Diversity = 1.4456e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 5.043005e+02	FES = 58100	$\mu_F = 0.556$	μ_{CR}
= 0.086 Diversity = 1.4461e+01	$k_t = [3, 4, 3, 2]$		
Generation 590: Best = 5.042846e+02	FES = 59100	$\mu_F = 0.477$	μ_{CR}
= 0.094 Diversity = 1.4444e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 5.042846e+02	FES = 60100	$\mu_F = 0.438$	μ_{CR}
= 0.095 Diversity = 1.4410e+01	$k_t = [4, 2, 4, 3]$		
Generation 610: Best = 5.042846e+02	FES = 61100	$\mu_F = 0.472$	μ_{CR}
= 0.119 Diversity = 1.4410e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 5.042846e+02	FES = 62100	$\mu_F = 0.436$	μ_{CR}
= 0.121 Diversity = 1.4393e+01	$k_t = [2, 4, 4, 3]$		
Generation 630: Best = 5.042846e+02	FES = 63100	$\mu_F = 0.455$	μ_{CR}
= 0.106 Diversity = 1.4395e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 5.042846e+02	FES = 64100	$\mu_F = 0.402$	μ_{CR}
= 0.107 Diversity = 1.4395e+01	$k_t = [3, 3, 2, 3]$		
Generation 650: Best = 5.042846e+02	FES = 65100	$\mu_F = 0.431$	μ_{CR}
= 0.082 Diversity = 1.4355e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 5.024828e+02	FES = 66100	$\mu_F = 0.434$	μ_{CR}
= 0.106 Diversity = 1.4358e+01	$k_t = [3, 4, 2, 3]$		
Generation 670: Best = 5.024828e+02	FES = 67100	$\mu_F = 0.340$	μ_{CR}
= 0.121 Diversity = 1.4361e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 5.024828e+02	FES = 68100	$\mu_F = 0.373$	μ_{CR}
= 0.117 Diversity = 1.4365e+01	$k_t = [4, 3, 3, 4]$		
Generation 690: Best = 5.024828e+02	FES = 69100	$\mu_F = 0.379$	μ_{CR}
= 0.139 Diversity = 1.4352e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 5.024828e+02	FES = 70100	$\mu_F = 0.266$	μ_{CR}
= 0.113 Diversity = 1.4351e+01	$k_t = [2, 3, 2, 3]$		
Generation 710: Best = 5.024828e+02	FES = 71100	$\mu_F = 0.250$	μ_{CR}

= 0.110	Diversity = 1.4352e+01	k_t = [1, 1, 1, 1]	μ_F = 0.266	μ_{CR}
Generation 720:	Best = 5.024828e+02	FES = 72100		
= 0.098	Diversity = 1.4359e+01	k_t = [4, 3, 3, 3]	μ_F = 0.235	μ_{CR}
Generation 730:	Best = 5.024828e+02	FES = 73100		
= 0.080	Diversity = 1.4357e+01	k_t = [1, 1, 1, 1]	μ_F = 0.206	μ_{CR}
Generation 740:	Best = 5.024828e+02	FES = 74100		
= 0.095	Diversity = 1.4350e+01	k_t = [2, 4, 2, 2]	μ_F = 0.257	μ_{CR}
Generation 750:	Best = 5.024828e+02	FES = 75100		
= 0.096	Diversity = 1.4347e+01	k_t = [1, 1, 1, 1]	μ_F = 0.334	μ_{CR}
Generation 760:	Best = 5.024828e+02	FES = 76100		
= 0.133	Diversity = 1.4134e+01	k_t = [4, 2, 4, 4]	μ_F = 0.265	μ_{CR}
Generation 770:	Best = 5.023802e+02	FES = 77100		
= 0.137	Diversity = 1.4138e+01	k_t = [1, 1, 1, 1]	μ_F = 0.251	μ_{CR}
Generation 780:	Best = 5.023802e+02	FES = 78100		
= 0.115	Diversity = 1.4134e+01	k_t = [2, 2, 2, 3]	μ_F = 0.188	μ_{CR}
Generation 790:	Best = 5.023802e+02	FES = 79100		
= 0.100	Diversity = 1.4130e+01	k_t = [1, 1, 1, 1]	μ_F = 0.253	μ_{CR}
Generation 800:	Best = 5.023802e+02	FES = 80100		
= 0.104	Diversity = 1.4136e+01	k_t = [4, 3, 4, 4]	μ_F = 0.280	μ_{CR}
Generation 810:	Best = 5.023802e+02	FES = 81100		
= 0.127	Diversity = 1.4124e+01	k_t = [1, 1, 1, 1]	μ_F = 0.219	μ_{CR}
Generation 820:	Best = 5.023802e+02	FES = 82100		
= 0.129	Diversity = 1.4086e+01	k_t = [2, 2, 3, 3]	μ_F = 0.166	μ_{CR}
Generation 830:	Best = 5.023802e+02	FES = 83100		
= 0.158	Diversity = 1.4084e+01	k_t = [1, 1, 1, 1]	μ_F = 0.266	μ_{CR}
Generation 840:	Best = 5.023802e+02	FES = 84100		
= 0.132	Diversity = 1.4081e+01	k_t = [4, 4, 3, 3]	μ_F = 0.237	μ_{CR}
Generation 850:	Best = 5.023802e+02	FES = 85100		
= 0.125	Diversity = 1.4079e+01	k_t = [1, 1, 1, 1]	μ_F = 0.219	μ_{CR}
Generation 860:	Best = 5.023802e+02	FES = 86100		
= 0.109	Diversity = 1.4075e+01	k_t = [3, 3, 2, 3]	μ_F = 0.213	μ_{CR}
Generation 870:	Best = 5.023802e+02	FES = 87100		
= 0.118	Diversity = 1.4073e+01	k_t = [1, 1, 1, 1]	μ_F = 0.260	μ_{CR}
Generation 880:	Best = 5.023802e+02	FES = 88100		
= 0.129	Diversity = 1.3987e+01	k_t = [3, 4, 3, 4]	μ_F = 0.289	μ_{CR}
Generation 890:	Best = 5.023026e+02	FES = 89100		
= 0.098	Diversity = 1.3991e+01	k_t = [1, 1, 1, 1]	μ_F = 0.323	μ_{CR}
Generation 900:	Best = 5.023026e+02	FES = 90100		
= 0.129	Diversity = 1.3753e+01	k_t = [4, 3, 3, 3]	μ_F = 0.221	μ_{CR}
Generation 910:	Best = 5.023026e+02	FES = 91100		
= 0.130	Diversity = 1.3666e+01	k_t = [1, 1, 1, 1]	μ_F = 0.173	μ_{CR}
Generation 920:	Best = 5.023026e+02	FES = 92100		
= 0.105	Diversity = 1.3664e+01	k_t = [3, 2, 3, 3]	μ_F = 0.175	μ_{CR}
Generation 930:	Best = 5.023026e+02	FES = 93100		
= 0.085	Diversity = 1.3662e+01	k_t = [1, 1, 1, 1]	μ_F = 0.228	μ_{CR}
Generation 940:	Best = 5.023026e+02	FES = 94100		
= 0.106	Diversity = 1.3630e+01	k_t = [4, 2, 3, 4]	μ_F = 0.209	μ_{CR}
Generation 950:	Best = 5.022566e+02	FES = 95100		
= 0.106	Diversity = 1.3629e+01	k_t = [1, 1, 1, 1]		

Generation 960: Best = 5.022566e+02 FES = 96100 μ_F = 0.233 μ_{CR}
= 0.116 Diversity = 1.3673e+01 k_t = [2, 2, 2, 4]
Generation 970: Best = 5.022566e+02 FES = 97100 μ_F = 0.221 μ_{CR}
= 0.103 Diversity = 1.3678e+01 k_t = [1, 1, 1, 1]
Generation 980: Best = 5.022566e+02 FES = 98100 μ_F = 0.171 μ_{CR}
= 0.089 Diversity = 1.3665e+01 k_t = [3, 4, 2, 3]
Generation 990: Best = 5.022566e+02 FES = 99100 μ_F = 0.199 μ_{CR}
= 0.088 Diversity = 1.3663e+01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 5.022566e+02

Total Function Evals = 100000

Total Successful Updates = 3456

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Results:

Best Fitness : 5.022566e+02

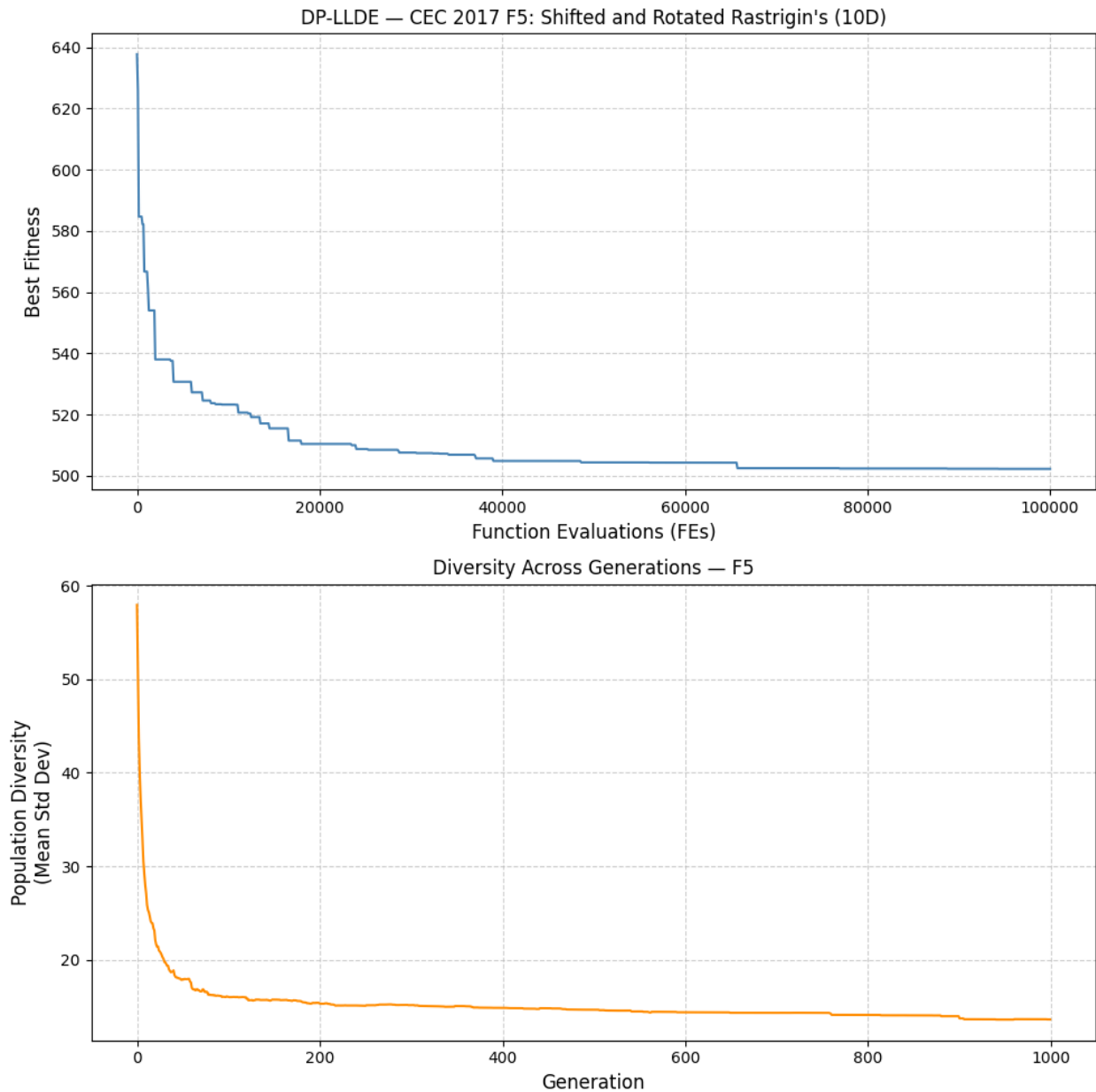
Global Optimum : 5.000000e+02

Error (|best-optimum|) : 2.256606e+00

Best Solution (5 dim) : [-17.1176507 56.16282262 -29.83522857 -
66.82340462 26.61671319]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F5.png

```
# =====
# F6 – Shifted and Rotated Schaffer's F7
# =====
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and
Rotated Schaffer's F7")
```

```
=====
=====
DP-LLDE on CEC 2017 F6 – Single Run
=====
```

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Function : Shifted and Rotated Schaffer's F7 (10D)

Bounds : [-100, 100]

Global optimum : 600.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 7.140452e+02

Generation 10: Best = 6.287927e+02 FES = 1100 $\mu_F = 0.575$ $\mu_{CR} = 0.346$ Diversity = 2.6238e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 6.079716e+02 FES = 2100 $\mu_F = 0.619$ $\mu_{CR} = 0.343$ Diversity = 1.1591e+01 $k_t = [3, 4, 2, 3]$

Generation 30: Best = 6.039505e+02 FES = 3100 $\mu_F = 0.658$ $\mu_{CR} = 0.346$ Diversity = 5.0554e+00 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 6.019566e+02 FES = 4100 $\mu_F = 0.700$ $\mu_{CR} = 0.343$ Diversity = 2.8815e+00 $k_t = [3, 3, 2, 4]$

Generation 50: Best = 6.013890e+02 FES = 5100 $\mu_F = 0.718$ $\mu_{CR} = 0.337$ Diversity = 1.6939e+00 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 6.007958e+02 FES = 6100 $\mu_F = 0.719$ $\mu_{CR} = 0.324$ Diversity = 1.1129e+00 $k_t = [2, 4, 2, 2]$

Generation 70: Best = 6.007287e+02 FES = 7100 $\mu_F = 0.703$ $\mu_{CR} = 0.301$ Diversity = 9.9868e-01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 6.004306e+02 FES = 8100 $\mu_F = 0.710$ $\mu_{CR} = 0.285$ Diversity = 8.1248e-01 $k_t = [2, 4, 3, 3]$

Generation 90: Best = 6.003992e+02 FES = 9100 $\mu_F = 0.726$ $\mu_{CR} = 0.252$ Diversity = 7.0312e-01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 6.002917e+02 FES = 10100 $\mu_F = 0.733$ $\mu_{CR} = 0.231$ Diversity = 5.4764e-01 $k_t = [2, 3, 2, 4]$

Generation 110: Best = 6.002033e+02 FES = 11100 $\mu_F = 0.730$ $\mu_{CR} = 0.216$ Diversity = 4.5748e-01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 6.002033e+02 FES = 12100 $\mu_F = 0.724$ $\mu_{CR} = 0.207$ Diversity = 3.6792e-01 $k_t = [3, 3, 3, 4]$

Generation 130: Best = 6.001554e+02 FES = 13100 $\mu_F = 0.725$ $\mu_{CR} = 0.183$ Diversity = 3.0947e-01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 6.001537e+02 FES = 14100 $\mu_F = 0.703$ $\mu_{CR} = 0.157$ Diversity = 2.5617e-01 $k_t = [2, 4, 2, 3]$

Generation 150: Best = 6.001382e+02 FES = 15100 $\mu_F = 0.708$ $\mu_{CR} = 0.151$ Diversity = 2.4460e-01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 6.000946e+02 FES = 16100 $\mu_F = 0.702$ $\mu_{CR} = 0.141$ Diversity = 2.1104e-01 $k_t = [3, 4, 3, 2]$

Generation 170: Best = 6.000946e+02 FES = 17100 $\mu_F = 0.711$ $\mu_{CR} = 0.124$ Diversity = 1.9055e-01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 6.000701e+02 FES = 18100 $\mu_F = 0.756$ $\mu_{CR} = 0.139$ Diversity = 1.7824e-01 $k_t = [2, 2, 4, 2]$

Generation 190: Best = 6.000654e+02 FES = 19100 $\mu_F = 0.759$ $\mu_{CR} = 0.136$ Diversity = 1.7511e-01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 6.000412e+02 FES = 20100 $\mu_F = 0.749$ $\mu_{CR} = 0.122$ Diversity = 1.2726e-01 $k_t = [4, 3, 3, 3]$

Generation 210: Best = 6.000374e+02 FES = 21100 $\mu_F = 0.764$ $\mu_{CR} = 0.125$ Diversity = 1.0930e-01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 6.000322e+02 FES = 22100 $\mu_F = 0.743$ μ_{CR}

= 0.123	Diversity = 8.9368e-02	k_t = [4, 3, 4, 4]	μ_F = 0.745	μ_{CR}
Generation 230:	Best = 6.000207e+02	FES = 23100		
= 0.127	Diversity = 6.6128e-02	k_t = [1, 1, 1, 1]	μ_F = 0.766	μ_{CR}
Generation 240:	Best = 6.000190e+02	FES = 24100		
= 0.125	Diversity = 6.1171e-02	k_t = [3, 4, 3, 2]	μ_F = 0.750	μ_{CR}
Generation 250:	Best = 6.000182e+02	FES = 25100		
= 0.125	Diversity = 5.5357e-02	k_t = [1, 1, 1, 1]	μ_F = 0.750	μ_{CR}
Generation 260:	Best = 6.000143e+02	FES = 26100		
= 0.118	Diversity = 4.5499e-02	k_t = [3, 3, 2, 2]	μ_F = 0.744	μ_{CR}
Generation 270:	Best = 6.000107e+02	FES = 27100		
= 0.130	Diversity = 4.5056e-02	k_t = [1, 1, 1, 1]	μ_F = 0.728	μ_{CR}
Generation 280:	Best = 6.000097e+02	FES = 28100		
= 0.119	Diversity = 3.1147e-02	k_t = [3, 2, 3, 4]	μ_F = 0.735	μ_{CR}
Generation 290:	Best = 6.000053e+02	FES = 29100		
= 0.121	Diversity = 2.3563e-02	k_t = [1, 1, 1, 1]	μ_F = 0.720	μ_{CR}
Generation 300:	Best = 6.000035e+02	FES = 30100		
= 0.120	Diversity = 1.5353e-02	k_t = [2, 3, 2, 4]	μ_F = 0.738	μ_{CR}
Generation 310:	Best = 6.000035e+02	FES = 31100		
= 0.107	Diversity = 1.1906e-02	k_t = [1, 1, 1, 1]	μ_F = 0.749	μ_{CR}
Generation 320:	Best = 6.000024e+02	FES = 32100		
= 0.122	Diversity = 9.5599e-03	k_t = [4, 3, 2, 3]	μ_F = 0.730	μ_{CR}
Generation 330:	Best = 6.000022e+02	FES = 33100		
= 0.140	Diversity = 8.0659e-03	k_t = [1, 1, 1, 1]	μ_F = 0.735	μ_{CR}
Generation 340:	Best = 6.000017e+02	FES = 34100		
= 0.152	Diversity = 6.7178e-03	k_t = [3, 2, 2, 4]	μ_F = 0.730	μ_{CR}
Generation 350:	Best = 6.000017e+02	FES = 35100		
= 0.156	Diversity = 4.7920e-03	k_t = [1, 1, 1, 1]	μ_F = 0.739	μ_{CR}
Generation 360:	Best = 6.000017e+02	FES = 36100		
= 0.137	Diversity = 3.3541e-03	k_t = [2, 2, 4, 4]	μ_F = 0.752	μ_{CR}
Generation 370:	Best = 6.000014e+02	FES = 37100		
= 0.142	Diversity = 3.1017e-03	k_t = [1, 1, 1, 1]	μ_F = 0.762	μ_{CR}
Generation 380:	Best = 6.000010e+02	FES = 38100		
= 0.142	Diversity = 3.0383e-03	k_t = [4, 3, 3, 3]	μ_F = 0.780	μ_{CR}
Generation 390:	Best = 6.000008e+02	FES = 39100		
= 0.130	Diversity = 2.8802e-03	k_t = [1, 1, 1, 1]	μ_F = 0.787	μ_{CR}
Generation 400:	Best = 6.000007e+02	FES = 40100		
= 0.138	Diversity = 2.2685e-03	k_t = [4, 2, 3, 4]	μ_F = 0.779	μ_{CR}
Generation 410:	Best = 6.000007e+02	FES = 41100		
= 0.136	Diversity = 1.7308e-03	k_t = [1, 1, 1, 1]	μ_F = 0.768	μ_{CR}
Generation 420:	Best = 6.000006e+02	FES = 42100		
= 0.131	Diversity = 1.4894e-03	k_t = [3, 3, 4, 2]	μ_F = 0.768	μ_{CR}
Generation 430:	Best = 6.000006e+02	FES = 43100		
= 0.132	Diversity = 1.4407e-03	k_t = [1, 1, 1, 1]	μ_F = 0.783	μ_{CR}
Generation 440:	Best = 6.000003e+02	FES = 44100		
= 0.130	Diversity = 1.2265e-03	k_t = [3, 4, 4, 4]	μ_F = 0.762	μ_{CR}
Generation 450:	Best = 6.000003e+02	FES = 45100		
= 0.111	Diversity = 1.0436e-03	k_t = [1, 1, 1, 1]	μ_F = 0.780	μ_{CR}
Generation 460:	Best = 6.000003e+02	FES = 46100		
= 0.127	Diversity = 9.4985e-04	k_t = [4, 2, 3, 3]		

Generation 470: Best = 6.000002e+02	FES = 47100	$\mu_F = 0.778$	μ_{CR}
= 0.130 Diversity = 9.5868e-04 $k_t = [1, 1, 1, 1]$			
Generation 480: Best = 6.000002e+02	FES = 48100	$\mu_F = 0.743$	μ_{CR}
= 0.125 Diversity = 6.8587e-04 $k_t = [2, 3, 2, 3]$			
Generation 490: Best = 6.000002e+02	FES = 49100	$\mu_F = 0.750$	μ_{CR}
= 0.129 Diversity = 5.8396e-04 $k_t = [1, 1, 1, 1]$			
Generation 500: Best = 6.000001e+02	FES = 50100	$\mu_F = 0.734$	μ_{CR}
= 0.129 Diversity = 4.6668e-04 $k_t = [2, 3, 3, 2]$			
Generation 510: Best = 6.000001e+02	FES = 51100	$\mu_F = 0.739$	μ_{CR}
= 0.131 Diversity = 4.2075e-04 $k_t = [1, 1, 1, 1]$			
Generation 520: Best = 6.000001e+02	FES = 52100	$\mu_F = 0.739$	μ_{CR}
= 0.120 Diversity = 3.9288e-04 $k_t = [4, 4, 3, 2]$			
Generation 530: Best = 6.000001e+02	FES = 53100	$\mu_F = 0.737$	μ_{CR}
= 0.120 Diversity = 3.6749e-04 $k_t = [1, 1, 1, 1]$			
Generation 540: Best = 6.000001e+02	FES = 54100	$\mu_F = 0.735$	μ_{CR}
= 0.128 Diversity = 3.0928e-04 $k_t = [2, 2, 4, 3]$			
Generation 550: Best = 6.000001e+02	FES = 55100	$\mu_F = 0.745$	μ_{CR}
= 0.120 Diversity = 3.0389e-04 $k_t = [1, 1, 1, 1]$			
Generation 560: Best = 6.000000e+02	FES = 56100	$\mu_F = 0.746$	μ_{CR}
= 0.123 Diversity = 2.6261e-04 $k_t = [2, 2, 3, 2]$			
Generation 570: Best = 6.000000e+02	FES = 57100	$\mu_F = 0.727$	μ_{CR}
= 0.119 Diversity = 2.0674e-04 $k_t = [1, 1, 1, 1]$			
Generation 580: Best = 6.000000e+02	FES = 58100	$\mu_F = 0.734$	μ_{CR}
= 0.133 Diversity = 1.9121e-04 $k_t = [2, 3, 3, 2]$			
Generation 590: Best = 6.000000e+02	FES = 59100	$\mu_F = 0.740$	μ_{CR}
= 0.123 Diversity = 2.1257e-04 $k_t = [1, 1, 1, 1]$			
Generation 600: Best = 6.000000e+02	FES = 60100	$\mu_F = 0.741$	μ_{CR}
= 0.114 Diversity = 1.6220e-04 $k_t = [2, 4, 2, 4]$			
Generation 610: Best = 6.000000e+02	FES = 61100	$\mu_F = 0.723$	μ_{CR}
= 0.115 Diversity = 1.4272e-04 $k_t = [1, 1, 1, 1]$			
Generation 620: Best = 6.000000e+02	FES = 62100	$\mu_F = 0.729$	μ_{CR}
= 0.135 Diversity = 9.5915e-05 $k_t = [3, 4, 4, 3]$			
Generation 630: Best = 6.000000e+02	FES = 63100	$\mu_F = 0.751$	μ_{CR}
= 0.120 Diversity = 7.5140e-05 $k_t = [1, 1, 1, 1]$			
Generation 640: Best = 6.000000e+02	FES = 64100	$\mu_F = 0.765$	μ_{CR}
= 0.114 Diversity = 4.9707e-05 $k_t = [3, 4, 2, 4]$			
Generation 650: Best = 6.000000e+02	FES = 65100	$\mu_F = 0.774$	μ_{CR}
= 0.116 Diversity = 3.1276e-05 $k_t = [1, 1, 1, 1]$			
Generation 660: Best = 6.000000e+02	FES = 66100	$\mu_F = 0.785$	μ_{CR}
= 0.118 Diversity = 2.3676e-05 $k_t = [4, 4, 2, 4]$			
Generation 670: Best = 6.000000e+02	FES = 67100	$\mu_F = 0.770$	μ_{CR}
= 0.116 Diversity = 2.1976e-05 $k_t = [1, 1, 1, 1]$			
Generation 680: Best = 6.000000e+02	FES = 68100	$\mu_F = 0.755$	μ_{CR}
= 0.110 Diversity = 1.7944e-05 $k_t = [4, 3, 3, 4]$			
Generation 690: Best = 6.000000e+02	FES = 69100	$\mu_F = 0.714$	μ_{CR}
= 0.122 Diversity = 1.6199e-05 $k_t = [1, 1, 1, 1]$			
Generation 700: Best = 6.000000e+02	FES = 70100	$\mu_F = 0.729$	μ_{CR}
= 0.117 Diversity = 1.4318e-05 $k_t = [2, 4, 4, 3]$			
Generation 710: Best = 6.000000e+02	FES = 71100	$\mu_F = 0.743$	μ_{CR}

= 0.124	Diversity = 1.3973e-05	k_t = [1, 1, 1, 1]	μ_F = 0.728	μ_{CR}
Generation 720:	Best = 6.000000e+02	FES = 72100		
= 0.131	Diversity = 1.1619e-05	k_t = [3, 2, 3, 4]	μ_F = 0.724	μ_{CR}
Generation 730:	Best = 6.000000e+02	FES = 73100		
= 0.132	Diversity = 1.0836e-05	k_t = [1, 1, 1, 1]	μ_F = 0.733	μ_{CR}
Generation 740:	Best = 6.000000e+02	FES = 74100		
= 0.144	Diversity = 1.0463e-05	k_t = [3, 2, 2, 4]	μ_F = 0.721	μ_{CR}
Generation 750:	Best = 6.000000e+02	FES = 75100		
= 0.134	Diversity = 6.4789e-06	k_t = [1, 1, 1, 1]	μ_F = 0.720	μ_{CR}
Generation 760:	Best = 6.000000e+02	FES = 76100		
= 0.129	Diversity = 5.6271e-06	k_t = [3, 2, 3, 2]	μ_F = 0.714	μ_{CR}
Generation 770:	Best = 6.000000e+02	FES = 77100		
= 0.134	Diversity = 5.1034e-06	k_t = [1, 1, 1, 1]	μ_F = 0.722	μ_{CR}
Generation 780:	Best = 6.000000e+02	FES = 78100		
= 0.142	Diversity = 4.3107e-06	k_t = [3, 2, 3, 2]	μ_F = 0.723	μ_{CR}
Generation 790:	Best = 6.000000e+02	FES = 79100		
= 0.138	Diversity = 3.3151e-06	k_t = [1, 1, 1, 1]	μ_F = 0.720	μ_{CR}
Generation 800:	Best = 6.000000e+02	FES = 80100		
= 0.146	Diversity = 2.2978e-06	k_t = [2, 4, 3, 4]	μ_F = 0.741	μ_{CR}
Generation 810:	Best = 6.000000e+02	FES = 81100		
= 0.151	Diversity = 1.0735e-06	k_t = [1, 1, 1, 1]	μ_F = 0.759	μ_{CR}
Generation 820:	Best = 6.000000e+02	FES = 82100		
= 0.160	Diversity = 3.6552e-07	k_t = [2, 3, 2, 4]	μ_F = 0.763	μ_{CR}
Generation 830:	Best = 6.000000e+02	FES = 83100		
= 0.167	Diversity = 2.2704e-07	k_t = [1, 1, 1, 1]	μ_F = 0.757	μ_{CR}
Generation 840:	Best = 6.000000e+02	FES = 84100		
= 0.160	Diversity = 1.9030e-07	k_t = [4, 3, 3, 2]	μ_F = 0.740	μ_{CR}
Generation 850:	Best = 6.000000e+02	FES = 85100		
= 0.164	Diversity = 1.6040e-07	k_t = [1, 1, 1, 1]	μ_F = 0.727	μ_{CR}
Generation 860:	Best = 6.000000e+02	FES = 86100		
= 0.162	Diversity = 1.3643e-07	k_t = [3, 3, 3, 4]	μ_F = 0.717	μ_{CR}
Generation 870:	Best = 6.000000e+02	FES = 87100		
= 0.155	Diversity = 1.3887e-07	k_t = [1, 1, 1, 1]	μ_F = 0.692	μ_{CR}
Generation 880:	Best = 6.000000e+02	FES = 88100		
= 0.165	Diversity = 1.3757e-07	k_t = [3, 3, 2, 2]	μ_F = 0.676	μ_{CR}
Generation 890:	Best = 6.000000e+02	FES = 89100		
= 0.167	Diversity = 1.3459e-07	k_t = [1, 1, 1, 1]	μ_F = 0.674	μ_{CR}
Generation 900:	Best = 6.000000e+02	FES = 90100		
= 0.138	Diversity = 1.4495e-07	k_t = [2, 3, 3, 2]	μ_F = 0.668	μ_{CR}
Generation 910:	Best = 6.000000e+02	FES = 91100		
= 0.161	Diversity = 1.2840e-07	k_t = [1, 1, 1, 1]	μ_F = 0.675	μ_{CR}
Generation 920:	Best = 6.000000e+02	FES = 92100		
= 0.154	Diversity = 1.2223e-07	k_t = [3, 3, 4, 2]	μ_F = 0.665	μ_{CR}
Generation 930:	Best = 6.000000e+02	FES = 93100		
= 0.148	Diversity = 1.2022e-07	k_t = [1, 1, 1, 1]	μ_F = 0.614	μ_{CR}
Generation 940:	Best = 6.000000e+02	FES = 94100		
= 0.118	Diversity = 1.1897e-07	k_t = [4, 4, 2, 4]	μ_F = 0.613	μ_{CR}
Generation 950:	Best = 6.000000e+02	FES = 95100		
= 0.136	Diversity = 1.1728e-07	k_t = [1, 1, 1, 1]		

```
Generation 960: Best = 6.000000e+02 FES = 96100  $\mu_F$  = 0.634  $\mu_{CR}$ 
= 0.111 Diversity = 1.0775e-07  $k_t$  = [2, 2, 2, 2]
Generation 970: Best = 6.000000e+02 FES = 97100  $\mu_F$  = 0.637  $\mu_{CR}$ 
= 0.121 Diversity = 1.1770e-07  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 6.000000e+02 FES = 98100  $\mu_F$  = 0.690  $\mu_{CR}$ 
= 0.121 Diversity = 1.2941e-07  $k_t$  = [4, 2, 3, 4]
Generation 990: Best = 6.000000e+02 FES = 99100  $\mu_F$  = 0.617  $\mu_{CR}$ 
= 0.113 Diversity = 1.2624e-07  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 6.000000e+02

Total Function Evals = 100000

Total Successful Updates = 15039

```
=====
=====
```

Results:

Best Fitness : 6.000000e+02

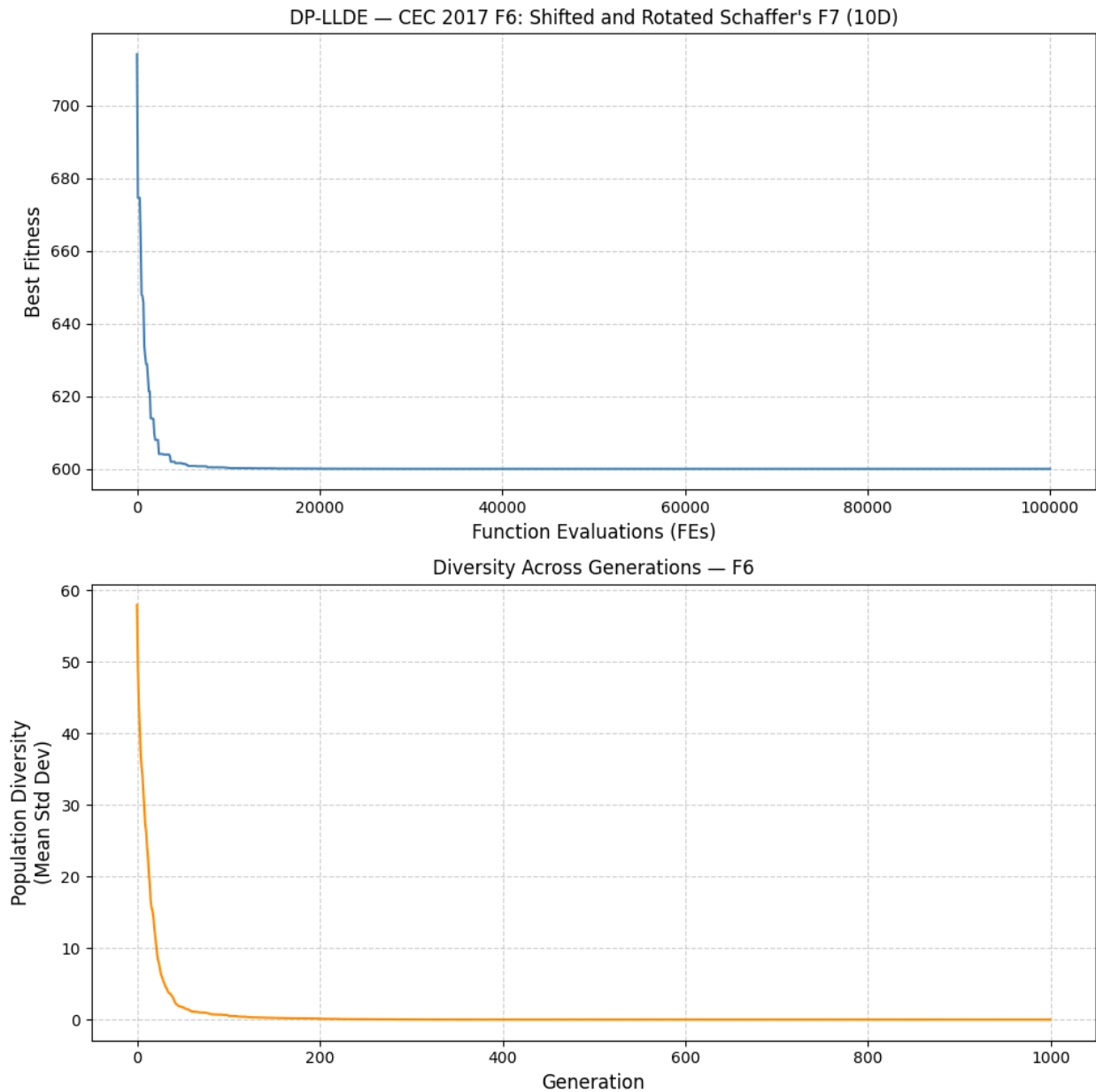
Global Optimum : 6.000000e+02

Error (|best-optimum|) : 6.101186e-08

Best Solution (5 dim) : [79.08939239 -24.57277764 -34.67392694 -
44.68162056 75.99787976]

D^0 (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F6.png

```
# =====
# F7 – Shifted and Rotated Lunacek Bi-Rastrigin
# =====
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and
Rotated Lunacek Bi-Rastrigin's")
```

```
=====
=====
DP-LLDE on CEC 2017 F7 – Single Run
=====
```

=====

Function : Shifted and Rotated Lunacek Bi-Rastrigin's (10D)

Bounds : [-100, 100]

Global optimum : 700.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 1.202264e+03

Generation 10:	Best = 7.953679e+02	FES = 1100	$\mu_F = 0.566$	μ_{CR}
= 0.350 Diversity = 1.5858e+01 $k_t = [1, 1, 1, 1]$				
Generation 20:	Best = 7.656406e+02	FES = 2100	$\mu_F = 0.602$	μ_{CR}
= 0.328 Diversity = 9.0948e+00 $k_t = [3, 4, 2, 3]$				
Generation 30:	Best = 7.567199e+02	FES = 3100	$\mu_F = 0.658$	μ_{CR}
= 0.302 Diversity = 7.7685e+00 $k_t = [1, 1, 1, 1]$				
Generation 40:	Best = 7.532509e+02	FES = 4100	$\mu_F = 0.636$	μ_{CR}
= 0.260 Diversity = 7.2760e+00 $k_t = [4, 2, 2, 3]$				
Generation 50:	Best = 7.446048e+02	FES = 5100	$\mu_F = 0.675$	μ_{CR}
= 0.239 Diversity = 7.0905e+00 $k_t = [1, 1, 1, 1]$				
Generation 60:	Best = 7.355680e+02	FES = 6100	$\mu_F = 0.659$	μ_{CR}
= 0.189 Diversity = 6.5390e+00 $k_t = [3, 3, 2, 3]$				
Generation 70:	Best = 7.355680e+02	FES = 7100	$\mu_F = 0.676$	μ_{CR}
= 0.158 Diversity = 6.1700e+00 $k_t = [1, 1, 1, 1]$				
Generation 80:	Best = 7.330654e+02	FES = 8100	$\mu_F = 0.673$	μ_{CR}
= 0.145 Diversity = 6.0583e+00 $k_t = [3, 3, 4, 3]$				
Generation 90:	Best = 7.326860e+02	FES = 9100	$\mu_F = 0.706$	μ_{CR}
= 0.154 Diversity = 6.0580e+00 $k_t = [1, 1, 1, 1]$				
Generation 100:	Best = 7.326860e+02	FES = 10100	$\mu_F = 0.737$	μ_{CR}
= 0.122 Diversity = 6.0181e+00 $k_t = [4, 2, 3, 4]$				
Generation 110:	Best = 7.207531e+02	FES = 11100	$\mu_F = 0.631$	μ_{CR}
= 0.093 Diversity = 5.9538e+00 $k_t = [1, 1, 1, 1]$				
Generation 120:	Best = 7.207531e+02	FES = 12100	$\mu_F = 0.638$	μ_{CR}
= 0.088 Diversity = 5.8451e+00 $k_t = [3, 4, 2, 2]$				
Generation 130:	Best = 7.207531e+02	FES = 13100	$\mu_F = 0.598$	μ_{CR}
= 0.075 Diversity = 5.7858e+00 $k_t = [1, 1, 1, 1]$				
Generation 140:	Best = 7.207531e+02	FES = 14100	$\mu_F = 0.556$	μ_{CR}
= 0.086 Diversity = 5.7361e+00 $k_t = [4, 4, 3, 2]$				
Generation 150:	Best = 7.207531e+02	FES = 15100	$\mu_F = 0.563$	μ_{CR}
= 0.078 Diversity = 5.7390e+00 $k_t = [1, 1, 1, 1]$				
Generation 160:	Best = 7.205494e+02	FES = 16100	$\mu_F = 0.591$	μ_{CR}
= 0.085 Diversity = 5.6811e+00 $k_t = [3, 2, 2, 3]$				
Generation 170:	Best = 7.205494e+02	FES = 17100	$\mu_F = 0.529$	μ_{CR}
= 0.070 Diversity = 5.6585e+00 $k_t = [1, 1, 1, 1]$				
Generation 180:	Best = 7.205494e+02	FES = 18100	$\mu_F = 0.578$	μ_{CR}
= 0.077 Diversity = 5.5043e+00 $k_t = [4, 4, 2, 3]$				
Generation 190:	Best = 7.205077e+02	FES = 19100	$\mu_F = 0.542$	μ_{CR}
= 0.072 Diversity = 5.4535e+00 $k_t = [1, 1, 1, 1]$				
Generation 200:	Best = 7.205077e+02	FES = 20100	$\mu_F = 0.541$	μ_{CR}
= 0.086 Diversity = 5.2615e+00 $k_t = [4, 3, 4, 4]$				
Generation 210:	Best = 7.205077e+02	FES = 21100	$\mu_F = 0.580$	μ_{CR}
= 0.090 Diversity = 5.2281e+00 $k_t = [1, 1, 1, 1]$				
Generation 220:	Best = 7.205077e+02	FES = 22100	$\mu_F = 0.641$	μ_{CR}

= 0.091	Diversity = 5.1615e+00	k_t = [2, 2, 4, 4]		
Generation 230:	Best = 7.195228e+02	FES = 23100	μ_F = 0.545	μ_{CR}
= 0.091	Diversity = 5.1196e+00	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 7.181223e+02	FES = 24100	μ_F = 0.571	μ_{CR}
= 0.109	Diversity = 5.1094e+00	k_t = [3, 4, 3, 3]		
Generation 250:	Best = 7.181223e+02	FES = 25100	μ_F = 0.493	μ_{CR}
= 0.103	Diversity = 5.1031e+00	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 7.181223e+02	FES = 26100	μ_F = 0.552	μ_{CR}
= 0.102	Diversity = 5.0502e+00	k_t = [4, 4, 2, 4]		
Generation 270:	Best = 7.181223e+02	FES = 27100	μ_F = 0.513	μ_{CR}
= 0.089	Diversity = 5.0413e+00	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 7.180216e+02	FES = 28100	μ_F = 0.490	μ_{CR}
= 0.077	Diversity = 4.9992e+00	k_t = [3, 4, 3, 3]		
Generation 290:	Best = 7.177566e+02	FES = 29100	μ_F = 0.437	μ_{CR}
= 0.082	Diversity = 4.9656e+00	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 7.171002e+02	FES = 30100	μ_F = 0.474	μ_{CR}
= 0.067	Diversity = 4.9594e+00	k_t = [4, 3, 3, 3]		
Generation 310:	Best = 7.167826e+02	FES = 31100	μ_F = 0.509	μ_{CR}
= 0.078	Diversity = 4.9455e+00	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 7.162891e+02	FES = 32100	μ_F = 0.548	μ_{CR}
= 0.079	Diversity = 4.9074e+00	k_t = [2, 2, 2, 2]		
Generation 330:	Best = 7.162891e+02	FES = 33100	μ_F = 0.476	μ_{CR}
= 0.083	Diversity = 4.8633e+00	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 7.162891e+02	FES = 34100	μ_F = 0.497	μ_{CR}
= 0.090	Diversity = 4.8582e+00	k_t = [3, 3, 3, 2]		
Generation 350:	Best = 7.162891e+02	FES = 35100	μ_F = 0.444	μ_{CR}
= 0.089	Diversity = 4.8566e+00	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 7.161921e+02	FES = 36100	μ_F = 0.443	μ_{CR}
= 0.088	Diversity = 4.8388e+00	k_t = [4, 4, 4, 3]		
Generation 370:	Best = 7.161921e+02	FES = 37100	μ_F = 0.404	μ_{CR}
= 0.089	Diversity = 4.8391e+00	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 7.161813e+02	FES = 38100	μ_F = 0.467	μ_{CR}
= 0.072	Diversity = 4.8312e+00	k_t = [2, 4, 3, 4]		
Generation 390:	Best = 7.161813e+02	FES = 39100	μ_F = 0.516	μ_{CR}
= 0.067	Diversity = 4.8300e+00	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 7.161813e+02	FES = 40100	μ_F = 0.501	μ_{CR}
= 0.059	Diversity = 4.8163e+00	k_t = [3, 2, 4, 3]		
Generation 410:	Best = 7.161403e+02	FES = 41100	μ_F = 0.522	μ_{CR}
= 0.068	Diversity = 4.7575e+00	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 7.161403e+02	FES = 42100	μ_F = 0.549	μ_{CR}
= 0.062	Diversity = 4.7311e+00	k_t = [4, 3, 2, 4]		
Generation 430:	Best = 7.161403e+02	FES = 43100	μ_F = 0.410	μ_{CR}
= 0.085	Diversity = 4.7306e+00	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 7.153862e+02	FES = 44100	μ_F = 0.415	μ_{CR}
= 0.092	Diversity = 4.7064e+00	k_t = [3, 3, 3, 4]		
Generation 450:	Best = 7.153862e+02	FES = 45100	μ_F = 0.388	μ_{CR}
= 0.085	Diversity = 4.7065e+00	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 7.153686e+02	FES = 46100	μ_F = 0.399	μ_{CR}
= 0.087	Diversity = 4.7053e+00	k_t = [2, 2, 4, 2]		

Generation 470: Best = 7.139183e+02	FES = 47100	$\mu_F = 0.447$	μ_{CR}
= 0.110 Diversity = 4.7071e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 7.139183e+02	FES = 48100	$\mu_F = 0.537$	μ_{CR}
= 0.116 Diversity = 4.7018e+00	$k_t = [4, 2, 4, 4]$		
Generation 490: Best = 7.139183e+02	FES = 49100	$\mu_F = 0.475$	μ_{CR}
= 0.116 Diversity = 4.6826e+00	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 7.139183e+02	FES = 50100	$\mu_F = 0.453$	μ_{CR}
= 0.126 Diversity = 4.6829e+00	$k_t = [3, 4, 3, 2]$		
Generation 510: Best = 7.139183e+02	FES = 51100	$\mu_F = 0.443$	μ_{CR}
= 0.113 Diversity = 4.6694e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 7.139183e+02	FES = 52100	$\mu_F = 0.432$	μ_{CR}
= 0.111 Diversity = 4.6696e+00	$k_t = [4, 3, 3, 3]$		
Generation 530: Best = 7.139183e+02	FES = 53100	$\mu_F = 0.355$	μ_{CR}
= 0.107 Diversity = 4.6596e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 7.139183e+02	FES = 54100	$\mu_F = 0.279$	μ_{CR}
= 0.082 Diversity = 4.6528e+00	$k_t = [4, 3, 3, 3]$		
Generation 550: Best = 7.131686e+02	FES = 55100	$\mu_F = 0.354$	μ_{CR}
= 0.094 Diversity = 4.6585e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 7.130908e+02	FES = 56100	$\mu_F = 0.283$	μ_{CR}
= 0.098 Diversity = 4.6575e+00	$k_t = [2, 3, 2, 3]$		
Generation 570: Best = 7.130908e+02	FES = 57100	$\mu_F = 0.260$	μ_{CR}
= 0.121 Diversity = 4.6394e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 7.130908e+02	FES = 58100	$\mu_F = 0.273$	μ_{CR}
= 0.125 Diversity = 4.6514e+00	$k_t = [2, 2, 3, 4]$		
Generation 590: Best = 7.127154e+02	FES = 59100	$\mu_F = 0.250$	μ_{CR}
= 0.101 Diversity = 4.6512e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 7.127154e+02	FES = 60100	$\mu_F = 0.245$	μ_{CR}
= 0.109 Diversity = 4.6463e+00	$k_t = [4, 3, 4, 2]$		
Generation 610: Best = 7.124979e+02	FES = 61100	$\mu_F = 0.274$	μ_{CR}
= 0.120 Diversity = 4.6441e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 7.124979e+02	FES = 62100	$\mu_F = 0.335$	μ_{CR}
= 0.113 Diversity = 4.6450e+00	$k_t = [3, 4, 2, 2]$		
Generation 630: Best = 7.124979e+02	FES = 63100	$\mu_F = 0.316$	μ_{CR}
= 0.084 Diversity = 4.6390e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 7.124979e+02	FES = 64100	$\mu_F = 0.274$	μ_{CR}
= 0.090 Diversity = 4.6393e+00	$k_t = [2, 2, 3, 3]$		
Generation 650: Best = 7.124979e+02	FES = 65100	$\mu_F = 0.273$	μ_{CR}
= 0.077 Diversity = 4.6379e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 7.124680e+02	FES = 66100	$\mu_F = 0.233$	μ_{CR}
= 0.080 Diversity = 4.6370e+00	$k_t = [2, 3, 3, 3]$		
Generation 670: Best = 7.124621e+02	FES = 67100	$\mu_F = 0.273$	μ_{CR}
= 0.091 Diversity = 4.6331e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 7.124359e+02	FES = 68100	$\mu_F = 0.210$	μ_{CR}
= 0.090 Diversity = 4.6322e+00	$k_t = [3, 4, 3, 3]$		
Generation 690: Best = 7.124359e+02	FES = 69100	$\mu_F = 0.212$	μ_{CR}
= 0.085 Diversity = 4.6297e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 7.124244e+02	FES = 70100	$\mu_F = 0.186$	μ_{CR}
= 0.100 Diversity = 4.6261e+00	$k_t = [3, 3, 3, 2]$		
Generation 710: Best = 7.124163e+02	FES = 71100	$\mu_F = 0.133$	μ_{CR}

= 0.095	Diversity = 4.6249e+00	k_t = [1, 1, 1, 1]	μ_F = 0.135	μ_{CR}
Generation 720:	Best = 7.123792e+02	FES = 72100		
= 0.126	Diversity = 4.6222e+00	k_t = [3, 2, 4, 3]	μ_F = 0.175	μ_{CR}
Generation 730:	Best = 7.123792e+02	FES = 73100		
= 0.120	Diversity = 4.6132e+00	k_t = [1, 1, 1, 1]	μ_F = 0.123	μ_{CR}
Generation 740:	Best = 7.122640e+02	FES = 74100		
= 0.087	Diversity = 4.6098e+00	k_t = [3, 2, 2, 2]	μ_F = 0.136	μ_{CR}
Generation 750:	Best = 7.122640e+02	FES = 75100		
= 0.091	Diversity = 4.6081e+00	k_t = [1, 1, 1, 1]	μ_F = 0.222	μ_{CR}
Generation 760:	Best = 7.122096e+02	FES = 76100		
= 0.089	Diversity = 4.6063e+00	k_t = [4, 2, 4, 4]	μ_F = 0.215	μ_{CR}
Generation 770:	Best = 7.122096e+02	FES = 77100		
= 0.094	Diversity = 4.6136e+00	k_t = [1, 1, 1, 1]	μ_F = 0.238	μ_{CR}
Generation 780:	Best = 7.122096e+02	FES = 78100		
= 0.087	Diversity = 4.6062e+00	k_t = [2, 4, 2, 3]	μ_F = 0.224	μ_{CR}
Generation 790:	Best = 7.121915e+02	FES = 79100		
= 0.083	Diversity = 4.6060e+00	k_t = [1, 1, 1, 1]	μ_F = 0.281	μ_{CR}
Generation 800:	Best = 7.120625e+02	FES = 80100		
= 0.070	Diversity = 4.6045e+00	k_t = [3, 2, 4, 4]	μ_F = 0.249	μ_{CR}
Generation 810:	Best = 7.120625e+02	FES = 81100		
= 0.079	Diversity = 4.6049e+00	k_t = [1, 1, 1, 1]	μ_F = 0.184	μ_{CR}
Generation 820:	Best = 7.120331e+02	FES = 82100		
= 0.084	Diversity = 4.6036e+00	k_t = [2, 2, 3, 3]	μ_F = 0.198	μ_{CR}
Generation 830:	Best = 7.120331e+02	FES = 83100		
= 0.077	Diversity = 4.6027e+00	k_t = [1, 1, 1, 1]	μ_F = 0.257	μ_{CR}
Generation 840:	Best = 7.120331e+02	FES = 84100		
= 0.083	Diversity = 4.6074e+00	k_t = [3, 4, 2, 4]	μ_F = 0.234	μ_{CR}
Generation 850:	Best = 7.120331e+02	FES = 85100		
= 0.068	Diversity = 4.6102e+00	k_t = [1, 1, 1, 1]	μ_F = 0.197	μ_{CR}
Generation 860:	Best = 7.120331e+02	FES = 86100		
= 0.069	Diversity = 4.6119e+00	k_t = [4, 3, 4, 3]	μ_F = 0.217	μ_{CR}
Generation 870:	Best = 7.120331e+02	FES = 87100		
= 0.063	Diversity = 4.6106e+00	k_t = [1, 1, 1, 1]	μ_F = 0.176	μ_{CR}
Generation 880:	Best = 7.120331e+02	FES = 88100		
= 0.090	Diversity = 4.6116e+00	k_t = [2, 3, 4, 3]	μ_F = 0.216	μ_{CR}
Generation 890:	Best = 7.120331e+02	FES = 89100		
= 0.087	Diversity = 4.6117e+00	k_t = [1, 1, 1, 1]	μ_F = 0.232	μ_{CR}
Generation 900:	Best = 7.120331e+02	FES = 90100		
= 0.093	Diversity = 4.5987e+00	k_t = [2, 3, 4, 4]	μ_F = 0.210	μ_{CR}
Generation 910:	Best = 7.120331e+02	FES = 91100		
= 0.075	Diversity = 4.5983e+00	k_t = [1, 1, 1, 1]	μ_F = 0.256	μ_{CR}
Generation 920:	Best = 7.120331e+02	FES = 92100		
= 0.073	Diversity = 4.5984e+00	k_t = [4, 2, 3, 2]	μ_F = 0.249	μ_{CR}
Generation 930:	Best = 7.120331e+02	FES = 93100		
= 0.092	Diversity = 4.5995e+00	k_t = [1, 1, 1, 1]	μ_F = 0.184	μ_{CR}
Generation 940:	Best = 7.120331e+02	FES = 94100		
= 0.093	Diversity = 4.5980e+00	k_t = [3, 2, 2, 3]	μ_F = 0.228	μ_{CR}
Generation 950:	Best = 7.120331e+02	FES = 95100		
= 0.079	Diversity = 4.5992e+00	k_t = [1, 1, 1, 1]		


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Generation 960: Best = 7.120331e+02  FES = 96100   $\mu_F$  = 0.125   $\mu_{CR}$ 
= 0.096  Diversity = 4.5991e+00  k_t = [3, 4, 4, 2]
Generation 970: Best = 7.120331e+02  FES = 97100   $\mu_F$  = 0.129   $\mu_{CR}$ 
= 0.085  Diversity = 4.5987e+00  k_t = [1, 1, 1, 1]
Generation 980: Best = 7.120331e+02  FES = 98100   $\mu_F$  = 0.120   $\mu_{CR}$ 
= 0.089  Diversity = 4.5977e+00  k_t = [4, 3, 4, 2]
Generation 990: Best = 7.120331e+02  FES = 99100   $\mu_F$  = 0.090   $\mu_{CR}$ 
= 0.077  Diversity = 4.5943e+00  k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 7.120259e+02

Total Function Evals = 100000

Total Successful Updates = 3727

```

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```

Results:

Best Fitness : 7.120259e+02

Global Optimum : 7.000000e+02

Error (|best-optimum|) : 1.202587e+01

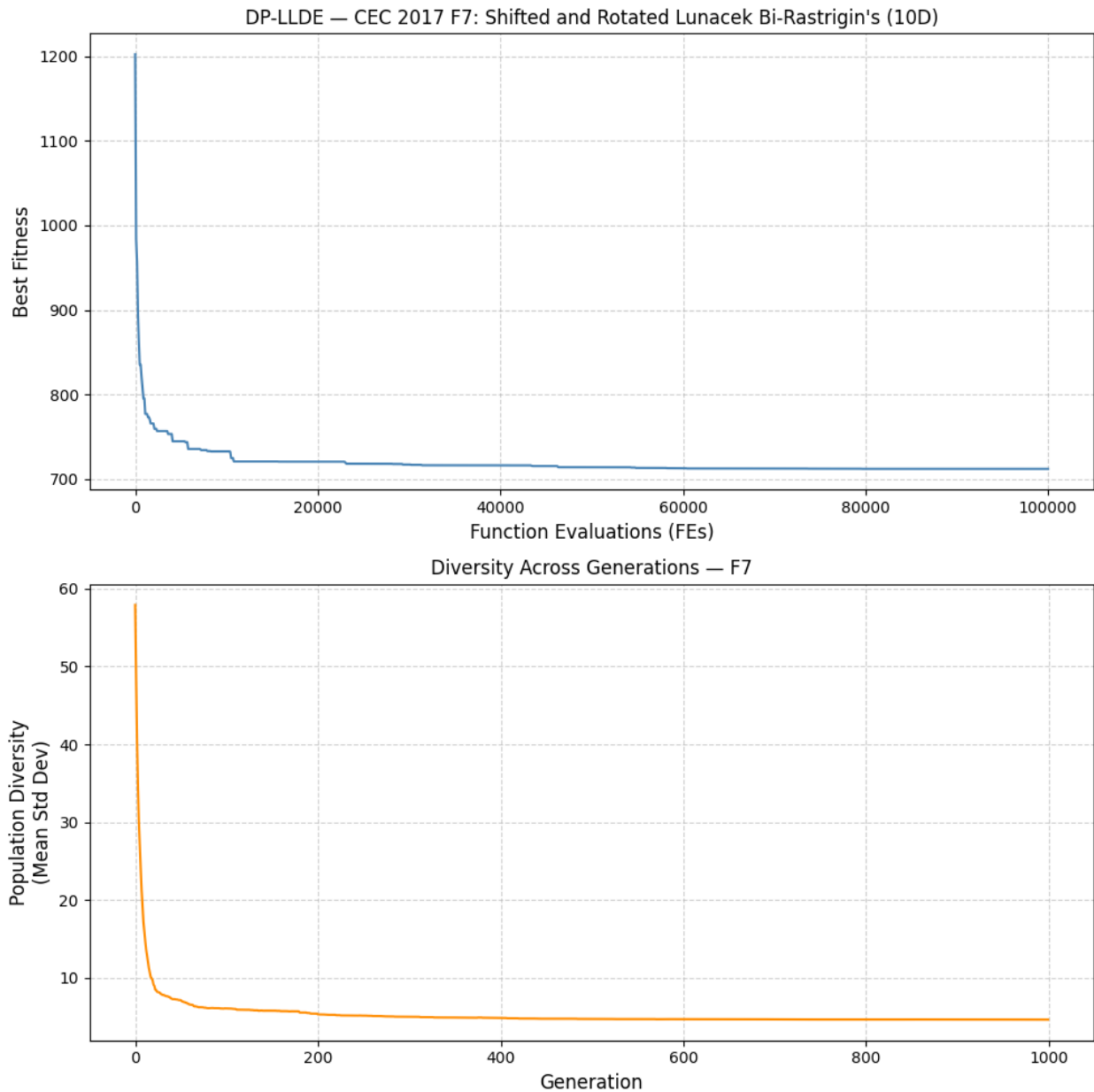
Best Solution (5 dim) : [-20.21392177 15.34441907 -21.98717224 -
7.6467485 10.55163608]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F7.png

```
# =====
# F8 – Shifted and Rotated Non-Continuous Rastrigin
# =====
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and
Rotated Non-Continuous Rastrigin's")
```

```
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=====
DP-LLDE on CEC 2017 F8 – Single Run
=====
```

```

=====
Function      : Shifted and Rotated Non-Continuous Rastrigin's (10D)
Bounds       : [-100, 100]
Global optimum : 800.0
Initial  $D^*(0)$  = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 9.505966e+02
Generation 10: Best = 8.531139e+02 FES = 1100  $\mu_F$  = 0.562  $\mu_{CR}$ 
= 0.351 Diversity = 2.7522e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 8.475953e+02 FES = 2100  $\mu_F$  = 0.604  $\mu_{CR}$ 
= 0.334 Diversity = 2.1184e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 8.286643e+02 FES = 3100  $\mu_F$  = 0.629  $\mu_{CR}$ 
= 0.276 Diversity = 1.9001e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 8.251394e+02 FES = 4100  $\mu_F$  = 0.680  $\mu_{CR}$ 
= 0.253 Diversity = 1.7988e+01  $k_t$  = [3, 3, 4, 4]
Generation 50: Best = 8.171760e+02 FES = 5100  $\mu_F$  = 0.745  $\mu_{CR}$ 
= 0.202 Diversity = 1.7878e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 8.171760e+02 FES = 6100  $\mu_F$  = 0.752  $\mu_{CR}$ 
= 0.160 Diversity = 1.7264e+01  $k_t$  = [2, 4, 2, 2]
Generation 70: Best = 8.171760e+02 FES = 7100  $\mu_F$  = 0.773  $\mu_{CR}$ 
= 0.134 Diversity = 1.7093e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 8.171760e+02 FES = 8100  $\mu_F$  = 0.791  $\mu_{CR}$ 
= 0.125 Diversity = 1.6789e+01  $k_t$  = [4, 2, 2, 3]
Generation 90: Best = 8.171760e+02 FES = 9100  $\mu_F$  = 0.777  $\mu_{CR}$ 
= 0.123 Diversity = 1.6784e+01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 8.171760e+02 FES = 10100  $\mu_F$  = 0.801  $\mu_{CR}$ 
= 0.119 Diversity = 1.6742e+01  $k_t$  = [3, 4, 3, 2]
Generation 110: Best = 8.170666e+02 FES = 11100  $\mu_F$  = 0.764  $\mu_{CR}$ 
= 0.130 Diversity = 1.6610e+01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 8.170666e+02 FES = 12100  $\mu_F$  = 0.809  $\mu_{CR}$ 
= 0.112 Diversity = 1.6675e+01  $k_t$  = [4, 3, 2, 4]
Generation 130: Best = 8.169965e+02 FES = 13100  $\mu_F$  = 0.724  $\mu_{CR}$ 
= 0.096 Diversity = 1.6413e+01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 8.169965e+02 FES = 14100  $\mu_F$  = 0.674  $\mu_{CR}$ 
= 0.091 Diversity = 1.6395e+01  $k_t$  = [2, 4, 4, 2]
Generation 150: Best = 8.120891e+02 FES = 15100  $\mu_F$  = 0.698  $\mu_{CR}$ 
= 0.090 Diversity = 1.5904e+01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 8.106361e+02 FES = 16100  $\mu_F$  = 0.680  $\mu_{CR}$ 
= 0.073 Diversity = 1.5644e+01  $k_t$  = [3, 3, 3, 4]
Generation 170: Best = 8.106361e+02 FES = 17100  $\mu_F$  = 0.671  $\mu_{CR}$ 
= 0.075 Diversity = 1.5772e+01  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 8.099386e+02 FES = 18100  $\mu_F$  = 0.664  $\mu_{CR}$ 
= 0.076 Diversity = 1.5646e+01  $k_t$  = [3, 2, 3, 3]
Generation 190: Best = 8.098567e+02 FES = 19100  $\mu_F$  = 0.636  $\mu_{CR}$ 
= 0.096 Diversity = 1.5647e+01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 8.098567e+02 FES = 20100  $\mu_F$  = 0.635  $\mu_{CR}$ 
= 0.113 Diversity = 1.5623e+01  $k_t$  = [4, 2, 4, 2]
Generation 210: Best = 8.096839e+02 FES = 21100  $\mu_F$  = 0.688  $\mu_{CR}$ 
= 0.090 Diversity = 1.5564e+01  $k_t$  = [1, 1, 1, 1]
Generation 220: Best = 8.096839e+02 FES = 22100  $\mu_F$  = 0.669  $\mu_{CR}$ 

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= 0.090	Diversity = 1.5522e+01	k_t = [2, 4, 4, 3]		
Generation 230:	Best = 8.096839e+02	FES = 23100	$\mu_F = 0.653$	μ_{CR}
= 0.078	Diversity = 1.5485e+01	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 8.096839e+02	FES = 24100	$\mu_F = 0.678$	μ_{CR}
= 0.073	Diversity = 1.5386e+01	k_t = [3, 4, 3, 3]		
Generation 250:	Best = 8.096839e+02	FES = 25100	$\mu_F = 0.635$	μ_{CR}
= 0.079	Diversity = 1.5314e+01	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 8.094992e+02	FES = 26100	$\mu_F = 0.669$	μ_{CR}
= 0.089	Diversity = 1.5405e+01	k_t = [4, 4, 4, 2]		
Generation 270:	Best = 8.094992e+02	FES = 27100	$\mu_F = 0.735$	μ_{CR}
= 0.073	Diversity = 1.5338e+01	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 8.094992e+02	FES = 28100	$\mu_F = 0.734$	μ_{CR}
= 0.079	Diversity = 1.5317e+01	k_t = [3, 4, 2, 3]		
Generation 290:	Best = 8.081328e+02	FES = 29100	$\mu_F = 0.665$	μ_{CR}
= 0.087	Diversity = 1.5314e+01	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 8.081328e+02	FES = 30100	$\mu_F = 0.655$	μ_{CR}
= 0.081	Diversity = 1.5543e+01	k_t = [4, 4, 3, 2]		
Generation 310:	Best = 8.074505e+02	FES = 31100	$\mu_F = 0.636$	μ_{CR}
= 0.076	Diversity = 1.5550e+01	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 8.074505e+02	FES = 32100	$\mu_F = 0.682$	μ_{CR}
= 0.080	Diversity = 1.5410e+01	k_t = [2, 4, 3, 2]		
Generation 330:	Best = 8.074505e+02	FES = 33100	$\mu_F = 0.610$	μ_{CR}
= 0.078	Diversity = 1.5314e+01	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 8.073111e+02	FES = 34100	$\mu_F = 0.619$	μ_{CR}
= 0.066	Diversity = 1.5186e+01	k_t = [4, 4, 4, 2]		
Generation 350:	Best = 8.073111e+02	FES = 35100	$\mu_F = 0.670$	μ_{CR}
= 0.083	Diversity = 1.5200e+01	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 8.073111e+02	FES = 36100	$\mu_F = 0.582$	μ_{CR}
= 0.074	Diversity = 1.5130e+01	k_t = [4, 4, 4, 4]		
Generation 370:	Best = 8.073111e+02	FES = 37100	$\mu_F = 0.690$	μ_{CR}
= 0.082	Diversity = 1.5111e+01	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 8.073051e+02	FES = 38100	$\mu_F = 0.721$	μ_{CR}
= 0.078	Diversity = 1.5203e+01	k_t = [3, 2, 2, 3]		
Generation 390:	Best = 8.069111e+02	FES = 39100	$\mu_F = 0.682$	μ_{CR}
= 0.073	Diversity = 1.5152e+01	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 8.069111e+02	FES = 40100	$\mu_F = 0.614$	μ_{CR}
= 0.079	Diversity = 1.5047e+01	k_t = [4, 3, 4, 4]		
Generation 410:	Best = 8.069111e+02	FES = 41100	$\mu_F = 0.583$	μ_{CR}
= 0.076	Diversity = 1.5063e+01	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 8.069111e+02	FES = 42100	$\mu_F = 0.561$	μ_{CR}
= 0.084	Diversity = 1.4890e+01	k_t = [2, 3, 4, 2]		
Generation 430:	Best = 8.068736e+02	FES = 43100	$\mu_F = 0.517$	μ_{CR}
= 0.097	Diversity = 1.4892e+01	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 8.064757e+02	FES = 44100	$\mu_F = 0.541$	μ_{CR}
= 0.101	Diversity = 1.4717e+01	k_t = [2, 4, 3, 3]		
Generation 450:	Best = 8.064757e+02	FES = 45100	$\mu_F = 0.559$	μ_{CR}
= 0.115	Diversity = 1.4719e+01	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 8.064296e+02	FES = 46100	$\mu_F = 0.574$	μ_{CR}
= 0.094	Diversity = 1.4681e+01	k_t = [3, 3, 3, 2]		

Generation 470: Best = 8.035969e+02	FES = 47100	$\mu_F = 0.499$	μ_{CR}
= 0.086 Diversity = 1.4532e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 8.035969e+02	FES = 48100	$\mu_F = 0.466$	μ_{CR}
= 0.095 Diversity = 1.4518e+01	$k_t = [3, 4, 4, 2]$		
Generation 490: Best = 8.035969e+02	FES = 49100	$\mu_F = 0.438$	μ_{CR}
= 0.102 Diversity = 1.4495e+01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 8.035969e+02	FES = 50100	$\mu_F = 0.441$	μ_{CR}
= 0.074 Diversity = 1.4497e+01	$k_t = [2, 4, 4, 2]$		
Generation 510: Best = 8.035969e+02	FES = 51100	$\mu_F = 0.298$	μ_{CR}
= 0.088 Diversity = 1.4431e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 8.035969e+02	FES = 52100	$\mu_F = 0.320$	μ_{CR}
= 0.100 Diversity = 1.4429e+01	$k_t = [3, 4, 2, 4]$		
Generation 530: Best = 8.035969e+02	FES = 53100	$\mu_F = 0.242$	μ_{CR}
= 0.118 Diversity = 1.4417e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 8.035969e+02	FES = 54100	$\mu_F = 0.263$	μ_{CR}
= 0.108 Diversity = 1.4398e+01	$k_t = [3, 2, 3, 3]$		
Generation 550: Best = 8.035969e+02	FES = 55100	$\mu_F = 0.265$	μ_{CR}
= 0.079 Diversity = 1.4382e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 8.035810e+02	FES = 56100	$\mu_F = 0.321$	μ_{CR}
= 0.087 Diversity = 1.4294e+01	$k_t = [3, 3, 3, 4]$		
Generation 570: Best = 8.035810e+02	FES = 57100	$\mu_F = 0.229$	μ_{CR}
= 0.079 Diversity = 1.4286e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 8.035810e+02	FES = 58100	$\mu_F = 0.224$	μ_{CR}
= 0.092 Diversity = 1.4277e+01	$k_t = [4, 2, 3, 2]$		
Generation 590: Best = 8.035810e+02	FES = 59100	$\mu_F = 0.214$	μ_{CR}
= 0.082 Diversity = 1.4251e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 8.035810e+02	FES = 60100	$\mu_F = 0.215$	μ_{CR}
= 0.090 Diversity = 1.4177e+01	$k_t = [3, 4, 4, 2]$		
Generation 610: Best = 8.034288e+02	FES = 61100	$\mu_F = 0.294$	μ_{CR}
= 0.090 Diversity = 1.4160e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 8.034288e+02	FES = 62100	$\mu_F = 0.321$	μ_{CR}
= 0.093 Diversity = 1.4147e+01	$k_t = [4, 2, 2, 2]$		
Generation 630: Best = 8.033658e+02	FES = 63100	$\mu_F = 0.372$	μ_{CR}
= 0.089 Diversity = 1.4142e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 8.033658e+02	FES = 64100	$\mu_F = 0.391$	μ_{CR}
= 0.084 Diversity = 1.4090e+01	$k_t = [2, 3, 4, 4]$		
Generation 650: Best = 8.033658e+02	FES = 65100	$\mu_F = 0.396$	μ_{CR}
= 0.078 Diversity = 1.4066e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 8.033658e+02	FES = 66100	$\mu_F = 0.353$	μ_{CR}
= 0.073 Diversity = 1.3888e+01	$k_t = [2, 3, 2, 4]$		
Generation 670: Best = 8.033658e+02	FES = 67100	$\mu_F = 0.346$	μ_{CR}
= 0.067 Diversity = 1.3891e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 8.032753e+02	FES = 68100	$\mu_F = 0.315$	μ_{CR}
= 0.085 Diversity = 1.3884e+01	$k_t = [4, 4, 2, 2]$		
Generation 690: Best = 8.032753e+02	FES = 69100	$\mu_F = 0.276$	μ_{CR}
= 0.090 Diversity = 1.3878e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 8.032753e+02	FES = 70100	$\mu_F = 0.294$	μ_{CR}
= 0.089 Diversity = 1.3828e+01	$k_t = [2, 2, 2, 3]$		
Generation 710: Best = 8.031921e+02	FES = 71100	$\mu_F = 0.314$	μ_{CR}

= 0.087	Diversity = 1.3827e+01	k_t = [1, 1, 1, 1]	μ_F = 0.253	μ_{CR}
Generation 720:	Best = 8.031921e+02	FES = 72100		
= 0.081	Diversity = 1.3837e+01	k_t = [2, 2, 2, 3]	μ_F = 0.199	μ_{CR}
Generation 730:	Best = 8.031921e+02	FES = 73100		
= 0.073	Diversity = 1.3830e+01	k_t = [1, 1, 1, 1]	μ_F = 0.183	μ_{CR}
Generation 740:	Best = 8.031921e+02	FES = 74100		
= 0.068	Diversity = 1.3824e+01	k_t = [2, 2, 2, 2]	μ_F = 0.180	μ_{CR}
Generation 750:	Best = 8.031921e+02	FES = 75100		
= 0.090	Diversity = 1.3824e+01	k_t = [1, 1, 1, 1]	μ_F = 0.179	μ_{CR}
Generation 760:	Best = 8.031921e+02	FES = 76100		
= 0.108	Diversity = 1.3817e+01	k_t = [3, 4, 4, 3]	μ_F = 0.169	μ_{CR}
Generation 770:	Best = 8.031704e+02	FES = 77100		
= 0.101	Diversity = 1.3753e+01	k_t = [1, 1, 1, 1]	μ_F = 0.253	μ_{CR}
Generation 780:	Best = 8.031508e+02	FES = 78100		
= 0.084	Diversity = 1.3712e+01	k_t = [4, 3, 4, 4]	μ_F = 0.274	μ_{CR}
Generation 790:	Best = 8.031503e+02	FES = 79100		
= 0.082	Diversity = 1.3708e+01	k_t = [1, 1, 1, 1]	μ_F = 0.251	μ_{CR}
Generation 800:	Best = 8.031503e+02	FES = 80100		
= 0.058	Diversity = 1.3662e+01	k_t = [4, 4, 2, 3]	μ_F = 0.232	μ_{CR}
Generation 810:	Best = 8.031503e+02	FES = 81100		
= 0.073	Diversity = 1.3661e+01	k_t = [1, 1, 1, 1]	μ_F = 0.302	μ_{CR}
Generation 820:	Best = 8.031503e+02	FES = 82100		
= 0.066	Diversity = 1.3661e+01	k_t = [2, 2, 2, 2]	μ_F = 0.272	μ_{CR}
Generation 830:	Best = 8.031501e+02	FES = 83100		
= 0.080	Diversity = 1.3623e+01	k_t = [1, 1, 1, 1]	μ_F = 0.204	μ_{CR}
Generation 840:	Best = 8.031501e+02	FES = 84100		
= 0.109	Diversity = 1.3621e+01	k_t = [2, 2, 3, 2]	μ_F = 0.182	μ_{CR}
Generation 850:	Best = 8.031352e+02	FES = 85100		
= 0.084	Diversity = 1.3622e+01	k_t = [1, 1, 1, 1]	μ_F = 0.221	μ_{CR}
Generation 860:	Best = 8.031027e+02	FES = 86100		
= 0.092	Diversity = 1.3620e+01	k_t = [2, 2, 4, 3]	μ_F = 0.197	μ_{CR}
Generation 870:	Best = 8.030723e+02	FES = 87100		
= 0.080	Diversity = 1.3614e+01	k_t = [1, 1, 1, 1]	μ_F = 0.164	μ_{CR}
Generation 880:	Best = 8.030723e+02	FES = 88100		
= 0.074	Diversity = 1.3615e+01	k_t = [4, 2, 4, 3]	μ_F = 0.215	μ_{CR}
Generation 890:	Best = 8.030723e+02	FES = 89100		
= 0.093	Diversity = 1.3585e+01	k_t = [1, 1, 1, 1]	μ_F = 0.214	μ_{CR}
Generation 900:	Best = 8.030723e+02	FES = 90100		
= 0.100	Diversity = 1.3572e+01	k_t = [4, 2, 2, 3]	μ_F = 0.179	μ_{CR}
Generation 910:	Best = 8.030723e+02	FES = 91100		
= 0.084	Diversity = 1.3571e+01	k_t = [1, 1, 1, 1]	μ_F = 0.233	μ_{CR}
Generation 920:	Best = 8.030723e+02	FES = 92100		
= 0.077	Diversity = 1.3456e+01	k_t = [4, 3, 2, 4]	μ_F = 0.217	μ_{CR}
Generation 930:	Best = 8.030705e+02	FES = 93100		
= 0.070	Diversity = 1.3457e+01	k_t = [1, 1, 1, 1]	μ_F = 0.269	μ_{CR}
Generation 940:	Best = 8.030705e+02	FES = 94100		
= 0.066	Diversity = 1.3352e+01	k_t = [2, 2, 2, 4]	μ_F = 0.346	μ_{CR}
Generation 950:	Best = 8.030705e+02	FES = 95100		
= 0.069	Diversity = 1.3349e+01	k_t = [1, 1, 1, 1]		

Generation 960: Best = 8.030705e+02 FES = 96100 μ_F = 0.272 μ_{CR}
= 0.074 Diversity = 1.3348e+01 k_t = [2, 2, 2, 2]
Generation 970: Best = 8.030705e+02 FES = 97100 μ_F = 0.265 μ_{CR}
= 0.071 Diversity = 1.3343e+01 k_t = [1, 1, 1, 1]
Generation 980: Best = 8.030705e+02 FES = 98100 μ_F = 0.254 μ_{CR}
= 0.071 Diversity = 1.3342e+01 k_t = [2, 4, 2, 4]
Generation 990: Best = 8.030705e+02 FES = 99100 μ_F = 0.241 μ_{CR}
= 0.061 Diversity = 1.3340e+01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 8.030705e+02

Total Function Evals = 100000

Total Successful Updates = 3441

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Results:

Best Fitness : 8.030705e+02

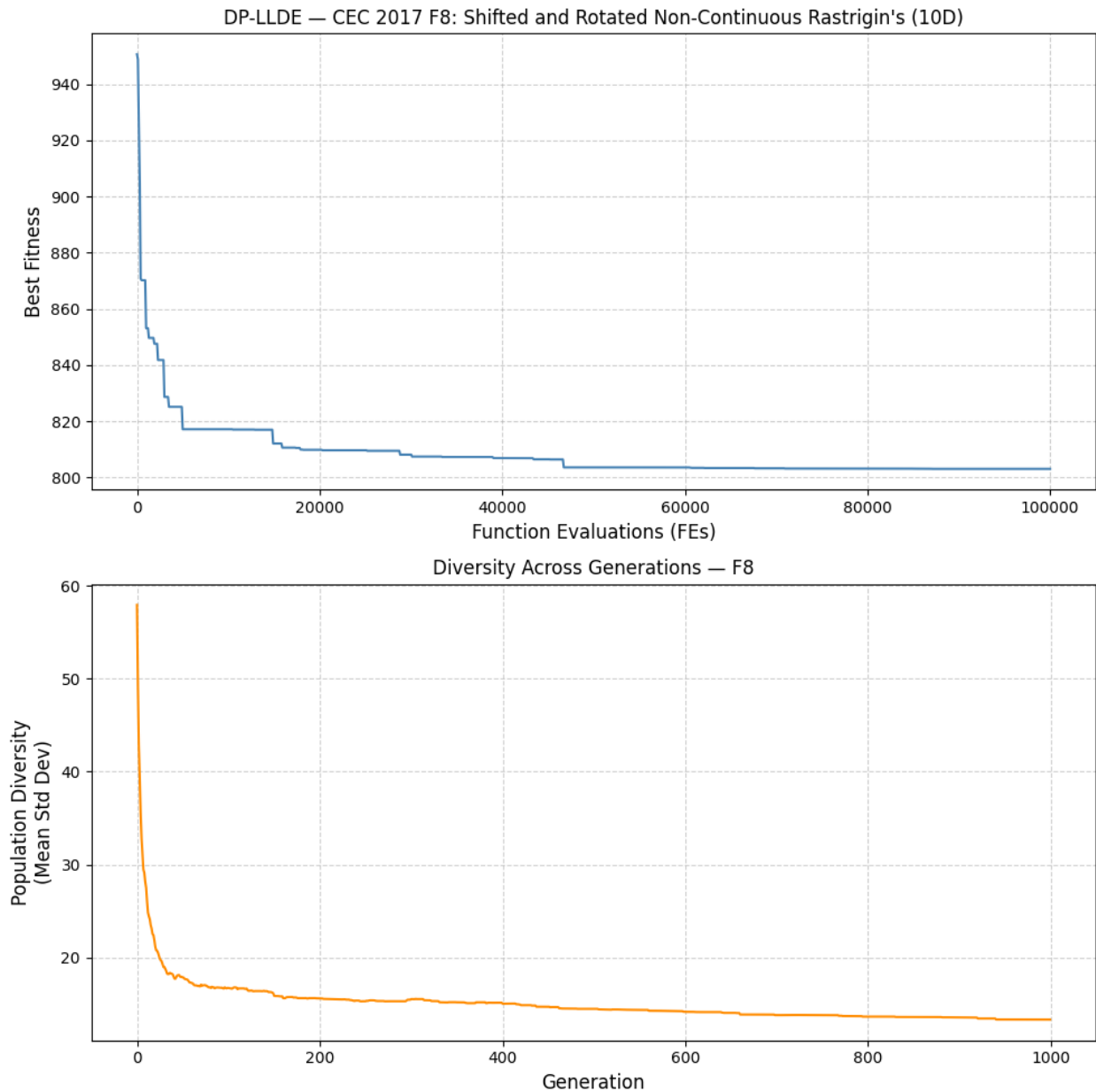
Global Optimum : 8.000000e+02

Error (|best-optimum|) : 3.070500e+00

Best Solution (5 dim) : [32.02582565 -55.93603147 36.08865964 -
47.69840492 -33.86572154]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F8.png

```
# =====
# F9 – Shifted and Rotated Levy
# =====
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and
Rotated Levy')
```

```
=====
=====
DP-LLDE on CEC 2017 F9 – Single Run
=====
```


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Function : Shifted and Rotated Levy (10D)

Bounds : [-100, 100]

Global optimum : 900.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = $5.7928e+00$

Generation 0: Best Fitness = $6.112557e+03$

Generation 10: Best = $1.141316e+03$ FES = 1100 $\mu_F = 0.567$ $\mu_{CR} = 0.350$ Diversity = $2.4170e+01$ $k_t = [1, 1, 1, 1]$

Generation 20: Best = $9.319786e+02$ FES = 2100 $\mu_F = 0.618$ $\mu_{CR} = 0.351$ Diversity = $8.4492e+00$ $k_t = [3, 4, 2, 3]$

Generation 30: Best = $9.051376e+02$ FES = 3100 $\mu_F = 0.661$ $\mu_{CR} = 0.343$ Diversity = $2.8887e+00$ $k_t = [1, 1, 1, 1]$

Generation 40: Best = $9.009171e+02$ FES = 4100 $\mu_F = 0.674$ $\mu_{CR} = 0.343$ Diversity = $1.1437e+00$ $k_t = [3, 3, 2, 4]$

Generation 50: Best = $9.003172e+02$ FES = 5100 $\mu_F = 0.664$ $\mu_{CR} = 0.341$ Diversity = $7.5003e-01$ $k_t = [1, 1, 1, 1]$

Generation 60: Best = $9.001825e+02$ FES = 6100 $\mu_F = 0.665$ $\mu_{CR} = 0.336$ Diversity = $4.1028e-01$ $k_t = [2, 2, 4, 4]$

Generation 70: Best = $9.000840e+02$ FES = 7100 $\mu_F = 0.664$ $\mu_{CR} = 0.333$ Diversity = $2.8919e-01$ $k_t = [1, 1, 1, 1]$

Generation 80: Best = $9.000276e+02$ FES = 8100 $\mu_F = 0.661$ $\mu_{CR} = 0.318$ Diversity = $1.5542e-01$ $k_t = [3, 3, 2, 4]$

Generation 90: Best = $9.000092e+02$ FES = 9100 $\mu_F = 0.678$ $\mu_{CR} = 0.313$ Diversity = $8.2945e-02$ $k_t = [1, 1, 1, 1]$

Generation 100: Best = $9.000031e+02$ FES = 10100 $\mu_F = 0.687$ $\mu_{CR} = 0.313$ Diversity = $4.1099e-02$ $k_t = [3, 3, 3, 4]$

Generation 110: Best = $9.000009e+02$ FES = 11100 $\mu_F = 0.696$ $\mu_{CR} = 0.307$ Diversity = $2.3501e-02$ $k_t = [1, 1, 1, 1]$

Generation 120: Best = $9.000003e+02$ FES = 12100 $\mu_F = 0.713$ $\mu_{CR} = 0.310$ Diversity = $1.4525e-02$ $k_t = [2, 2, 2, 2]$

Generation 130: Best = $9.000001e+02$ FES = 13100 $\mu_F = 0.715$ $\mu_{CR} = 0.302$ Diversity = $9.0613e-03$ $k_t = [1, 1, 1, 1]$

Generation 140: Best = $9.000000e+02$ FES = 14100 $\mu_F = 0.706$ $\mu_{CR} = 0.282$ Diversity = $5.2580e-03$ $k_t = [4, 2, 2, 3]$

Generation 150: Best = $9.000000e+02$ FES = 15100 $\mu_F = 0.713$ $\mu_{CR} = 0.262$ Diversity = $3.0322e-03$ $k_t = [1, 1, 1, 1]$

Generation 160: Best = $9.000000e+02$ FES = 16100 $\mu_F = 0.712$ $\mu_{CR} = 0.245$ Diversity = $1.6210e-03$ $k_t = [2, 4, 2, 4]$

Generation 170: Best = $9.000000e+02$ FES = 17100 $\mu_F = 0.712$ $\mu_{CR} = 0.241$ Diversity = $8.9177e-04$ $k_t = [1, 1, 1, 1]$

Generation 180: Best = $9.000000e+02$ FES = 18100 $\mu_F = 0.731$ $\mu_{CR} = 0.235$ Diversity = $5.1676e-04$ $k_t = [3, 3, 3, 2]$

Generation 190: Best = $9.000000e+02$ FES = 19100 $\mu_F = 0.737$ $\mu_{CR} = 0.232$ Diversity = $2.9133e-04$ $k_t = [1, 1, 1, 1]$

Generation 200: Best = $9.000000e+02$ FES = 20100 $\mu_F = 0.745$ $\mu_{CR} = 0.230$ Diversity = $1.5381e-04$ $k_t = [3, 3, 4, 2]$

Generation 210: Best = $9.000000e+02$ FES = 21100 $\mu_F = 0.735$ $\mu_{CR} = 0.230$ Diversity = $8.3186e-05$ $k_t = [1, 1, 1, 1]$

Generation 220: Best = $9.000000e+02$ FES = 22100 $\mu_F = 0.736$ μ_{CR}

= 0.231	Diversity = 4.1786e-05	k_t = [4, 3, 2, 3]	μ_F = 0.739	μ_{CR}
Generation 230:	Best = 9.000000e+02	FES = 23100		
= 0.215	Diversity = 2.6158e-05	k_t = [1, 1, 1, 1]	μ_F = 0.738	μ_{CR}
Generation 240:	Best = 9.000000e+02	FES = 24100		
= 0.209	Diversity = 1.2841e-05	k_t = [4, 3, 4, 4]	μ_F = 0.753	μ_{CR}
Generation 250:	Best = 9.000000e+02	FES = 25100		
= 0.204	Diversity = 7.5545e-06	k_t = [1, 1, 1, 1]	μ_F = 0.750	μ_{CR}
Generation 260:	Best = 9.000000e+02	FES = 26100		
= 0.197	Diversity = 4.8378e-06	k_t = [4, 4, 2, 3]	μ_F = 0.756	μ_{CR}
Generation 270:	Best = 9.000000e+02	FES = 27100		
= 0.201	Diversity = 2.5647e-06	k_t = [1, 1, 1, 1]	μ_F = 0.752	μ_{CR}
Generation 280:	Best = 9.000000e+02	FES = 28100		
= 0.184	Diversity = 1.3136e-06	k_t = [2, 2, 4, 4]	μ_F = 0.751	μ_{CR}
Generation 290:	Best = 9.000000e+02	FES = 29100		
= 0.182	Diversity = 6.7855e-07	k_t = [1, 1, 1, 1]	μ_F = 0.754	μ_{CR}
Generation 300:	Best = 9.000000e+02	FES = 30100		
= 0.167	Diversity = 4.0206e-07	k_t = [2, 3, 3, 2]	μ_F = 0.762	μ_{CR}
Generation 310:	Best = 9.000000e+02	FES = 31100		
= 0.163	Diversity = 2.6220e-07	k_t = [1, 1, 1, 1]	μ_F = 0.776	μ_{CR}
Generation 320:	Best = 9.000000e+02	FES = 32100		
= 0.154	Diversity = 1.8478e-07	k_t = [2, 4, 3, 2]	μ_F = 0.783	μ_{CR}
Generation 330:	Best = 9.000000e+02	FES = 33100		
= 0.157	Diversity = 1.6805e-07	k_t = [1, 1, 1, 1]	μ_F = 0.793	μ_{CR}
Generation 340:	Best = 9.000000e+02	FES = 34100		
= 0.156	Diversity = 1.3667e-07	k_t = [2, 4, 2, 3]	μ_F = 0.805	μ_{CR}
Generation 350:	Best = 9.000000e+02	FES = 35100		
= 0.159	Diversity = 1.1846e-07	k_t = [1, 1, 1, 1]	μ_F = 0.815	μ_{CR}
Generation 360:	Best = 9.000000e+02	FES = 36100		
= 0.164	Diversity = 9.3164e-08	k_t = [2, 4, 3, 2]	μ_F = 0.817	μ_{CR}
Generation 370:	Best = 9.000000e+02	FES = 37100		
= 0.162	Diversity = 9.0797e-08	k_t = [1, 1, 1, 1]	μ_F = 0.817	μ_{CR}
Generation 380:	Best = 9.000000e+02	FES = 38100		
= 0.162	Diversity = 8.4472e-08	k_t = [3, 2, 2, 2]	μ_F = 0.816	μ_{CR}
Generation 390:	Best = 9.000000e+02	FES = 39100		
= 0.158	Diversity = 8.3113e-08	k_t = [1, 1, 1, 1]	μ_F = 0.809	μ_{CR}
Generation 400:	Best = 9.000000e+02	FES = 40100		
= 0.160	Diversity = 8.3414e-08	k_t = [2, 4, 4, 2]	μ_F = 0.813	μ_{CR}
Generation 410:	Best = 9.000000e+02	FES = 41100		
= 0.154	Diversity = 7.7766e-08	k_t = [1, 1, 1, 1]	μ_F = 0.821	μ_{CR}
Generation 420:	Best = 9.000000e+02	FES = 42100		
= 0.159	Diversity = 8.0633e-08	k_t = [3, 2, 3, 2]	μ_F = 0.818	μ_{CR}
Generation 430:	Best = 9.000000e+02	FES = 43100		
= 0.157	Diversity = 8.2702e-08	k_t = [1, 1, 1, 1]	μ_F = 0.818	μ_{CR}
Generation 440:	Best = 9.000000e+02	FES = 44100		
= 0.156	Diversity = 8.0615e-08	k_t = [2, 2, 4, 3]	μ_F = 0.807	μ_{CR}
Generation 450:	Best = 9.000000e+02	FES = 45100		
= 0.150	Diversity = 8.0961e-08	k_t = [1, 1, 1, 1]	μ_F = 0.795	μ_{CR}
Generation 460:	Best = 9.000000e+02	FES = 46100		
= 0.142	Diversity = 7.6253e-08	k_t = [4, 4, 2, 2]		

Generation 470: Best = 9.000000e+02	FES = 47100	$\mu_F = 0.791$	μ_{CR}
= 0.136 Diversity = 7.3570e-08	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 9.000000e+02	FES = 48100	$\mu_F = 0.807$	μ_{CR}
= 0.142 Diversity = 7.1190e-08	$k_t = [4, 4, 3, 3]$		
Generation 490: Best = 9.000000e+02	FES = 49100	$\mu_F = 0.811$	μ_{CR}
= 0.142 Diversity = 6.7661e-08	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 9.000000e+02	FES = 50100	$\mu_F = 0.815$	μ_{CR}
= 0.145 Diversity = 6.5742e-08	$k_t = [2, 4, 2, 3]$		
Generation 510: Best = 9.000000e+02	FES = 51100	$\mu_F = 0.818$	μ_{CR}
= 0.137 Diversity = 7.5212e-08	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 9.000000e+02	FES = 52100	$\mu_F = 0.830$	μ_{CR}
= 0.136 Diversity = 7.9989e-08	$k_t = [3, 2, 4, 2]$		
Generation 530: Best = 9.000000e+02	FES = 53100	$\mu_F = 0.816$	μ_{CR}
= 0.148 Diversity = 8.6617e-08	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 9.000000e+02	FES = 54100	$\mu_F = 0.816$	μ_{CR}
= 0.141 Diversity = 8.8771e-08	$k_t = [3, 2, 3, 4]$		
Generation 550: Best = 9.000000e+02	FES = 55100	$\mu_F = 0.818$	μ_{CR}
= 0.141 Diversity = 9.4579e-08	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 9.000000e+02	FES = 56100	$\mu_F = 0.831$	μ_{CR}
= 0.138 Diversity = 9.8169e-08	$k_t = [2, 4, 4, 3]$		
Generation 570: Best = 9.000000e+02	FES = 57100	$\mu_F = 0.819$	μ_{CR}
= 0.144 Diversity = 9.2612e-08	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 9.000000e+02	FES = 58100	$\mu_F = 0.822$	μ_{CR}
= 0.140 Diversity = 8.9679e-08	$k_t = [3, 2, 4, 3]$		
Generation 590: Best = 9.000000e+02	FES = 59100	$\mu_F = 0.814$	μ_{CR}
= 0.140 Diversity = 8.5170e-08	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 9.000000e+02	FES = 60100	$\mu_F = 0.813$	μ_{CR}
= 0.150 Diversity = 8.1625e-08	$k_t = [2, 2, 2, 4]$		
Generation 610: Best = 9.000000e+02	FES = 61100	$\mu_F = 0.816$	μ_{CR}
= 0.145 Diversity = 8.6237e-08	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 9.000000e+02	FES = 62100	$\mu_F = 0.806$	μ_{CR}
= 0.151 Diversity = 7.7432e-08	$k_t = [2, 2, 4, 3]$		
Generation 630: Best = 9.000000e+02	FES = 63100	$\mu_F = 0.807$	μ_{CR}
= 0.146 Diversity = 7.4995e-08	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 9.000000e+02	FES = 64100	$\mu_F = 0.818$	μ_{CR}
= 0.150 Diversity = 6.9645e-08	$k_t = [4, 2, 2, 3]$		
Generation 650: Best = 9.000000e+02	FES = 65100	$\mu_F = 0.805$	μ_{CR}
= 0.147 Diversity = 6.8643e-08	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 9.000000e+02	FES = 66100	$\mu_F = 0.815$	μ_{CR}
= 0.138 Diversity = 6.4405e-08	$k_t = [4, 2, 2, 3]$		
Generation 670: Best = 9.000000e+02	FES = 67100	$\mu_F = 0.812$	μ_{CR}
= 0.133 Diversity = 6.3589e-08	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 9.000000e+02	FES = 68100	$\mu_F = 0.821$	μ_{CR}
= 0.126 Diversity = 6.2235e-08	$k_t = [4, 2, 4, 4]$		
Generation 690: Best = 9.000000e+02	FES = 69100	$\mu_F = 0.817$	μ_{CR}
= 0.124 Diversity = 6.0669e-08	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 9.000000e+02	FES = 70100	$\mu_F = 0.822$	μ_{CR}
= 0.122 Diversity = 6.2779e-08	$k_t = [2, 3, 3, 3]$		
Generation 710: Best = 9.000000e+02	FES = 71100	$\mu_F = 0.823$	μ_{CR}

= 0.125	Diversity = 6.7530e-08	k_t = [1, 1, 1, 1]	μ_F = 0.814	μ_{CR}
Generation 720:	Best = 9.000000e+02	FES = 72100		
= 0.130	Diversity = 6.4627e-08	k_t = [3, 3, 4, 2]	μ_F = 0.817	μ_{CR}
Generation 730:	Best = 9.000000e+02	FES = 73100		
= 0.134	Diversity = 6.6671e-08	k_t = [1, 1, 1, 1]	μ_F = 0.816	μ_{CR}
Generation 740:	Best = 9.000000e+02	FES = 74100		
= 0.139	Diversity = 6.3698e-08	k_t = [2, 3, 2, 3]	μ_F = 0.808	μ_{CR}
Generation 750:	Best = 9.000000e+02	FES = 75100		
= 0.146	Diversity = 6.7715e-08	k_t = [1, 1, 1, 1]	μ_F = 0.819	μ_{CR}
Generation 760:	Best = 9.000000e+02	FES = 76100		
= 0.146	Diversity = 6.6483e-08	k_t = [4, 3, 2, 2]	μ_F = 0.829	μ_{CR}
Generation 770:	Best = 9.000000e+02	FES = 77100		
= 0.145	Diversity = 7.2184e-08	k_t = [1, 1, 1, 1]	μ_F = 0.818	μ_{CR}
Generation 780:	Best = 9.000000e+02	FES = 78100		
= 0.145	Diversity = 6.7616e-08	k_t = [3, 3, 4, 3]	μ_F = 0.818	μ_{CR}
Generation 790:	Best = 9.000000e+02	FES = 79100		
= 0.149	Diversity = 6.5121e-08	k_t = [1, 1, 1, 1]	μ_F = 0.821	μ_{CR}
Generation 800:	Best = 9.000000e+02	FES = 80100		
= 0.147	Diversity = 6.2231e-08	k_t = [2, 2, 2, 3]	μ_F = 0.819	μ_{CR}
Generation 810:	Best = 9.000000e+02	FES = 81100		
= 0.150	Diversity = 6.6126e-08	k_t = [1, 1, 1, 1]	μ_F = 0.820	μ_{CR}
Generation 820:	Best = 9.000000e+02	FES = 82100		
= 0.148	Diversity = 6.3183e-08	k_t = [2, 4, 3, 2]	μ_F = 0.814	μ_{CR}
Generation 830:	Best = 9.000000e+02	FES = 83100		
= 0.147	Diversity = 6.9282e-08	k_t = [1, 1, 1, 1]	μ_F = 0.816	μ_{CR}
Generation 840:	Best = 9.000000e+02	FES = 84100		
= 0.148	Diversity = 7.0458e-08	k_t = [2, 2, 2, 3]	μ_F = 0.811	μ_{CR}
Generation 850:	Best = 9.000000e+02	FES = 85100		
= 0.152	Diversity = 7.4680e-08	k_t = [1, 1, 1, 1]	μ_F = 0.825	μ_{CR}
Generation 860:	Best = 9.000000e+02	FES = 86100		
= 0.154	Diversity = 7.2130e-08	k_t = [3, 4, 2, 2]	μ_F = 0.829	μ_{CR}
Generation 870:	Best = 9.000000e+02	FES = 87100		
= 0.146	Diversity = 7.1232e-08	k_t = [1, 1, 1, 1]	μ_F = 0.835	μ_{CR}
Generation 880:	Best = 9.000000e+02	FES = 88100		
= 0.144	Diversity = 6.9646e-08	k_t = [4, 4, 3, 3]	μ_F = 0.827	μ_{CR}
Generation 890:	Best = 9.000000e+02	FES = 89100		
= 0.136	Diversity = 6.9374e-08	k_t = [1, 1, 1, 1]	μ_F = 0.814	μ_{CR}
Generation 900:	Best = 9.000000e+02	FES = 90100		
= 0.142	Diversity = 6.4672e-08	k_t = [3, 2, 4, 3]	μ_F = 0.821	μ_{CR}
Generation 910:	Best = 9.000000e+02	FES = 91100		
= 0.146	Diversity = 6.5990e-08	k_t = [1, 1, 1, 1]	μ_F = 0.826	μ_{CR}
Generation 920:	Best = 9.000000e+02	FES = 92100		
= 0.147	Diversity = 6.9661e-08	k_t = [4, 3, 2, 4]	μ_F = 0.831	μ_{CR}
Generation 930:	Best = 9.000000e+02	FES = 93100		
= 0.150	Diversity = 7.6494e-08	k_t = [1, 1, 1, 1]	μ_F = 0.824	μ_{CR}
Generation 940:	Best = 9.000000e+02	FES = 94100		
= 0.153	Diversity = 7.4034e-08	k_t = [2, 4, 3, 3]	μ_F = 0.815	μ_{CR}
Generation 950:	Best = 9.000000e+02	FES = 95100		
= 0.152	Diversity = 7.5858e-08	k_t = [1, 1, 1, 1]		

```

Generation 960: Best = 9.000000e+02 FES = 96100  $\mu_F$  = 0.809  $\mu_{CR}$ 
= 0.152 Diversity = 7.2262e-08  $k_t$  = [2, 4, 3, 4]
Generation 970: Best = 9.000000e+02 FES = 97100  $\mu_F$  = 0.822  $\mu_{CR}$ 
= 0.152 Diversity = 7.1986e-08  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 9.000000e+02 FES = 98100  $\mu_F$  = 0.815  $\mu_{CR}$ 
= 0.152 Diversity = 6.6847e-08  $k_t$  = [4, 3, 3, 4]
Generation 990: Best = 9.000000e+02 FES = 99100  $\mu_F$  = 0.811  $\mu_{CR}$ 
= 0.158 Diversity = 6.4897e-08  $k_t$  = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 9.000000e+02

Total Function Evals = 100000

Total Successful Updates = 44485

```

=====
=====

```

Results:

Best Fitness : 9.000000e+02

Global Optimum : 9.000000e+02

Error (|best-optimum|) : 0.000000e+00

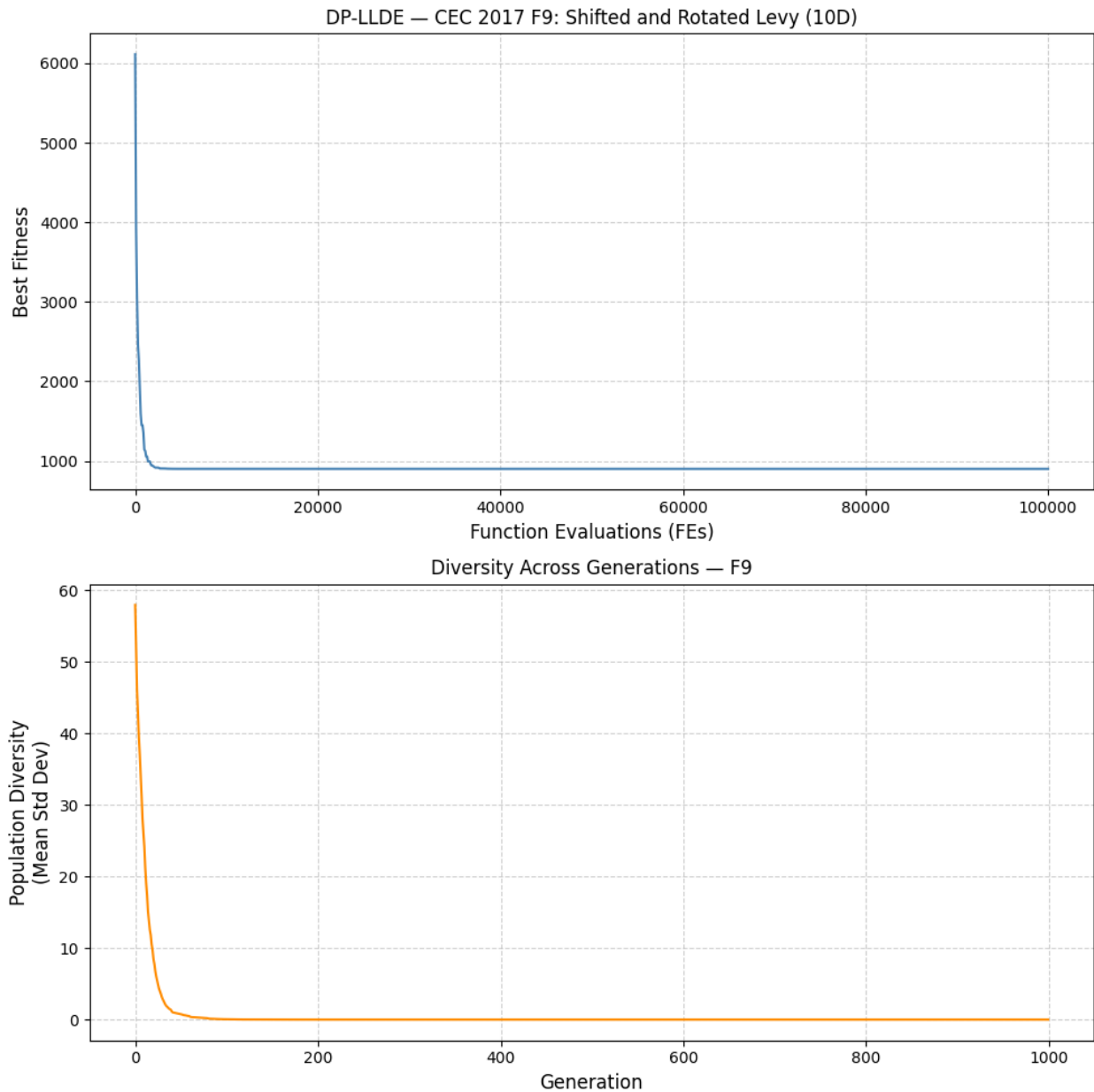
Best Solution (5 dim) : [-23.59314013 0.12032433 20.83264061
52.54992616 -18.22787194]

D^0 (init. diversity): 5.7928e+01

```

=====
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```



Graph saved as convergence_F9.png

```
# =====
# F10 – Shifted and Rotated Schwefel
# =====
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and
Rotated Schwefel's")
```

```
=====
=====
DP-LLDE on CEC 2017 F10 – Single Run
=====
```

=====

Function : Shifted and Rotated Schwefel's (10D)

Bounds : [-100, 100]

Global optimum : 1000.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 3.749402e+03

Generation 10: Best = 3.090221e+03 FES = 1100 $\mu_F = 0.597$ μ_{CR}

= 0.332 Diversity = 4.7508e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 2.657155e+03 FES = 2100 $\mu_F = 0.662$ μ_{CR}

= 0.320 Diversity = 4.8669e+01 $k_t = [3, 4, 2, 3]$

Generation 30: Best = 2.236696e+03 FES = 3100 $\mu_F = 0.672$ μ_{CR}

= 0.276 Diversity = 4.7241e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 2.236696e+03 FES = 4100 $\mu_F = 0.730$ μ_{CR}

= 0.223 Diversity = 4.6825e+01 $k_t = [4, 4, 3, 4]$

Generation 50: Best = 2.236696e+03 FES = 5100 $\mu_F = 0.728$ μ_{CR}

= 0.167 Diversity = 4.5803e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 2.204884e+03 FES = 6100 $\mu_F = 0.723$ μ_{CR}

= 0.149 Diversity = 4.5706e+01 $k_t = [3, 4, 4, 2]$

Generation 70: Best = 2.180874e+03 FES = 7100 $\mu_F = 0.748$ μ_{CR}

= 0.118 Diversity = 4.5861e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.180874e+03 FES = 8100 $\mu_F = 0.759$ μ_{CR}

= 0.119 Diversity = 4.4917e+01 $k_t = [4, 3, 4, 3]$

Generation 90: Best = 2.180874e+03 FES = 9100 $\mu_F = 0.720$ μ_{CR}

= 0.132 Diversity = 4.4988e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.142627e+03 FES = 10100 $\mu_F = 0.766$ μ_{CR}

= 0.114 Diversity = 4.4762e+01 $k_t = [4, 4, 4, 3]$

Generation 110: Best = 2.036947e+03 FES = 11100 $\mu_F = 0.811$ μ_{CR}

= 0.102 Diversity = 4.5651e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.036947e+03 FES = 12100 $\mu_F = 0.769$ μ_{CR}

= 0.078 Diversity = 4.4944e+01 $k_t = [2, 3, 2, 3]$

Generation 130: Best = 2.036947e+03 FES = 13100 $\mu_F = 0.710$ μ_{CR}

= 0.076 Diversity = 4.4338e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.012609e+03 FES = 14100 $\mu_F = 0.620$ μ_{CR}

= 0.078 Diversity = 4.3739e+01 $k_t = [2, 2, 2, 4]$

Generation 150: Best = 1.806470e+03 FES = 15100 $\mu_F = 0.682$ μ_{CR}

= 0.080 Diversity = 4.3871e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 1.806470e+03 FES = 16100 $\mu_F = 0.695$ μ_{CR}

= 0.078 Diversity = 4.3697e+01 $k_t = [4, 2, 4, 3]$

Generation 170: Best = 1.751588e+03 FES = 17100 $\mu_F = 0.715$ μ_{CR}

= 0.080 Diversity = 4.3503e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 1.751588e+03 FES = 18100 $\mu_F = 0.692$ μ_{CR}

= 0.069 Diversity = 4.3509e+01 $k_t = [2, 4, 4, 4]$

Generation 190: Best = 1.751588e+03 FES = 19100 $\mu_F = 0.716$ μ_{CR}

= 0.092 Diversity = 4.3429e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 1.751588e+03 FES = 20100 $\mu_F = 0.726$ μ_{CR}

= 0.084 Diversity = 4.4034e+01 $k_t = [2, 4, 4, 4]$

Generation 210: Best = 1.751588e+03 FES = 21100 $\mu_F = 0.666$ μ_{CR}

= 0.084 Diversity = 4.3488e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 1.751588e+03 FES = 22100 $\mu_F = 0.680$ μ_{CR}

= 0.078	Diversity = 4.3520e+01	k_t = [3, 2, 2, 2]	μ_F = 0.625	μ_{CR}
Generation 230:	Best = 1.751588e+03	FES = 23100		
= 0.084	Diversity = 4.3488e+01	k_t = [1, 1, 1, 1]	μ_F = 0.594	μ_{CR}
Generation 240:	Best = 1.751588e+03	FES = 24100		
= 0.113	Diversity = 4.3216e+01	k_t = [3, 4, 4, 4]	μ_F = 0.647	μ_{CR}
Generation 250:	Best = 1.751588e+03	FES = 25100		
= 0.110	Diversity = 4.2699e+01	k_t = [1, 1, 1, 1]	μ_F = 0.593	μ_{CR}
Generation 260:	Best = 1.751588e+03	FES = 26100		
= 0.112	Diversity = 4.2456e+01	k_t = [3, 3, 4, 3]	μ_F = 0.531	μ_{CR}
Generation 270:	Best = 1.750233e+03	FES = 27100		
= 0.092	Diversity = 4.2275e+01	k_t = [1, 1, 1, 1]	μ_F = 0.536	μ_{CR}
Generation 280:	Best = 1.557417e+03	FES = 28100		
= 0.080	Diversity = 4.2150e+01	k_t = [2, 4, 4, 4]	μ_F = 0.574	μ_{CR}
Generation 290:	Best = 1.427111e+03	FES = 29100		
= 0.067	Diversity = 4.1990e+01	k_t = [1, 1, 1, 1]	μ_F = 0.506	μ_{CR}
Generation 300:	Best = 1.304411e+03	FES = 30100		
= 0.078	Diversity = 4.2026e+01	k_t = [2, 2, 2, 3]	μ_F = 0.511	μ_{CR}
Generation 310:	Best = 1.304411e+03	FES = 31100		
= 0.080	Diversity = 4.1870e+01	k_t = [1, 1, 1, 1]	μ_F = 0.409	μ_{CR}
Generation 320:	Best = 1.304411e+03	FES = 32100		
= 0.089	Diversity = 4.1872e+01	k_t = [2, 2, 3, 3]	μ_F = 0.390	μ_{CR}
Generation 330:	Best = 1.304411e+03	FES = 33100		
= 0.101	Diversity = 4.1840e+01	k_t = [1, 1, 1, 1]	μ_F = 0.351	μ_{CR}
Generation 340:	Best = 1.304411e+03	FES = 34100		
= 0.096	Diversity = 4.1756e+01	k_t = [2, 2, 2, 4]	μ_F = 0.364	μ_{CR}
Generation 350:	Best = 1.304411e+03	FES = 35100		
= 0.088	Diversity = 4.1635e+01	k_t = [1, 1, 1, 1]	μ_F = 0.328	μ_{CR}
Generation 360:	Best = 1.304411e+03	FES = 36100		
= 0.110	Diversity = 4.1664e+01	k_t = [3, 3, 2, 2]	μ_F = 0.346	μ_{CR}
Generation 370:	Best = 1.304411e+03	FES = 37100		
= 0.093	Diversity = 4.1403e+01	k_t = [1, 1, 1, 1]	μ_F = 0.305	μ_{CR}
Generation 380:	Best = 1.304411e+03	FES = 38100		
= 0.105	Diversity = 4.1469e+01	k_t = [2, 4, 4, 4]	μ_F = 0.270	μ_{CR}
Generation 390:	Best = 1.304411e+03	FES = 39100		
= 0.097	Diversity = 4.1473e+01	k_t = [1, 1, 1, 1]	μ_F = 0.340	μ_{CR}
Generation 400:	Best = 1.263513e+03	FES = 40100		
= 0.099	Diversity = 4.1436e+01	k_t = [4, 4, 3, 3]	μ_F = 0.290	μ_{CR}
Generation 410:	Best = 1.263513e+03	FES = 41100		
= 0.099	Diversity = 4.1397e+01	k_t = [1, 1, 1, 1]	μ_F = 0.359	μ_{CR}
Generation 420:	Best = 1.263513e+03	FES = 42100		
= 0.118	Diversity = 4.1357e+01	k_t = [4, 4, 2, 3]	μ_F = 0.378	μ_{CR}
Generation 430:	Best = 1.263513e+03	FES = 43100		
= 0.084	Diversity = 4.1381e+01	k_t = [1, 1, 1, 1]	μ_F = 0.468	μ_{CR}
Generation 440:	Best = 1.263513e+03	FES = 44100		
= 0.088	Diversity = 4.1280e+01	k_t = [2, 2, 3, 3]	μ_F = 0.493	μ_{CR}
Generation 450:	Best = 1.263513e+03	FES = 45100		
= 0.089	Diversity = 4.1284e+01	k_t = [1, 1, 1, 1]	μ_F = 0.387	μ_{CR}
Generation 460:	Best = 1.263513e+03	FES = 46100		
= 0.103	Diversity = 4.1169e+01	k_t = [3, 4, 4, 3]		

Generation 470: Best = 1.263513e+03	FES = 47100	$\mu_F = 0.391$	μ_{CR}
= 0.127 Diversity = 4.1420e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.263513e+03	FES = 48100	$\mu_F = 0.435$	μ_{CR}
= 0.117 Diversity = 4.1281e+01	$k_t = [2, 4, 3, 4]$		
Generation 490: Best = 1.255508e+03	FES = 49100	$\mu_F = 0.326$	μ_{CR}
= 0.104 Diversity = 4.1258e+01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 1.240923e+03	FES = 50100	$\mu_F = 0.385$	μ_{CR}
= 0.116 Diversity = 4.1328e+01	$k_t = [3, 3, 4, 4]$		
Generation 510: Best = 1.240923e+03	FES = 51100	$\mu_F = 0.371$	μ_{CR}
= 0.100 Diversity = 4.0981e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.219024e+03	FES = 52100	$\mu_F = 0.312$	μ_{CR}
= 0.094 Diversity = 4.0984e+01	$k_t = [4, 4, 4, 2]$		
Generation 530: Best = 1.206851e+03	FES = 53100	$\mu_F = 0.261$	μ_{CR}
= 0.096 Diversity = 4.1027e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.206851e+03	FES = 54100	$\mu_F = 0.309$	μ_{CR}
= 0.085 Diversity = 4.1138e+01	$k_t = [4, 2, 3, 4]$		
Generation 550: Best = 1.183208e+03	FES = 55100	$\mu_F = 0.297$	μ_{CR}
= 0.090 Diversity = 4.1128e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.183208e+03	FES = 56100	$\mu_F = 0.260$	μ_{CR}
= 0.115 Diversity = 4.1399e+01	$k_t = [4, 3, 2, 4]$		
Generation 570: Best = 1.183208e+03	FES = 57100	$\mu_F = 0.265$	μ_{CR}
= 0.117 Diversity = 4.1398e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.179652e+03	FES = 58100	$\mu_F = 0.243$	μ_{CR}
= 0.109 Diversity = 4.1402e+01	$k_t = [3, 4, 2, 2]$		
Generation 590: Best = 1.122112e+03	FES = 59100	$\mu_F = 0.234$	μ_{CR}
= 0.124 Diversity = 4.1286e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.115008e+03	FES = 60100	$\mu_F = 0.334$	μ_{CR}
= 0.115 Diversity = 4.1282e+01	$k_t = [3, 3, 4, 2]$		
Generation 610: Best = 1.115008e+03	FES = 61100	$\mu_F = 0.290$	μ_{CR}
= 0.125 Diversity = 4.1289e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.105611e+03	FES = 62100	$\mu_F = 0.286$	μ_{CR}
= 0.115 Diversity = 4.1293e+01	$k_t = [4, 2, 4, 2]$		
Generation 630: Best = 1.105611e+03	FES = 63100	$\mu_F = 0.259$	μ_{CR}
= 0.123 Diversity = 4.1290e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.105611e+03	FES = 64100	$\mu_F = 0.240$	μ_{CR}
= 0.093 Diversity = 4.1290e+01	$k_t = [3, 4, 2, 2]$		
Generation 650: Best = 1.105611e+03	FES = 65100	$\mu_F = 0.232$	μ_{CR}
= 0.092 Diversity = 4.1292e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.105611e+03	FES = 66100	$\mu_F = 0.245$	μ_{CR}
= 0.104 Diversity = 4.1100e+01	$k_t = [4, 2, 4, 4]$		
Generation 670: Best = 1.105611e+03	FES = 67100	$\mu_F = 0.248$	μ_{CR}
= 0.097 Diversity = 4.1102e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.105611e+03	FES = 68100	$\mu_F = 0.272$	μ_{CR}
= 0.079 Diversity = 4.1093e+01	$k_t = [3, 2, 3, 2]$		
Generation 690: Best = 1.082848e+03	FES = 69100	$\mu_F = 0.219$	μ_{CR}
= 0.073 Diversity = 4.1092e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.048891e+03	FES = 70100	$\mu_F = 0.174$	μ_{CR}
= 0.065 Diversity = 4.1054e+01	$k_t = [2, 4, 4, 2]$		
Generation 710: Best = 1.048891e+03	FES = 71100	$\mu_F = 0.159$	μ_{CR}

= 0.065	Diversity = 4.1039e+01	k_t = [1, 1, 1, 1]		
Generation 720:	Best = 1.048891e+03	FES = 72100	$\mu_F = 0.142$	μ_{CR}
= 0.075	Diversity = 4.1028e+01	k_t = [3, 3, 3, 2]		
Generation 730:	Best = 1.048891e+03	FES = 73100	$\mu_F = 0.110$	μ_{CR}
= 0.091	Diversity = 4.1029e+01	k_t = [1, 1, 1, 1]		
Generation 740:	Best = 1.048891e+03	FES = 74100	$\mu_F = 0.091$	μ_{CR}
= 0.096	Diversity = 4.0999e+01	k_t = [3, 2, 3, 3]		
Generation 750:	Best = 1.048078e+03	FES = 75100	$\mu_F = 0.205$	μ_{CR}
= 0.105	Diversity = 4.0994e+01	k_t = [1, 1, 1, 1]		
Generation 760:	Best = 1.048078e+03	FES = 76100	$\mu_F = 0.158$	μ_{CR}
= 0.085	Diversity = 4.0980e+01	k_t = [4, 2, 2, 2]		
Generation 770:	Best = 1.048078e+03	FES = 77100	$\mu_F = 0.127$	μ_{CR}
= 0.096	Diversity = 4.0963e+01	k_t = [1, 1, 1, 1]		
Generation 780:	Best = 1.048078e+03	FES = 78100	$\mu_F = 0.106$	μ_{CR}
= 0.101	Diversity = 4.0945e+01	k_t = [3, 4, 4, 2]		
Generation 790:	Best = 1.048078e+03	FES = 79100	$\mu_F = 0.223$	μ_{CR}
= 0.067	Diversity = 4.0948e+01	k_t = [1, 1, 1, 1]		
Generation 800:	Best = 1.048078e+03	FES = 80100	$\mu_F = 0.223$	μ_{CR}
= 0.084	Diversity = 4.0977e+01	k_t = [4, 3, 3, 2]		
Generation 810:	Best = 1.048078e+03	FES = 81100	$\mu_F = 0.230$	μ_{CR}
= 0.108	Diversity = 4.0978e+01	k_t = [1, 1, 1, 1]		
Generation 820:	Best = 1.043658e+03	FES = 82100	$\mu_F = 0.220$	μ_{CR}
= 0.077	Diversity = 4.0982e+01	k_t = [3, 3, 2, 2]		
Generation 830:	Best = 1.043658e+03	FES = 83100	$\mu_F = 0.224$	μ_{CR}
= 0.086	Diversity = 4.0984e+01	k_t = [1, 1, 1, 1]		
Generation 840:	Best = 1.043658e+03	FES = 84100	$\mu_F = 0.277$	μ_{CR}
= 0.099	Diversity = 4.0639e+01	k_t = [3, 2, 4, 4]		
Generation 850:	Best = 1.043658e+03	FES = 85100	$\mu_F = 0.317$	μ_{CR}
= 0.100	Diversity = 4.0559e+01	k_t = [1, 1, 1, 1]		
Generation 860:	Best = 1.043658e+03	FES = 86100	$\mu_F = 0.360$	μ_{CR}
= 0.087	Diversity = 4.0557e+01	k_t = [2, 2, 4, 2]		
Generation 870:	Best = 1.043658e+03	FES = 87100	$\mu_F = 0.298$	μ_{CR}
= 0.102	Diversity = 4.0557e+01	k_t = [1, 1, 1, 1]		
Generation 880:	Best = 1.043658e+03	FES = 88100	$\mu_F = 0.292$	μ_{CR}
= 0.109	Diversity = 4.0566e+01	k_t = [3, 2, 2, 4]		
Generation 890:	Best = 1.043658e+03	FES = 89100	$\mu_F = 0.301$	μ_{CR}
= 0.098	Diversity = 4.0568e+01	k_t = [1, 1, 1, 1]		
Generation 900:	Best = 1.043658e+03	FES = 90100	$\mu_F = 0.317$	μ_{CR}
= 0.100	Diversity = 4.0425e+01	k_t = [4, 2, 3, 4]		
Generation 910:	Best = 1.043658e+03	FES = 91100	$\mu_F = 0.330$	μ_{CR}
= 0.102	Diversity = 4.0410e+01	k_t = [1, 1, 1, 1]		
Generation 920:	Best = 1.043658e+03	FES = 92100	$\mu_F = 0.441$	μ_{CR}
= 0.088	Diversity = 4.0794e+01	k_t = [3, 4, 2, 4]		
Generation 930:	Best = 1.042824e+03	FES = 93100	$\mu_F = 0.451$	μ_{CR}
= 0.070	Diversity = 4.0690e+01	k_t = [1, 1, 1, 1]		
Generation 940:	Best = 1.042824e+03	FES = 94100	$\mu_F = 0.563$	μ_{CR}
= 0.073	Diversity = 4.0416e+01	k_t = [4, 3, 2, 4]		
Generation 950:	Best = 1.042824e+03	FES = 95100	$\mu_F = 0.512$	μ_{CR}
= 0.067	Diversity = 4.0417e+01	k_t = [1, 1, 1, 1]		

```

Generation 960: Best = 1.042824e+03 FES = 96100  $\mu_F$  = 0.449  $\mu_{CR}$ 
= 0.075 Diversity = 4.0640e+01  $k_t$  = [4, 3, 2, 3]
Generation 970: Best = 1.042824e+03 FES = 97100  $\mu_F$  = 0.331  $\mu_{CR}$ 
= 0.072 Diversity = 4.0629e+01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.041861e+03 FES = 98100  $\mu_F$  = 0.345  $\mu_{CR}$ 
= 0.058 Diversity = 4.0283e+01  $k_t$  = [3, 3, 2, 3]
Generation 990: Best = 1.041861e+03 FES = 99100  $\mu_F$  = 0.367  $\mu_{CR}$ 
= 0.077 Diversity = 4.0282e+01  $k_t$  = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 1.041861e+03

Total Function Evals = 100000

Total Successful Updates = 2523

```

=====
=====

```

Results:

Best Fitness : 1.041861e+03

Global Optimum : 1.000000e+03

Error (|best-optimum|) : 4.186088e+01

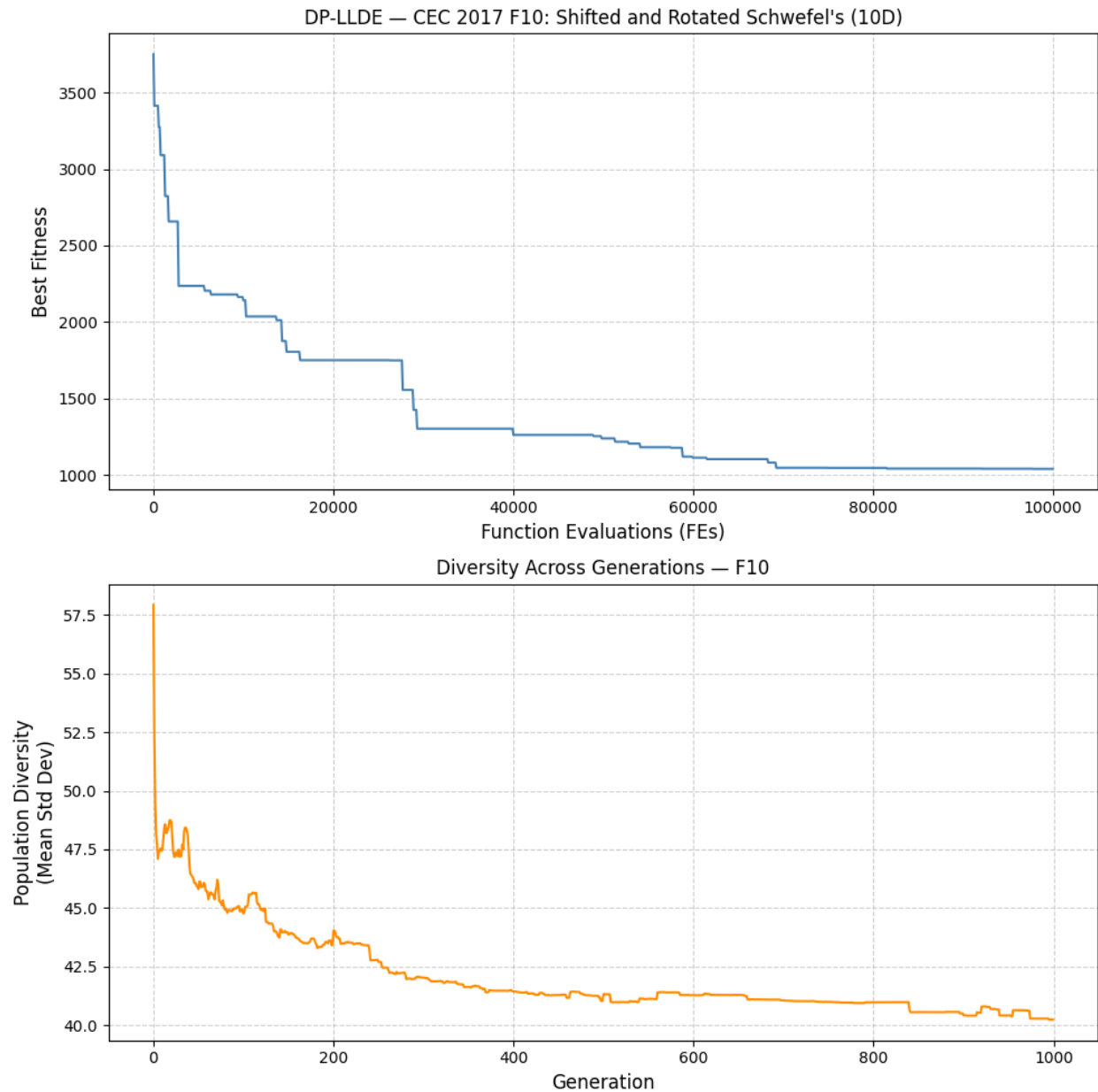
Best Solution (5 dim) : [-21.86193909 -48.71800106 44.58845229
57.78940841 98.04204166]

$D^0(0)$ (init. diversity): 5.7928e+01

```

=====
=====

```



Graph saved as convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
# =====  
# F11 – Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)  
# =====  
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid:  
Zakharov; Rosenbrock; Rastrigin')
```

DP-LLDE on CEC 2017 F11 – Single Run

```

Function      : Hybrid: Zakharov; Rosenbrock; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 1100.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.292232e+04
Generation 10: Best = 1.271871e+03 FES = 1100  $\mu_F$  = 0.588  $\mu_{CR}$ 
= 0.351 Diversity = 3.0353e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 1.134212e+03 FES = 2100  $\mu_F$  = 0.635  $\mu_{CR}$ 
= 0.351 Diversity = 1.5183e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 1.118996e+03 FES = 3100  $\mu_F$  = 0.655  $\mu_{CR}$ 
= 0.345 Diversity = 1.0068e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 1.108991e+03 FES = 4100  $\mu_F$  = 0.678  $\mu_{CR}$ 
= 0.329 Diversity = 8.7750e+00  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 1.108991e+03 FES = 5100  $\mu_F$  = 0.690  $\mu_{CR}$ 
= 0.289 Diversity = 8.3305e+00  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 1.108437e+03 FES = 6100  $\mu_F$  = 0.707  $\mu_{CR}$ 
= 0.268 Diversity = 7.9392e+00  $k_t$  = [2, 2, 4, 2]
Generation 70: Best = 1.108437e+03 FES = 7100  $\mu_F$  = 0.726  $\mu_{CR}$ 
= 0.246 Diversity = 7.7497e+00  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 1.108206e+03 FES = 8100  $\mu_F$  = 0.747  $\mu_{CR}$ 
= 0.208 Diversity = 7.4304e+00  $k_t$  = [3, 3, 4, 4]
Generation 90: Best = 1.107276e+03 FES = 9100  $\mu_F$  = 0.759  $\mu_{CR}$ 
= 0.172 Diversity = 7.1124e+00  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 1.106608e+03 FES = 10100  $\mu_F$  = 0.731  $\mu_{CR}$ 
= 0.160 Diversity = 7.2031e+00  $k_t$  = [4, 2, 2, 3]
Generation 110: Best = 1.103787e+03 FES = 11100  $\mu_F$  = 0.708  $\mu_{CR}$ 
= 0.130 Diversity = 7.1606e+00  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 1.103787e+03 FES = 12100  $\mu_F$  = 0.709  $\mu_{CR}$ 
= 0.125 Diversity = 7.0031e+00  $k_t$  = [2, 4, 2, 4]
Generation 130: Best = 1.103787e+03 FES = 13100  $\mu_F$  = 0.694  $\mu_{CR}$ 
= 0.124 Diversity = 6.8187e+00  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 1.103241e+03 FES = 14100  $\mu_F$  = 0.730  $\mu_{CR}$ 
= 0.120 Diversity = 6.6938e+00  $k_t$  = [4, 3, 2, 4]
Generation 150: Best = 1.102984e+03 FES = 15100  $\mu_F$  = 0.723  $\mu_{CR}$ 
= 0.121 Diversity = 6.5067e+00  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 1.102984e+03 FES = 16100  $\mu_F$  = 0.735  $\mu_{CR}$ 
= 0.118 Diversity = 6.3301e+00  $k_t$  = [4, 4, 2, 3]
Generation 170: Best = 1.102883e+03 FES = 17100  $\mu_F$  = 0.724  $\mu_{CR}$ 
= 0.114 Diversity = 6.2294e+00  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 1.102883e+03 FES = 18100  $\mu_F$  = 0.706  $\mu_{CR}$ 
= 0.117 Diversity = 6.1919e+00  $k_t$  = [3, 2, 2, 3]
Generation 190: Best = 1.102523e+03 FES = 19100  $\mu_F$  = 0.702  $\mu_{CR}$ 
= 0.122 Diversity = 6.0944e+00  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 1.102495e+03 FES = 20100  $\mu_F$  = 0.728  $\mu_{CR}$ 
= 0.123 Diversity = 6.0015e+00  $k_t$  = [3, 2, 2, 3]

```

Generation 210: Best = 1.102495e+03	FES = 21100	$\mu_F = 0.701$	μ_{CR}
= 0.104 Diversity = 5.9204e+00	$k_t = [1, 1, 1, 1]$		
Generation 220: Best = 1.102495e+03	FES = 22100	$\mu_F = 0.717$	μ_{CR}
= 0.113 Diversity = 5.7684e+00	$k_t = [3, 2, 4, 4]$		
Generation 230: Best = 1.102289e+03	FES = 23100	$\mu_F = 0.671$	μ_{CR}
= 0.130 Diversity = 5.7140e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.102289e+03	FES = 24100	$\mu_F = 0.676$	μ_{CR}
= 0.128 Diversity = 5.6388e+00	$k_t = [2, 2, 3, 4]$		
Generation 250: Best = 1.102253e+03	FES = 25100	$\mu_F = 0.670$	μ_{CR}
= 0.134 Diversity = 5.3482e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.102243e+03	FES = 26100	$\mu_F = 0.740$	μ_{CR}
= 0.138 Diversity = 5.1365e+00	$k_t = [3, 2, 3, 4]$		
Generation 270: Best = 1.102243e+03	FES = 27100	$\mu_F = 0.754$	μ_{CR}
= 0.131 Diversity = 5.0124e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.102243e+03	FES = 28100	$\mu_F = 0.748$	μ_{CR}
= 0.129 Diversity = 4.9514e+00	$k_t = [2, 4, 2, 3]$		
Generation 290: Best = 1.102154e+03	FES = 29100	$\mu_F = 0.780$	μ_{CR}
= 0.114 Diversity = 4.9270e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.102154e+03	FES = 30100	$\mu_F = 0.784$	μ_{CR}
= 0.120 Diversity = 4.6605e+00	$k_t = [3, 2, 3, 4]$		
Generation 310: Best = 1.102154e+03	FES = 31100	$\mu_F = 0.794$	μ_{CR}
= 0.130 Diversity = 4.6344e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.102074e+03	FES = 32100	$\mu_F = 0.802$	μ_{CR}
= 0.102 Diversity = 4.4467e+00	$k_t = [2, 3, 2, 4]$		
Generation 330: Best = 1.102054e+03	FES = 33100	$\mu_F = 0.767$	μ_{CR}
= 0.096 Diversity = 4.2896e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.101918e+03	FES = 34100	$\mu_F = 0.808$	μ_{CR}
= 0.099 Diversity = 4.1534e+00	$k_t = [3, 2, 3, 4]$		
Generation 350: Best = 1.101847e+03	FES = 35100	$\mu_F = 0.818$	μ_{CR}
= 0.111 Diversity = 4.0017e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.101826e+03	FES = 36100	$\mu_F = 0.828$	μ_{CR}
= 0.127 Diversity = 4.0095e+00	$k_t = [4, 3, 2, 2]$		
Generation 370: Best = 1.101707e+03	FES = 37100	$\mu_F = 0.825$	μ_{CR}
= 0.128 Diversity = 4.0179e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.101339e+03	FES = 38100	$\mu_F = 0.840$	μ_{CR}
= 0.127 Diversity = 3.8673e+00	$k_t = [3, 4, 2, 4]$		
Generation 390: Best = 1.101319e+03	FES = 39100	$\mu_F = 0.845$	μ_{CR}
= 0.120 Diversity = 3.4335e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.101229e+03	FES = 40100	$\mu_F = 0.861$	μ_{CR}
= 0.140 Diversity = 2.4282e+00	$k_t = [4, 4, 4, 3]$		
Generation 410: Best = 1.100868e+03	FES = 41100	$\mu_F = 0.858$	μ_{CR}
= 0.146 Diversity = 1.2596e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.100690e+03	FES = 42100	$\mu_F = 0.864$	μ_{CR}
= 0.156 Diversity = 7.2941e-01	$k_t = [3, 4, 4, 3]$		
Generation 430: Best = 1.100618e+03	FES = 43100	$\mu_F = 0.848$	μ_{CR}
= 0.159 Diversity = 5.0640e-01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.100503e+03	FES = 44100	$\mu_F = 0.845$	μ_{CR}
= 0.159 Diversity = 4.0824e-01	$k_t = [2, 3, 2, 3]$		
Generation 450: Best = 1.100445e+03	FES = 45100	$\mu_F = 0.839$	μ_{CR}
= 0.158 Diversity = 4.1521e-01	$k_t = [1, 1, 1, 1]$		

Generation 460: Best = 1.100316e+03	FES = 46100	$\mu_F = 0.834$	μ_{CR}
= 0.151 Diversity = 3.8530e-01	$k_t = [2, 3, 3, 4]$		
Generation 470: Best = 1.100216e+03	FES = 47100	$\mu_F = 0.836$	μ_{CR}
= 0.147 Diversity = 5.2153e-01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.100192e+03	FES = 48100	$\mu_F = 0.835$	μ_{CR}
= 0.155 Diversity = 4.5190e-01	$k_t = [4, 2, 3, 3]$		
Generation 490: Best = 1.100146e+03	FES = 49100	$\mu_F = 0.783$	μ_{CR}
= 0.148 Diversity = 4.2358e-01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 1.100122e+03	FES = 50100	$\mu_F = 0.777$	μ_{CR}
= 0.127 Diversity = 3.8812e-01	$k_t = [4, 3, 2, 4]$		
Generation 510: Best = 1.100091e+03	FES = 51100	$\mu_F = 0.760$	μ_{CR}
= 0.119 Diversity = 3.4623e-01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.100083e+03	FES = 52100	$\mu_F = 0.762$	μ_{CR}
= 0.116 Diversity = 3.4818e-01	$k_t = [2, 2, 3, 4]$		
Generation 530: Best = 1.100065e+03	FES = 53100	$\mu_F = 0.756$	μ_{CR}
= 0.116 Diversity = 2.8405e-01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.100045e+03	FES = 54100	$\mu_F = 0.763$	μ_{CR}
= 0.120 Diversity = 3.0959e-01	$k_t = [2, 4, 3, 3]$		
Generation 550: Best = 1.100028e+03	FES = 55100	$\mu_F = 0.765$	μ_{CR}
= 0.127 Diversity = 3.4039e-01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.100027e+03	FES = 56100	$\mu_F = 0.766$	μ_{CR}
= 0.124 Diversity = 3.0554e-01	$k_t = [2, 3, 2, 2]$		
Generation 570: Best = 1.100026e+03	FES = 57100	$\mu_F = 0.762$	μ_{CR}
= 0.128 Diversity = 2.6275e-01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.100013e+03	FES = 58100	$\mu_F = 0.780$	μ_{CR}
= 0.130 Diversity = 2.5256e-01	$k_t = [4, 4, 2, 4]$		
Generation 590: Best = 1.100013e+03	FES = 59100	$\mu_F = 0.769$	μ_{CR}
= 0.127 Diversity = 2.6254e-01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.100007e+03	FES = 60100	$\mu_F = 0.782$	μ_{CR}
= 0.132 Diversity = 2.5251e-01	$k_t = [2, 4, 3, 4]$		
Generation 610: Best = 1.100004e+03	FES = 61100	$\mu_F = 0.794$	μ_{CR}
= 0.129 Diversity = 2.9783e-01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.100004e+03	FES = 62100	$\mu_F = 0.798$	μ_{CR}
= 0.117 Diversity = 2.4427e-01	$k_t = [3, 4, 3, 3]$		
Generation 630: Best = 1.100004e+03	FES = 63100	$\mu_F = 0.789$	μ_{CR}
= 0.119 Diversity = 1.8898e-01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.100003e+03	FES = 64100	$\mu_F = 0.798$	μ_{CR}
= 0.110 Diversity = 1.7436e-01	$k_t = [4, 2, 3, 2]$		
Generation 650: Best = 1.100002e+03	FES = 65100	$\mu_F = 0.795$	μ_{CR}
= 0.115 Diversity = 1.4606e-01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.100002e+03	FES = 66100	$\mu_F = 0.785$	μ_{CR}
= 0.114 Diversity = 1.2892e-01	$k_t = [4, 4, 2, 2]$		
Generation 670: Best = 1.100001e+03	FES = 67100	$\mu_F = 0.773$	μ_{CR}
= 0.111 Diversity = 1.4930e-01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.100001e+03	FES = 68100	$\mu_F = 0.766$	μ_{CR}
= 0.115 Diversity = 1.2212e-01	$k_t = [3, 3, 3, 3]$		
Generation 690: Best = 1.100001e+03	FES = 69100	$\mu_F = 0.775$	μ_{CR}
= 0.111 Diversity = 1.3962e-01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.100001e+03	FES = 70100	$\mu_F = 0.770$	μ_{CR}
= 0.107 Diversity = 1.3888e-01	$k_t = [4, 3, 4, 3]$		

Generation 710: Best = 1.100001e+03	FES = 71100	$\mu_F = 0.772$	μ_{CR}
= 0.096 Diversity = 1.0171e-01 $k_t = [1, 1, 1, 1]$			
Generation 720: Best = 1.100000e+03	FES = 72100	$\mu_F = 0.787$	μ_{CR}
= 0.104 Diversity = 7.9745e-02 $k_t = [4, 3, 4, 4]$			
Generation 730: Best = 1.100000e+03	FES = 73100	$\mu_F = 0.789$	μ_{CR}
= 0.110 Diversity = 6.1305e-02 $k_t = [1, 1, 1, 1]$			
Generation 740: Best = 1.100000e+03	FES = 74100	$\mu_F = 0.792$	μ_{CR}
= 0.114 Diversity = 5.5109e-02 $k_t = [4, 2, 3, 4]$			
Generation 750: Best = 1.100000e+03	FES = 75100	$\mu_F = 0.799$	μ_{CR}
= 0.121 Diversity = 5.0450e-02 $k_t = [1, 1, 1, 1]$			
Generation 760: Best = 1.100000e+03	FES = 76100	$\mu_F = 0.803$	μ_{CR}
= 0.114 Diversity = 5.0421e-02 $k_t = [2, 4, 3, 4]$			
Generation 770: Best = 1.100000e+03	FES = 77100	$\mu_F = 0.818$	μ_{CR}
= 0.122 Diversity = 4.0483e-02 $k_t = [1, 1, 1, 1]$			
Generation 780: Best = 1.100000e+03	FES = 78100	$\mu_F = 0.802$	μ_{CR}
= 0.130 Diversity = 3.2572e-02 $k_t = [3, 2, 4, 3]$			
Generation 790: Best = 1.100000e+03	FES = 79100	$\mu_F = 0.799$	μ_{CR}
= 0.133 Diversity = 2.5582e-02 $k_t = [1, 1, 1, 1]$			
Generation 800: Best = 1.100000e+03	FES = 80100	$\mu_F = 0.784$	μ_{CR}
= 0.120 Diversity = 2.0404e-02 $k_t = [2, 4, 2, 3]$			
Generation 810: Best = 1.100000e+03	FES = 81100	$\mu_F = 0.792$	μ_{CR}
= 0.119 Diversity = 1.6693e-02 $k_t = [1, 1, 1, 1]$			
Generation 820: Best = 1.100000e+03	FES = 82100	$\mu_F = 0.781$	μ_{CR}
= 0.116 Diversity = 1.2349e-02 $k_t = [3, 4, 2, 2]$			
Generation 830: Best = 1.100000e+03	FES = 83100	$\mu_F = 0.782$	μ_{CR}
= 0.114 Diversity = 9.4407e-03 $k_t = [1, 1, 1, 1]$			
Generation 840: Best = 1.100000e+03	FES = 84100	$\mu_F = 0.784$	μ_{CR}
= 0.114 Diversity = 8.6472e-03 $k_t = [3, 2, 3, 2]$			
Generation 850: Best = 1.100000e+03	FES = 85100	$\mu_F = 0.798$	μ_{CR}
= 0.113 Diversity = 6.5623e-03 $k_t = [1, 1, 1, 1]$			
Generation 860: Best = 1.100000e+03	FES = 86100	$\mu_F = 0.787$	μ_{CR}
= 0.110 Diversity = 4.4106e-03 $k_t = [3, 4, 3, 3]$			
Generation 870: Best = 1.100000e+03	FES = 87100	$\mu_F = 0.783$	μ_{CR}
= 0.110 Diversity = 3.7165e-03 $k_t = [1, 1, 1, 1]$			
Generation 880: Best = 1.100000e+03	FES = 88100	$\mu_F = 0.785$	μ_{CR}
= 0.118 Diversity = 2.6157e-03 $k_t = [2, 4, 3, 4]$			
Generation 890: Best = 1.100000e+03	FES = 89100	$\mu_F = 0.794$	μ_{CR}
= 0.133 Diversity = 2.0850e-03 $k_t = [1, 1, 1, 1]$			
Generation 900: Best = 1.100000e+03	FES = 90100	$\mu_F = 0.810$	μ_{CR}
= 0.135 Diversity = 1.5943e-03 $k_t = [4, 3, 2, 4]$			
Generation 910: Best = 1.100000e+03	FES = 91100	$\mu_F = 0.796$	μ_{CR}
= 0.127 Diversity = 1.2720e-03 $k_t = [1, 1, 1, 1]$			
Generation 920: Best = 1.100000e+03	FES = 92100	$\mu_F = 0.794$	μ_{CR}
= 0.115 Diversity = 9.1272e-04 $k_t = [2, 4, 3, 2]$			
Generation 930: Best = 1.100000e+03	FES = 93100	$\mu_F = 0.799$	μ_{CR}
= 0.110 Diversity = 8.2168e-04 $k_t = [1, 1, 1, 1]$			
Generation 940: Best = 1.100000e+03	FES = 94100	$\mu_F = 0.812$	μ_{CR}
= 0.110 Diversity = 5.8622e-04 $k_t = [4, 3, 2, 2]$			
Generation 950: Best = 1.100000e+03	FES = 95100	$\mu_F = 0.810$	μ_{CR}
= 0.102 Diversity = 4.3890e-04 $k_t = [1, 1, 1, 1]$			


```
Generation 960: Best = 1.100000e+03 FES = 96100  $\mu_F$  = 0.805  $\mu_{CR}$ 
= 0.102 Diversity = 3.8068e-04  $k_t$  = [2, 4, 4, 4]
Generation 970: Best = 1.100000e+03 FES = 97100  $\mu_F$  = 0.797  $\mu_{CR}$ 
= 0.106 Diversity = 2.6296e-04  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.100000e+03 FES = 98100  $\mu_F$  = 0.770  $\mu_{CR}$ 
= 0.109 Diversity = 1.6173e-04  $k_t$  = [4, 3, 4, 3]
Generation 990: Best = 1.100000e+03 FES = 99100  $\mu_F$  = 0.763  $\mu_{CR}$ 
= 0.114 Diversity = 1.4594e-04  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 1.100000e+03

Total Function Evals = 100000

Total Successful Updates = 18452

```
=====
=====
```

Results:

Best Fitness : 1.100000e+03

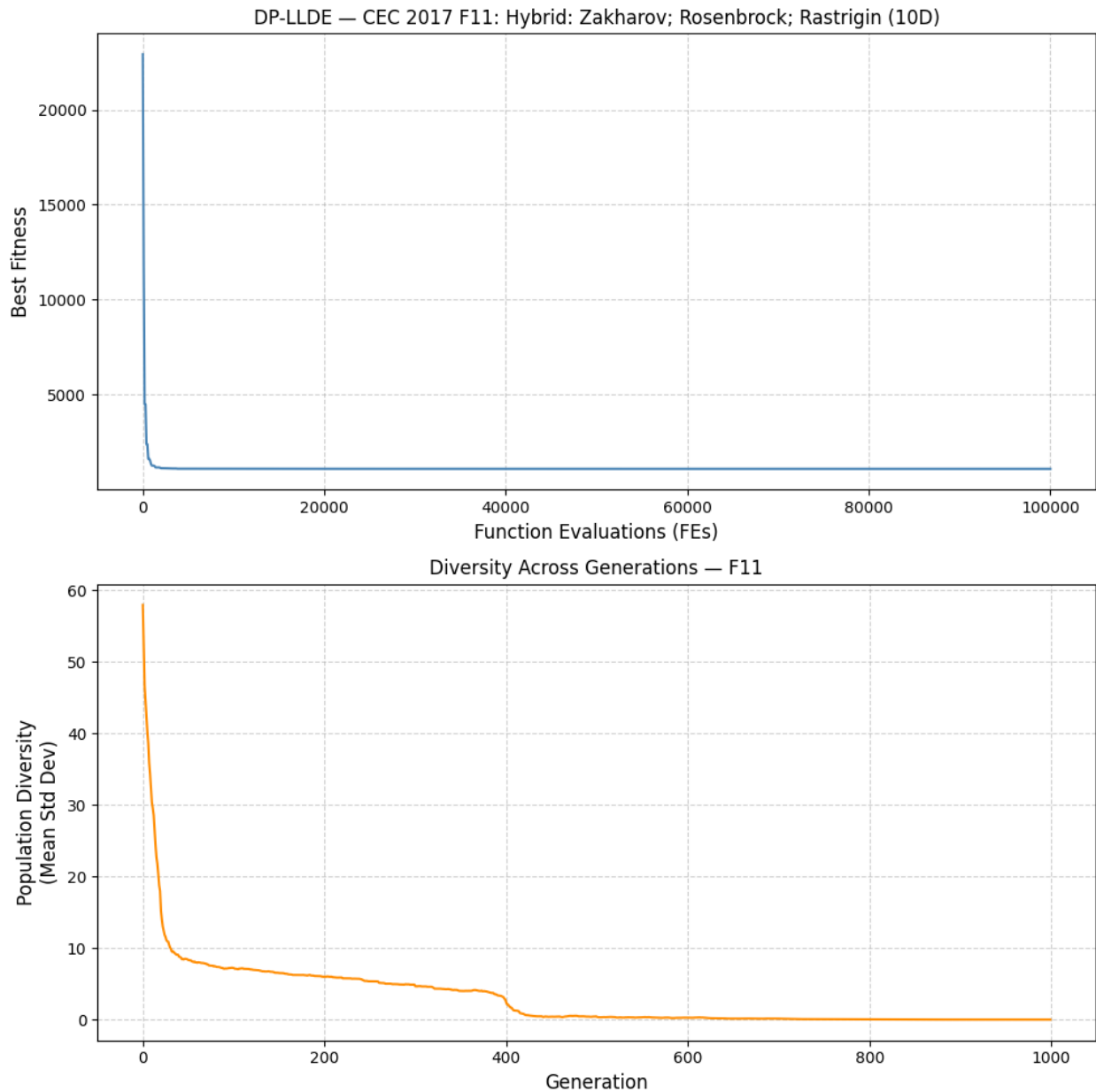
Global Optimum : 1.100000e+03

Error (|best-optimum|) : 8.412826e-11

Best Solution (5 dim) : [-24.75258849 -16.78695993 -58.62265609
9.08308829 19.51774179]

D^(0) (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F11.png

```
# =====  
# F12 – Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)  
# =====  
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid:  
Elliptic; Schwefel; Bent Cigar')
```

```
=====
```

DP-LLDE on CEC 2017 F12 – Single Run

```
=====
```

=====

Function : Hybrid: Elliptic; Schwefel; Bent Cigar (10D)

Bounds : [-100, 100]

Global optimum : 1200.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = $5.7928e+00$

Generation 0: Best Fitness = $7.052719e+09$

Generation 10: Best = $1.627111e+08$ FES = 1100 $\mu_F = 0.573$ $\mu_{CR} = 0.351$ Diversity = $2.8978e+01$ $k_t = [1, 1, 1, 1]$

Generation 20: Best = $1.917464e+07$ FES = 2100 $\mu_F = 0.637$ $\mu_{CR} = 0.348$ Diversity = $2.1528e+01$ $k_t = [3, 4, 2, 3]$

Generation 30: Best = $2.601435e+06$ FES = 3100 $\mu_F = 0.671$ $\mu_{CR} = 0.337$ Diversity = $1.9530e+01$ $k_t = [1, 1, 1, 1]$

Generation 40: Best = $3.248931e+05$ FES = 4100 $\mu_F = 0.707$ $\mu_{CR} = 0.340$ Diversity = $1.7305e+01$ $k_t = [3, 3, 2, 4]$

Generation 50: Best = $9.309877e+04$ FES = 5100 $\mu_F = 0.745$ $\mu_{CR} = 0.275$ Diversity = $1.7237e+01$ $k_t = [1, 1, 1, 1]$

Generation 60: Best = $3.565290e+04$ FES = 6100 $\mu_F = 0.745$ $\mu_{CR} = 0.264$ Diversity = $1.7742e+01$ $k_t = [3, 3, 2, 4]$

Generation 70: Best = $2.005953e+04$ FES = 7100 $\mu_F = 0.754$ $\mu_{CR} = 0.259$ Diversity = $1.3791e+01$ $k_t = [1, 1, 1, 1]$

Generation 80: Best = $1.454209e+04$ FES = 8100 $\mu_F = 0.777$ $\mu_{CR} = 0.219$ Diversity = $1.2135e+01$ $k_t = [4, 4, 2, 4]$

Generation 90: Best = $1.165461e+04$ FES = 9100 $\mu_F = 0.776$ $\mu_{CR} = 0.199$ Diversity = $1.1679e+01$ $k_t = [1, 1, 1, 1]$

Generation 100: Best = $1.008554e+04$ FES = 10100 $\mu_F = 0.788$ $\mu_{CR} = 0.186$ Diversity = $1.1242e+01$ $k_t = [4, 2, 2, 3]$

Generation 110: Best = $8.151984e+03$ FES = 11100 $\mu_F = 0.788$ $\mu_{CR} = 0.153$ Diversity = $1.2027e+01$ $k_t = [1, 1, 1, 1]$

Generation 120: Best = $7.330858e+03$ FES = 12100 $\mu_F = 0.821$ $\mu_{CR} = 0.126$ Diversity = $1.1691e+01$ $k_t = [2, 4, 2, 4]$

Generation 130: Best = $6.802897e+03$ FES = 13100 $\mu_F = 0.814$ $\mu_{CR} = 0.120$ Diversity = $1.0958e+01$ $k_t = [1, 1, 1, 1]$

Generation 140: Best = $6.106865e+03$ FES = 14100 $\mu_F = 0.825$ $\mu_{CR} = 0.097$ Diversity = $9.6725e+00$ $k_t = [4, 3, 3, 4]$

Generation 150: Best = $5.274377e+03$ FES = 15100 $\mu_F = 0.822$ $\mu_{CR} = 0.090$ Diversity = $8.9900e+00$ $k_t = [1, 1, 1, 1]$

Generation 160: Best = $4.152990e+03$ FES = 16100 $\mu_F = 0.829$ $\mu_{CR} = 0.082$ Diversity = $8.7067e+00$ $k_t = [2, 3, 2, 3]$

Generation 170: Best = $3.468085e+03$ FES = 17100 $\mu_F = 0.841$ $\mu_{CR} = 0.073$ Diversity = $8.4634e+00$ $k_t = [1, 1, 1, 1]$

Generation 180: Best = $3.468085e+03$ FES = 18100 $\mu_F = 0.854$ $\mu_{CR} = 0.088$ Diversity = $8.9472e+00$ $k_t = [2, 3, 3, 3]$

Generation 190: Best = $3.210681e+03$ FES = 19100 $\mu_F = 0.873$ $\mu_{CR} = 0.094$ Diversity = $1.1603e+01$ $k_t = [1, 1, 1, 1]$

Generation 200: Best = $2.510921e+03$ FES = 20100 $\mu_F = 0.868$ $\mu_{CR} = 0.081$ Diversity = $1.4286e+01$ $k_t = [2, 4, 4, 3]$

Generation 210: Best = $2.510921e+03$ FES = 21100 $\mu_F = 0.854$ $\mu_{CR} = 0.091$ Diversity = $1.5227e+01$ $k_t = [1, 1, 1, 1]$

Generation 220: Best = $2.239711e+03$ FES = 22100 $\mu_F = 0.843$ μ_{CR}

= 0.075	Diversity = 1.6547e+01	k_t = [3, 4, 4, 2]		
Generation 230:	Best = 2.123812e+03	FES = 23100	$\mu_F = 0.840$	μ_{CR}
= 0.077	Diversity = 1.5557e+01	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 1.839492e+03	FES = 24100	$\mu_F = 0.833$	μ_{CR}
= 0.075	Diversity = 1.5460e+01	k_t = [4, 4, 3, 3]		
Generation 250:	Best = 1.839492e+03	FES = 25100	$\mu_F = 0.836$	μ_{CR}
= 0.098	Diversity = 1.6269e+01	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 1.839492e+03	FES = 26100	$\mu_F = 0.841$	μ_{CR}
= 0.090	Diversity = 1.6725e+01	k_t = [4, 3, 4, 3]		
Generation 270:	Best = 1.697378e+03	FES = 27100	$\mu_F = 0.851$	μ_{CR}
= 0.102	Diversity = 1.7661e+01	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 1.697378e+03	FES = 28100	$\mu_F = 0.812$	μ_{CR}
= 0.112	Diversity = 1.7046e+01	k_t = [4, 3, 4, 3]		
Generation 290:	Best = 1.697378e+03	FES = 29100	$\mu_F = 0.818$	μ_{CR}
= 0.113	Diversity = 1.6566e+01	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 1.519064e+03	FES = 30100	$\mu_F = 0.813$	μ_{CR}
= 0.096	Diversity = 1.6639e+01	k_t = [3, 3, 4, 2]		
Generation 310:	Best = 1.484407e+03	FES = 31100	$\mu_F = 0.795$	μ_{CR}
= 0.085	Diversity = 1.6707e+01	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 1.484407e+03	FES = 32100	$\mu_F = 0.782$	μ_{CR}
= 0.082	Diversity = 1.6654e+01	k_t = [2, 3, 2, 3]		
Generation 330:	Best = 1.484407e+03	FES = 33100	$\mu_F = 0.777$	μ_{CR}
= 0.062	Diversity = 1.6029e+01	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 1.480291e+03	FES = 34100	$\mu_F = 0.789$	μ_{CR}
= 0.100	Diversity = 1.5221e+01	k_t = [2, 4, 4, 4]		
Generation 350:	Best = 1.480291e+03	FES = 35100	$\mu_F = 0.749$	μ_{CR}
= 0.101	Diversity = 1.4602e+01	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 1.480291e+03	FES = 36100	$\mu_F = 0.610$	μ_{CR}
= 0.099	Diversity = 1.4517e+01	k_t = [3, 3, 3, 2]		
Generation 370:	Best = 1.480291e+03	FES = 37100	$\mu_F = 0.599$	μ_{CR}
= 0.085	Diversity = 1.4145e+01	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 1.402095e+03	FES = 38100	$\mu_F = 0.574$	μ_{CR}
= 0.083	Diversity = 1.3624e+01	k_t = [4, 2, 2, 3]		
Generation 390:	Best = 1.402095e+03	FES = 39100	$\mu_F = 0.598$	μ_{CR}
= 0.110	Diversity = 1.3006e+01	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 1.363954e+03	FES = 40100	$\mu_F = 0.600$	μ_{CR}
= 0.110	Diversity = 1.2709e+01	k_t = [4, 3, 4, 3]		
Generation 410:	Best = 1.363954e+03	FES = 41100	$\mu_F = 0.622$	μ_{CR}
= 0.105	Diversity = 1.2526e+01	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 1.363954e+03	FES = 42100	$\mu_F = 0.679$	μ_{CR}
= 0.097	Diversity = 1.1722e+01	k_t = [2, 3, 2, 3]		
Generation 430:	Best = 1.312984e+03	FES = 43100	$\mu_F = 0.639$	μ_{CR}
= 0.114	Diversity = 1.1662e+01	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 1.312984e+03	FES = 44100	$\mu_F = 0.601$	μ_{CR}
= 0.101	Diversity = 1.1385e+01	k_t = [4, 3, 2, 4]		
Generation 450:	Best = 1.301347e+03	FES = 45100	$\mu_F = 0.574$	μ_{CR}
= 0.087	Diversity = 1.0691e+01	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 1.301347e+03	FES = 46100	$\mu_F = 0.677$	μ_{CR}
= 0.080	Diversity = 1.0241e+01	k_t = [3, 4, 2, 3]		

Generation 470: Best = 1.301347e+03	FES = 47100	$\mu_F = 0.602$	μ_{CR}
= 0.077 Diversity = 1.0044e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.301347e+03	FES = 48100	$\mu_F = 0.628$	μ_{CR}
= 0.068 Diversity = 9.8680e+00	$k_t = [4, 2, 4, 3]$		
Generation 490: Best = 1.301347e+03	FES = 49100	$\mu_F = 0.598$	μ_{CR}
= 0.072 Diversity = 9.4545e+00	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 1.301347e+03	FES = 50100	$\mu_F = 0.579$	μ_{CR}
= 0.073 Diversity = 8.9083e+00	$k_t = [3, 3, 4, 3]$		
Generation 510: Best = 1.270986e+03	FES = 51100	$\mu_F = 0.646$	μ_{CR}
= 0.070 Diversity = 8.3723e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.270986e+03	FES = 52100	$\mu_F = 0.705$	μ_{CR}
= 0.086 Diversity = 7.6136e+00	$k_t = [2, 2, 2, 4]$		
Generation 530: Best = 1.270986e+03	FES = 53100	$\mu_F = 0.649$	μ_{CR}
= 0.107 Diversity = 7.3753e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.270986e+03	FES = 54100	$\mu_F = 0.702$	μ_{CR}
= 0.113 Diversity = 6.9504e+00	$k_t = [4, 2, 4, 4]$		
Generation 550: Best = 1.270986e+03	FES = 55100	$\mu_F = 0.681$	μ_{CR}
= 0.103 Diversity = 6.7719e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.269984e+03	FES = 56100	$\mu_F = 0.723$	μ_{CR}
= 0.102 Diversity = 6.5981e+00	$k_t = [3, 4, 2, 3]$		
Generation 570: Best = 1.268549e+03	FES = 57100	$\mu_F = 0.716$	μ_{CR}
= 0.098 Diversity = 6.4988e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.268549e+03	FES = 58100	$\mu_F = 0.743$	μ_{CR}
= 0.100 Diversity = 6.4528e+00	$k_t = [4, 2, 3, 3]$		
Generation 590: Best = 1.268549e+03	FES = 59100	$\mu_F = 0.749$	μ_{CR}
= 0.093 Diversity = 6.4111e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.268549e+03	FES = 60100	$\mu_F = 0.728$	μ_{CR}
= 0.091 Diversity = 6.2509e+00	$k_t = [4, 2, 2, 2]$		
Generation 610: Best = 1.268549e+03	FES = 61100	$\mu_F = 0.706$	μ_{CR}
= 0.092 Diversity = 6.0981e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.251297e+03	FES = 62100	$\mu_F = 0.726$	μ_{CR}
= 0.109 Diversity = 6.1192e+00	$k_t = [3, 3, 2, 3]$		
Generation 630: Best = 1.251297e+03	FES = 63100	$\mu_F = 0.599$	μ_{CR}
= 0.092 Diversity = 6.0621e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.250669e+03	FES = 64100	$\mu_F = 0.534$	μ_{CR}
= 0.080 Diversity = 5.9867e+00	$k_t = [4, 3, 3, 2]$		
Generation 650: Best = 1.229548e+03	FES = 65100	$\mu_F = 0.585$	μ_{CR}
= 0.102 Diversity = 5.8497e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.229548e+03	FES = 66100	$\mu_F = 0.680$	μ_{CR}
= 0.086 Diversity = 5.4229e+00	$k_t = [3, 4, 3, 3]$		
Generation 670: Best = 1.229548e+03	FES = 67100	$\mu_F = 0.640$	μ_{CR}
= 0.088 Diversity = 5.3862e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.229548e+03	FES = 68100	$\mu_F = 0.552$	μ_{CR}
= 0.069 Diversity = 5.3859e+00	$k_t = [2, 2, 4, 2]$		
Generation 690: Best = 1.229548e+03	FES = 69100	$\mu_F = 0.635$	μ_{CR}
= 0.077 Diversity = 5.3278e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.229548e+03	FES = 70100	$\mu_F = 0.660$	μ_{CR}
= 0.073 Diversity = 4.3838e+00	$k_t = [4, 3, 4, 3]$		
Generation 710: Best = 1.229548e+03	FES = 71100	$\mu_F = 0.623$	μ_{CR}

= 0.077	Diversity = 4.3496e+00	k_t = [1, 1, 1, 1]	μ_F = 0.553	μ_{CR}
Generation 720:	Best = 1.229026e+03	FES = 72100		
= 0.080	Diversity = 4.1351e+00	k_t = [4, 2, 2, 2]	μ_F = 0.615	μ_{CR}
Generation 730:	Best = 1.229026e+03	FES = 73100		
= 0.085	Diversity = 3.9984e+00	k_t = [1, 1, 1, 1]	μ_F = 0.647	μ_{CR}
Generation 740:	Best = 1.229026e+03	FES = 74100		
= 0.070	Diversity = 3.9772e+00	k_t = [4, 4, 2, 2]	μ_F = 0.580	μ_{CR}
Generation 750:	Best = 1.229026e+03	FES = 75100		
= 0.072	Diversity = 3.9144e+00	k_t = [1, 1, 1, 1]	μ_F = 0.498	μ_{CR}
Generation 760:	Best = 1.222908e+03	FES = 76100		
= 0.090	Diversity = 3.9141e+00	k_t = [3, 3, 4, 2]	μ_F = 0.416	μ_{CR}
Generation 770:	Best = 1.222908e+03	FES = 77100		
= 0.094	Diversity = 3.9062e+00	k_t = [1, 1, 1, 1]	μ_F = 0.440	μ_{CR}
Generation 780:	Best = 1.222603e+03	FES = 78100		
= 0.081	Diversity = 3.9279e+00	k_t = [3, 3, 3, 4]	μ_F = 0.375	μ_{CR}
Generation 790:	Best = 1.222603e+03	FES = 79100		
= 0.078	Diversity = 3.9047e+00	k_t = [1, 1, 1, 1]	μ_F = 0.403	μ_{CR}
Generation 800:	Best = 1.222603e+03	FES = 80100		
= 0.067	Diversity = 3.8996e+00	k_t = [2, 4, 4, 3]	μ_F = 0.409	μ_{CR}
Generation 810:	Best = 1.222603e+03	FES = 81100		
= 0.076	Diversity = 3.8895e+00	k_t = [1, 1, 1, 1]	μ_F = 0.393	μ_{CR}
Generation 820:	Best = 1.222603e+03	FES = 82100		
= 0.073	Diversity = 3.8620e+00	k_t = [2, 4, 3, 3]	μ_F = 0.463	μ_{CR}
Generation 830:	Best = 1.222603e+03	FES = 83100		
= 0.092	Diversity = 3.7607e+00	k_t = [1, 1, 1, 1]	μ_F = 0.388	μ_{CR}
Generation 840:	Best = 1.220503e+03	FES = 84100		
= 0.077	Diversity = 3.7622e+00	k_t = [2, 2, 3, 3]	μ_F = 0.347	μ_{CR}
Generation 850:	Best = 1.220503e+03	FES = 85100		
= 0.091	Diversity = 3.7369e+00	k_t = [1, 1, 1, 1]	μ_F = 0.336	μ_{CR}
Generation 860:	Best = 1.220503e+03	FES = 86100		
= 0.099	Diversity = 3.7279e+00	k_t = [2, 4, 4, 3]	μ_F = 0.335	μ_{CR}
Generation 870:	Best = 1.218469e+03	FES = 87100		
= 0.092	Diversity = 3.7265e+00	k_t = [1, 1, 1, 1]	μ_F = 0.461	μ_{CR}
Generation 880:	Best = 1.218469e+03	FES = 88100		
= 0.078	Diversity = 3.4756e+00	k_t = [4, 4, 2, 4]	μ_F = 0.539	μ_{CR}
Generation 890:	Best = 1.218469e+03	FES = 89100		
= 0.084	Diversity = 3.3816e+00	k_t = [1, 1, 1, 1]	μ_F = 0.518	μ_{CR}
Generation 900:	Best = 1.218469e+03	FES = 90100		
= 0.097	Diversity = 3.3065e+00	k_t = [2, 3, 4, 4]	μ_F = 0.506	μ_{CR}
Generation 910:	Best = 1.218469e+03	FES = 91100		
= 0.109	Diversity = 3.3100e+00	k_t = [1, 1, 1, 1]	μ_F = 0.532	μ_{CR}
Generation 920:	Best = 1.218469e+03	FES = 92100		
= 0.108	Diversity = 3.2983e+00	k_t = [3, 2, 3, 3]	μ_F = 0.530	μ_{CR}
Generation 930:	Best = 1.218469e+03	FES = 93100		
= 0.116	Diversity = 3.2006e+00	k_t = [1, 1, 1, 1]	μ_F = 0.477	μ_{CR}
Generation 940:	Best = 1.218469e+03	FES = 94100		
= 0.112	Diversity = 3.1020e+00	k_t = [2, 4, 4, 3]	μ_F = 0.471	μ_{CR}
Generation 950:	Best = 1.218469e+03	FES = 95100		
= 0.099	Diversity = 3.0913e+00	k_t = [1, 1, 1, 1]		

```
Generation 960: Best = 1.218469e+03 FES = 96100  $\mu_F$  = 0.432  $\mu_{CR}$ 
= 0.097 Diversity = 2.9666e+00  $k_t$  = [3, 3, 4, 2]
Generation 970: Best = 1.218469e+03 FES = 97100  $\mu_F$  = 0.391  $\mu_{CR}$ 
= 0.092 Diversity = 2.9635e+00  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.218469e+03 FES = 98100  $\mu_F$  = 0.361  $\mu_{CR}$ 
= 0.103 Diversity = 2.9707e+00  $k_t$  = [2, 4, 3, 2]
Generation 990: Best = 1.218469e+03 FES = 99100  $\mu_F$  = 0.236  $\mu_{CR}$ 
= 0.096 Diversity = 2.9556e+00  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 1.218469e+03

Total Function Evals = 100000

Total Successful Updates = 6446

```
=====
=====
```

Results:

Best Fitness : 1.218469e+03

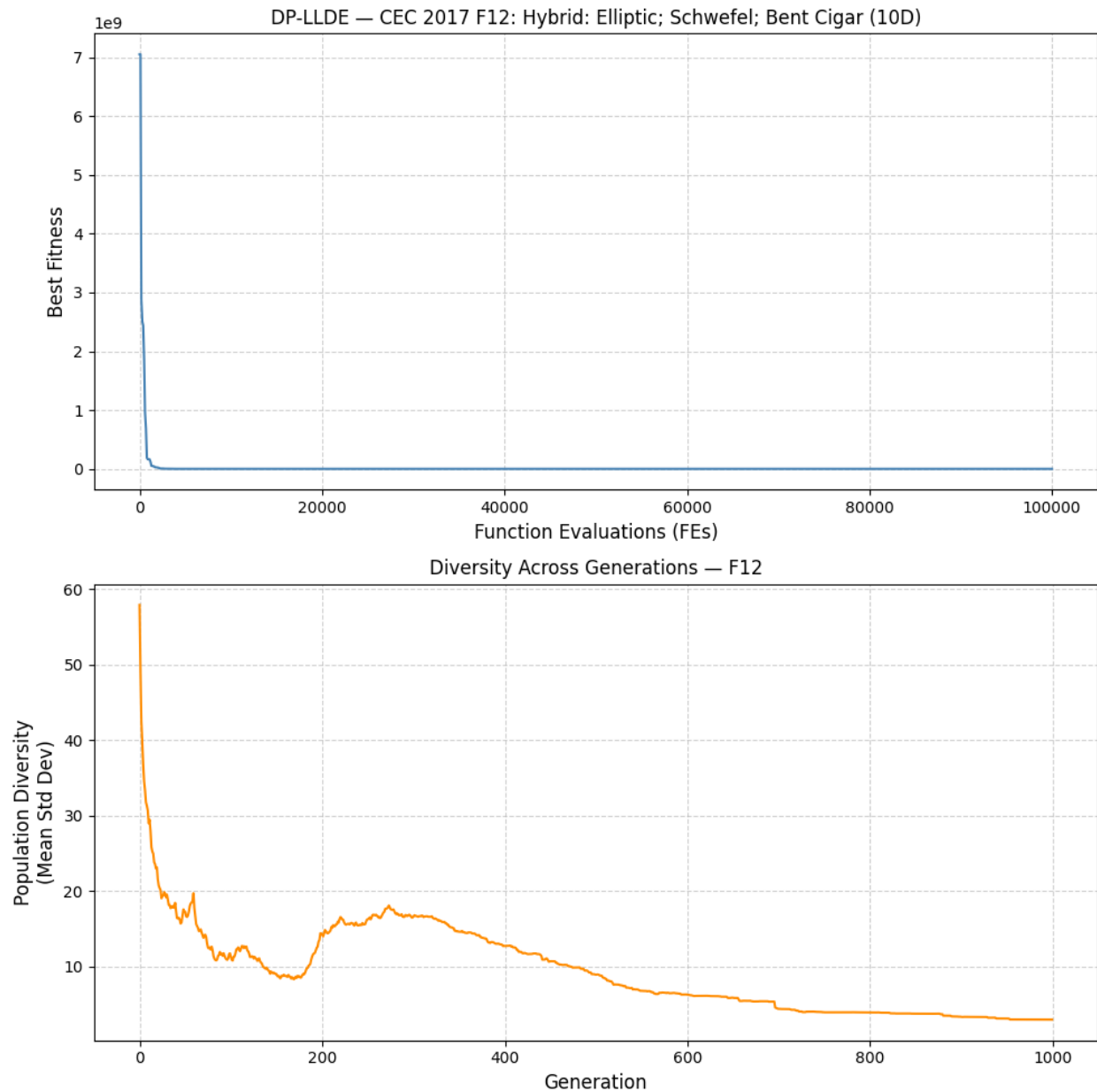
Global Optimum : 1.200000e+03

Error (|best-optimum|) : 1.846903e+01

Best Solution (5 dim) : [66.91177829 23.65883265 -86.82677329
40.44736044 -32.94865397]

D^(0) (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F12.png

```
# =====
# F13 – Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# =====
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent
Cigar; Rosenbrock; Bi-Rastrigin')
```

```
=====
DP-LLDE on CEC 2017 F13 – Single Run
=====
```



```

=====
Function      : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 1300.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 1.212051e+09
Generation 10: Best = 1.045708e+06 FES = 1100  $\mu_F$  = 0.584  $\mu_{CR}$ 
= 0.355 Diversity = 3.2362e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 3.799438e+04 FES = 2100  $\mu_F$  = 0.642  $\mu_{CR}$ 
= 0.355 Diversity = 2.8379e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 7.581503e+03 FES = 3100  $\mu_F$  = 0.683  $\mu_{CR}$ 
= 0.343 Diversity = 2.8756e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 2.871346e+03 FES = 4100  $\mu_F$  = 0.730  $\mu_{CR}$ 
= 0.345 Diversity = 2.7411e+01  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 1.772135e+03 FES = 5100  $\mu_F$  = 0.754  $\mu_{CR}$ 
= 0.357 Diversity = 2.5668e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 1.469520e+03 FES = 6100  $\mu_F$  = 0.766  $\mu_{CR}$ 
= 0.352 Diversity = 2.0091e+01  $k_t$  = [2, 2, 4, 4]
Generation 70: Best = 1.356039e+03 FES = 7100  $\mu_F$  = 0.775  $\mu_{CR}$ 
= 0.341 Diversity = 1.3483e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 1.323341e+03 FES = 8100  $\mu_F$  = 0.774  $\mu_{CR}$ 
= 0.332 Diversity = 9.8047e+00  $k_t$  = [3, 2, 3, 2]
Generation 90: Best = 1.323341e+03 FES = 9100  $\mu_F$  = 0.788  $\mu_{CR}$ 
= 0.306 Diversity = 8.6026e+00  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 1.321813e+03 FES = 10100  $\mu_F$  = 0.786  $\mu_{CR}$ 
= 0.278 Diversity = 7.7309e+00  $k_t$  = [2, 3, 4, 4]
Generation 110: Best = 1.318595e+03 FES = 11100  $\mu_F$  = 0.800  $\mu_{CR}$ 
= 0.258 Diversity = 7.2646e+00  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 1.314261e+03 FES = 12100  $\mu_F$  = 0.780  $\mu_{CR}$ 
= 0.203 Diversity = 6.7657e+00  $k_t$  = [2, 2, 2, 2]
Generation 130: Best = 1.306044e+03 FES = 13100  $\mu_F$  = 0.769  $\mu_{CR}$ 
= 0.153 Diversity = 6.6637e+00  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 1.306044e+03 FES = 14100  $\mu_F$  = 0.688  $\mu_{CR}$ 
= 0.139 Diversity = 6.2794e+00  $k_t$  = [3, 3, 2, 3]
Generation 150: Best = 1.305802e+03 FES = 15100  $\mu_F$  = 0.723  $\mu_{CR}$ 
= 0.116 Diversity = 5.5416e+00  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 1.305802e+03 FES = 16100  $\mu_F$  = 0.728  $\mu_{CR}$ 
= 0.104 Diversity = 5.2013e+00  $k_t$  = [3, 2, 2, 4]
Generation 170: Best = 1.305802e+03 FES = 17100  $\mu_F$  = 0.747  $\mu_{CR}$ 
= 0.085 Diversity = 5.1613e+00  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 1.305713e+03 FES = 18100  $\mu_F$  = 0.767  $\mu_{CR}$ 
= 0.086 Diversity = 4.7962e+00  $k_t$  = [2, 2, 2, 3]
Generation 190: Best = 1.305465e+03 FES = 19100  $\mu_F$  = 0.779  $\mu_{CR}$ 
= 0.090 Diversity = 4.7982e+00  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 1.305465e+03 FES = 20100  $\mu_F$  = 0.753  $\mu_{CR}$ 
= 0.101 Diversity = 4.6379e+00  $k_t$  = [4, 3, 4, 2]
Generation 210: Best = 1.304004e+03 FES = 21100  $\mu_F$  = 0.779  $\mu_{CR}$ 
= 0.090 Diversity = 4.5465e+00  $k_t$  = [1, 1, 1, 1]
Generation 220: Best = 1.303828e+03 FES = 22100  $\mu_F$  = 0.781  $\mu_{CR}$ 

```

= 0.086	Diversity = 4.3442e+00	k_t = [3, 4, 2, 2]	μ_F = 0.780	μ_{CR}
Generation 230:	Best = 1.303828e+03	FES = 23100		
= 0.092	Diversity = 4.2418e+00	k_t = [1, 1, 1, 1]	μ_F = 0.784	μ_{CR}
Generation 240:	Best = 1.303720e+03	FES = 24100		
= 0.092	Diversity = 4.0490e+00	k_t = [3, 2, 4, 2]	μ_F = 0.781	μ_{CR}
Generation 250:	Best = 1.303618e+03	FES = 25100		
= 0.095	Diversity = 3.7255e+00	k_t = [1, 1, 1, 1]	μ_F = 0.785	μ_{CR}
Generation 260:	Best = 1.303403e+03	FES = 26100		
= 0.100	Diversity = 3.6441e+00	k_t = [4, 2, 4, 3]	μ_F = 0.796	μ_{CR}
Generation 270:	Best = 1.303403e+03	FES = 27100		
= 0.091	Diversity = 3.6518e+00	k_t = [1, 1, 1, 1]	μ_F = 0.797	μ_{CR}
Generation 280:	Best = 1.303321e+03	FES = 28100		
= 0.078	Diversity = 3.2238e+00	k_t = [4, 4, 3, 4]	μ_F = 0.824	μ_{CR}
Generation 290:	Best = 1.303321e+03	FES = 29100		
= 0.078	Diversity = 3.1578e+00	k_t = [1, 1, 1, 1]	μ_F = 0.811	μ_{CR}
Generation 300:	Best = 1.303284e+03	FES = 30100		
= 0.093	Diversity = 2.9050e+00	k_t = [3, 4, 4, 4]	μ_F = 0.832	μ_{CR}
Generation 310:	Best = 1.303284e+03	FES = 31100		
= 0.090	Diversity = 2.9132e+00	k_t = [1, 1, 1, 1]	μ_F = 0.815	μ_{CR}
Generation 320:	Best = 1.303142e+03	FES = 32100		
= 0.092	Diversity = 2.9238e+00	k_t = [2, 4, 4, 3]	μ_F = 0.801	μ_{CR}
Generation 330:	Best = 1.303142e+03	FES = 33100		
= 0.098	Diversity = 2.6297e+00	k_t = [1, 1, 1, 1]	μ_F = 0.767	μ_{CR}
Generation 340:	Best = 1.303142e+03	FES = 34100		
= 0.101	Diversity = 2.5797e+00	k_t = [3, 4, 2, 4]	μ_F = 0.767	μ_{CR}
Generation 350:	Best = 1.303122e+03	FES = 35100		
= 0.101	Diversity = 2.6314e+00	k_t = [1, 1, 1, 1]	μ_F = 0.768	μ_{CR}
Generation 360:	Best = 1.303109e+03	FES = 36100		
= 0.098	Diversity = 2.5856e+00	k_t = [2, 3, 3, 3]	μ_F = 0.724	μ_{CR}
Generation 370:	Best = 1.303064e+03	FES = 37100		
= 0.094	Diversity = 2.5615e+00	k_t = [1, 1, 1, 1]	μ_F = 0.687	μ_{CR}
Generation 380:	Best = 1.303064e+03	FES = 38100		
= 0.097	Diversity = 2.5607e+00	k_t = [2, 2, 2, 2]	μ_F = 0.610	μ_{CR}
Generation 390:	Best = 1.303064e+03	FES = 39100		
= 0.111	Diversity = 2.5608e+00	k_t = [1, 1, 1, 1]	μ_F = 0.636	μ_{CR}
Generation 400:	Best = 1.303064e+03	FES = 40100		
= 0.115	Diversity = 2.4773e+00	k_t = [3, 4, 2, 2]	μ_F = 0.644	μ_{CR}
Generation 410:	Best = 1.302948e+03	FES = 41100		
= 0.102	Diversity = 2.4688e+00	k_t = [1, 1, 1, 1]	μ_F = 0.707	μ_{CR}
Generation 420:	Best = 1.302948e+03	FES = 42100		
= 0.118	Diversity = 2.4117e+00	k_t = [4, 3, 2, 4]	μ_F = 0.696	μ_{CR}
Generation 430:	Best = 1.302919e+03	FES = 43100		
= 0.098	Diversity = 2.3919e+00	k_t = [1, 1, 1, 1]	μ_F = 0.755	μ_{CR}
Generation 440:	Best = 1.302458e+03	FES = 44100		
= 0.087	Diversity = 2.3417e+00	k_t = [3, 2, 4, 4]	μ_F = 0.791	μ_{CR}
Generation 450:	Best = 1.302427e+03	FES = 45100		
= 0.089	Diversity = 2.3611e+00	k_t = [1, 1, 1, 1]	μ_F = 0.807	μ_{CR}
Generation 460:	Best = 1.302406e+03	FES = 46100		
= 0.086	Diversity = 2.2509e+00	k_t = [4, 4, 3, 3]		

Generation 470: Best = 1.302402e+03	FES = 47100	$\mu_F = 0.777$	μ_{CR}
= 0.085 Diversity = 2.2973e+00 $k_t = [1, 1, 1, 1]$			
Generation 480: Best = 1.302402e+03	FES = 48100	$\mu_F = 0.785$	μ_{CR}
= 0.095 Diversity = 2.2761e+00 $k_t = [4, 3, 2, 3]$			
Generation 490: Best = 1.302402e+03	FES = 49100	$\mu_F = 0.780$	μ_{CR}
= 0.094 Diversity = 2.2357e+00 $k_t = [1, 1, 1, 1]$			
Generation 500: Best = 1.301766e+03	FES = 50100	$\mu_F = 0.767$	μ_{CR}
= 0.067 Diversity = 2.2177e+00 $k_t = [3, 4, 2, 3]$			
Generation 510: Best = 1.301766e+03	FES = 51100	$\mu_F = 0.786$	μ_{CR}
= 0.083 Diversity = 2.1991e+00 $k_t = [1, 1, 1, 1]$			
Generation 520: Best = 1.301766e+03	FES = 52100	$\mu_F = 0.775$	μ_{CR}
= 0.081 Diversity = 2.1746e+00 $k_t = [3, 2, 4, 4]$			
Generation 530: Best = 1.301750e+03	FES = 53100	$\mu_F = 0.791$	μ_{CR}
= 0.094 Diversity = 2.1351e+00 $k_t = [1, 1, 1, 1]$			
Generation 540: Best = 1.301724e+03	FES = 54100	$\mu_F = 0.797$	μ_{CR}
= 0.083 Diversity = 2.0955e+00 $k_t = [2, 3, 3, 2]$			
Generation 550: Best = 1.301691e+03	FES = 55100	$\mu_F = 0.809$	μ_{CR}
= 0.094 Diversity = 2.0731e+00 $k_t = [1, 1, 1, 1]$			
Generation 560: Best = 1.301675e+03	FES = 56100	$\mu_F = 0.813$	μ_{CR}
= 0.101 Diversity = 2.0861e+00 $k_t = [2, 3, 4, 3]$			
Generation 570: Best = 1.301656e+03	FES = 57100	$\mu_F = 0.813$	μ_{CR}
= 0.088 Diversity = 2.0619e+00 $k_t = [1, 1, 1, 1]$			
Generation 580: Best = 1.301656e+03	FES = 58100	$\mu_F = 0.842$	μ_{CR}
= 0.095 Diversity = 2.0790e+00 $k_t = [3, 3, 2, 2]$			
Generation 590: Best = 1.301371e+03	FES = 59100	$\mu_F = 0.862$	μ_{CR}
= 0.099 Diversity = 2.0336e+00 $k_t = [1, 1, 1, 1]$			
Generation 600: Best = 1.301309e+03	FES = 60100	$\mu_F = 0.880$	μ_{CR}
= 0.085 Diversity = 1.9956e+00 $k_t = [3, 4, 2, 2]$			
Generation 610: Best = 1.301055e+03	FES = 61100	$\mu_F = 0.845$	μ_{CR}
= 0.112 Diversity = 1.9176e+00 $k_t = [1, 1, 1, 1]$			
Generation 620: Best = 1.301055e+03	FES = 62100	$\mu_F = 0.860$	μ_{CR}
= 0.098 Diversity = 1.8216e+00 $k_t = [4, 2, 4, 4]$			
Generation 630: Best = 1.300883e+03	FES = 63100	$\mu_F = 0.816$	μ_{CR}
= 0.099 Diversity = 1.7219e+00 $k_t = [1, 1, 1, 1]$			
Generation 640: Best = 1.300808e+03	FES = 64100	$\mu_F = 0.831$	μ_{CR}
= 0.109 Diversity = 1.6146e+00 $k_t = [4, 3, 2, 2]$			
Generation 650: Best = 1.300554e+03	FES = 65100	$\mu_F = 0.831$	μ_{CR}
= 0.110 Diversity = 1.6348e+00 $k_t = [1, 1, 1, 1]$			
Generation 660: Best = 1.300554e+03	FES = 66100	$\mu_F = 0.825$	μ_{CR}
= 0.122 Diversity = 1.5595e+00 $k_t = [4, 3, 4, 4]$			
Generation 670: Best = 1.300549e+03	FES = 67100	$\mu_F = 0.841$	μ_{CR}
= 0.112 Diversity = 1.5081e+00 $k_t = [1, 1, 1, 1]$			
Generation 680: Best = 1.300500e+03	FES = 68100	$\mu_F = 0.854$	μ_{CR}
= 0.103 Diversity = 1.4321e+00 $k_t = [2, 3, 2, 3]$			
Generation 690: Best = 1.300500e+03	FES = 69100	$\mu_F = 0.885$	μ_{CR}
= 0.107 Diversity = 1.3884e+00 $k_t = [1, 1, 1, 1]$			
Generation 700: Best = 1.300500e+03	FES = 70100	$\mu_F = 0.866$	μ_{CR}
= 0.102 Diversity = 1.3645e+00 $k_t = [3, 3, 2, 4]$			
Generation 710: Best = 1.300408e+03	FES = 71100	$\mu_F = 0.871$	μ_{CR}

= 0.096	Diversity = 1.3345e+00	k_t = [1, 1, 1, 1]	μ_F = 0.865	μ_{CR}
Generation 720:	Best = 1.300373e+03	FES = 72100		
= 0.101	Diversity = 1.3174e+00	k_t = [2, 2, 4, 4]	μ_F = 0.860	μ_{CR}
Generation 730:	Best = 1.300311e+03	FES = 73100		
= 0.097	Diversity = 9.9591e-01	k_t = [1, 1, 1, 1]	μ_F = 0.838	μ_{CR}
Generation 740:	Best = 1.300234e+03	FES = 74100		
= 0.102	Diversity = 9.1998e-01	k_t = [2, 3, 3, 2]	μ_F = 0.816	μ_{CR}
Generation 750:	Best = 1.300176e+03	FES = 75100		
= 0.103	Diversity = 8.1204e-01	k_t = [1, 1, 1, 1]	μ_F = 0.804	μ_{CR}
Generation 760:	Best = 1.300104e+03	FES = 76100		
= 0.107	Diversity = 6.9852e-01	k_t = [4, 3, 4, 2]	μ_F = 0.792	μ_{CR}
Generation 770:	Best = 1.300060e+03	FES = 77100		
= 0.103	Diversity = 7.0934e-01	k_t = [1, 1, 1, 1]	μ_F = 0.760	μ_{CR}
Generation 780:	Best = 1.300059e+03	FES = 78100		
= 0.119	Diversity = 5.4718e-01	k_t = [4, 2, 3, 4]	μ_F = 0.793	μ_{CR}
Generation 790:	Best = 1.300035e+03	FES = 79100		
= 0.120	Diversity = 6.4945e-01	k_t = [1, 1, 1, 1]	μ_F = 0.796	μ_{CR}
Generation 800:	Best = 1.300035e+03	FES = 80100		
= 0.125	Diversity = 5.5504e-01	k_t = [3, 2, 4, 4]	μ_F = 0.781	μ_{CR}
Generation 810:	Best = 1.300019e+03	FES = 81100		
= 0.105	Diversity = 5.2797e-01	k_t = [1, 1, 1, 1]	μ_F = 0.778	μ_{CR}
Generation 820:	Best = 1.300019e+03	FES = 82100		
= 0.100	Diversity = 5.2439e-01	k_t = [3, 3, 4, 3]	μ_F = 0.774	μ_{CR}
Generation 830:	Best = 1.300012e+03	FES = 83100		
= 0.095	Diversity = 4.5264e-01	k_t = [1, 1, 1, 1]	μ_F = 0.778	μ_{CR}
Generation 840:	Best = 1.300012e+03	FES = 84100		
= 0.101	Diversity = 4.0628e-01	k_t = [3, 2, 4, 4]	μ_F = 0.762	μ_{CR}
Generation 850:	Best = 1.300007e+03	FES = 85100		
= 0.096	Diversity = 3.3406e-01	k_t = [1, 1, 1, 1]	μ_F = 0.761	μ_{CR}
Generation 860:	Best = 1.300007e+03	FES = 86100		
= 0.111	Diversity = 3.4378e-01	k_t = [3, 3, 2, 4]	μ_F = 0.755	μ_{CR}
Generation 870:	Best = 1.300007e+03	FES = 87100		
= 0.117	Diversity = 3.6017e-01	k_t = [1, 1, 1, 1]	μ_F = 0.767	μ_{CR}
Generation 880:	Best = 1.300007e+03	FES = 88100		
= 0.115	Diversity = 3.5970e-01	k_t = [3, 2, 2, 2]	μ_F = 0.767	μ_{CR}
Generation 890:	Best = 1.300006e+03	FES = 89100		
= 0.100	Diversity = 2.8986e-01	k_t = [1, 1, 1, 1]	μ_F = 0.768	μ_{CR}
Generation 900:	Best = 1.300006e+03	FES = 90100		
= 0.101	Diversity = 2.0496e-01	k_t = [4, 2, 3, 4]	μ_F = 0.769	μ_{CR}
Generation 910:	Best = 1.300006e+03	FES = 91100		
= 0.102	Diversity = 1.7385e-01	k_t = [1, 1, 1, 1]	μ_F = 0.770	μ_{CR}
Generation 920:	Best = 1.300006e+03	FES = 92100		
= 0.095	Diversity = 1.6264e-01	k_t = [3, 3, 4, 4]	μ_F = 0.770	μ_{CR}
Generation 930:	Best = 1.300006e+03	FES = 93100		
= 0.094	Diversity = 1.4691e-01	k_t = [1, 1, 1, 1]	μ_F = 0.782	μ_{CR}
Generation 940:	Best = 1.300006e+03	FES = 94100		
= 0.088	Diversity = 1.5745e-01	k_t = [2, 2, 2, 2]	μ_F = 0.777	μ_{CR}
Generation 950:	Best = 1.300006e+03	FES = 95100		
= 0.084	Diversity = 1.5672e-01	k_t = [1, 1, 1, 1]		

```

Generation 960: Best = 1.300006e+03 FES = 96100  $\mu_F$  = 0.773  $\mu_{CR}$ 
= 0.084 Diversity = 1.4554e-01  $k_t$  = [4, 2, 4, 2]
Generation 970: Best = 1.300006e+03 FES = 97100  $\mu_F$  = 0.766  $\mu_{CR}$ 
= 0.081 Diversity = 1.2887e-01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.300006e+03 FES = 98100  $\mu_F$  = 0.773  $\mu_{CR}$ 
= 0.089 Diversity = 1.0972e-01  $k_t$  = [2, 4, 3, 2]
Generation 990: Best = 1.300006e+03 FES = 99100  $\mu_F$  = 0.766  $\mu_{CR}$ 
= 0.104 Diversity = 1.0123e-01  $k_t$  = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 1.300005e+03

Total Function Evals = 100000

Total Successful Updates = 11274

```

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```

Results:

Best Fitness : 1.300005e+03

Global Optimum : 1.300000e+03

Error (|best-optimum|) : 5.376836e-03

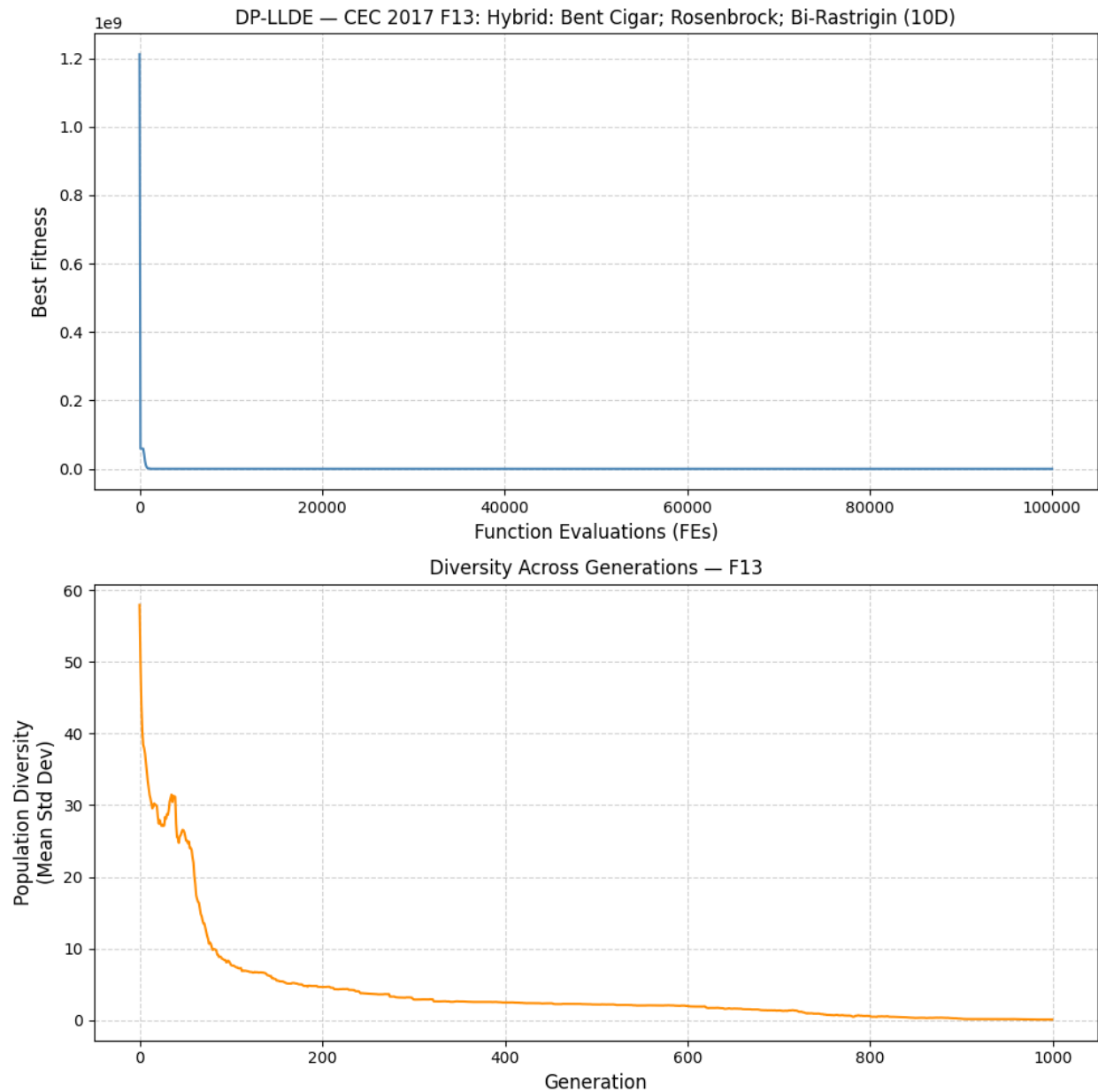
Best Solution (5 dim) : [4.57478136 -71.12263854 -6.25069245 -
14.83525892 25.13689495]

$D^{(0)}$ (init. diversity): 5.7928e+01

```

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Graph saved as convergence_F13.png

```
# =====
# F14 – Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# =====
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid:
Elliptic; Ackley; Schaffer; Rastrigin')
```

```
=====
DP-LLDE on CEC 2017 F14 – Single Run
=====
```

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Function : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (10D)

Bounds : [-100, 100]

Global optimum : 1400.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = $5.7928e+00$

Generation 0: Best Fitness = $2.917787e+04$

Generation 10: Best = $1.075449e+04$ FES = 1100 $\mu_F = 0.584$ $\mu_{CR} = 0.354$ Diversity = $4.0088e+01$ $k_t = [1, 1, 1, 1]$

Generation 20: Best = $1.665269e+03$ FES = 2100 $\mu_F = 0.650$ $\mu_{CR} = 0.349$ Diversity = $3.6596e+01$ $k_t = [3, 4, 2, 3]$

Generation 30: Best = $1.489428e+03$ FES = 3100 $\mu_F = 0.698$ $\mu_{CR} = 0.351$ Diversity = $2.9082e+01$ $k_t = [1, 1, 1, 1]$

Generation 40: Best = $1.444248e+03$ FES = 4100 $\mu_F = 0.732$ $\mu_{CR} = 0.359$ Diversity = $1.9936e+01$ $k_t = [4, 4, 3, 4]$

Generation 50: Best = $1.440402e+03$ FES = 5100 $\mu_F = 0.751$ $\mu_{CR} = 0.366$ Diversity = $1.7377e+01$ $k_t = [1, 1, 1, 1]$

Generation 60: Best = $1.434822e+03$ FES = 6100 $\mu_F = 0.771$ $\mu_{CR} = 0.327$ Diversity = $1.5720e+01$ $k_t = [2, 2, 4, 2]$

Generation 70: Best = $1.427961e+03$ FES = 7100 $\mu_F = 0.796$ $\mu_{CR} = 0.320$ Diversity = $1.5140e+01$ $k_t = [1, 1, 1, 1]$

Generation 80: Best = $1.420224e+03$ FES = 8100 $\mu_F = 0.801$ $\mu_{CR} = 0.289$ Diversity = $1.5143e+01$ $k_t = [4, 4, 2, 4]$

Generation 90: Best = $1.420224e+03$ FES = 9100 $\mu_F = 0.800$ $\mu_{CR} = 0.258$ Diversity = $1.5168e+01$ $k_t = [1, 1, 1, 1]$

Generation 100: Best = $1.420214e+03$ FES = 10100 $\mu_F = 0.805$ $\mu_{CR} = 0.199$ Diversity = $1.5687e+01$ $k_t = [3, 2, 2, 2]$

Generation 110: Best = $1.420214e+03$ FES = 11100 $\mu_F = 0.821$ $\mu_{CR} = 0.169$ Diversity = $1.5861e+01$ $k_t = [1, 1, 1, 1]$

Generation 120: Best = $1.418507e+03$ FES = 12100 $\mu_F = 0.797$ $\mu_{CR} = 0.145$ Diversity = $1.5567e+01$ $k_t = [4, 4, 4, 3]$

Generation 130: Best = $1.418507e+03$ FES = 13100 $\mu_F = 0.780$ $\mu_{CR} = 0.127$ Diversity = $1.5887e+01$ $k_t = [1, 1, 1, 1]$

Generation 140: Best = $1.411947e+03$ FES = 14100 $\mu_F = 0.762$ $\mu_{CR} = 0.115$ Diversity = $1.6210e+01$ $k_t = [2, 2, 2, 3]$

Generation 150: Best = $1.411947e+03$ FES = 15100 $\mu_F = 0.765$ $\mu_{CR} = 0.120$ Diversity = $1.5819e+01$ $k_t = [1, 1, 1, 1]$

Generation 160: Best = $1.411235e+03$ FES = 16100 $\mu_F = 0.783$ $\mu_{CR} = 0.118$ Diversity = $1.5305e+01$ $k_t = [3, 2, 2, 4]$

Generation 170: Best = $1.411235e+03$ FES = 17100 $\mu_F = 0.828$ $\mu_{CR} = 0.117$ Diversity = $1.5031e+01$ $k_t = [1, 1, 1, 1]$

Generation 180: Best = $1.411235e+03$ FES = 18100 $\mu_F = 0.822$ $\mu_{CR} = 0.117$ Diversity = $1.4516e+01$ $k_t = [3, 2, 4, 3]$

Generation 190: Best = $1.411235e+03$ FES = 19100 $\mu_F = 0.825$ $\mu_{CR} = 0.120$ Diversity = $1.3997e+01$ $k_t = [1, 1, 1, 1]$

Generation 200: Best = $1.409687e+03$ FES = 20100 $\mu_F = 0.846$ $\mu_{CR} = 0.116$ Diversity = $1.3086e+01$ $k_t = [4, 3, 3, 3]$

Generation 210: Best = $1.409687e+03$ FES = 21100 $\mu_F = 0.860$ $\mu_{CR} = 0.101$ Diversity = $1.2358e+01$ $k_t = [1, 1, 1, 1]$

Generation 220: Best = $1.409687e+03$ FES = 22100 $\mu_F = 0.880$ μ_{CR}

= 0.120	Diversity = 1.2474e+01	k_t = [3, 2, 3, 3]	μ_F = 0.862	μ_{CR}
Generation 230:	Best = 1.409541e+03	FES = 23100		
= 0.108	Diversity = 1.1995e+01	k_t = [1, 1, 1, 1]	μ_F = 0.872	μ_{CR}
Generation 240:	Best = 1.407837e+03	FES = 24100		
= 0.100	Diversity = 1.1528e+01	k_t = [4, 3, 4, 2]	μ_F = 0.862	μ_{CR}
Generation 250:	Best = 1.406627e+03	FES = 25100		
= 0.103	Diversity = 1.0881e+01	k_t = [1, 1, 1, 1]	μ_F = 0.861	μ_{CR}
Generation 260:	Best = 1.404686e+03	FES = 26100		
= 0.102	Diversity = 9.6829e+00	k_t = [3, 3, 2, 2]	μ_F = 0.849	μ_{CR}
Generation 270:	Best = 1.403750e+03	FES = 27100		
= 0.101	Diversity = 9.0089e+00	k_t = [1, 1, 1, 1]	μ_F = 0.847	μ_{CR}
Generation 280:	Best = 1.403507e+03	FES = 28100		
= 0.105	Diversity = 7.5931e+00	k_t = [2, 2, 4, 4]	μ_F = 0.861	μ_{CR}
Generation 290:	Best = 1.403507e+03	FES = 29100		
= 0.098	Diversity = 7.1263e+00	k_t = [1, 1, 1, 1]	μ_F = 0.828	μ_{CR}
Generation 300:	Best = 1.402507e+03	FES = 30100		
= 0.103	Diversity = 6.9377e+00	k_t = [2, 2, 4, 2]	μ_F = 0.837	μ_{CR}
Generation 310:	Best = 1.402294e+03	FES = 31100		
= 0.104	Diversity = 6.7994e+00	k_t = [1, 1, 1, 1]	μ_F = 0.828	μ_{CR}
Generation 320:	Best = 1.402211e+03	FES = 32100		
= 0.100	Diversity = 5.0507e+00	k_t = [3, 2, 3, 3]	μ_F = 0.855	μ_{CR}
Generation 330:	Best = 1.402024e+03	FES = 33100		
= 0.111	Diversity = 4.9011e+00	k_t = [1, 1, 1, 1]	μ_F = 0.863	μ_{CR}
Generation 340:	Best = 1.401812e+03	FES = 34100		
= 0.125	Diversity = 4.5973e+00	k_t = [2, 4, 2, 4]	μ_F = 0.872	μ_{CR}
Generation 350:	Best = 1.401755e+03	FES = 35100		
= 0.128	Diversity = 4.3852e+00	k_t = [1, 1, 1, 1]	μ_F = 0.869	μ_{CR}
Generation 360:	Best = 1.401694e+03	FES = 36100		
= 0.126	Diversity = 4.0751e+00	k_t = [3, 4, 4, 4]	μ_F = 0.850	μ_{CR}
Generation 370:	Best = 1.401694e+03	FES = 37100		
= 0.120	Diversity = 4.1076e+00	k_t = [1, 1, 1, 1]	μ_F = 0.844	μ_{CR}
Generation 380:	Best = 1.400880e+03	FES = 38100		
= 0.115	Diversity = 4.0974e+00	k_t = [2, 2, 3, 3]	μ_F = 0.845	μ_{CR}
Generation 390:	Best = 1.400852e+03	FES = 39100		
= 0.110	Diversity = 4.0139e+00	k_t = [1, 1, 1, 1]	μ_F = 0.839	μ_{CR}
Generation 400:	Best = 1.400852e+03	FES = 40100		
= 0.097	Diversity = 3.7775e+00	k_t = [3, 2, 2, 4]	μ_F = 0.807	μ_{CR}
Generation 410:	Best = 1.400706e+03	FES = 41100		
= 0.097	Diversity = 3.7157e+00	k_t = [1, 1, 1, 1]	μ_F = 0.823	μ_{CR}
Generation 420:	Best = 1.400654e+03	FES = 42100		
= 0.089	Diversity = 3.6852e+00	k_t = [3, 2, 3, 2]	μ_F = 0.790	μ_{CR}
Generation 430:	Best = 1.400630e+03	FES = 43100		
= 0.095	Diversity = 3.7294e+00	k_t = [1, 1, 1, 1]	μ_F = 0.786	μ_{CR}
Generation 440:	Best = 1.400282e+03	FES = 44100		
= 0.096	Diversity = 3.6910e+00	k_t = [4, 3, 3, 3]	μ_F = 0.799	μ_{CR}
Generation 450:	Best = 1.400282e+03	FES = 45100		
= 0.113	Diversity = 3.6542e+00	k_t = [1, 1, 1, 1]	μ_F = 0.789	μ_{CR}
Generation 460:	Best = 1.400251e+03	FES = 46100		
= 0.109	Diversity = 3.6891e+00	k_t = [3, 3, 4, 2]		

Generation 470: Best = 1.400251e+03	FES = 47100	$\mu_F = 0.830$	μ_{CR}
= 0.119 Diversity = 3.6363e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.400250e+03	FES = 48100	$\mu_F = 0.839$	μ_{CR}
= 0.127 Diversity = 3.6058e+00	$k_t = [2, 4, 3, 3]$		
Generation 490: Best = 1.400250e+03	FES = 49100	$\mu_F = 0.825$	μ_{CR}
= 0.106 Diversity = 3.5566e+00	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 1.400239e+03	FES = 50100	$\mu_F = 0.844$	μ_{CR}
= 0.088 Diversity = 3.6498e+00	$k_t = [2, 3, 2, 4]$		
Generation 510: Best = 1.400238e+03	FES = 51100	$\mu_F = 0.836$	μ_{CR}
= 0.091 Diversity = 3.6504e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.400163e+03	FES = 52100	$\mu_F = 0.837$	μ_{CR}
= 0.092 Diversity = 3.4239e+00	$k_t = [4, 3, 3, 4]$		
Generation 530: Best = 1.400118e+03	FES = 53100	$\mu_F = 0.822$	μ_{CR}
= 0.093 Diversity = 3.4137e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.400113e+03	FES = 54100	$\mu_F = 0.799$	μ_{CR}
= 0.096 Diversity = 3.4149e+00	$k_t = [2, 4, 2, 2]$		
Generation 550: Best = 1.400113e+03	FES = 55100	$\mu_F = 0.793$	μ_{CR}
= 0.106 Diversity = 3.3996e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.400113e+03	FES = 56100	$\mu_F = 0.801$	μ_{CR}
= 0.108 Diversity = 3.3874e+00	$k_t = [3, 3, 4, 3]$		
Generation 570: Best = 1.400105e+03	FES = 57100	$\mu_F = 0.798$	μ_{CR}
= 0.113 Diversity = 3.3055e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.400102e+03	FES = 58100	$\mu_F = 0.783$	μ_{CR}
= 0.103 Diversity = 3.3312e+00	$k_t = [2, 4, 2, 2]$		
Generation 590: Best = 1.400102e+03	FES = 59100	$\mu_F = 0.821$	μ_{CR}
= 0.107 Diversity = 3.2911e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.400102e+03	FES = 60100	$\mu_F = 0.768$	μ_{CR}
= 0.101 Diversity = 3.2966e+00	$k_t = [2, 2, 2, 3]$		
Generation 610: Best = 1.400102e+03	FES = 61100	$\mu_F = 0.742$	μ_{CR}
= 0.089 Diversity = 3.2436e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.400095e+03	FES = 62100	$\mu_F = 0.791$	μ_{CR}
= 0.087 Diversity = 3.2236e+00	$k_t = [3, 2, 4, 2]$		
Generation 630: Best = 1.400095e+03	FES = 63100	$\mu_F = 0.794$	μ_{CR}
= 0.084 Diversity = 3.2060e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.400095e+03	FES = 64100	$\mu_F = 0.762$	μ_{CR}
= 0.080 Diversity = 3.1922e+00	$k_t = [4, 2, 3, 2]$		
Generation 650: Best = 1.400095e+03	FES = 65100	$\mu_F = 0.744$	μ_{CR}
= 0.079 Diversity = 3.1909e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.400095e+03	FES = 66100	$\mu_F = 0.754$	μ_{CR}
= 0.072 Diversity = 3.1922e+00	$k_t = [3, 3, 4, 4]$		
Generation 670: Best = 1.400095e+03	FES = 67100	$\mu_F = 0.757$	μ_{CR}
= 0.091 Diversity = 3.1572e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.400095e+03	FES = 68100	$\mu_F = 0.732$	μ_{CR}
= 0.089 Diversity = 3.1432e+00	$k_t = [3, 3, 4, 2]$		
Generation 690: Best = 1.400091e+03	FES = 69100	$\mu_F = 0.723$	μ_{CR}
= 0.105 Diversity = 3.1257e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.400091e+03	FES = 70100	$\mu_F = 0.790$	μ_{CR}
= 0.107 Diversity = 2.9676e+00	$k_t = [4, 4, 2, 4]$		
Generation 710: Best = 1.400088e+03	FES = 71100	$\mu_F = 0.797$	μ_{CR}

= 0.100	Diversity = 2.9669e+00	k_t = [1, 1, 1, 1]	μ_F = 0.786	μ_{CR}
Generation 720:	Best = 1.400088e+03	FES = 72100		
= 0.082	Diversity = 2.9350e+00	k_t = [2, 4, 4, 3]	μ_F = 0.777	μ_{CR}
Generation 730:	Best = 1.400047e+03	FES = 73100		
= 0.109	Diversity = 2.9032e+00	k_t = [1, 1, 1, 1]	μ_F = 0.793	μ_{CR}
Generation 740:	Best = 1.400021e+03	FES = 74100		
= 0.114	Diversity = 2.8979e+00	k_t = [4, 4, 4, 2]	μ_F = 0.805	μ_{CR}
Generation 750:	Best = 1.400013e+03	FES = 75100		
= 0.107	Diversity = 2.8961e+00	k_t = [1, 1, 1, 1]	μ_F = 0.813	μ_{CR}
Generation 760:	Best = 1.400005e+03	FES = 76100		
= 0.120	Diversity = 2.8948e+00	k_t = [4, 2, 3, 3]	μ_F = 0.781	μ_{CR}
Generation 770:	Best = 1.400005e+03	FES = 77100		
= 0.120	Diversity = 2.8927e+00	k_t = [1, 1, 1, 1]	μ_F = 0.772	μ_{CR}
Generation 780:	Best = 1.400004e+03	FES = 78100		
= 0.118	Diversity = 2.7738e+00	k_t = [2, 3, 3, 4]	μ_F = 0.768	μ_{CR}
Generation 790:	Best = 1.400004e+03	FES = 79100		
= 0.113	Diversity = 2.7513e+00	k_t = [1, 1, 1, 1]	μ_F = 0.751	μ_{CR}
Generation 800:	Best = 1.400003e+03	FES = 80100		
= 0.127	Diversity = 2.6964e+00	k_t = [3, 2, 3, 4]	μ_F = 0.785	μ_{CR}
Generation 810:	Best = 1.400002e+03	FES = 81100		
= 0.129	Diversity = 2.6640e+00	k_t = [1, 1, 1, 1]	μ_F = 0.813	μ_{CR}
Generation 820:	Best = 1.400001e+03	FES = 82100		
= 0.125	Diversity = 2.6584e+00	k_t = [2, 4, 2, 2]	μ_F = 0.781	μ_{CR}
Generation 830:	Best = 1.400001e+03	FES = 83100		
= 0.135	Diversity = 2.6297e+00	k_t = [1, 1, 1, 1]	μ_F = 0.802	μ_{CR}
Generation 840:	Best = 1.400001e+03	FES = 84100		
= 0.124	Diversity = 2.6180e+00	k_t = [2, 3, 4, 3]	μ_F = 0.817	μ_{CR}
Generation 850:	Best = 1.400001e+03	FES = 85100		
= 0.121	Diversity = 2.5883e+00	k_t = [1, 1, 1, 1]	μ_F = 0.809	μ_{CR}
Generation 860:	Best = 1.400001e+03	FES = 86100		
= 0.112	Diversity = 2.5873e+00	k_t = [3, 4, 2, 2]	μ_F = 0.835	μ_{CR}
Generation 870:	Best = 1.400001e+03	FES = 87100		
= 0.110	Diversity = 2.5872e+00	k_t = [1, 1, 1, 1]	μ_F = 0.815	μ_{CR}
Generation 880:	Best = 1.400001e+03	FES = 88100		
= 0.118	Diversity = 2.4008e+00	k_t = [2, 4, 2, 4]	μ_F = 0.823	μ_{CR}
Generation 890:	Best = 1.400001e+03	FES = 89100		
= 0.120	Diversity = 2.3432e+00	k_t = [1, 1, 1, 1]	μ_F = 0.813	μ_{CR}
Generation 900:	Best = 1.400001e+03	FES = 90100		
= 0.129	Diversity = 2.3422e+00	k_t = [3, 2, 4, 2]	μ_F = 0.836	μ_{CR}
Generation 910:	Best = 1.400000e+03	FES = 91100		
= 0.131	Diversity = 2.3410e+00	k_t = [1, 1, 1, 1]	μ_F = 0.833	μ_{CR}
Generation 920:	Best = 1.400000e+03	FES = 92100		
= 0.141	Diversity = 2.0837e+00	k_t = [4, 2, 2, 4]	μ_F = 0.845	μ_{CR}
Generation 930:	Best = 1.400000e+03	FES = 93100		
= 0.142	Diversity = 2.0298e+00	k_t = [1, 1, 1, 1]	μ_F = 0.838	μ_{CR}
Generation 940:	Best = 1.400000e+03	FES = 94100		
= 0.148	Diversity = 1.8971e+00	k_t = [2, 4, 2, 4]	μ_F = 0.819	μ_{CR}
Generation 950:	Best = 1.400000e+03	FES = 95100		
= 0.144	Diversity = 1.8318e+00	k_t = [1, 1, 1, 1]		

```

Generation 960: Best = 1.400000e+03 FES = 96100  $\mu_F$  = 0.818  $\mu_{CR}$ 
= 0.150 Diversity = 1.7431e+00  $k_t$  = [2, 2, 4, 2]
Generation 970: Best = 1.400000e+03 FES = 97100  $\mu_F$  = 0.820  $\mu_{CR}$ 
= 0.153 Diversity = 1.7426e+00  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.400000e+03 FES = 98100  $\mu_F$  = 0.816  $\mu_{CR}$ 
= 0.151 Diversity = 1.7234e+00  $k_t$  = [4, 2, 3, 2]
Generation 990: Best = 1.400000e+03 FES = 99100  $\mu_F$  = 0.818  $\mu_{CR}$ 
= 0.145 Diversity = 1.7234e+00  $k_t$  = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 1.400000e+03

Total Function Evals = 100000

Total Successful Updates = 9286

```

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Results:

Best Fitness : 1.400000e+03

Global Optimum : 1.400000e+03

Error (|best-optimum|) : 1.726725e-05

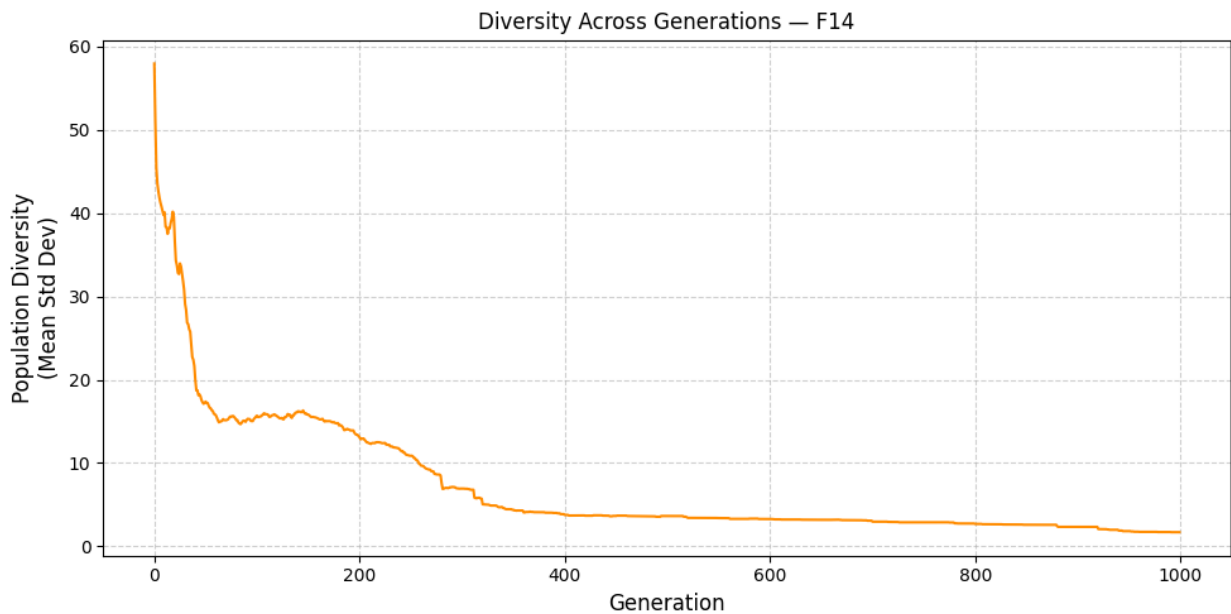
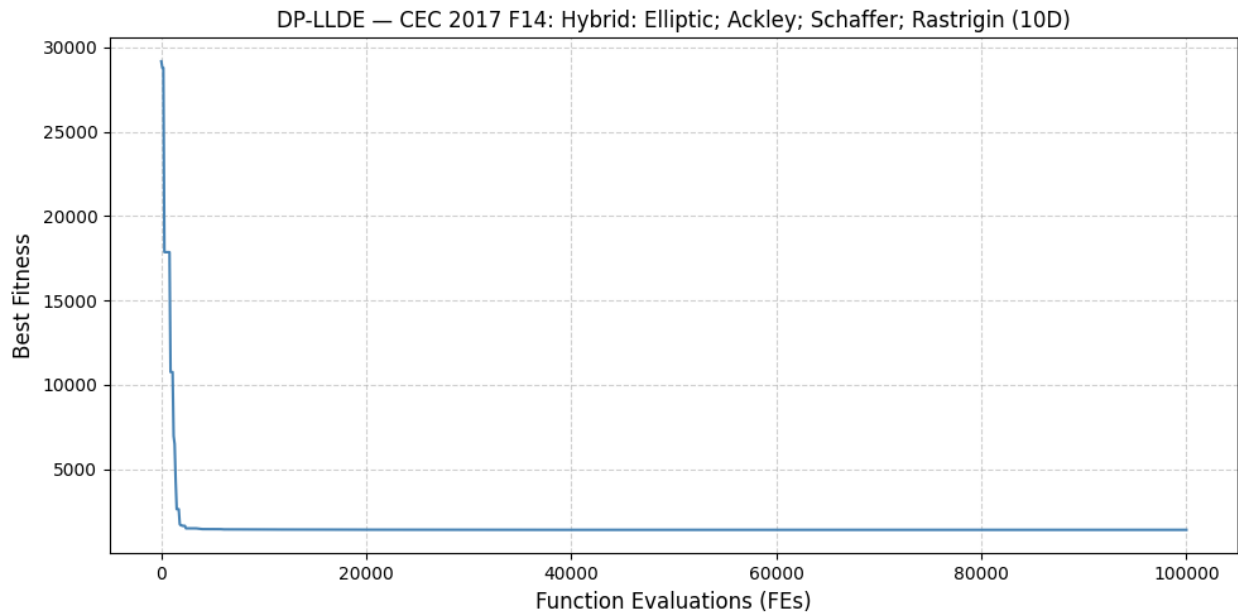
Best Solution (5 dim) : [-36.61541071 -70.77608521 66.34143737 -
39.92138189 61.67148926]

$D^0(0)$ (init. diversity): 5.7928e+01

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```



Graph saved as convergence_F14.png

```
# =====
# F15 – Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# =====
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent
Cigar; HGBat; Rastrigin; Rosenbrock')
```

```
=====
=====
DP-LLDE on CEC 2017 F15 – Single Run
=====
```

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Function : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock
(10D)

Bounds : [-100, 100]

Global optimum : 1500.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 3.114909e+07

Generation 10:	Best = 1.716556e+04	FES = 1100	$\mu_F = 0.595$	μ_{CR}
= 0.352 Diversity = 3.8041e+01 $k_t = [1, 1, 1, 1]$				
Generation 20:	Best = 4.154817e+03	FES = 2100	$\mu_F = 0.661$	μ_{CR}
= 0.351 Diversity = 3.0781e+01 $k_t = [3, 4, 2, 3]$				
Generation 30:	Best = 1.566638e+03	FES = 3100	$\mu_F = 0.708$	μ_{CR}
= 0.352 Diversity = 2.6521e+01 $k_t = [1, 1, 1, 1]$				
Generation 40:	Best = 1.538772e+03	FES = 4100	$\mu_F = 0.735$	μ_{CR}
= 0.362 Diversity = 1.8624e+01 $k_t = [4, 4, 3, 4]$				
Generation 50:	Best = 1.514788e+03	FES = 5100	$\mu_F = 0.760$	μ_{CR}
= 0.364 Diversity = 1.1408e+01 $k_t = [1, 1, 1, 1]$				
Generation 60:	Best = 1.507662e+03	FES = 6100	$\mu_F = 0.764$	μ_{CR}
= 0.360 Diversity = 9.0641e+00 $k_t = [2, 4, 4, 4]$				
Generation 70:	Best = 1.507662e+03	FES = 7100	$\mu_F = 0.769$	μ_{CR}
= 0.360 Diversity = 8.1496e+00 $k_t = [1, 1, 1, 1]$				
Generation 80:	Best = 1.507130e+03	FES = 8100	$\mu_F = 0.743$	μ_{CR}
= 0.323 Diversity = 7.6065e+00 $k_t = [3, 3, 2, 3]$				
Generation 90:	Best = 1.506329e+03	FES = 9100	$\mu_F = 0.755$	μ_{CR}
= 0.282 Diversity = 6.8809e+00 $k_t = [1, 1, 1, 1]$				
Generation 100:	Best = 1.504734e+03	FES = 10100	$\mu_F = 0.732$	μ_{CR}
= 0.263 Diversity = 6.8987e+00 $k_t = [4, 3, 4, 2]$				
Generation 110:	Best = 1.503816e+03	FES = 11100	$\mu_F = 0.768$	μ_{CR}
= 0.238 Diversity = 6.6797e+00 $k_t = [1, 1, 1, 1]$				
Generation 120:	Best = 1.503441e+03	FES = 12100	$\mu_F = 0.797$	μ_{CR}
= 0.202 Diversity = 6.3818e+00 $k_t = [3, 3, 4, 4]$				
Generation 130:	Best = 1.503415e+03	FES = 13100	$\mu_F = 0.805$	μ_{CR}
= 0.176 Diversity = 6.1569e+00 $k_t = [1, 1, 1, 1]$				
Generation 140:	Best = 1.503306e+03	FES = 14100	$\mu_F = 0.839$	μ_{CR}
= 0.151 Diversity = 5.9630e+00 $k_t = [4, 4, 4, 3]$				
Generation 150:	Best = 1.502733e+03	FES = 15100	$\mu_F = 0.818$	μ_{CR}
= 0.149 Diversity = 5.9408e+00 $k_t = [1, 1, 1, 1]$				
Generation 160:	Best = 1.502733e+03	FES = 16100	$\mu_F = 0.813$	μ_{CR}
= 0.130 Diversity = 5.7171e+00 $k_t = [3, 2, 3, 3]$				
Generation 170:	Best = 1.502710e+03	FES = 17100	$\mu_F = 0.830$	μ_{CR}
= 0.140 Diversity = 5.5951e+00 $k_t = [1, 1, 1, 1]$				
Generation 180:	Best = 1.502710e+03	FES = 18100	$\mu_F = 0.841$	μ_{CR}
= 0.127 Diversity = 5.6386e+00 $k_t = [2, 4, 3, 3]$				
Generation 190:	Best = 1.502710e+03	FES = 19100	$\mu_F = 0.843$	μ_{CR}
= 0.123 Diversity = 5.5900e+00 $k_t = [1, 1, 1, 1]$				
Generation 200:	Best = 1.502565e+03	FES = 20100	$\mu_F = 0.863$	μ_{CR}
= 0.092 Diversity = 5.2387e+00 $k_t = [2, 3, 4, 4]$				
Generation 210:	Best = 1.501979e+03	FES = 21100	$\mu_F = 0.859$	μ_{CR}
= 0.103 Diversity = 5.2618e+00 $k_t = [1, 1, 1, 1]$				

Generation 220: Best = 1.501763e+03	FES = 22100	$\mu_F = 0.787$	μ_{CR}
= 0.105 Diversity = 5.1446e+00	$k_t = [3, 4, 4, 2]$		
Generation 230: Best = 1.501697e+03	FES = 23100	$\mu_F = 0.811$	μ_{CR}
= 0.119 Diversity = 5.1551e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.501690e+03	FES = 24100	$\mu_F = 0.862$	μ_{CR}
= 0.118 Diversity = 4.9677e+00	$k_t = [4, 3, 3, 4]$		
Generation 250: Best = 1.501542e+03	FES = 25100	$\mu_F = 0.855$	μ_{CR}
= 0.108 Diversity = 4.8464e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.501424e+03	FES = 26100	$\mu_F = 0.859$	μ_{CR}
= 0.108 Diversity = 4.6446e+00	$k_t = [2, 2, 2, 4]$		
Generation 270: Best = 1.501424e+03	FES = 27100	$\mu_F = 0.821$	μ_{CR}
= 0.110 Diversity = 4.4856e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.501315e+03	FES = 28100	$\mu_F = 0.825$	μ_{CR}
= 0.119 Diversity = 4.2419e+00	$k_t = [2, 4, 4, 3]$		
Generation 290: Best = 1.501076e+03	FES = 29100	$\mu_F = 0.818$	μ_{CR}
= 0.130 Diversity = 4.1040e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.500995e+03	FES = 30100	$\mu_F = 0.843$	μ_{CR}
= 0.131 Diversity = 3.9439e+00	$k_t = [3, 2, 3, 3]$		
Generation 310: Best = 1.500950e+03	FES = 31100	$\mu_F = 0.880$	μ_{CR}
= 0.129 Diversity = 3.4311e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.500800e+03	FES = 32100	$\mu_F = 0.878$	μ_{CR}
= 0.129 Diversity = 2.4297e+00	$k_t = [3, 3, 3, 2]$		
Generation 330: Best = 1.500800e+03	FES = 33100	$\mu_F = 0.863$	μ_{CR}
= 0.136 Diversity = 1.8159e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.500527e+03	FES = 34100	$\mu_F = 0.839$	μ_{CR}
= 0.136 Diversity = 1.6898e+00	$k_t = [2, 2, 4, 3]$		
Generation 350: Best = 1.500467e+03	FES = 35100	$\mu_F = 0.830$	μ_{CR}
= 0.131 Diversity = 1.6594e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.500444e+03	FES = 36100	$\mu_F = 0.814$	μ_{CR}
= 0.125 Diversity = 1.5355e+00	$k_t = [4, 3, 2, 4]$		
Generation 370: Best = 1.500380e+03	FES = 37100	$\mu_F = 0.798$	μ_{CR}
= 0.126 Diversity = 1.6789e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.500380e+03	FES = 38100	$\mu_F = 0.822$	μ_{CR}
= 0.113 Diversity = 1.7891e+00	$k_t = [3, 2, 2, 2]$		
Generation 390: Best = 1.500316e+03	FES = 39100	$\mu_F = 0.831$	μ_{CR}
= 0.111 Diversity = 1.9374e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.500256e+03	FES = 40100	$\mu_F = 0.829$	μ_{CR}
= 0.121 Diversity = 2.0130e+00	$k_t = [3, 3, 3, 2]$		
Generation 410: Best = 1.500244e+03	FES = 41100	$\mu_F = 0.813$	μ_{CR}
= 0.105 Diversity = 2.0832e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.500224e+03	FES = 42100	$\mu_F = 0.799$	μ_{CR}
= 0.106 Diversity = 2.1333e+00	$k_t = [3, 4, 3, 3]$		
Generation 430: Best = 1.500189e+03	FES = 43100	$\mu_F = 0.817$	μ_{CR}
= 0.093 Diversity = 2.0825e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.500182e+03	FES = 44100	$\mu_F = 0.780$	μ_{CR}
= 0.107 Diversity = 2.0663e+00	$k_t = [4, 4, 2, 2]$		
Generation 450: Best = 1.500178e+03	FES = 45100	$\mu_F = 0.808$	μ_{CR}
= 0.096 Diversity = 2.0084e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.500176e+03	FES = 46100	$\mu_F = 0.802$	μ_{CR}

= 0.100	Diversity = 1.9192e+00	k_t = [4, 2, 3, 2]		
Generation 470:	Best = 1.500176e+03	FES = 47100	$\mu_F = 0.788$	μ_{CR}
= 0.089	Diversity = 1.7356e+00	k_t = [1, 1, 1, 1]		
Generation 480:	Best = 1.500176e+03	FES = 48100	$\mu_F = 0.758$	μ_{CR}
= 0.090	Diversity = 1.6309e+00	k_t = [2, 3, 2, 2]		
Generation 490:	Best = 1.500176e+03	FES = 49100	$\mu_F = 0.711$	μ_{CR}
= 0.099	Diversity = 1.4381e+00	k_t = [1, 1, 1, 1]		
Generation 500:	Best = 1.500153e+03	FES = 50100	$\mu_F = 0.708$	μ_{CR}
= 0.107	Diversity = 1.3387e+00	k_t = [4, 3, 3, 4]		
Generation 510:	Best = 1.500152e+03	FES = 51100	$\mu_F = 0.712$	μ_{CR}
= 0.115	Diversity = 1.2803e+00	k_t = [1, 1, 1, 1]		
Generation 520:	Best = 1.500152e+03	FES = 52100	$\mu_F = 0.746$	μ_{CR}
= 0.112	Diversity = 1.2781e+00	k_t = [4, 3, 4, 2]		
Generation 530:	Best = 1.500152e+03	FES = 53100	$\mu_F = 0.758$	μ_{CR}
= 0.114	Diversity = 1.2544e+00	k_t = [1, 1, 1, 1]		
Generation 540:	Best = 1.500152e+03	FES = 54100	$\mu_F = 0.765$	μ_{CR}
= 0.126	Diversity = 1.2294e+00	k_t = [2, 4, 3, 3]		
Generation 550:	Best = 1.500147e+03	FES = 55100	$\mu_F = 0.754$	μ_{CR}
= 0.113	Diversity = 1.2411e+00	k_t = [1, 1, 1, 1]		
Generation 560:	Best = 1.500130e+03	FES = 56100	$\mu_F = 0.741$	μ_{CR}
= 0.120	Diversity = 1.1974e+00	k_t = [2, 3, 2, 2]		
Generation 570:	Best = 1.500115e+03	FES = 57100	$\mu_F = 0.757$	μ_{CR}
= 0.118	Diversity = 1.2093e+00	k_t = [1, 1, 1, 1]		
Generation 580:	Best = 1.500115e+03	FES = 58100	$\mu_F = 0.725$	μ_{CR}
= 0.119	Diversity = 1.2158e+00	k_t = [3, 2, 4, 4]		
Generation 590:	Best = 1.500105e+03	FES = 59100	$\mu_F = 0.730$	μ_{CR}
= 0.117	Diversity = 1.1274e+00	k_t = [1, 1, 1, 1]		
Generation 600:	Best = 1.500096e+03	FES = 60100	$\mu_F = 0.749$	μ_{CR}
= 0.116	Diversity = 1.1311e+00	k_t = [2, 3, 2, 2]		
Generation 610:	Best = 1.500096e+03	FES = 61100	$\mu_F = 0.798$	μ_{CR}
= 0.110	Diversity = 1.1147e+00	k_t = [1, 1, 1, 1]		
Generation 620:	Best = 1.500077e+03	FES = 62100	$\mu_F = 0.768$	μ_{CR}
= 0.118	Diversity = 1.0502e+00	k_t = [3, 4, 4, 3]		
Generation 630:	Best = 1.500077e+03	FES = 63100	$\mu_F = 0.804$	μ_{CR}
= 0.111	Diversity = 1.0111e+00	k_t = [1, 1, 1, 1]		
Generation 640:	Best = 1.500077e+03	FES = 64100	$\mu_F = 0.810$	μ_{CR}
= 0.109	Diversity = 9.8579e-01	k_t = [2, 4, 4, 4]		
Generation 650:	Best = 1.500077e+03	FES = 65100	$\mu_F = 0.812$	μ_{CR}
= 0.105	Diversity = 1.0224e+00	k_t = [1, 1, 1, 1]		
Generation 660:	Best = 1.500077e+03	FES = 66100	$\mu_F = 0.826$	μ_{CR}
= 0.101	Diversity = 1.0369e+00	k_t = [3, 3, 3, 3]		
Generation 670:	Best = 1.500076e+03	FES = 67100	$\mu_F = 0.841$	μ_{CR}
= 0.101	Diversity = 9.9991e-01	k_t = [1, 1, 1, 1]		
Generation 680:	Best = 1.500076e+03	FES = 68100	$\mu_F = 0.839$	μ_{CR}
= 0.109	Diversity = 9.7668e-01	k_t = [4, 4, 4, 4]		
Generation 690:	Best = 1.500065e+03	FES = 69100	$\mu_F = 0.808$	μ_{CR}
= 0.112	Diversity = 9.1230e-01	k_t = [1, 1, 1, 1]		
Generation 700:	Best = 1.500065e+03	FES = 70100	$\mu_F = 0.838$	μ_{CR}
= 0.111	Diversity = 8.7703e-01	k_t = [4, 3, 4, 3]		

Generation 710: Best = 1.500065e+03	FES = 71100	$\mu_F = 0.826$	μ_{CR}
= 0.091 Diversity = 8.6233e-01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.500054e+03	FES = 72100	$\mu_F = 0.825$	μ_{CR}
= 0.119 Diversity = 7.8597e-01	$k_t = [3, 2, 4, 4]$		
Generation 730: Best = 1.500054e+03	FES = 73100	$\mu_F = 0.794$	μ_{CR}
= 0.110 Diversity = 7.5957e-01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.500038e+03	FES = 74100	$\mu_F = 0.768$	μ_{CR}
= 0.100 Diversity = 7.2306e-01	$k_t = [4, 2, 4, 4]$		
Generation 750: Best = 1.500038e+03	FES = 75100	$\mu_F = 0.781$	μ_{CR}
= 0.097 Diversity = 7.2007e-01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.500038e+03	FES = 76100	$\mu_F = 0.789$	μ_{CR}
= 0.100 Diversity = 7.0669e-01	$k_t = [3, 4, 4, 2]$		
Generation 770: Best = 1.500038e+03	FES = 77100	$\mu_F = 0.785$	μ_{CR}
= 0.110 Diversity = 7.2883e-01	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 1.500038e+03	FES = 78100	$\mu_F = 0.791$	μ_{CR}
= 0.112 Diversity = 7.4033e-01	$k_t = [2, 2, 4, 3]$		
Generation 790: Best = 1.500036e+03	FES = 79100	$\mu_F = 0.825$	μ_{CR}
= 0.109 Diversity = 7.2063e-01	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 1.500036e+03	FES = 80100	$\mu_F = 0.803$	μ_{CR}
= 0.096 Diversity = 7.0228e-01	$k_t = [3, 3, 3, 2]$		
Generation 810: Best = 1.500036e+03	FES = 81100	$\mu_F = 0.814$	μ_{CR}
= 0.089 Diversity = 7.0620e-01	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 1.500013e+03	FES = 82100	$\mu_F = 0.814$	μ_{CR}
= 0.105 Diversity = 6.8132e-01	$k_t = [4, 2, 2, 4]$		
Generation 830: Best = 1.500013e+03	FES = 83100	$\mu_F = 0.790$	μ_{CR}
= 0.112 Diversity = 6.4112e-01	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 1.500013e+03	FES = 84100	$\mu_F = 0.808$	μ_{CR}
= 0.096 Diversity = 6.1342e-01	$k_t = [2, 4, 2, 3]$		
Generation 850: Best = 1.500013e+03	FES = 85100	$\mu_F = 0.825$	μ_{CR}
= 0.111 Diversity = 5.9257e-01	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 1.500013e+03	FES = 86100	$\mu_F = 0.789$	μ_{CR}
= 0.105 Diversity = 6.0108e-01	$k_t = [4, 4, 4, 2]$		
Generation 870: Best = 1.500013e+03	FES = 87100	$\mu_F = 0.799$	μ_{CR}
= 0.105 Diversity = 6.2312e-01	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 1.500011e+03	FES = 88100	$\mu_F = 0.799$	μ_{CR}
= 0.111 Diversity = 6.4447e-01	$k_t = [2, 3, 2, 2]$		
Generation 890: Best = 1.500009e+03	FES = 89100	$\mu_F = 0.806$	μ_{CR}
= 0.112 Diversity = 6.2555e-01	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 1.500009e+03	FES = 90100	$\mu_F = 0.814$	μ_{CR}
= 0.100 Diversity = 7.2250e-01	$k_t = [2, 3, 3, 3]$		
Generation 910: Best = 1.500009e+03	FES = 91100	$\mu_F = 0.804$	μ_{CR}
= 0.097 Diversity = 6.6972e-01	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 1.500009e+03	FES = 92100	$\mu_F = 0.809$	μ_{CR}
= 0.099 Diversity = 6.4442e-01	$k_t = [4, 2, 4, 2]$		
Generation 930: Best = 1.500008e+03	FES = 93100	$\mu_F = 0.802$	μ_{CR}
= 0.097 Diversity = 6.3329e-01	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 1.500003e+03	FES = 94100	$\mu_F = 0.782$	μ_{CR}
= 0.096 Diversity = 6.3022e-01	$k_t = [2, 3, 3, 2]$		
Generation 950: Best = 1.500002e+03	FES = 95100	$\mu_F = 0.791$	μ_{CR}


```

= 0.091 Diversity = 6.4974e-01 k_t = [1, 1, 1, 1]
Generation 960: Best = 1.500002e+03 FES = 96100  $\mu_F$  = 0.796  $\mu_{CR}$ 
= 0.077 Diversity = 5.7892e-01 k_t = [4, 2, 2, 4]
Generation 970: Best = 1.500002e+03 FES = 97100  $\mu_F$  = 0.814  $\mu_{CR}$ 
= 0.085 Diversity = 5.3756e-01 k_t = [1, 1, 1, 1]
Generation 980: Best = 1.500002e+03 FES = 98100  $\mu_F$  = 0.821  $\mu_{CR}$ 
= 0.090 Diversity = 5.7235e-01 k_t = [2, 4, 4, 3]
Generation 990: Best = 1.500002e+03 FES = 99100  $\mu_F$  = 0.808  $\mu_{CR}$ 
= 0.088 Diversity = 5.5640e-01 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

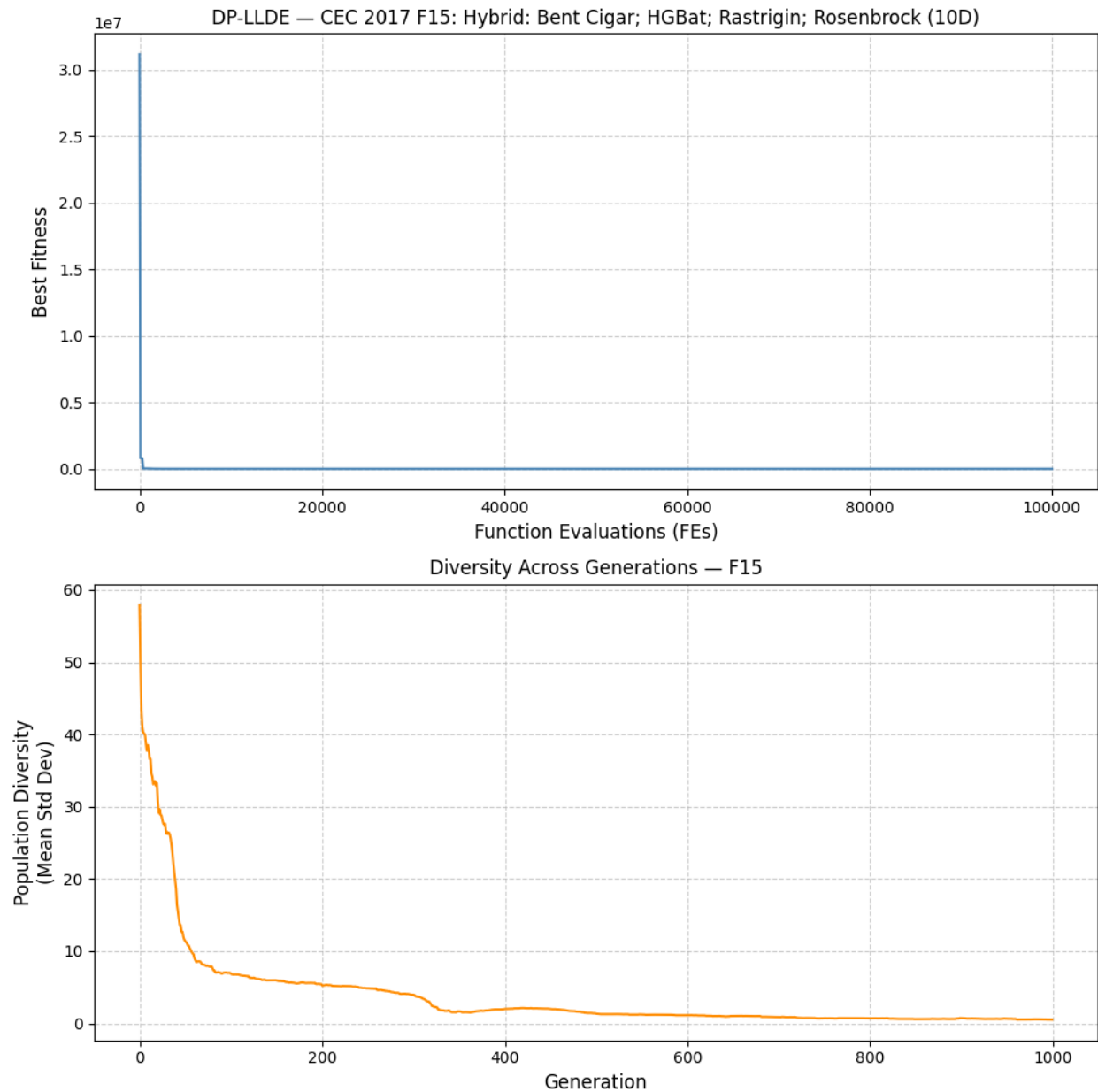
Best Fitness           = 1.500002e+03
Total Function Evals   = 100000
Total Successful Updates = 9849

```

```

=====
=====
Results:
  Best Fitness           : 1.500002e+03
  Global Optimum         : 1.500000e+03
  Error (|best-optimum|) : 2.265975e-03
  Best Solution (5 dim)  : [ 5.34079306  2.02640126  65.49400292 -
42.02743575  73.0824889 ]
  D^(0) (init. diversity): 5.7928e+01
=====
=====

```



Graph saved as convergence_F15.png

```
# =====
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# =====
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid:
Schaffer; HGBat; Rosenbrock; Schwefel')
```

```
=====
DP-LLDE on CEC 2017 F16 – Single Run
=====
```

=====

Function : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (10D)

Bounds : [-100, 100]

Global optimum : 1600.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = $5.7928e+00$

Generation 0: Best Fitness = $2.512498e+03$

Generation 10: Best = $2.092900e+03$ FES = 1100 $\mu_F = 0.583$ μ_{CR}

= 0.331 Diversity = $3.5424e+01$ $k_t = [1, 1, 1, 1]$

Generation 20: Best = $1.795796e+03$ FES = 2100 $\mu_F = 0.652$ μ_{CR}

= 0.316 Diversity = $3.2154e+01$ $k_t = [2, 4, 2, 2]$

Generation 30: Best = $1.702964e+03$ FES = 3100 $\mu_F = 0.702$ μ_{CR}

= 0.293 Diversity = $3.1532e+01$ $k_t = [1, 1, 1, 1]$

Generation 40: Best = $1.656168e+03$ FES = 4100 $\mu_F = 0.716$ μ_{CR}

= 0.265 Diversity = $2.9411e+01$ $k_t = [3, 4, 2, 3]$

Generation 50: Best = $1.651521e+03$ FES = 5100 $\mu_F = 0.739$ μ_{CR}

= 0.252 Diversity = $2.9683e+01$ $k_t = [1, 1, 1, 1]$

Generation 60: Best = $1.651512e+03$ FES = 6100 $\mu_F = 0.755$ μ_{CR}

= 0.222 Diversity = $2.8320e+01$ $k_t = [3, 2, 4, 2]$

Generation 70: Best = $1.637368e+03$ FES = 7100 $\mu_F = 0.779$ μ_{CR}

= 0.202 Diversity = $2.7630e+01$ $k_t = [1, 1, 1, 1]$

Generation 80: Best = $1.630846e+03$ FES = 8100 $\mu_F = 0.789$ μ_{CR}

= 0.190 Diversity = $2.6962e+01$ $k_t = [4, 3, 4, 2]$

Generation 90: Best = $1.627424e+03$ FES = 9100 $\mu_F = 0.797$ μ_{CR}

= 0.169 Diversity = $2.7216e+01$ $k_t = [1, 1, 1, 1]$

Generation 100: Best = $1.619418e+03$ FES = 10100 $\mu_F = 0.802$ μ_{CR}

= 0.146 Diversity = $2.6312e+01$ $k_t = [2, 4, 3, 4]$

Generation 110: Best = $1.608614e+03$ FES = 11100 $\mu_F = 0.818$ μ_{CR}

= 0.135 Diversity = $2.5394e+01$ $k_t = [1, 1, 1, 1]$

Generation 120: Best = $1.608614e+03$ FES = 12100 $\mu_F = 0.829$ μ_{CR}

= 0.132 Diversity = $2.4499e+01$ $k_t = [2, 2, 2, 3]$

Generation 130: Best = $1.607841e+03$ FES = 13100 $\mu_F = 0.834$ μ_{CR}

= 0.126 Diversity = $2.4139e+01$ $k_t = [1, 1, 1, 1]$

Generation 140: Best = $1.607094e+03$ FES = 14100 $\mu_F = 0.814$ μ_{CR}

= 0.125 Diversity = $2.2714e+01$ $k_t = [4, 4, 2, 3]$

Generation 150: Best = $1.607094e+03$ FES = 15100 $\mu_F = 0.821$ μ_{CR}

= 0.130 Diversity = $2.1374e+01$ $k_t = [1, 1, 1, 1]$

Generation 160: Best = $1.604977e+03$ FES = 16100 $\mu_F = 0.829$ μ_{CR}

= 0.136 Diversity = $1.9202e+01$ $k_t = [3, 2, 4, 4]$

Generation 170: Best = $1.603569e+03$ FES = 17100 $\mu_F = 0.846$ μ_{CR}

= 0.129 Diversity = $1.7580e+01$ $k_t = [1, 1, 1, 1]$

Generation 180: Best = $1.602962e+03$ FES = 18100 $\mu_F = 0.853$ μ_{CR}

= 0.140 Diversity = $1.6684e+01$ $k_t = [3, 2, 4, 3]$

Generation 190: Best = $1.602387e+03$ FES = 19100 $\mu_F = 0.860$ μ_{CR}

= 0.138 Diversity = $1.5812e+01$ $k_t = [1, 1, 1, 1]$

Generation 200: Best = $1.601904e+03$ FES = 20100 $\mu_F = 0.867$ μ_{CR}

= 0.137 Diversity = $1.4873e+01$ $k_t = [3, 4, 3, 4]$

Generation 210: Best = $1.601684e+03$ FES = 21100 $\mu_F = 0.873$ μ_{CR}

= 0.148 Diversity = $1.5282e+01$ $k_t = [1, 1, 1, 1]$

Generation 220: Best = $1.601283e+03$ FES = 22100 $\mu_F = 0.879$ μ_{CR}

= 0.162	Diversity = 1.3362e+01	k_t = [4, 4, 2, 2]	μ_F = 0.885	μ_{CR}
Generation 230:	Best = 1.601280e+03	FES = 23100		
= 0.181	Diversity = 1.3087e+01	k_t = [1, 1, 1, 1]	μ_F = 0.857	μ_{CR}
Generation 240:	Best = 1.601242e+03	FES = 24100		
= 0.176	Diversity = 9.4934e+00	k_t = [3, 4, 4, 2]	μ_F = 0.852	μ_{CR}
Generation 250:	Best = 1.601181e+03	FES = 25100		
= 0.150	Diversity = 7.1807e+00	k_t = [1, 1, 1, 1]	μ_F = 0.842	μ_{CR}
Generation 260:	Best = 1.600999e+03	FES = 26100		
= 0.161	Diversity = 5.3305e+00	k_t = [3, 4, 4, 2]	μ_F = 0.836	μ_{CR}
Generation 270:	Best = 1.600999e+03	FES = 27100		
= 0.143	Diversity = 2.3484e+00	k_t = [1, 1, 1, 1]	μ_F = 0.819	μ_{CR}
Generation 280:	Best = 1.600999e+03	FES = 28100		
= 0.133	Diversity = 2.0973e+00	k_t = [2, 3, 4, 3]	μ_F = 0.792	μ_{CR}
Generation 290:	Best = 1.600937e+03	FES = 29100		
= 0.136	Diversity = 1.9146e+00	k_t = [1, 1, 1, 1]	μ_F = 0.780	μ_{CR}
Generation 300:	Best = 1.600927e+03	FES = 30100		
= 0.144	Diversity = 1.6958e+00	k_t = [4, 2, 4, 2]	μ_F = 0.786	μ_{CR}
Generation 310:	Best = 1.600862e+03	FES = 31100		
= 0.130	Diversity = 1.6835e+00	k_t = [1, 1, 1, 1]	μ_F = 0.798	μ_{CR}
Generation 320:	Best = 1.600862e+03	FES = 32100		
= 0.116	Diversity = 1.5705e+00	k_t = [3, 4, 3, 3]	μ_F = 0.809	μ_{CR}
Generation 330:	Best = 1.600862e+03	FES = 33100		
= 0.110	Diversity = 1.4944e+00	k_t = [1, 1, 1, 1]	μ_F = 0.812	μ_{CR}
Generation 340:	Best = 1.600862e+03	FES = 34100		
= 0.101	Diversity = 1.4381e+00	k_t = [2, 4, 2, 3]	μ_F = 0.814	μ_{CR}
Generation 350:	Best = 1.600844e+03	FES = 35100		
= 0.103	Diversity = 1.3991e+00	k_t = [1, 1, 1, 1]	μ_F = 0.809	μ_{CR}
Generation 360:	Best = 1.600843e+03	FES = 36100		
= 0.117	Diversity = 1.3171e+00	k_t = [4, 4, 2, 4]	μ_F = 0.828	μ_{CR}
Generation 370:	Best = 1.600817e+03	FES = 37100		
= 0.127	Diversity = 1.3816e+00	k_t = [1, 1, 1, 1]	μ_F = 0.813	μ_{CR}
Generation 380:	Best = 1.600817e+03	FES = 38100		
= 0.112	Diversity = 1.3698e+00	k_t = [2, 3, 4, 3]	μ_F = 0.821	μ_{CR}
Generation 390:	Best = 1.600817e+03	FES = 39100		
= 0.097	Diversity = 1.4255e+00	k_t = [1, 1, 1, 1]	μ_F = 0.850	μ_{CR}
Generation 400:	Best = 1.600812e+03	FES = 40100		
= 0.108	Diversity = 1.4407e+00	k_t = [3, 3, 3, 3]	μ_F = 0.812	μ_{CR}
Generation 410:	Best = 1.600687e+03	FES = 41100		
= 0.107	Diversity = 1.5185e+00	k_t = [1, 1, 1, 1]	μ_F = 0.810	μ_{CR}
Generation 420:	Best = 1.600687e+03	FES = 42100		
= 0.100	Diversity = 1.4892e+00	k_t = [4, 2, 4, 4]	μ_F = 0.776	μ_{CR}
Generation 430:	Best = 1.600687e+03	FES = 43100		
= 0.119	Diversity = 1.5035e+00	k_t = [1, 1, 1, 1]	μ_F = 0.787	μ_{CR}
Generation 440:	Best = 1.600687e+03	FES = 44100		
= 0.103	Diversity = 1.4613e+00	k_t = [3, 2, 4, 4]	μ_F = 0.811	μ_{CR}
Generation 450:	Best = 1.600687e+03	FES = 45100		
= 0.100	Diversity = 1.4567e+00	k_t = [1, 1, 1, 1]	μ_F = 0.751	μ_{CR}
Generation 460:	Best = 1.600687e+03	FES = 46100		
= 0.119	Diversity = 1.4500e+00	k_t = [3, 2, 2, 3]		

Generation 470: Best = 1.600686e+03	FES = 47100	$\mu_F = 0.783$	μ_{CR}
= 0.112 Diversity = 1.4352e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.600675e+03	FES = 48100	$\mu_F = 0.796$	μ_{CR}
= 0.104 Diversity = 1.4430e+00	$k_t = [2, 2, 2, 2]$		
Generation 490: Best = 1.600675e+03	FES = 49100	$\mu_F = 0.823$	μ_{CR}
= 0.110 Diversity = 1.4466e+00	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 1.600675e+03	FES = 50100	$\mu_F = 0.815$	μ_{CR}
= 0.096 Diversity = 1.4247e+00	$k_t = [3, 2, 2, 2]$		
Generation 510: Best = 1.600673e+03	FES = 51100	$\mu_F = 0.836$	μ_{CR}
= 0.093 Diversity = 1.4569e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.600672e+03	FES = 52100	$\mu_F = 0.822$	μ_{CR}
= 0.099 Diversity = 1.4525e+00	$k_t = [2, 2, 4, 3]$		
Generation 530: Best = 1.600672e+03	FES = 53100	$\mu_F = 0.811$	μ_{CR}
= 0.102 Diversity = 1.4804e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.600671e+03	FES = 54100	$\mu_F = 0.770$	μ_{CR}
= 0.099 Diversity = 1.5337e+00	$k_t = [3, 2, 4, 4]$		
Generation 550: Best = 1.600671e+03	FES = 55100	$\mu_F = 0.784$	μ_{CR}
= 0.102 Diversity = 1.5177e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.600671e+03	FES = 56100	$\mu_F = 0.810$	μ_{CR}
= 0.105 Diversity = 1.5150e+00	$k_t = [4, 4, 2, 3]$		
Generation 570: Best = 1.600646e+03	FES = 57100	$\mu_F = 0.808$	μ_{CR}
= 0.095 Diversity = 1.5313e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.600646e+03	FES = 58100	$\mu_F = 0.824$	μ_{CR}
= 0.107 Diversity = 1.5540e+00	$k_t = [4, 3, 4, 4]$		
Generation 590: Best = 1.600646e+03	FES = 59100	$\mu_F = 0.826$	μ_{CR}
= 0.102 Diversity = 1.5294e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.600646e+03	FES = 60100	$\mu_F = 0.813$	μ_{CR}
= 0.092 Diversity = 1.5289e+00	$k_t = [2, 4, 3, 2]$		
Generation 610: Best = 1.600636e+03	FES = 61100	$\mu_F = 0.810$	μ_{CR}
= 0.098 Diversity = 1.5412e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.600629e+03	FES = 62100	$\mu_F = 0.827$	μ_{CR}
= 0.092 Diversity = 1.5280e+00	$k_t = [3, 4, 2, 2]$		
Generation 630: Best = 1.600629e+03	FES = 63100	$\mu_F = 0.806$	μ_{CR}
= 0.095 Diversity = 1.5172e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.600629e+03	FES = 64100	$\mu_F = 0.793$	μ_{CR}
= 0.091 Diversity = 1.5315e+00	$k_t = [3, 2, 3, 2]$		
Generation 650: Best = 1.600629e+03	FES = 65100	$\mu_F = 0.796$	μ_{CR}
= 0.078 Diversity = 1.5340e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.600629e+03	FES = 66100	$\mu_F = 0.808$	μ_{CR}
= 0.091 Diversity = 1.4860e+00	$k_t = [3, 4, 3, 4]$		
Generation 670: Best = 1.600629e+03	FES = 67100	$\mu_F = 0.774$	μ_{CR}
= 0.087 Diversity = 1.4873e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.600628e+03	FES = 68100	$\mu_F = 0.753$	μ_{CR}
= 0.097 Diversity = 1.5057e+00	$k_t = [3, 3, 2, 2]$		
Generation 690: Best = 1.600620e+03	FES = 69100	$\mu_F = 0.724$	μ_{CR}
= 0.114 Diversity = 1.5380e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.600616e+03	FES = 70100	$\mu_F = 0.719$	μ_{CR}
= 0.103 Diversity = 1.5152e+00	$k_t = [3, 4, 2, 4]$		
Generation 710: Best = 1.600616e+03	FES = 71100	$\mu_F = 0.707$	μ_{CR}

= 0.094	Diversity = 1.5040e+00	k_t = [1, 1, 1, 1]	μ_F = 0.733	μ_{CR}
Generation 720:	Best = 1.600592e+03	FES = 72100		
= 0.101	Diversity = 1.5047e+00	k_t = [3, 3, 3, 3]	μ_F = 0.730	μ_{CR}
Generation 730:	Best = 1.600592e+03	FES = 73100		
= 0.096	Diversity = 1.5063e+00	k_t = [1, 1, 1, 1]	μ_F = 0.777	μ_{CR}
Generation 740:	Best = 1.600592e+03	FES = 74100		
= 0.098	Diversity = 1.5182e+00	k_t = [4, 2, 4, 2]	μ_F = 0.803	μ_{CR}
Generation 750:	Best = 1.600592e+03	FES = 75100		
= 0.098	Diversity = 1.5175e+00	k_t = [1, 1, 1, 1]	μ_F = 0.836	μ_{CR}
Generation 760:	Best = 1.600592e+03	FES = 76100		
= 0.090	Diversity = 1.4953e+00	k_t = [3, 2, 2, 4]	μ_F = 0.837	μ_{CR}
Generation 770:	Best = 1.600592e+03	FES = 77100		
= 0.103	Diversity = 1.5078e+00	k_t = [1, 1, 1, 1]	μ_F = 0.847	μ_{CR}
Generation 780:	Best = 1.600592e+03	FES = 78100		
= 0.106	Diversity = 1.4960e+00	k_t = [2, 3, 2, 4]	μ_F = 0.818	μ_{CR}
Generation 790:	Best = 1.600592e+03	FES = 79100		
= 0.098	Diversity = 1.4987e+00	k_t = [1, 1, 1, 1]	μ_F = 0.855	μ_{CR}
Generation 800:	Best = 1.600586e+03	FES = 80100		
= 0.096	Diversity = 1.5137e+00	k_t = [4, 2, 3, 3]	μ_F = 0.852	μ_{CR}
Generation 810:	Best = 1.600580e+03	FES = 81100		
= 0.097	Diversity = 1.4800e+00	k_t = [1, 1, 1, 1]	μ_F = 0.812	μ_{CR}
Generation 820:	Best = 1.600580e+03	FES = 82100		
= 0.103	Diversity = 1.4814e+00	k_t = [3, 2, 3, 3]	μ_F = 0.827	μ_{CR}
Generation 830:	Best = 1.600579e+03	FES = 83100		
= 0.115	Diversity = 1.4950e+00	k_t = [1, 1, 1, 1]	μ_F = 0.805	μ_{CR}
Generation 840:	Best = 1.600579e+03	FES = 84100		
= 0.113	Diversity = 1.4802e+00	k_t = [3, 4, 4, 2]	μ_F = 0.809	μ_{CR}
Generation 850:	Best = 1.600579e+03	FES = 85100		
= 0.088	Diversity = 1.4512e+00	k_t = [1, 1, 1, 1]	μ_F = 0.820	μ_{CR}
Generation 860:	Best = 1.600566e+03	FES = 86100		
= 0.099	Diversity = 1.4146e+00	k_t = [2, 3, 2, 2]	μ_F = 0.811	μ_{CR}
Generation 870:	Best = 1.600566e+03	FES = 87100		
= 0.107	Diversity = 1.3695e+00	k_t = [1, 1, 1, 1]	μ_F = 0.811	μ_{CR}
Generation 880:	Best = 1.600566e+03	FES = 88100		
= 0.089	Diversity = 1.2923e+00	k_t = [4, 4, 2, 4]	μ_F = 0.818	μ_{CR}
Generation 890:	Best = 1.600566e+03	FES = 89100		
= 0.094	Diversity = 1.3032e+00	k_t = [1, 1, 1, 1]	μ_F = 0.837	μ_{CR}
Generation 900:	Best = 1.600566e+03	FES = 90100		
= 0.108	Diversity = 1.2912e+00	k_t = [3, 2, 2, 3]	μ_F = 0.838	μ_{CR}
Generation 910:	Best = 1.600566e+03	FES = 91100		
= 0.090	Diversity = 1.2207e+00	k_t = [1, 1, 1, 1]	μ_F = 0.825	μ_{CR}
Generation 920:	Best = 1.600566e+03	FES = 92100		
= 0.095	Diversity = 1.2247e+00	k_t = [3, 4, 4, 3]	μ_F = 0.820	μ_{CR}
Generation 930:	Best = 1.600566e+03	FES = 93100		
= 0.092	Diversity = 1.1138e+00	k_t = [1, 1, 1, 1]	μ_F = 0.818	μ_{CR}
Generation 940:	Best = 1.600551e+03	FES = 94100		
= 0.086	Diversity = 1.0616e+00	k_t = [2, 3, 2, 3]	μ_F = 0.822	μ_{CR}
Generation 950:	Best = 1.600551e+03	FES = 95100		
= 0.095	Diversity = 1.0557e+00	k_t = [1, 1, 1, 1]		

```
Generation 960: Best = 1.600549e+03 FES = 96100  $\mu_F$  = 0.843  $\mu_{CR}$ 
= 0.092 Diversity = 1.0234e+00  $k_t$  = [3, 2, 2, 2]
Generation 970: Best = 1.600549e+03 FES = 97100  $\mu_F$  = 0.805  $\mu_{CR}$ 
= 0.084 Diversity = 9.8979e-01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.600541e+03 FES = 98100  $\mu_F$  = 0.803  $\mu_{CR}$ 
= 0.087 Diversity = 1.0001e+00  $k_t$  = [3, 3, 3, 4]
Generation 990: Best = 1.600541e+03 FES = 99100  $\mu_F$  = 0.808  $\mu_{CR}$ 
= 0.101 Diversity = 9.5515e-01  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 1.600541e+03

Total Function Evals = 100000

Total Successful Updates = 9813

```
=====
=====
```

Results:

Best Fitness : 1.600541e+03

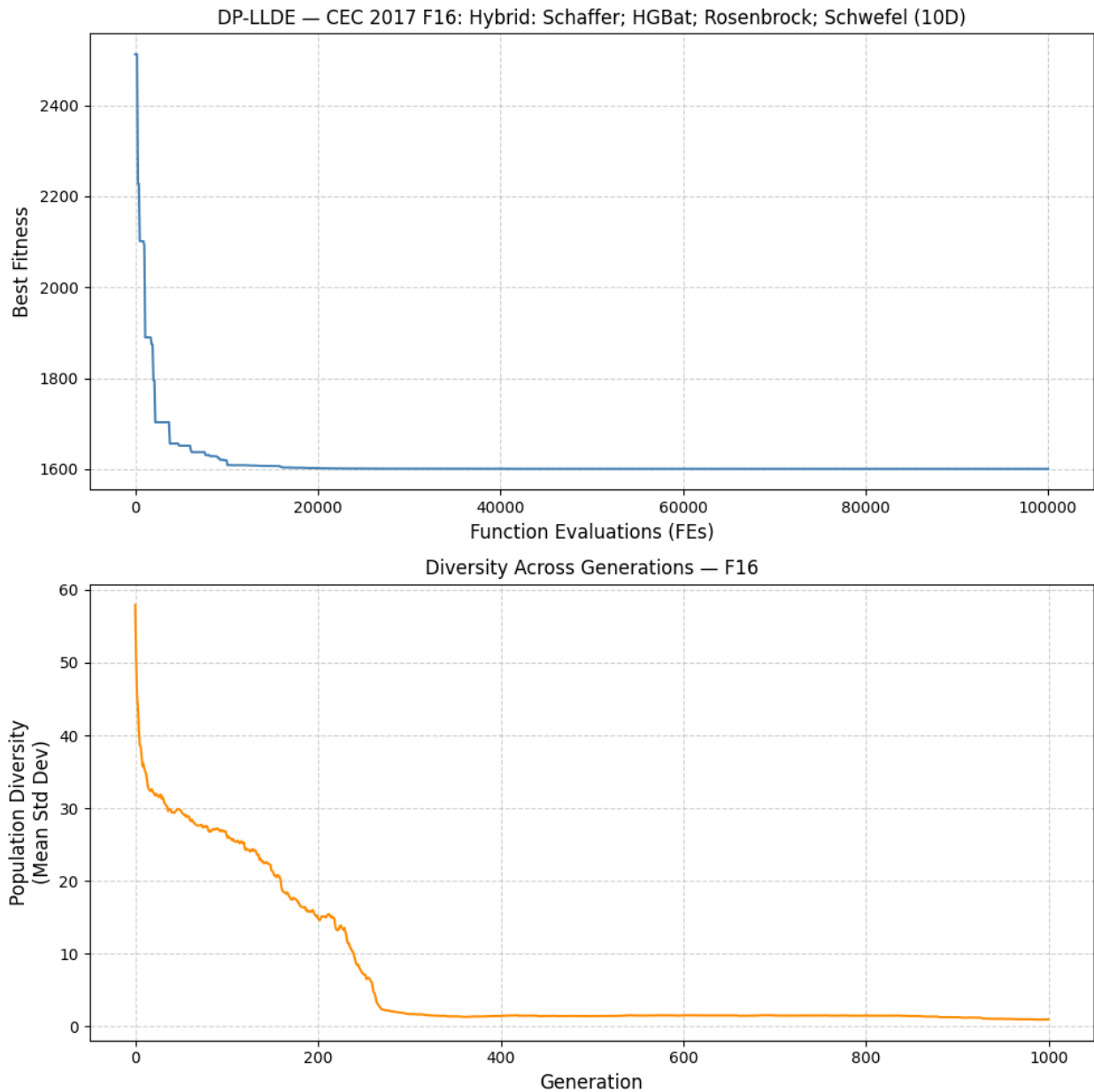
Global Optimum : 1.600000e+03

Error (|best-optimum|) : 5.408353e-01

Best Solution (5 dim) : [31.46657414 50.74861162 44.72941921 -
64.06915013 -39.75655086]

D^0 (init. diversity): 5.7928e+01

```
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```



Graph saved as convergence_F16.png

```
# =====
# F17 — Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock;
# Schwefel; Rastrigin)
# =====
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid:
Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin')
```

DP-LLDE on CEC 2017 F17 — Single Run


```

=====
Function      : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock;
Schwefel; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 1700.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.471803e+03
Generation 10: Best = 1.909686e+03 FES = 1100  $\mu_F$  = 0.576  $\mu_{CR}$ 
= 0.344 Diversity = 3.4412e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 1.805206e+03 FES = 2100  $\mu_F$  = 0.630  $\mu_{CR}$ 
= 0.335 Diversity = 3.0982e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 1.791016e+03 FES = 3100  $\mu_F$  = 0.674  $\mu_{CR}$ 
= 0.306 Diversity = 2.9771e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 1.768535e+03 FES = 4100  $\mu_F$  = 0.730  $\mu_{CR}$ 
= 0.286 Diversity = 2.9770e+01  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 1.768535e+03 FES = 5100  $\mu_F$  = 0.768  $\mu_{CR}$ 
= 0.260 Diversity = 3.0645e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 1.752140e+03 FES = 6100  $\mu_F$  = 0.785  $\mu_{CR}$ 
= 0.220 Diversity = 2.9833e+01  $k_t$  = [3, 3, 4, 3]
Generation 70: Best = 1.750809e+03 FES = 7100  $\mu_F$  = 0.821  $\mu_{CR}$ 
= 0.197 Diversity = 2.9104e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 1.750714e+03 FES = 8100  $\mu_F$  = 0.830  $\mu_{CR}$ 
= 0.175 Diversity = 2.8688e+01  $k_t$  = [2, 2, 3, 2]
Generation 90: Best = 1.750433e+03 FES = 9100  $\mu_F$  = 0.832  $\mu_{CR}$ 
= 0.152 Diversity = 2.8027e+01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 1.745466e+03 FES = 10100  $\mu_F$  = 0.842  $\mu_{CR}$ 
= 0.137 Diversity = 2.7721e+01  $k_t$  = [4, 4, 3, 3]
Generation 110: Best = 1.744376e+03 FES = 11100  $\mu_F$  = 0.839  $\mu_{CR}$ 
= 0.137 Diversity = 2.6526e+01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 1.728556e+03 FES = 12100  $\mu_F$  = 0.840  $\mu_{CR}$ 
= 0.126 Diversity = 2.5604e+01  $k_t$  = [3, 3, 2, 3]
Generation 130: Best = 1.728556e+03 FES = 13100  $\mu_F$  = 0.836  $\mu_{CR}$ 
= 0.123 Diversity = 2.4804e+01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 1.728556e+03 FES = 14100  $\mu_F$  = 0.845  $\mu_{CR}$ 
= 0.120 Diversity = 2.3667e+01  $k_t$  = [3, 3, 2, 2]
Generation 150: Best = 1.726274e+03 FES = 15100  $\mu_F$  = 0.830  $\mu_{CR}$ 
= 0.113 Diversity = 2.2690e+01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 1.721116e+03 FES = 16100  $\mu_F$  = 0.824  $\mu_{CR}$ 
= 0.109 Diversity = 2.0440e+01  $k_t$  = [3, 3, 3, 4]
Generation 170: Best = 1.719020e+03 FES = 17100  $\mu_F$  = 0.830  $\mu_{CR}$ 
= 0.101 Diversity = 2.0384e+01  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 1.717438e+03 FES = 18100  $\mu_F$  = 0.860  $\mu_{CR}$ 
= 0.099 Diversity = 1.9862e+01  $k_t$  = [3, 4, 2, 4]
Generation 190: Best = 1.712820e+03 FES = 19100  $\mu_F$  = 0.846  $\mu_{CR}$ 
= 0.095 Diversity = 1.9450e+01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 1.710971e+03 FES = 20100  $\mu_F$  = 0.868  $\mu_{CR}$ 
= 0.101 Diversity = 1.9201e+01  $k_t$  = [4, 3, 2, 3]
Generation 210: Best = 1.705188e+03 FES = 21100  $\mu_F$  = 0.835  $\mu_{CR}$ 

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= 0.100	Diversity = 1.8846e+01	k_t = [1, 1, 1, 1]	μ_F = 0.855	μ_{CR}
Generation 220:	Best = 1.702883e+03	FES = 22100		
= 0.104	Diversity = 1.7655e+01	k_t = [4, 2, 3, 4]	μ_F = 0.854	μ_{CR}
Generation 230:	Best = 1.702883e+03	FES = 23100		
= 0.102	Diversity = 1.6741e+01	k_t = [1, 1, 1, 1]	μ_F = 0.855	μ_{CR}
Generation 240:	Best = 1.702883e+03	FES = 24100		
= 0.104	Diversity = 1.6781e+01	k_t = [3, 2, 4, 2]	μ_F = 0.827	μ_{CR}
Generation 250:	Best = 1.702863e+03	FES = 25100		
= 0.099	Diversity = 1.6728e+01	k_t = [1, 1, 1, 1]	μ_F = 0.857	μ_{CR}
Generation 260:	Best = 1.702863e+03	FES = 26100		
= 0.096	Diversity = 1.6603e+01	k_t = [2, 2, 3, 2]	μ_F = 0.872	μ_{CR}
Generation 270:	Best = 1.702863e+03	FES = 27100		
= 0.101	Diversity = 1.6494e+01	k_t = [1, 1, 1, 1]	μ_F = 0.866	μ_{CR}
Generation 280:	Best = 1.702863e+03	FES = 28100		
= 0.090	Diversity = 1.6285e+01	k_t = [4, 3, 2, 2]	μ_F = 0.863	μ_{CR}
Generation 290:	Best = 1.702863e+03	FES = 29100		
= 0.102	Diversity = 1.6311e+01	k_t = [1, 1, 1, 1]	μ_F = 0.886	μ_{CR}
Generation 300:	Best = 1.702863e+03	FES = 30100		
= 0.101	Diversity = 1.5983e+01	k_t = [4, 3, 4, 4]	μ_F = 0.887	μ_{CR}
Generation 310:	Best = 1.702863e+03	FES = 31100		
= 0.099	Diversity = 1.5728e+01	k_t = [1, 1, 1, 1]	μ_F = 0.881	μ_{CR}
Generation 320:	Best = 1.702863e+03	FES = 32100		
= 0.089	Diversity = 1.5668e+01	k_t = [4, 4, 3, 3]	μ_F = 0.873	μ_{CR}
Generation 330:	Best = 1.702863e+03	FES = 33100		
= 0.100	Diversity = 1.4674e+01	k_t = [1, 1, 1, 1]	μ_F = 0.850	μ_{CR}
Generation 340:	Best = 1.702842e+03	FES = 34100		
= 0.109	Diversity = 1.4501e+01	k_t = [2, 4, 4, 4]	μ_F = 0.839	μ_{CR}
Generation 350:	Best = 1.702842e+03	FES = 35100		
= 0.092	Diversity = 1.4481e+01	k_t = [1, 1, 1, 1]	μ_F = 0.868	μ_{CR}
Generation 360:	Best = 1.702842e+03	FES = 36100		
= 0.084	Diversity = 1.4185e+01	k_t = [2, 4, 2, 4]	μ_F = 0.861	μ_{CR}
Generation 370:	Best = 1.702842e+03	FES = 37100		
= 0.080	Diversity = 1.3979e+01	k_t = [1, 1, 1, 1]	μ_F = 0.847	μ_{CR}
Generation 380:	Best = 1.702842e+03	FES = 38100		
= 0.069	Diversity = 1.4018e+01	k_t = [3, 2, 3, 2]	μ_F = 0.869	μ_{CR}
Generation 390:	Best = 1.702842e+03	FES = 39100		
= 0.080	Diversity = 1.3891e+01	k_t = [1, 1, 1, 1]	μ_F = 0.877	μ_{CR}
Generation 400:	Best = 1.702842e+03	FES = 40100		
= 0.080	Diversity = 1.3783e+01	k_t = [4, 3, 2, 2]	μ_F = 0.886	μ_{CR}
Generation 410:	Best = 1.702842e+03	FES = 41100		
= 0.078	Diversity = 1.3802e+01	k_t = [1, 1, 1, 1]	μ_F = 0.901	μ_{CR}
Generation 420:	Best = 1.702842e+03	FES = 42100		
= 0.083	Diversity = 1.3537e+01	k_t = [2, 2, 2, 2]	μ_F = 0.889	μ_{CR}
Generation 430:	Best = 1.702837e+03	FES = 43100		
= 0.082	Diversity = 1.3406e+01	k_t = [1, 1, 1, 1]	μ_F = 0.897	μ_{CR}
Generation 440:	Best = 1.702797e+03	FES = 44100		
= 0.083	Diversity = 1.2812e+01	k_t = [4, 4, 4, 4]	μ_F = 0.892	μ_{CR}
Generation 450:	Best = 1.701978e+03	FES = 45100		
= 0.070	Diversity = 1.2633e+01	k_t = [1, 1, 1, 1]		

Generation 460: Best = 1.701935e+03	FES = 46100	$\mu_F = 0.865$	μ_{CR}
= 0.075 Diversity = 1.2610e+01	$k_t = [4, 4, 2, 4]$		
Generation 470: Best = 1.701935e+03	FES = 47100	$\mu_F = 0.874$	μ_{CR}
= 0.090 Diversity = 1.2347e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.701933e+03	FES = 48100	$\mu_F = 0.868$	μ_{CR}
= 0.090 Diversity = 1.2392e+01	$k_t = [4, 3, 2, 3]$		
Generation 490: Best = 1.701926e+03	FES = 49100	$\mu_F = 0.859$	μ_{CR}
= 0.071 Diversity = 1.2422e+01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 1.701926e+03	FES = 50100	$\mu_F = 0.883$	μ_{CR}
= 0.084 Diversity = 1.1864e+01	$k_t = [2, 4, 2, 4]$		
Generation 510: Best = 1.701926e+03	FES = 51100	$\mu_F = 0.875$	μ_{CR}
= 0.092 Diversity = 1.1803e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.701926e+03	FES = 52100	$\mu_F = 0.862$	μ_{CR}
= 0.097 Diversity = 1.1685e+01	$k_t = [2, 3, 4, 3]$		
Generation 530: Best = 1.701914e+03	FES = 53100	$\mu_F = 0.863$	μ_{CR}
= 0.103 Diversity = 1.1676e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.701914e+03	FES = 54100	$\mu_F = 0.873$	μ_{CR}
= 0.109 Diversity = 1.1105e+01	$k_t = [2, 3, 4, 4]$		
Generation 550: Best = 1.701914e+03	FES = 55100	$\mu_F = 0.883$	μ_{CR}
= 0.117 Diversity = 1.1016e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.701914e+03	FES = 56100	$\mu_F = 0.878$	μ_{CR}
= 0.093 Diversity = 1.1099e+01	$k_t = [4, 2, 4, 2]$		
Generation 570: Best = 1.701914e+03	FES = 57100	$\mu_F = 0.888$	μ_{CR}
= 0.086 Diversity = 1.1023e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.701911e+03	FES = 58100	$\mu_F = 0.908$	μ_{CR}
= 0.079 Diversity = 1.0949e+01	$k_t = [2, 2, 2, 4]$		
Generation 590: Best = 1.701911e+03	FES = 59100	$\mu_F = 0.891$	μ_{CR}
= 0.087 Diversity = 1.0983e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.701775e+03	FES = 60100	$\mu_F = 0.893$	μ_{CR}
= 0.085 Diversity = 1.1044e+01	$k_t = [4, 2, 3, 3]$		
Generation 610: Best = 1.701775e+03	FES = 61100	$\mu_F = 0.882$	μ_{CR}
= 0.090 Diversity = 1.1038e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.701766e+03	FES = 62100	$\mu_F = 0.898$	μ_{CR}
= 0.095 Diversity = 1.0862e+01	$k_t = [4, 2, 2, 4]$		
Generation 630: Best = 1.701765e+03	FES = 63100	$\mu_F = 0.855$	μ_{CR}
= 0.090 Diversity = 1.0858e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.701760e+03	FES = 64100	$\mu_F = 0.875$	μ_{CR}
= 0.092 Diversity = 1.0918e+01	$k_t = [2, 2, 4, 2]$		
Generation 650: Best = 1.701711e+03	FES = 65100	$\mu_F = 0.873$	μ_{CR}
= 0.079 Diversity = 1.0999e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.701711e+03	FES = 66100	$\mu_F = 0.888$	μ_{CR}
= 0.082 Diversity = 1.1220e+01	$k_t = [2, 2, 3, 2]$		
Generation 670: Best = 1.701579e+03	FES = 67100	$\mu_F = 0.912$	μ_{CR}
= 0.081 Diversity = 1.1212e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.701579e+03	FES = 68100	$\mu_F = 0.893$	μ_{CR}
= 0.079 Diversity = 1.0902e+01	$k_t = [3, 2, 4, 4]$		
Generation 690: Best = 1.701565e+03	FES = 69100	$\mu_F = 0.888$	μ_{CR}
= 0.079 Diversity = 1.0913e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.701564e+03	FES = 70100	$\mu_F = 0.889$	μ_{CR}

= 0.088	Diversity = 1.0703e+01	k_t = [4, 2, 4, 4]		
Generation 710:	Best = 1.701542e+03	FES = 71100	$\mu_F = 0.891$	μ_{CR}
= 0.091	Diversity = 1.0581e+01	k_t = [1, 1, 1, 1]		
Generation 720:	Best = 1.701542e+03	FES = 72100	$\mu_F = 0.895$	μ_{CR}
= 0.083	Diversity = 1.0455e+01	k_t = [2, 4, 3, 3]		
Generation 730:	Best = 1.701525e+03	FES = 73100	$\mu_F = 0.894$	μ_{CR}
= 0.097	Diversity = 1.0437e+01	k_t = [1, 1, 1, 1]		
Generation 740:	Best = 1.701524e+03	FES = 74100	$\mu_F = 0.881$	μ_{CR}
= 0.105	Diversity = 1.0449e+01	k_t = [2, 3, 4, 3]		
Generation 750:	Best = 1.701438e+03	FES = 75100	$\mu_F = 0.900$	μ_{CR}
= 0.100	Diversity = 1.0431e+01	k_t = [1, 1, 1, 1]		
Generation 760:	Best = 1.700809e+03	FES = 76100	$\mu_F = 0.869$	μ_{CR}
= 0.086	Diversity = 1.0463e+01	k_t = [2, 2, 4, 3]		
Generation 770:	Best = 1.700805e+03	FES = 77100	$\mu_F = 0.846$	μ_{CR}
= 0.098	Diversity = 1.0462e+01	k_t = [1, 1, 1, 1]		
Generation 780:	Best = 1.700805e+03	FES = 78100	$\mu_F = 0.852$	μ_{CR}
= 0.092	Diversity = 1.0507e+01	k_t = [3, 3, 4, 3]		
Generation 790:	Best = 1.700800e+03	FES = 79100	$\mu_F = 0.852$	μ_{CR}
= 0.086	Diversity = 1.0506e+01	k_t = [1, 1, 1, 1]		
Generation 800:	Best = 1.700764e+03	FES = 80100	$\mu_F = 0.847$	μ_{CR}
= 0.082	Diversity = 1.0347e+01	k_t = [2, 3, 4, 4]		
Generation 810:	Best = 1.700762e+03	FES = 81100	$\mu_F = 0.786$	μ_{CR}
= 0.081	Diversity = 1.0344e+01	k_t = [1, 1, 1, 1]		
Generation 820:	Best = 1.700761e+03	FES = 82100	$\mu_F = 0.853$	μ_{CR}
= 0.075	Diversity = 1.0203e+01	k_t = [3, 2, 2, 4]		
Generation 830:	Best = 1.700761e+03	FES = 83100	$\mu_F = 0.841$	μ_{CR}
= 0.077	Diversity = 1.0163e+01	k_t = [1, 1, 1, 1]		
Generation 840:	Best = 1.700560e+03	FES = 84100	$\mu_F = 0.807$	μ_{CR}
= 0.089	Diversity = 1.0144e+01	k_t = [4, 2, 3, 3]		
Generation 850:	Best = 1.700560e+03	FES = 85100	$\mu_F = 0.798$	μ_{CR}
= 0.082	Diversity = 1.0144e+01	k_t = [1, 1, 1, 1]		
Generation 860:	Best = 1.700560e+03	FES = 86100	$\mu_F = 0.792$	μ_{CR}
= 0.085	Diversity = 1.0124e+01	k_t = [4, 2, 4, 2]		
Generation 870:	Best = 1.700560e+03	FES = 87100	$\mu_F = 0.781$	μ_{CR}
= 0.080	Diversity = 1.0125e+01	k_t = [1, 1, 1, 1]		
Generation 880:	Best = 1.700541e+03	FES = 88100	$\mu_F = 0.800$	μ_{CR}
= 0.078	Diversity = 1.0359e+01	k_t = [2, 3, 2, 4]		
Generation 890:	Best = 1.700541e+03	FES = 89100	$\mu_F = 0.853$	μ_{CR}
= 0.082	Diversity = 1.0501e+01	k_t = [1, 1, 1, 1]		
Generation 900:	Best = 1.700541e+03	FES = 90100	$\mu_F = 0.875$	μ_{CR}
= 0.107	Diversity = 1.0478e+01	k_t = [3, 3, 2, 2]		
Generation 910:	Best = 1.700540e+03	FES = 91100	$\mu_F = 0.864$	μ_{CR}
= 0.103	Diversity = 1.0488e+01	k_t = [1, 1, 1, 1]		
Generation 920:	Best = 1.700486e+03	FES = 92100	$\mu_F = 0.843$	μ_{CR}
= 0.097	Diversity = 1.0480e+01	k_t = [4, 4, 2, 3]		
Generation 930:	Best = 1.700486e+03	FES = 93100	$\mu_F = 0.862$	μ_{CR}
= 0.082	Diversity = 1.0439e+01	k_t = [1, 1, 1, 1]		
Generation 940:	Best = 1.700486e+03	FES = 94100	$\mu_F = 0.871$	μ_{CR}
= 0.087	Diversity = 1.0477e+01	k_t = [3, 2, 4, 2]		

```

Generation 950: Best = 1.700486e+03 FES = 95100  $\mu_F$  = 0.865  $\mu_{CR}$ 
= 0.084 Diversity = 1.0444e+01  $k_t$  = [1, 1, 1, 1]
Generation 960: Best = 1.700486e+03 FES = 96100  $\mu_F$  = 0.862  $\mu_{CR}$ 
= 0.104 Diversity = 1.0372e+01  $k_t$  = [4, 4, 2, 4]
Generation 970: Best = 1.700486e+03 FES = 97100  $\mu_F$  = 0.852  $\mu_{CR}$ 
= 0.083 Diversity = 1.0352e+01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.700438e+03 FES = 98100  $\mu_F$  = 0.864  $\mu_{CR}$ 
= 0.090 Diversity = 1.0233e+01  $k_t$  = [3, 4, 4, 4]
Generation 990: Best = 1.700438e+03 FES = 99100  $\mu_F$  = 0.862  $\mu_{CR}$ 
= 0.099 Diversity = 1.0147e+01  $k_t$  = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 1.700438e+03

Total Function Evals = 100000

Total Successful Updates = 6514

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Results:

Best Fitness : 1.700438e+03

Global Optimum : 1.700000e+03

Error (|best-optimum|) : 4.377342e-01

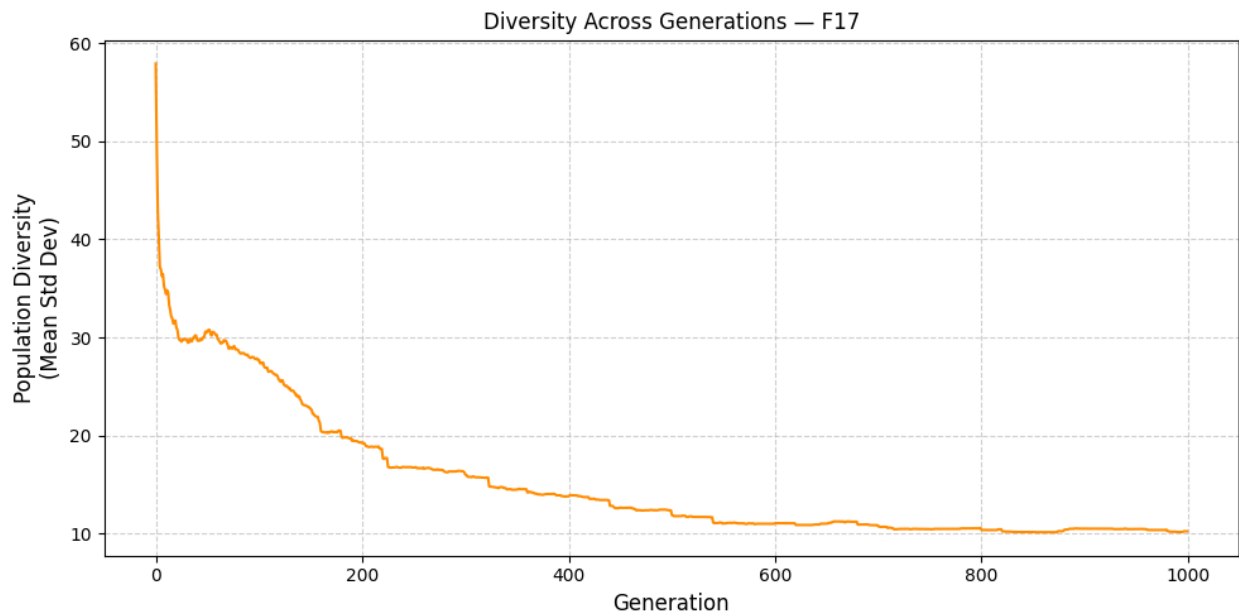
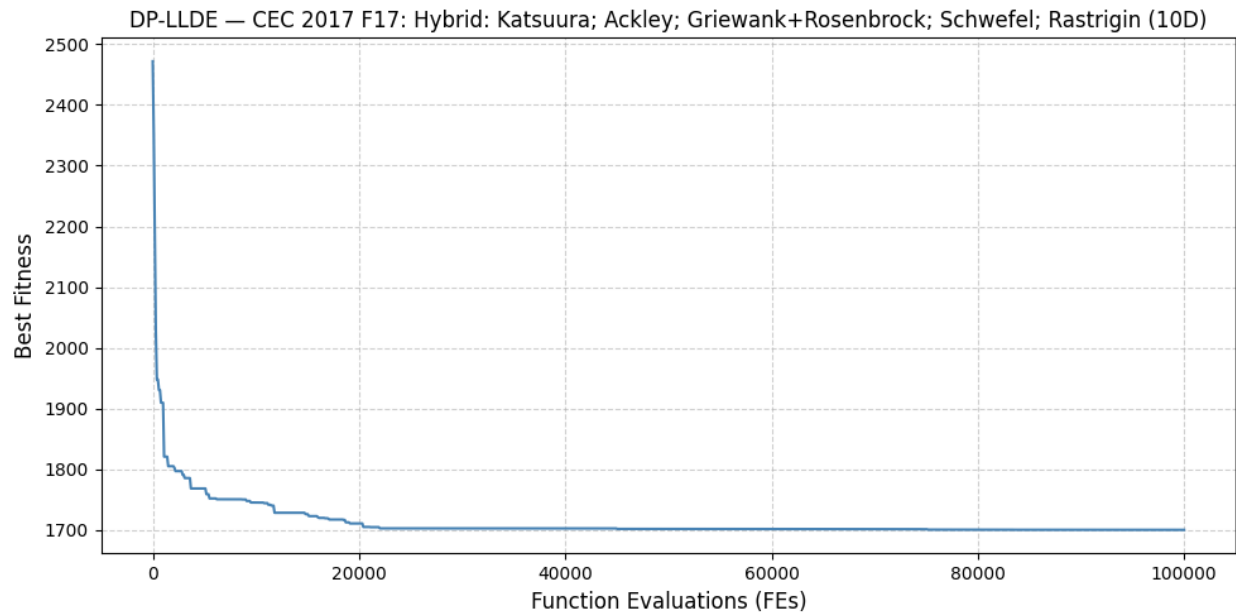
Best Solution (5 dim) : [-32.46121542 -7.16024458 -21.90881584
18.54121437 -53.05441659]

D^0 (init. diversity): 5.7928e+01

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```



Graph saved as convergence_F17.png

```
# =====
# F18 – Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid:
Elliptic; Ackley; Rastrigin; HGBat; Discus')
```

DP-LLDE on CEC 2017 F18 – Single Run

```

=====
Function      : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus
(10D)
Bounds       : [-100, 100]
Global optimum : 1800.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 6.135676e+08
Generation 10: Best = 6.173839e+04 FES = 1100  $\mu_F$  = 0.588  $\mu_{CR}$ 
= 0.354 Diversity = 3.2488e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 6.173839e+04 FES = 2100  $\mu_F$  = 0.645  $\mu_{CR}$ 
= 0.357 Diversity = 2.8199e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 6.015107e+03 FES = 3100  $\mu_F$  = 0.679  $\mu_{CR}$ 
= 0.349 Diversity = 2.8395e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 2.494801e+03 FES = 4100  $\mu_F$  = 0.721  $\mu_{CR}$ 
= 0.349 Diversity = 2.4456e+01  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 1.939301e+03 FES = 5100  $\mu_F$  = 0.759  $\mu_{CR}$ 
= 0.352 Diversity = 1.8976e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 1.855505e+03 FES = 6100  $\mu_F$  = 0.767  $\mu_{CR}$ 
= 0.333 Diversity = 1.6670e+01  $k_t$  = [2, 2, 4, 4]
Generation 70: Best = 1.832942e+03 FES = 7100  $\mu_F$  = 0.782  $\mu_{CR}$ 
= 0.322 Diversity = 1.4497e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 1.826405e+03 FES = 8100  $\mu_F$  = 0.771  $\mu_{CR}$ 
= 0.306 Diversity = 1.3496e+01  $k_t$  = [3, 2, 3, 2]
Generation 90: Best = 1.825704e+03 FES = 9100  $\mu_F$  = 0.781  $\mu_{CR}$ 
= 0.280 Diversity = 1.3144e+01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 1.821378e+03 FES = 10100  $\mu_F$  = 0.796  $\mu_{CR}$ 
= 0.251 Diversity = 1.3236e+01  $k_t$  = [2, 3, 4, 4]
Generation 110: Best = 1.818331e+03 FES = 11100  $\mu_F$  = 0.761  $\mu_{CR}$ 
= 0.255 Diversity = 1.2053e+01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 1.818271e+03 FES = 12100  $\mu_F$  = 0.788  $\mu_{CR}$ 
= 0.240 Diversity = 1.0431e+01  $k_t$  = [2, 2, 2, 2]
Generation 130: Best = 1.817350e+03 FES = 13100  $\mu_F$  = 0.802  $\mu_{CR}$ 
= 0.215 Diversity = 1.0395e+01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 1.813656e+03 FES = 14100  $\mu_F$  = 0.806  $\mu_{CR}$ 
= 0.195 Diversity = 9.0361e+00  $k_t$  = [2, 3, 3, 3]
Generation 150: Best = 1.804682e+03 FES = 15100  $\mu_F$  = 0.809  $\mu_{CR}$ 
= 0.176 Diversity = 7.5793e+00  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 1.804682e+03 FES = 16100  $\mu_F$  = 0.837  $\mu_{CR}$ 
= 0.177 Diversity = 5.6304e+00  $k_t$  = [4, 4, 3, 3]
Generation 170: Best = 1.804682e+03 FES = 17100  $\mu_F$  = 0.864  $\mu_{CR}$ 
= 0.171 Diversity = 4.0529e+00  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 1.803880e+03 FES = 18100  $\mu_F$  = 0.853  $\mu_{CR}$ 
= 0.172 Diversity = 2.5888e+00  $k_t$  = [4, 4, 3, 2]
Generation 190: Best = 1.803114e+03 FES = 19100  $\mu_F$  = 0.833  $\mu_{CR}$ 
= 0.169 Diversity = 1.9754e+00  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 1.802423e+03 FES = 20100  $\mu_F$  = 0.836  $\mu_{CR}$ 
= 0.153 Diversity = 2.0858e+00  $k_t$  = [3, 4, 2, 2]
Generation 210: Best = 1.801865e+03 FES = 21100  $\mu_F$  = 0.825  $\mu_{CR}$ 
= 0.156 Diversity = 1.4018e+00  $k_t$  = [1, 1, 1, 1]

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Generation 220: Best = 1.801735e+03	FES = 22100	$\mu_F = 0.792$	μ_{CR}
= 0.145 Diversity = 1.1819e+00	$k_t = [4, 4, 4, 4]$		
Generation 230: Best = 1.801588e+03	FES = 23100	$\mu_F = 0.785$	μ_{CR}
= 0.131 Diversity = 7.8966e-01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.801558e+03	FES = 24100	$\mu_F = 0.770$	μ_{CR}
= 0.140 Diversity = 6.7681e-01	$k_t = [2, 3, 2, 3]$		
Generation 250: Best = 1.801503e+03	FES = 25100	$\mu_F = 0.757$	μ_{CR}
= 0.128 Diversity = 7.0118e-01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.801396e+03	FES = 26100	$\mu_F = 0.758$	μ_{CR}
= 0.138 Diversity = 6.5804e-01	$k_t = [3, 3, 3, 4]$		
Generation 270: Best = 1.801396e+03	FES = 27100	$\mu_F = 0.767$	μ_{CR}
= 0.140 Diversity = 6.7904e-01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.801384e+03	FES = 28100	$\mu_F = 0.761$	μ_{CR}
= 0.145 Diversity = 6.9872e-01	$k_t = [2, 3, 4, 4]$		
Generation 290: Best = 1.801384e+03	FES = 29100	$\mu_F = 0.745$	μ_{CR}
= 0.132 Diversity = 7.1628e-01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.801372e+03	FES = 30100	$\mu_F = 0.737$	μ_{CR}
= 0.138 Diversity = 7.1289e-01	$k_t = [2, 2, 4, 4]$		
Generation 310: Best = 1.801372e+03	FES = 31100	$\mu_F = 0.739$	μ_{CR}
= 0.134 Diversity = 7.3151e-01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.801365e+03	FES = 32100	$\mu_F = 0.717$	μ_{CR}
= 0.125 Diversity = 7.5129e-01	$k_t = [4, 3, 3, 3]$		
Generation 330: Best = 1.801324e+03	FES = 33100	$\mu_F = 0.752$	μ_{CR}
= 0.119 Diversity = 7.8567e-01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.801324e+03	FES = 34100	$\mu_F = 0.773$	μ_{CR}
= 0.116 Diversity = 7.9085e-01	$k_t = [4, 4, 3, 3]$		
Generation 350: Best = 1.801287e+03	FES = 35100	$\mu_F = 0.800$	μ_{CR}
= 0.113 Diversity = 8.8789e-01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.801282e+03	FES = 36100	$\mu_F = 0.776$	μ_{CR}
= 0.119 Diversity = 9.6486e-01	$k_t = [3, 2, 3, 4]$		
Generation 370: Best = 1.801150e+03	FES = 37100	$\mu_F = 0.795$	μ_{CR}
= 0.115 Diversity = 1.0469e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.801150e+03	FES = 38100	$\mu_F = 0.814$	μ_{CR}
= 0.108 Diversity = 1.1105e+00	$k_t = [4, 2, 4, 3]$		
Generation 390: Best = 1.801150e+03	FES = 39100	$\mu_F = 0.831$	μ_{CR}
= 0.096 Diversity = 1.1470e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.801150e+03	FES = 40100	$\mu_F = 0.818$	μ_{CR}
= 0.105 Diversity = 1.1669e+00	$k_t = [3, 2, 2, 3]$		
Generation 410: Best = 1.801149e+03	FES = 41100	$\mu_F = 0.808$	μ_{CR}
= 0.113 Diversity = 1.2034e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.801037e+03	FES = 42100	$\mu_F = 0.819$	μ_{CR}
= 0.119 Diversity = 1.3742e+00	$k_t = [2, 2, 3, 2]$		
Generation 430: Best = 1.801037e+03	FES = 43100	$\mu_F = 0.836$	μ_{CR}
= 0.120 Diversity = 1.4425e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.801037e+03	FES = 44100	$\mu_F = 0.868$	μ_{CR}
= 0.118 Diversity = 1.4931e+00	$k_t = [3, 4, 2, 4]$		
Generation 450: Best = 1.801037e+03	FES = 45100	$\mu_F = 0.849$	μ_{CR}
= 0.120 Diversity = 1.4896e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.801037e+03	FES = 46100	$\mu_F = 0.852$	μ_{CR}

= 0.122	Diversity = 1.4707e+00	k_t = [2, 2, 3, 2]	μ_F = 0.841	μ_{CR}
Generation 470:	Best = 1.801037e+03	FES = 47100		
= 0.118	Diversity = 1.4236e+00	k_t = [1, 1, 1, 1]	μ_F = 0.834	μ_{CR}
Generation 480:	Best = 1.801037e+03	FES = 48100		
= 0.113	Diversity = 1.3561e+00	k_t = [3, 4, 2, 4]	μ_F = 0.833	μ_{CR}
Generation 490:	Best = 1.801037e+03	FES = 49100		
= 0.125	Diversity = 1.2633e+00	k_t = [1, 1, 1, 1]	μ_F = 0.830	μ_{CR}
Generation 500:	Best = 1.801025e+03	FES = 50100		
= 0.119	Diversity = 1.1572e+00	k_t = [4, 2, 2, 4]	μ_F = 0.847	μ_{CR}
Generation 510:	Best = 1.801025e+03	FES = 51100		
= 0.130	Diversity = 1.0235e+00	k_t = [1, 1, 1, 1]	μ_F = 0.815	μ_{CR}
Generation 520:	Best = 1.801025e+03	FES = 52100		
= 0.126	Diversity = 8.7647e-01	k_t = [4, 2, 2, 4]	μ_F = 0.795	μ_{CR}
Generation 530:	Best = 1.801025e+03	FES = 53100		
= 0.124	Diversity = 8.4653e-01	k_t = [1, 1, 1, 1]	μ_F = 0.804	μ_{CR}
Generation 540:	Best = 1.801025e+03	FES = 54100		
= 0.138	Diversity = 8.2465e-01	k_t = [4, 3, 4, 2]	μ_F = 0.793	μ_{CR}
Generation 550:	Best = 1.801025e+03	FES = 55100		
= 0.134	Diversity = 7.6543e-01	k_t = [1, 1, 1, 1]	μ_F = 0.813	μ_{CR}
Generation 560:	Best = 1.801025e+03	FES = 56100		
= 0.114	Diversity = 6.9063e-01	k_t = [3, 3, 3, 4]	μ_F = 0.823	μ_{CR}
Generation 570:	Best = 1.801025e+03	FES = 57100		
= 0.103	Diversity = 6.7925e-01	k_t = [1, 1, 1, 1]	μ_F = 0.798	μ_{CR}
Generation 580:	Best = 1.801025e+03	FES = 58100		
= 0.105	Diversity = 6.3208e-01	k_t = [3, 3, 2, 3]	μ_F = 0.792	μ_{CR}
Generation 590:	Best = 1.801023e+03	FES = 59100		
= 0.100	Diversity = 6.0128e-01	k_t = [1, 1, 1, 1]	μ_F = 0.776	μ_{CR}
Generation 600:	Best = 1.801021e+03	FES = 60100		
= 0.099	Diversity = 5.8807e-01	k_t = [4, 4, 3, 2]	μ_F = 0.775	μ_{CR}
Generation 610:	Best = 1.801021e+03	FES = 61100		
= 0.094	Diversity = 5.5419e-01	k_t = [1, 1, 1, 1]	μ_F = 0.769	μ_{CR}
Generation 620:	Best = 1.801021e+03	FES = 62100		
= 0.105	Diversity = 5.4407e-01	k_t = [2, 4, 2, 4]	μ_F = 0.760	μ_{CR}
Generation 630:	Best = 1.801021e+03	FES = 63100		
= 0.102	Diversity = 5.4095e-01	k_t = [1, 1, 1, 1]	μ_F = 0.749	μ_{CR}
Generation 640:	Best = 1.801021e+03	FES = 64100		
= 0.109	Diversity = 4.8625e-01	k_t = [4, 3, 3, 4]	μ_F = 0.783	μ_{CR}
Generation 650:	Best = 1.801021e+03	FES = 65100		
= 0.128	Diversity = 4.6591e-01	k_t = [1, 1, 1, 1]	μ_F = 0.784	μ_{CR}
Generation 660:	Best = 1.801021e+03	FES = 66100		
= 0.123	Diversity = 4.6072e-01	k_t = [4, 2, 3, 4]	μ_F = 0.794	μ_{CR}
Generation 670:	Best = 1.801021e+03	FES = 67100		
= 0.120	Diversity = 4.3388e-01	k_t = [1, 1, 1, 1]	μ_F = 0.788	μ_{CR}
Generation 680:	Best = 1.801021e+03	FES = 68100		
= 0.122	Diversity = 4.3179e-01	k_t = [2, 3, 2, 4]	μ_F = 0.809	μ_{CR}
Generation 690:	Best = 1.801006e+03	FES = 69100		
= 0.104	Diversity = 4.3177e-01	k_t = [1, 1, 1, 1]	μ_F = 0.805	μ_{CR}
Generation 700:	Best = 1.801006e+03	FES = 70100		
= 0.102	Diversity = 4.0066e-01	k_t = [4, 2, 4, 4]		

Generation 710: Best = 1.801006e+03	FES = 71100	$\mu_F = 0.820$	μ_{CR}
= 0.105 Diversity = 3.9758e-01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.801006e+03	FES = 72100	$\mu_F = 0.806$	μ_{CR}
= 0.103 Diversity = 3.8393e-01	$k_t = [2, 4, 2, 3]$		
Generation 730: Best = 1.801006e+03	FES = 73100	$\mu_F = 0.822$	μ_{CR}
= 0.099 Diversity = 3.6677e-01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.801006e+03	FES = 74100	$\mu_F = 0.833$	μ_{CR}
= 0.091 Diversity = 3.6713e-01	$k_t = [3, 2, 2, 2]$		
Generation 750: Best = 1.801006e+03	FES = 75100	$\mu_F = 0.782$	μ_{CR}
= 0.094 Diversity = 3.6199e-01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.801006e+03	FES = 76100	$\mu_F = 0.787$	μ_{CR}
= 0.098 Diversity = 3.5943e-01	$k_t = [4, 4, 2, 3]$		
Generation 770: Best = 1.801006e+03	FES = 77100	$\mu_F = 0.801$	μ_{CR}
= 0.102 Diversity = 3.4743e-01	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 1.801006e+03	FES = 78100	$\mu_F = 0.818$	μ_{CR}
= 0.103 Diversity = 3.4679e-01	$k_t = [2, 2, 2, 2]$		
Generation 790: Best = 1.801006e+03	FES = 79100	$\mu_F = 0.804$	μ_{CR}
= 0.115 Diversity = 3.4314e-01	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 1.801006e+03	FES = 80100	$\mu_F = 0.813$	μ_{CR}
= 0.123 Diversity = 3.4200e-01	$k_t = [4, 3, 3, 2]$		
Generation 810: Best = 1.801006e+03	FES = 81100	$\mu_F = 0.803$	μ_{CR}
= 0.116 Diversity = 3.4176e-01	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 1.801006e+03	FES = 82100	$\mu_F = 0.769$	μ_{CR}
= 0.106 Diversity = 3.4139e-01	$k_t = [3, 2, 2, 2]$		
Generation 830: Best = 1.801006e+03	FES = 83100	$\mu_F = 0.775$	μ_{CR}
= 0.106 Diversity = 3.2367e-01	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 1.801006e+03	FES = 84100	$\mu_F = 0.791$	μ_{CR}
= 0.114 Diversity = 3.2302e-01	$k_t = [4, 3, 4, 3]$		
Generation 850: Best = 1.801006e+03	FES = 85100	$\mu_F = 0.789$	μ_{CR}
= 0.110 Diversity = 3.2180e-01	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 1.801006e+03	FES = 86100	$\mu_F = 0.813$	μ_{CR}
= 0.097 Diversity = 3.1703e-01	$k_t = [3, 2, 2, 3]$		
Generation 870: Best = 1.801006e+03	FES = 87100	$\mu_F = 0.814$	μ_{CR}
= 0.093 Diversity = 3.1442e-01	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 1.801006e+03	FES = 88100	$\mu_F = 0.808$	μ_{CR}
= 0.104 Diversity = 3.1441e-01	$k_t = [2, 4, 3, 2]$		
Generation 890: Best = 1.801006e+03	FES = 89100	$\mu_F = 0.847$	μ_{CR}
= 0.105 Diversity = 3.0837e-01	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 1.801006e+03	FES = 90100	$\mu_F = 0.850$	μ_{CR}
= 0.090 Diversity = 3.0817e-01	$k_t = [4, 4, 3, 3]$		
Generation 910: Best = 1.801006e+03	FES = 91100	$\mu_F = 0.834$	μ_{CR}
= 0.091 Diversity = 3.0636e-01	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 1.801006e+03	FES = 92100	$\mu_F = 0.842$	μ_{CR}
= 0.090 Diversity = 3.0317e-01	$k_t = [4, 3, 3, 2]$		
Generation 930: Best = 1.801006e+03	FES = 93100	$\mu_F = 0.820$	μ_{CR}
= 0.109 Diversity = 3.0506e-01	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 1.801003e+03	FES = 94100	$\mu_F = 0.818$	μ_{CR}
= 0.114 Diversity = 3.0671e-01	$k_t = [3, 2, 4, 4]$		
Generation 950: Best = 1.801003e+03	FES = 95100	$\mu_F = 0.823$	μ_{CR}

```

= 0.108 Diversity = 3.0668e-01 k_t = [1, 1, 1, 1]
Generation 960: Best = 1.801003e+03 FES = 96100  $\mu_F$  = 0.772  $\mu_{CR}$ 
= 0.112 Diversity = 2.9594e-01 k_t = [2, 3, 2, 3]
Generation 970: Best = 1.801003e+03 FES = 97100  $\mu_F$  = 0.765  $\mu_{CR}$ 
= 0.126 Diversity = 2.9666e-01 k_t = [1, 1, 1, 1]
Generation 980: Best = 1.801003e+03 FES = 98100  $\mu_F$  = 0.781  $\mu_{CR}$ 
= 0.114 Diversity = 2.9214e-01 k_t = [3, 3, 3, 4]
Generation 990: Best = 1.800998e+03 FES = 99100  $\mu_F$  = 0.781  $\mu_{CR}$ 
= 0.120 Diversity = 2.7914e-01 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

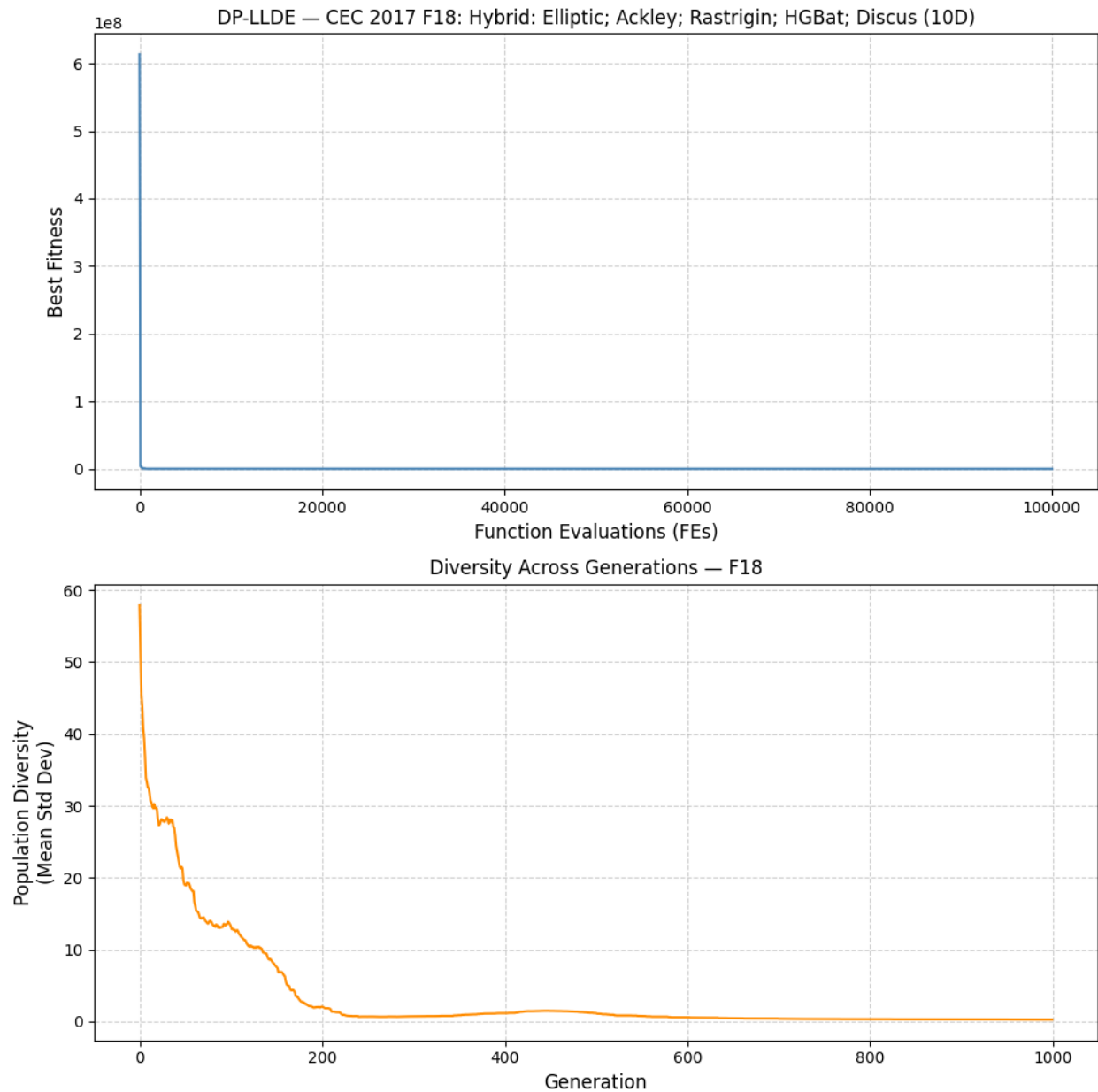
Best Fitness           = 1.800998e+03
Total Function Evals   = 100000
Total Successful Updates = 10428

```

```

=====
=====
Results:
  Best Fitness           : 1.800998e+03
  Global Optimum         : 1.800000e+03
  Error (|best-optimum|) : 9.980688e-01
  Best Solution (5 dim)  : [ 18.15058429   1.84791411   1.06235292 -
48.84219321 -11.41756835]
  D^(0) (init. diversity): 5.7928e+01
=====
=====

```



Graph saved as convergence_F18.png

```
# =====
# F19 – Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock;
# Weierstrass; Schaffer)
# =====
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent
Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer')
```

DP-LLDE on CEC 2017 F19 – Single Run

```

=====
Function      : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock;
Weierstrass; Schaffer (10D)
Bounds       : [-100, 100]
Global optimum : 1900.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.530389e+08
Generation 10: Best = 1.362757e+04 FES = 1100  $\mu_F$  = 0.588  $\mu_{CR}$ 
= 0.350 Diversity = 3.5008e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 2.425325e+03 FES = 2100  $\mu_F$  = 0.654  $\mu_{CR}$ 
= 0.340 Diversity = 3.1260e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 1.944905e+03 FES = 3100  $\mu_F$  = 0.702  $\mu_{CR}$ 
= 0.337 Diversity = 2.3865e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 1.913897e+03 FES = 4100  $\mu_F$  = 0.737  $\mu_{CR}$ 
= 0.346 Diversity = 2.0274e+01  $k_t$  = [4, 4, 3, 4]
Generation 50: Best = 1.907341e+03 FES = 5100  $\mu_F$  = 0.767  $\mu_{CR}$ 
= 0.348 Diversity = 1.7260e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 1.905922e+03 FES = 6100  $\mu_F$  = 0.776  $\mu_{CR}$ 
= 0.322 Diversity = 1.7172e+01  $k_t$  = [2, 3, 4, 3]
Generation 70: Best = 1.904296e+03 FES = 7100  $\mu_F$  = 0.818  $\mu_{CR}$ 
= 0.315 Diversity = 1.7707e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 1.904296e+03 FES = 8100  $\mu_F$  = 0.855  $\mu_{CR}$ 
= 0.271 Diversity = 1.8055e+01  $k_t$  = [4, 4, 2, 4]
Generation 90: Best = 1.903381e+03 FES = 9100  $\mu_F$  = 0.860  $\mu_{CR}$ 
= 0.242 Diversity = 1.8989e+01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 1.903187e+03 FES = 10100  $\mu_F$  = 0.874  $\mu_{CR}$ 
= 0.224 Diversity = 1.8645e+01  $k_t$  = [2, 2, 3, 4]
Generation 110: Best = 1.902578e+03 FES = 11100  $\mu_F$  = 0.875  $\mu_{CR}$ 
= 0.194 Diversity = 1.9067e+01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 1.901794e+03 FES = 12100  $\mu_F$  = 0.876  $\mu_{CR}$ 
= 0.175 Diversity = 1.8250e+01  $k_t$  = [2, 4, 3, 2]
Generation 130: Best = 1.901594e+03 FES = 13100  $\mu_F$  = 0.896  $\mu_{CR}$ 
= 0.156 Diversity = 1.7640e+01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 1.901537e+03 FES = 14100  $\mu_F$  = 0.894  $\mu_{CR}$ 
= 0.155 Diversity = 1.7235e+01  $k_t$  = [4, 2, 2, 2]
Generation 150: Best = 1.901260e+03 FES = 15100  $\mu_F$  = 0.885  $\mu_{CR}$ 
= 0.142 Diversity = 1.6673e+01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 1.901002e+03 FES = 16100  $\mu_F$  = 0.892  $\mu_{CR}$ 
= 0.125 Diversity = 1.5147e+01  $k_t$  = [2, 4, 2, 4]
Generation 170: Best = 1.900844e+03 FES = 17100  $\mu_F$  = 0.906  $\mu_{CR}$ 
= 0.111 Diversity = 1.5260e+01  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 1.900826e+03 FES = 18100  $\mu_F$  = 0.902  $\mu_{CR}$ 
= 0.112 Diversity = 1.4927e+01  $k_t$  = [3, 3, 4, 3]
Generation 190: Best = 1.900639e+03 FES = 19100  $\mu_F$  = 0.905  $\mu_{CR}$ 
= 0.110 Diversity = 1.4220e+01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 1.900547e+03 FES = 20100  $\mu_F$  = 0.887  $\mu_{CR}$ 
= 0.116 Diversity = 1.4163e+01  $k_t$  = [3, 4, 3, 3]
Generation 210: Best = 1.900438e+03 FES = 21100  $\mu_F$  = 0.890  $\mu_{CR}$ 

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= 0.128	Diversity = 1.3771e+01	k_t = [1, 1, 1, 1]		
Generation 220:	Best = 1.900345e+03	FES = 22100	$\mu_F = 0.889$	μ_{CR}
= 0.129	Diversity = 1.1905e+01	k_t = [3, 2, 4, 2]		
Generation 230:	Best = 1.900343e+03	FES = 23100	$\mu_F = 0.891$	μ_{CR}
= 0.125	Diversity = 1.0716e+01	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 1.900281e+03	FES = 24100	$\mu_F = 0.890$	μ_{CR}
= 0.126	Diversity = 1.0270e+01	k_t = [3, 2, 2, 2]		
Generation 250:	Best = 1.900219e+03	FES = 25100	$\mu_F = 0.890$	μ_{CR}
= 0.116	Diversity = 9.8296e+00	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 1.900216e+03	FES = 26100	$\mu_F = 0.877$	μ_{CR}
= 0.118	Diversity = 8.8742e+00	k_t = [3, 4, 4, 2]		
Generation 270:	Best = 1.900160e+03	FES = 27100	$\mu_F = 0.867$	μ_{CR}
= 0.121	Diversity = 8.1233e+00	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 1.900149e+03	FES = 28100	$\mu_F = 0.871$	μ_{CR}
= 0.112	Diversity = 7.7947e+00	k_t = [4, 4, 4, 2]		
Generation 290:	Best = 1.900143e+03	FES = 29100	$\mu_F = 0.886$	μ_{CR}
= 0.108	Diversity = 6.3447e+00	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 1.900123e+03	FES = 30100	$\mu_F = 0.890$	μ_{CR}
= 0.099	Diversity = 4.9001e+00	k_t = [4, 4, 2, 2]		
Generation 310:	Best = 1.900114e+03	FES = 31100	$\mu_F = 0.878$	μ_{CR}
= 0.104	Diversity = 4.6618e+00	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 1.900089e+03	FES = 32100	$\mu_F = 0.876$	μ_{CR}
= 0.108	Diversity = 4.6791e+00	k_t = [3, 4, 3, 2]		
Generation 330:	Best = 1.900082e+03	FES = 33100	$\mu_F = 0.865$	μ_{CR}
= 0.108	Diversity = 4.5101e+00	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 1.900051e+03	FES = 34100	$\mu_F = 0.856$	μ_{CR}
= 0.126	Diversity = 4.4287e+00	k_t = [2, 2, 2, 3]		
Generation 350:	Best = 1.900051e+03	FES = 35100	$\mu_F = 0.854$	μ_{CR}
= 0.110	Diversity = 4.2363e+00	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 1.900051e+03	FES = 36100	$\mu_F = 0.847$	μ_{CR}
= 0.104	Diversity = 2.3321e+00	k_t = [3, 4, 4, 4]		
Generation 370:	Best = 1.900035e+03	FES = 37100	$\mu_F = 0.828$	μ_{CR}
= 0.101	Diversity = 2.1561e+00	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 1.900028e+03	FES = 38100	$\mu_F = 0.808$	μ_{CR}
= 0.098	Diversity = 1.7635e+00	k_t = [2, 3, 3, 2]		
Generation 390:	Best = 1.900028e+03	FES = 39100	$\mu_F = 0.773$	μ_{CR}
= 0.105	Diversity = 1.7296e+00	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 1.900028e+03	FES = 40100	$\mu_F = 0.761$	μ_{CR}
= 0.109	Diversity = 1.6714e+00	k_t = [2, 4, 4, 3]		
Generation 410:	Best = 1.900025e+03	FES = 41100	$\mu_F = 0.741$	μ_{CR}
= 0.111	Diversity = 1.6267e+00	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 1.900017e+03	FES = 42100	$\mu_F = 0.756$	μ_{CR}
= 0.101	Diversity = 1.5433e+00	k_t = [3, 4, 4, 4]		
Generation 430:	Best = 1.900012e+03	FES = 43100	$\mu_F = 0.775$	μ_{CR}
= 0.117	Diversity = 1.5416e+00	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 1.900012e+03	FES = 44100	$\mu_F = 0.793$	μ_{CR}
= 0.113	Diversity = 1.5325e+00	k_t = [3, 4, 3, 2]		
Generation 450:	Best = 1.900012e+03	FES = 45100	$\mu_F = 0.798$	μ_{CR}
= 0.114	Diversity = 1.4658e+00	k_t = [1, 1, 1, 1]		

Generation 460: Best = 1.900012e+03	FES = 46100	$\mu_F = 0.802$	μ_{CR}
= 0.098 Diversity = 1.4504e+00	$k_t = [4, 4, 4, 3]$		
Generation 470: Best = 1.900012e+03	FES = 47100	$\mu_F = 0.772$	μ_{CR}
= 0.102 Diversity = 1.3771e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.900012e+03	FES = 48100	$\mu_F = 0.794$	μ_{CR}
= 0.114 Diversity = 1.3585e+00	$k_t = [2, 4, 4, 3]$		
Generation 490: Best = 1.900012e+03	FES = 49100	$\mu_F = 0.809$	μ_{CR}
= 0.112 Diversity = 1.3403e+00	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 1.900012e+03	FES = 50100	$\mu_F = 0.822$	μ_{CR}
= 0.108 Diversity = 1.2675e+00	$k_t = [4, 4, 4, 3]$		
Generation 510: Best = 1.900012e+03	FES = 51100	$\mu_F = 0.803$	μ_{CR}
= 0.110 Diversity = 1.2655e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.900012e+03	FES = 52100	$\mu_F = 0.805$	μ_{CR}
= 0.112 Diversity = 1.1886e+00	$k_t = [4, 3, 4, 2]$		
Generation 530: Best = 1.900011e+03	FES = 53100	$\mu_F = 0.831$	μ_{CR}
= 0.124 Diversity = 1.1617e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.900011e+03	FES = 54100	$\mu_F = 0.804$	μ_{CR}
= 0.125 Diversity = 1.0962e+00	$k_t = [4, 2, 3, 3]$		
Generation 550: Best = 1.900011e+03	FES = 55100	$\mu_F = 0.810$	μ_{CR}
= 0.120 Diversity = 1.0953e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.900011e+03	FES = 56100	$\mu_F = 0.780$	μ_{CR}
= 0.118 Diversity = 1.0830e+00	$k_t = [2, 3, 3, 2]$		
Generation 570: Best = 1.900010e+03	FES = 57100	$\mu_F = 0.781$	μ_{CR}
= 0.110 Diversity = 1.0809e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.900010e+03	FES = 58100	$\mu_F = 0.827$	μ_{CR}
= 0.099 Diversity = 9.3654e-01	$k_t = [2, 3, 2, 4]$		
Generation 590: Best = 1.900010e+03	FES = 59100	$\mu_F = 0.847$	μ_{CR}
= 0.109 Diversity = 9.1928e-01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.900010e+03	FES = 60100	$\mu_F = 0.863$	μ_{CR}
= 0.113 Diversity = 8.8732e-01	$k_t = [3, 4, 4, 3]$		
Generation 610: Best = 1.900010e+03	FES = 61100	$\mu_F = 0.869$	μ_{CR}
= 0.113 Diversity = 8.8342e-01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.900010e+03	FES = 62100	$\mu_F = 0.865$	μ_{CR}
= 0.119 Diversity = 8.6891e-01	$k_t = [3, 4, 4, 3]$		
Generation 630: Best = 1.900010e+03	FES = 63100	$\mu_F = 0.885$	μ_{CR}
= 0.112 Diversity = 8.5367e-01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.900010e+03	FES = 64100	$\mu_F = 0.881$	μ_{CR}
= 0.122 Diversity = 7.6671e-01	$k_t = [3, 3, 4, 3]$		
Generation 650: Best = 1.900010e+03	FES = 65100	$\mu_F = 0.880$	μ_{CR}
= 0.124 Diversity = 7.6369e-01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.900010e+03	FES = 66100	$\mu_F = 0.875$	μ_{CR}
= 0.118 Diversity = 7.0159e-01	$k_t = [4, 3, 4, 3]$		
Generation 670: Best = 1.900010e+03	FES = 67100	$\mu_F = 0.845$	μ_{CR}
= 0.121 Diversity = 7.0277e-01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.900010e+03	FES = 68100	$\mu_F = 0.838$	μ_{CR}
= 0.114 Diversity = 6.9874e-01	$k_t = [3, 2, 3, 2]$		
Generation 690: Best = 1.900010e+03	FES = 69100	$\mu_F = 0.832$	μ_{CR}
= 0.119 Diversity = 6.9427e-01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.900010e+03	FES = 70100	$\mu_F = 0.809$	μ_{CR}

= 0.120	Diversity = 6.9304e-01	k_t = [3, 4, 4, 2]		
Generation 710:	Best = 1.900010e+03	FES = 71100	$\mu_F = 0.824$	μ_{CR}
= 0.118	Diversity = 6.8541e-01	k_t = [1, 1, 1, 1]		
Generation 720:	Best = 1.900009e+03	FES = 72100	$\mu_F = 0.810$	μ_{CR}
= 0.115	Diversity = 6.7687e-01	k_t = [4, 2, 2, 3]		
Generation 730:	Best = 1.900007e+03	FES = 73100	$\mu_F = 0.808$	μ_{CR}
= 0.117	Diversity = 6.7855e-01	k_t = [1, 1, 1, 1]		
Generation 740:	Best = 1.900007e+03	FES = 74100	$\mu_F = 0.829$	μ_{CR}
= 0.119	Diversity = 6.7868e-01	k_t = [4, 2, 3, 2]		
Generation 750:	Best = 1.900007e+03	FES = 75100	$\mu_F = 0.841$	μ_{CR}
= 0.123	Diversity = 6.7518e-01	k_t = [1, 1, 1, 1]		
Generation 760:	Best = 1.900007e+03	FES = 76100	$\mu_F = 0.857$	μ_{CR}
= 0.120	Diversity = 5.9311e-01	k_t = [2, 4, 3, 4]		
Generation 770:	Best = 1.900007e+03	FES = 77100	$\mu_F = 0.827$	μ_{CR}
= 0.127	Diversity = 5.9409e-01	k_t = [1, 1, 1, 1]		
Generation 780:	Best = 1.900007e+03	FES = 78100	$\mu_F = 0.829$	μ_{CR}
= 0.118	Diversity = 5.9174e-01	k_t = [4, 2, 2, 2]		
Generation 790:	Best = 1.900007e+03	FES = 79100	$\mu_F = 0.837$	μ_{CR}
= 0.127	Diversity = 5.9179e-01	k_t = [1, 1, 1, 1]		
Generation 800:	Best = 1.900007e+03	FES = 80100	$\mu_F = 0.860$	μ_{CR}
= 0.116	Diversity = 5.7834e-01	k_t = [4, 4, 3, 4]		
Generation 810:	Best = 1.900007e+03	FES = 81100	$\mu_F = 0.864$	μ_{CR}
= 0.128	Diversity = 4.8683e-01	k_t = [1, 1, 1, 1]		
Generation 820:	Best = 1.900005e+03	FES = 82100	$\mu_F = 0.873$	μ_{CR}
= 0.125	Diversity = 4.1743e-01	k_t = [4, 3, 3, 4]		
Generation 830:	Best = 1.900005e+03	FES = 83100	$\mu_F = 0.863$	μ_{CR}
= 0.133	Diversity = 4.0726e-01	k_t = [1, 1, 1, 1]		
Generation 840:	Best = 1.900005e+03	FES = 84100	$\mu_F = 0.870$	μ_{CR}
= 0.128	Diversity = 3.9972e-01	k_t = [4, 2, 3, 2]		
Generation 850:	Best = 1.900005e+03	FES = 85100	$\mu_F = 0.859$	μ_{CR}
= 0.134	Diversity = 3.9448e-01	k_t = [1, 1, 1, 1]		
Generation 860:	Best = 1.900005e+03	FES = 86100	$\mu_F = 0.877$	μ_{CR}
= 0.135	Diversity = 3.9099e-01	k_t = [4, 4, 3, 3]		
Generation 870:	Best = 1.900005e+03	FES = 87100	$\mu_F = 0.857$	μ_{CR}
= 0.130	Diversity = 3.9120e-01	k_t = [1, 1, 1, 1]		
Generation 880:	Best = 1.900005e+03	FES = 88100	$\mu_F = 0.863$	μ_{CR}
= 0.114	Diversity = 3.8906e-01	k_t = [4, 2, 2, 3]		
Generation 890:	Best = 1.900005e+03	FES = 89100	$\mu_F = 0.862$	μ_{CR}
= 0.110	Diversity = 3.6833e-01	k_t = [1, 1, 1, 1]		
Generation 900:	Best = 1.900002e+03	FES = 90100	$\mu_F = 0.843$	μ_{CR}
= 0.119	Diversity = 3.6730e-01	k_t = [2, 2, 4, 4]		
Generation 910:	Best = 1.900002e+03	FES = 91100	$\mu_F = 0.845$	μ_{CR}
= 0.126	Diversity = 3.6951e-01	k_t = [1, 1, 1, 1]		
Generation 920:	Best = 1.900002e+03	FES = 92100	$\mu_F = 0.863$	μ_{CR}
= 0.116	Diversity = 3.2488e-01	k_t = [3, 4, 3, 4]		
Generation 930:	Best = 1.900002e+03	FES = 93100	$\mu_F = 0.858$	μ_{CR}
= 0.114	Diversity = 3.1809e-01	k_t = [1, 1, 1, 1]		
Generation 940:	Best = 1.900002e+03	FES = 94100	$\mu_F = 0.850$	μ_{CR}
= 0.113	Diversity = 3.1743e-01	k_t = [4, 3, 2, 2]		


```
Generation 950: Best = 1.900002e+03 FES = 95100  $\mu_F$  = 0.869  $\mu_{CR}$ 
= 0.110 Diversity = 3.1963e-01  $k_t$  = [1, 1, 1, 1]
Generation 960: Best = 1.900002e+03 FES = 96100  $\mu_F$  = 0.864  $\mu_{CR}$ 
= 0.104 Diversity = 3.1728e-01  $k_t$  = [3, 4, 4, 4]
Generation 970: Best = 1.900002e+03 FES = 97100  $\mu_F$  = 0.859  $\mu_{CR}$ 
= 0.104 Diversity = 3.1734e-01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 1.900002e+03 FES = 98100  $\mu_F$  = 0.860  $\mu_{CR}$ 
= 0.106 Diversity = 3.1589e-01  $k_t$  = [2, 3, 2, 3]
Generation 990: Best = 1.900002e+03 FES = 99100  $\mu_F$  = 0.848  $\mu_{CR}$ 
= 0.111 Diversity = 3.1237e-01  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 1.900002e+03

Total Function Evals = 100000

Total Successful Updates = 12337

```
=====
=====
```

Results:

Best Fitness : 1.900002e+03

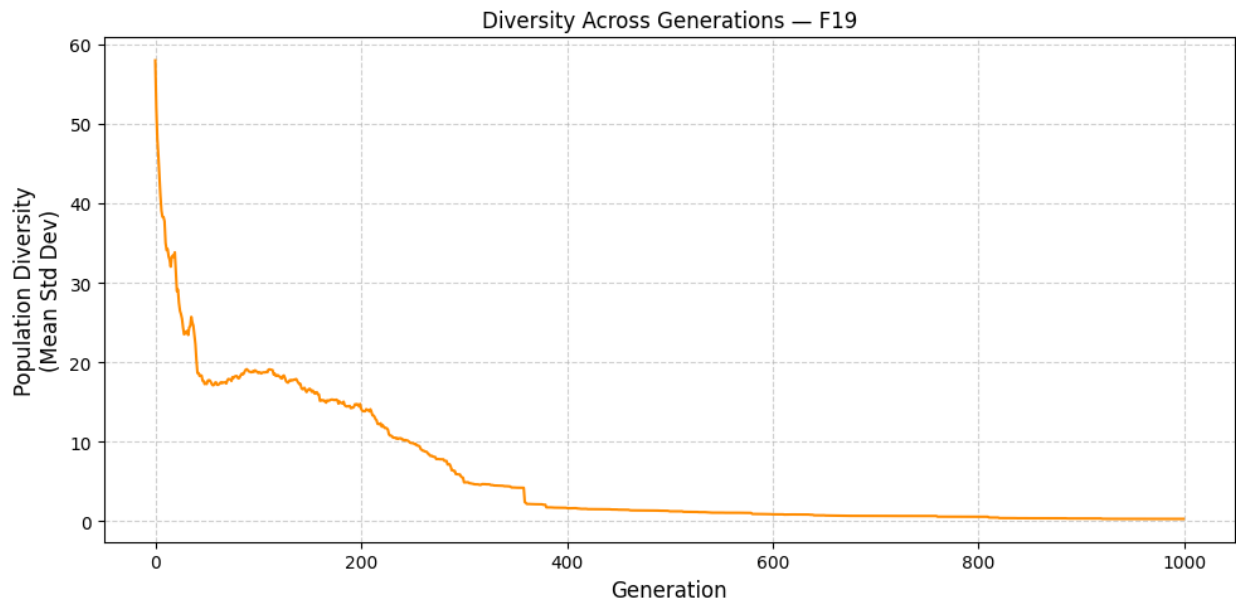
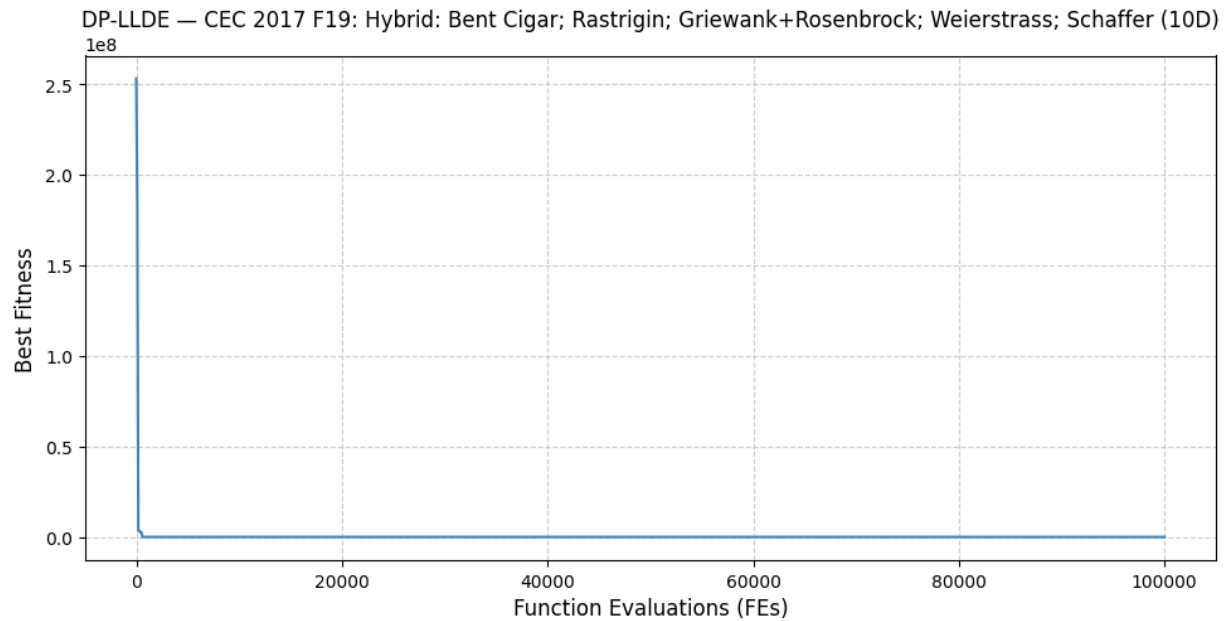
Global Optimum : 1.900000e+03

Error (|best-optimum|) : 2.205140e-03

Best Solution (5 dim) : [79.29102741 6.94739621 -35.04044538
35.2345538 25.85542325]

D^0 (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F19.png

```
# =====
# F20 – Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin;
# Schwefel; Schaffer)
# =====
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid:
HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer')
```

DP-LLDE on CEC 2017 F20 – Single Run

```

=====
Function      : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin;
Schwefel; Schaffer (10D)
Bounds       : [-100, 100]
Global optimum : 2000.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.391561e+03
Generation 10: Best = 2.225493e+03 FES = 1100  $\mu_F$  = 0.582  $\mu_{CR}$ 
= 0.328 Diversity = 4.4557e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 2.117427e+03 FES = 2100  $\mu_F$  = 0.660  $\mu_{CR}$ 
= 0.321 Diversity = 4.0237e+01  $k_t$  = [2, 4, 2, 2]
Generation 30: Best = 2.095462e+03 FES = 3100  $\mu_F$  = 0.728  $\mu_{CR}$ 
= 0.303 Diversity = 3.7668e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 2.059318e+03 FES = 4100  $\mu_F$  = 0.778  $\mu_{CR}$ 
= 0.302 Diversity = 3.5774e+01  $k_t$  = [3, 4, 2, 3]
Generation 50: Best = 2.042947e+03 FES = 5100  $\mu_F$  = 0.824  $\mu_{CR}$ 
= 0.294 Diversity = 3.3518e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 2.039710e+03 FES = 6100  $\mu_F$  = 0.848  $\mu_{CR}$ 
= 0.260 Diversity = 2.9903e+01  $k_t$  = [2, 3, 3, 4]
Generation 70: Best = 2.037479e+03 FES = 7100  $\mu_F$  = 0.863  $\mu_{CR}$ 
= 0.253 Diversity = 2.7338e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 2.032407e+03 FES = 8100  $\mu_F$  = 0.878  $\mu_{CR}$ 
= 0.234 Diversity = 2.5429e+01  $k_t$  = [3, 2, 4, 4]
Generation 90: Best = 2.031006e+03 FES = 9100  $\mu_F$  = 0.870  $\mu_{CR}$ 
= 0.232 Diversity = 2.4831e+01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 2.028643e+03 FES = 10100  $\mu_F$  = 0.883  $\mu_{CR}$ 
= 0.205 Diversity = 2.4819e+01  $k_t$  = [3, 4, 2, 3]
Generation 110: Best = 2.026598e+03 FES = 11100  $\mu_F$  = 0.885  $\mu_{CR}$ 
= 0.169 Diversity = 2.4047e+01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 2.023754e+03 FES = 12100  $\mu_F$  = 0.897  $\mu_{CR}$ 
= 0.154 Diversity = 2.3229e+01  $k_t$  = [3, 2, 3, 2]
Generation 130: Best = 2.023754e+03 FES = 13100  $\mu_F$  = 0.892  $\mu_{CR}$ 
= 0.138 Diversity = 2.3207e+01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 2.015947e+03 FES = 14100  $\mu_F$  = 0.902  $\mu_{CR}$ 
= 0.123 Diversity = 2.3025e+01  $k_t$  = [4, 2, 2, 3]
Generation 150: Best = 2.015901e+03 FES = 15100  $\mu_F$  = 0.913  $\mu_{CR}$ 
= 0.119 Diversity = 2.2018e+01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 2.015901e+03 FES = 16100  $\mu_F$  = 0.899  $\mu_{CR}$ 
= 0.113 Diversity = 2.1356e+01  $k_t$  = [2, 3, 2, 3]
Generation 170: Best = 2.010048e+03 FES = 17100  $\mu_F$  = 0.910  $\mu_{CR}$ 
= 0.115 Diversity = 2.0966e+01  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 2.009899e+03 FES = 18100  $\mu_F$  = 0.906  $\mu_{CR}$ 
= 0.109 Diversity = 2.0164e+01  $k_t$  = [3, 4, 2, 2]
Generation 190: Best = 2.008321e+03 FES = 19100  $\mu_F$  = 0.886  $\mu_{CR}$ 
= 0.106 Diversity = 1.9031e+01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 2.006746e+03 FES = 20100  $\mu_F$  = 0.886  $\mu_{CR}$ 
= 0.119 Diversity = 1.7187e+01  $k_t$  = [4, 4, 3, 4]
Generation 210: Best = 2.003463e+03 FES = 21100  $\mu_F$  = 0.883  $\mu_{CR}$ 

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= 0.127	Diversity = 1.6296e+01	k_t = [1, 1, 1, 1]		
Generation 220:	Best = 2.002650e+03	FES = 22100	$\mu_F = 0.894$	μ_{CR}
= 0.127	Diversity = 1.5028e+01	k_t = [3, 2, 4, 2]		
Generation 230:	Best = 2.001878e+03	FES = 23100	$\mu_F = 0.873$	μ_{CR}
= 0.129	Diversity = 1.4396e+01	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 2.001790e+03	FES = 24100	$\mu_F = 0.878$	μ_{CR}
= 0.134	Diversity = 1.3364e+01	k_t = [4, 2, 3, 4]		
Generation 250:	Best = 2.001223e+03	FES = 25100	$\mu_F = 0.883$	μ_{CR}
= 0.130	Diversity = 1.2069e+01	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 2.000868e+03	FES = 26100	$\mu_F = 0.885$	μ_{CR}
= 0.135	Diversity = 9.9190e+00	k_t = [3, 3, 2, 3]		
Generation 270:	Best = 2.000553e+03	FES = 27100	$\mu_F = 0.880$	μ_{CR}
= 0.136	Diversity = 9.6781e+00	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 2.000500e+03	FES = 28100	$\mu_F = 0.874$	μ_{CR}
= 0.146	Diversity = 9.2232e+00	k_t = [4, 2, 3, 2]		
Generation 290:	Best = 2.000475e+03	FES = 29100	$\mu_F = 0.873$	μ_{CR}
= 0.144	Diversity = 8.4433e+00	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 2.000281e+03	FES = 30100	$\mu_F = 0.880$	μ_{CR}
= 0.140	Diversity = 8.3136e+00	k_t = [3, 3, 4, 2]		
Generation 310:	Best = 2.000239e+03	FES = 31100	$\mu_F = 0.885$	μ_{CR}
= 0.143	Diversity = 7.9394e+00	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 2.000218e+03	FES = 32100	$\mu_F = 0.882$	μ_{CR}
= 0.143	Diversity = 7.8084e+00	k_t = [2, 2, 3, 2]		
Generation 330:	Best = 2.000154e+03	FES = 33100	$\mu_F = 0.882$	μ_{CR}
= 0.152	Diversity = 5.8510e+00	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 2.000126e+03	FES = 34100	$\mu_F = 0.879$	μ_{CR}
= 0.152	Diversity = 5.9584e+00	k_t = [4, 2, 4, 4]		
Generation 350:	Best = 2.000120e+03	FES = 35100	$\mu_F = 0.866$	μ_{CR}
= 0.154	Diversity = 5.9966e+00	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 2.000085e+03	FES = 36100	$\mu_F = 0.876$	μ_{CR}
= 0.152	Diversity = 5.6098e+00	k_t = [2, 2, 3, 2]		
Generation 370:	Best = 2.000083e+03	FES = 37100	$\mu_F = 0.861$	μ_{CR}
= 0.150	Diversity = 5.4709e+00	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 2.000031e+03	FES = 38100	$\mu_F = 0.868$	μ_{CR}
= 0.154	Diversity = 5.4400e+00	k_t = [2, 2, 3, 3]		
Generation 390:	Best = 2.000031e+03	FES = 39100	$\mu_F = 0.860$	μ_{CR}
= 0.163	Diversity = 5.5235e+00	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 2.000026e+03	FES = 40100	$\mu_F = 0.872$	μ_{CR}
= 0.162	Diversity = 5.3070e+00	k_t = [4, 3, 4, 3]		
Generation 410:	Best = 2.000025e+03	FES = 41100	$\mu_F = 0.872$	μ_{CR}
= 0.156	Diversity = 5.3856e+00	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 2.000020e+03	FES = 42100	$\mu_F = 0.860$	μ_{CR}
= 0.156	Diversity = 4.9897e+00	k_t = [2, 4, 3, 2]		
Generation 430:	Best = 2.000014e+03	FES = 43100	$\mu_F = 0.868$	μ_{CR}
= 0.163	Diversity = 4.8801e+00	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 2.000011e+03	FES = 44100	$\mu_F = 0.886$	μ_{CR}
= 0.167	Diversity = 4.7844e+00	k_t = [4, 3, 2, 4]		
Generation 450:	Best = 2.000009e+03	FES = 45100	$\mu_F = 0.883$	μ_{CR}
= 0.164	Diversity = 4.7870e+00	k_t = [1, 1, 1, 1]		

Generation 460: Best = 2.000008e+03	FES = 46100	$\mu_F = 0.872$	μ_{CR}
= 0.165 Diversity = 4.7790e+00	$k_t = [3, 2, 4, 4]$		
Generation 470: Best = 2.000006e+03	FES = 47100	$\mu_F = 0.881$	μ_{CR}
= 0.157 Diversity = 4.6919e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.000004e+03	FES = 48100	$\mu_F = 0.888$	μ_{CR}
= 0.157 Diversity = 4.6500e+00	$k_t = [3, 3, 4, 3]$		
Generation 490: Best = 2.000002e+03	FES = 49100	$\mu_F = 0.889$	μ_{CR}
= 0.161 Diversity = 4.6968e+00	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 2.000002e+03	FES = 50100	$\mu_F = 0.894$	μ_{CR}
= 0.158 Diversity = 4.8027e+00	$k_t = [4, 2, 2, 4]$		
Generation 510: Best = 2.000002e+03	FES = 51100	$\mu_F = 0.904$	μ_{CR}
= 0.170 Diversity = 4.9478e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.000001e+03	FES = 52100	$\mu_F = 0.893$	μ_{CR}
= 0.173 Diversity = 5.0285e+00	$k_t = [4, 3, 2, 4]$		
Generation 530: Best = 2.000001e+03	FES = 53100	$\mu_F = 0.888$	μ_{CR}
= 0.174 Diversity = 4.9091e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.000001e+03	FES = 54100	$\mu_F = 0.879$	μ_{CR}
= 0.174 Diversity = 5.0778e+00	$k_t = [4, 4, 2, 2]$		
Generation 550: Best = 2.000001e+03	FES = 55100	$\mu_F = 0.876$	μ_{CR}
= 0.169 Diversity = 5.0751e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.000001e+03	FES = 56100	$\mu_F = 0.882$	μ_{CR}
= 0.166 Diversity = 5.2774e+00	$k_t = [4, 3, 4, 4]$		
Generation 570: Best = 2.000000e+03	FES = 57100	$\mu_F = 0.890$	μ_{CR}
= 0.169 Diversity = 5.3281e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.000000e+03	FES = 58100	$\mu_F = 0.881$	μ_{CR}
= 0.160 Diversity = 5.4216e+00	$k_t = [4, 2, 3, 4]$		
Generation 590: Best = 2.000000e+03	FES = 59100	$\mu_F = 0.888$	μ_{CR}
= 0.164 Diversity = 5.6160e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.000000e+03	FES = 60100	$\mu_F = 0.889$	μ_{CR}
= 0.165 Diversity = 5.5841e+00	$k_t = [2, 4, 4, 2]$		
Generation 610: Best = 2.000000e+03	FES = 61100	$\mu_F = 0.882$	μ_{CR}
= 0.167 Diversity = 5.5030e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.000000e+03	FES = 62100	$\mu_F = 0.886$	μ_{CR}
= 0.160 Diversity = 5.5015e+00	$k_t = [2, 4, 2, 2]$		
Generation 630: Best = 2.000000e+03	FES = 63100	$\mu_F = 0.879$	μ_{CR}
= 0.148 Diversity = 5.4763e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.000000e+03	FES = 64100	$\mu_F = 0.891$	μ_{CR}
= 0.152 Diversity = 5.2279e+00	$k_t = [4, 3, 2, 2]$		
Generation 650: Best = 2.000000e+03	FES = 65100	$\mu_F = 0.881$	μ_{CR}
= 0.159 Diversity = 5.3958e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.000000e+03	FES = 66100	$\mu_F = 0.875$	μ_{CR}
= 0.152 Diversity = 5.3771e+00	$k_t = [2, 2, 3, 3]$		
Generation 670: Best = 2.000000e+03	FES = 67100	$\mu_F = 0.875$	μ_{CR}
= 0.146 Diversity = 5.3825e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.000000e+03	FES = 68100	$\mu_F = 0.886$	μ_{CR}
= 0.146 Diversity = 5.3415e+00	$k_t = [3, 3, 4, 2]$		
Generation 690: Best = 2.000000e+03	FES = 69100	$\mu_F = 0.883$	μ_{CR}
= 0.143 Diversity = 5.3593e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.000000e+03	FES = 70100	$\mu_F = 0.879$	μ_{CR}

= 0.146	Diversity = 5.3767e+00	k_t = [4, 3, 3, 2]		
Generation 710:	Best = 2.000000e+03	FES = 71100	$\mu_F = 0.879$	μ_{CR}
= 0.146	Diversity = 5.3961e+00	k_t = [1, 1, 1, 1]		
Generation 720:	Best = 2.000000e+03	FES = 72100	$\mu_F = 0.890$	μ_{CR}
= 0.146	Diversity = 5.4850e+00	k_t = [4, 2, 2, 2]		
Generation 730:	Best = 2.000000e+03	FES = 73100	$\mu_F = 0.892$	μ_{CR}
= 0.144	Diversity = 5.4139e+00	k_t = [1, 1, 1, 1]		
Generation 740:	Best = 2.000000e+03	FES = 74100	$\mu_F = 0.899$	μ_{CR}
= 0.144	Diversity = 5.2297e+00	k_t = [3, 2, 4, 4]		
Generation 750:	Best = 2.000000e+03	FES = 75100	$\mu_F = 0.888$	μ_{CR}
= 0.146	Diversity = 5.0484e+00	k_t = [1, 1, 1, 1]		
Generation 760:	Best = 2.000000e+03	FES = 76100	$\mu_F = 0.880$	μ_{CR}
= 0.141	Diversity = 5.1216e+00	k_t = [2, 4, 2, 3]		
Generation 770:	Best = 2.000000e+03	FES = 77100	$\mu_F = 0.882$	μ_{CR}
= 0.150	Diversity = 4.9884e+00	k_t = [1, 1, 1, 1]		
Generation 780:	Best = 2.000000e+03	FES = 78100	$\mu_F = 0.876$	μ_{CR}
= 0.139	Diversity = 4.8684e+00	k_t = [2, 3, 4, 3]		
Generation 790:	Best = 2.000000e+03	FES = 79100	$\mu_F = 0.857$	μ_{CR}
= 0.137	Diversity = 5.0027e+00	k_t = [1, 1, 1, 1]		
Generation 800:	Best = 2.000000e+03	FES = 80100	$\mu_F = 0.863$	μ_{CR}
= 0.137	Diversity = 5.0428e+00	k_t = [4, 4, 3, 3]		
Generation 810:	Best = 2.000000e+03	FES = 81100	$\mu_F = 0.868$	μ_{CR}
= 0.144	Diversity = 5.0499e+00	k_t = [1, 1, 1, 1]		
Generation 820:	Best = 2.000000e+03	FES = 82100	$\mu_F = 0.864$	μ_{CR}
= 0.139	Diversity = 5.2182e+00	k_t = [2, 3, 2, 4]		
Generation 830:	Best = 2.000000e+03	FES = 83100	$\mu_F = 0.864$	μ_{CR}
= 0.148	Diversity = 5.2895e+00	k_t = [1, 1, 1, 1]		
Generation 840:	Best = 2.000000e+03	FES = 84100	$\mu_F = 0.852$	μ_{CR}
= 0.151	Diversity = 5.3286e+00	k_t = [4, 4, 3, 4]		
Generation 850:	Best = 2.000000e+03	FES = 85100	$\mu_F = 0.844$	μ_{CR}
= 0.149	Diversity = 5.3839e+00	k_t = [1, 1, 1, 1]		
Generation 860:	Best = 2.000000e+03	FES = 86100	$\mu_F = 0.859$	μ_{CR}
= 0.143	Diversity = 5.3320e+00	k_t = [2, 3, 2, 2]		
Generation 870:	Best = 2.000000e+03	FES = 87100	$\mu_F = 0.852$	μ_{CR}
= 0.146	Diversity = 5.4299e+00	k_t = [1, 1, 1, 1]		
Generation 880:	Best = 2.000000e+03	FES = 88100	$\mu_F = 0.851$	μ_{CR}
= 0.148	Diversity = 5.5317e+00	k_t = [2, 3, 2, 3]		
Generation 890:	Best = 2.000000e+03	FES = 89100	$\mu_F = 0.843$	μ_{CR}
= 0.149	Diversity = 5.4961e+00	k_t = [1, 1, 1, 1]		
Generation 900:	Best = 2.000000e+03	FES = 90100	$\mu_F = 0.843$	μ_{CR}
= 0.156	Diversity = 5.6020e+00	k_t = [2, 3, 4, 3]		
Generation 910:	Best = 2.000000e+03	FES = 91100	$\mu_F = 0.830$	μ_{CR}
= 0.149	Diversity = 5.6767e+00	k_t = [1, 1, 1, 1]		
Generation 920:	Best = 2.000000e+03	FES = 92100	$\mu_F = 0.837$	μ_{CR}
= 0.157	Diversity = 5.5682e+00	k_t = [2, 3, 4, 3]		
Generation 930:	Best = 2.000000e+03	FES = 93100	$\mu_F = 0.848$	μ_{CR}
= 0.151	Diversity = 5.4129e+00	k_t = [1, 1, 1, 1]		
Generation 940:	Best = 2.000000e+03	FES = 94100	$\mu_F = 0.854$	μ_{CR}
= 0.150	Diversity = 5.3271e+00	k_t = [2, 4, 2, 2]		

```
Generation 950: Best = 2.000000e+03 FES = 95100  $\mu_F$  = 0.844  $\mu_{CR}$ 
= 0.150 Diversity = 5.3271e+00  $k_t$  = [1, 1, 1, 1]
Generation 960: Best = 2.000000e+03 FES = 96100  $\mu_F$  = 0.847  $\mu_{CR}$ 
= 0.145 Diversity = 5.2289e+00  $k_t$  = [3, 3, 2, 3]
Generation 970: Best = 2.000000e+03 FES = 97100  $\mu_F$  = 0.859  $\mu_{CR}$ 
= 0.141 Diversity = 5.0313e+00  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 2.000000e+03 FES = 98100  $\mu_F$  = 0.859  $\mu_{CR}$ 
= 0.150 Diversity = 4.8193e+00  $k_t$  = [2, 4, 4, 2]
Generation 990: Best = 2.000000e+03 FES = 99100  $\mu_F$  = 0.870  $\mu_{CR}$ 
= 0.150 Diversity = 4.7305e+00  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 2.000000e+03

Total Function Evals = 100000

Total Successful Updates = 21526

```
=====
=====
```

Results:

Best Fitness : 2.000000e+03

Global Optimum : 2.000000e+03

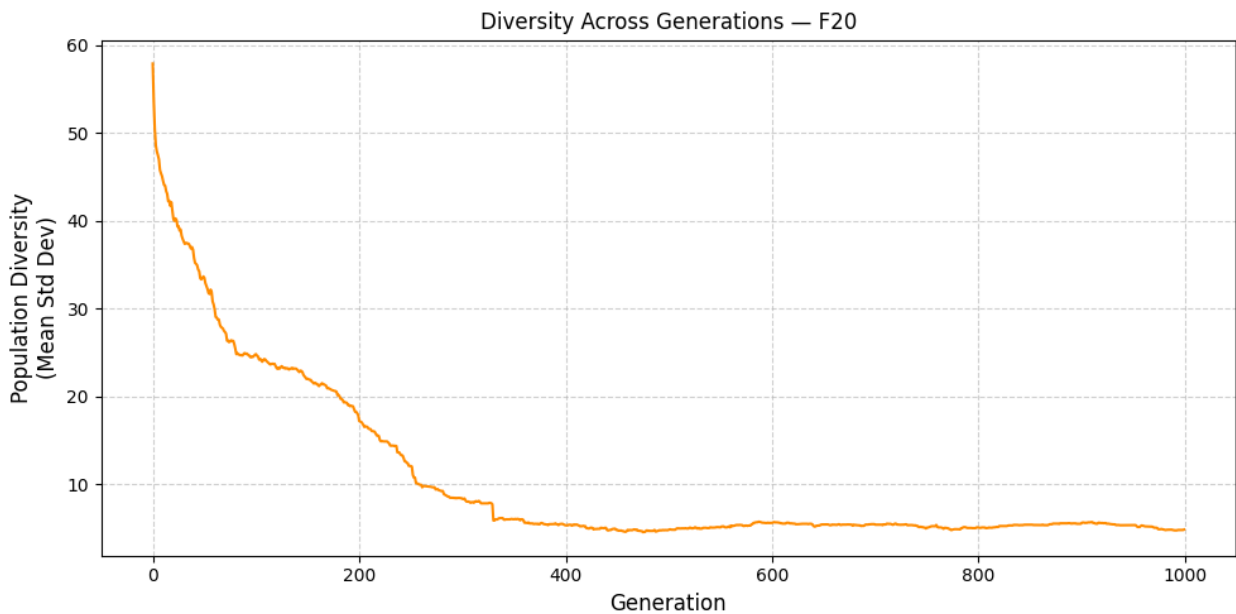
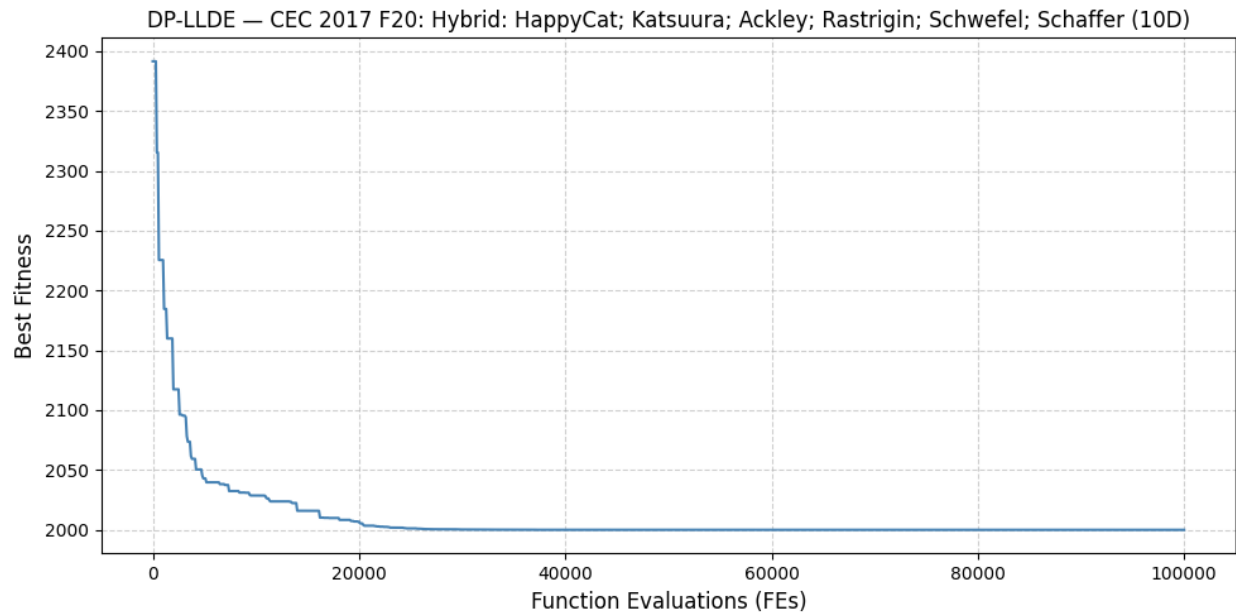
Error (|best-optimum|) : 2.547147e-05

Best Solution (5 dim) : [-30.59057464 -55.78803878 -29.57451965

61.04539565 69.96466587]

D^0 (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F20.png

6.4 Composite Functions (F21-F30)

```
# =====  
# F21 – Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)  
# =====  
run_and_plot('F21', make_wrapper(composition.f21), 2100.0,  
            'Composition: Rosenbrock; Elliptic; Rastrigin')
```


DP-LLDE on CEC 2017 F21 – Single Run

```

Function      : Composition: Rosenbrock; Elliptic; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 2100.0
Initial  $D^*(0)$  = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.446690e+03
Generation 10: Best = 2.310034e+03 FES = 1100  $\mu_F$  = 0.583  $\mu_{CR}$ 
= 0.347 Diversity = 3.2716e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 2.278833e+03 FES = 2100  $\mu_F$  = 0.622  $\mu_{CR}$ 
= 0.339 Diversity = 2.8158e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 2.237480e+03 FES = 3100  $\mu_F$  = 0.674  $\mu_{CR}$ 
= 0.287 Diversity = 2.7997e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 2.237480e+03 FES = 4100  $\mu_F$  = 0.690  $\mu_{CR}$ 
= 0.261 Diversity = 2.6939e+01  $k_t$  = [3, 3, 4, 4]
Generation 50: Best = 2.231671e+03 FES = 5100  $\mu_F$  = 0.702  $\mu_{CR}$ 
= 0.256 Diversity = 2.7479e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 2.227785e+03 FES = 6100  $\mu_F$  = 0.643  $\mu_{CR}$ 
= 0.202 Diversity = 2.7784e+01  $k_t$  = [3, 3, 2, 3]
Generation 70: Best = 2.226086e+03 FES = 7100  $\mu_F$  = 0.703  $\mu_{CR}$ 
= 0.215 Diversity = 2.9952e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 2.214754e+03 FES = 8100  $\mu_F$  = 0.709  $\mu_{CR}$ 
= 0.210 Diversity = 3.1849e+01  $k_t$  = [4, 3, 3, 4]
Generation 90: Best = 2.214754e+03 FES = 9100  $\mu_F$  = 0.750  $\mu_{CR}$ 
= 0.172 Diversity = 3.4527e+01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 2.212512e+03 FES = 10100  $\mu_F$  = 0.782  $\mu_{CR}$ 
= 0.147 Diversity = 3.7413e+01  $k_t$  = [3, 4, 3, 4]
Generation 110: Best = 2.211596e+03 FES = 11100  $\mu_F$  = 0.798  $\mu_{CR}$ 
= 0.152 Diversity = 2.4309e+01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 2.207026e+03 FES = 12100  $\mu_F$  = 0.806  $\mu_{CR}$ 
= 0.139 Diversity = 1.3492e+01  $k_t$  = [2, 4, 2, 4]
Generation 130: Best = 2.203422e+03 FES = 13100  $\mu_F$  = 0.800  $\mu_{CR}$ 
= 0.138 Diversity = 1.0794e+01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 2.202881e+03 FES = 14100  $\mu_F$  = 0.788  $\mu_{CR}$ 
= 0.125 Diversity = 9.1176e+00  $k_t$  = [3, 4, 3, 2]
Generation 150: Best = 2.201532e+03 FES = 15100  $\mu_F$  = 0.764  $\mu_{CR}$ 
= 0.126 Diversity = 6.3651e+00  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 2.200959e+03 FES = 16100  $\mu_F$  = 0.761  $\mu_{CR}$ 
= 0.139 Diversity = 3.9673e+00  $k_t$  = [2, 3, 2, 2]
Generation 170: Best = 2.200829e+03 FES = 17100  $\mu_F$  = 0.745  $\mu_{CR}$ 
= 0.136 Diversity = 2.5898e+00  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 2.200528e+03 FES = 18100  $\mu_F$  = 0.748  $\mu_{CR}$ 
= 0.124 Diversity = 1.7247e+00  $k_t$  = [3, 2, 2, 3]
Generation 190: Best = 2.200292e+03 FES = 19100  $\mu_F$  = 0.723  $\mu_{CR}$ 
= 0.124 Diversity = 1.1890e+00  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 2.200212e+03 FES = 20100  $\mu_F$  = 0.717  $\mu_{CR}$ 
= 0.128 Diversity = 9.8271e-01  $k_t$  = [3, 4, 3, 2]

```

Generation 210: Best = 2.200212e+03	FES = 21100	$\mu_F = 0.704$	μ_{CR}
= 0.137 Diversity = 7.8318e-01 $k_t = [1, 1, 1, 1]$			
Generation 220: Best = 2.200112e+03	FES = 22100	$\mu_F = 0.683$	μ_{CR}
= 0.137 Diversity = 5.6717e-01 $k_t = [4, 4, 4, 2]$			
Generation 230: Best = 2.200084e+03	FES = 23100	$\mu_F = 0.672$	μ_{CR}
= 0.145 Diversity = 4.0260e-01 $k_t = [1, 1, 1, 1]$			
Generation 240: Best = 2.200035e+03	FES = 24100	$\mu_F = 0.668$	μ_{CR}
= 0.151 Diversity = 2.7694e-01 $k_t = [4, 3, 4, 2]$			
Generation 250: Best = 2.200033e+03	FES = 25100	$\mu_F = 0.670$	μ_{CR}
= 0.170 Diversity = 1.6558e-01 $k_t = [1, 1, 1, 1]$			
Generation 260: Best = 2.200007e+03	FES = 26100	$\mu_F = 0.687$	μ_{CR}
= 0.183 Diversity = 8.5993e-02 $k_t = [3, 2, 4, 4]$			
Generation 270: Best = 2.200007e+03	FES = 27100	$\mu_F = 0.700$	μ_{CR}
= 0.182 Diversity = 4.2872e-02 $k_t = [1, 1, 1, 1]$			
Generation 280: Best = 2.200004e+03	FES = 28100	$\mu_F = 0.706$	μ_{CR}
= 0.192 Diversity = 1.9740e-02 $k_t = [3, 2, 3, 4]$			
Generation 290: Best = 2.200002e+03	FES = 29100	$\mu_F = 0.705$	μ_{CR}
= 0.201 Diversity = 1.0050e-02 $k_t = [1, 1, 1, 1]$			
Generation 300: Best = 2.200001e+03	FES = 30100	$\mu_F = 0.718$	μ_{CR}
= 0.200 Diversity = 5.0210e-03 $k_t = [2, 4, 2, 3]$			
Generation 310: Best = 2.200000e+03	FES = 31100	$\mu_F = 0.723$	μ_{CR}
= 0.199 Diversity = 2.6986e-03 $k_t = [1, 1, 1, 1]$			
Generation 320: Best = 2.200000e+03	FES = 32100	$\mu_F = 0.733$	μ_{CR}
= 0.201 Diversity = 1.2028e-03 $k_t = [2, 3, 2, 3]$			
Generation 330: Best = 2.200000e+03	FES = 33100	$\mu_F = 0.728$	μ_{CR}
= 0.204 Diversity = 6.4373e-04 $k_t = [1, 1, 1, 1]$			
Generation 340: Best = 2.200000e+03	FES = 34100	$\mu_F = 0.730$	μ_{CR}
= 0.208 Diversity = 3.3176e-04 $k_t = [2, 4, 4, 3]$			
Generation 350: Best = 2.200000e+03	FES = 35100	$\mu_F = 0.727$	μ_{CR}
= 0.208 Diversity = 1.7321e-04 $k_t = [1, 1, 1, 1]$			
Generation 360: Best = 2.200000e+03	FES = 36100	$\mu_F = 0.731$	μ_{CR}
= 0.199 Diversity = 7.3237e-05 $k_t = [4, 4, 2, 4]$			
Generation 370: Best = 2.200000e+03	FES = 37100	$\mu_F = 0.735$	μ_{CR}
= 0.197 Diversity = 3.8850e-05 $k_t = [1, 1, 1, 1]$			
Generation 380: Best = 2.200000e+03	FES = 38100	$\mu_F = 0.747$	μ_{CR}
= 0.196 Diversity = 1.9195e-05 $k_t = [2, 3, 4, 3]$			
Generation 390: Best = 2.200000e+03	FES = 39100	$\mu_F = 0.735$	μ_{CR}
= 0.201 Diversity = 1.0630e-05 $k_t = [1, 1, 1, 1]$			
Generation 400: Best = 2.200000e+03	FES = 40100	$\mu_F = 0.751$	μ_{CR}
= 0.212 Diversity = 5.4897e-06 $k_t = [4, 3, 3, 2]$			
Generation 410: Best = 2.200000e+03	FES = 41100	$\mu_F = 0.746$	μ_{CR}
= 0.215 Diversity = 2.9066e-06 $k_t = [1, 1, 1, 1]$			
Generation 420: Best = 2.200000e+03	FES = 42100	$\mu_F = 0.747$	μ_{CR}
= 0.212 Diversity = 1.3162e-06 $k_t = [3, 4, 3, 3]$			
Generation 430: Best = 2.200000e+03	FES = 43100	$\mu_F = 0.751$	μ_{CR}
= 0.209 Diversity = 6.9556e-07 $k_t = [1, 1, 1, 1]$			
Generation 440: Best = 2.200000e+03	FES = 44100	$\mu_F = 0.748$	μ_{CR}
= 0.210 Diversity = 3.1868e-07 $k_t = [2, 3, 3, 4]$			
Generation 450: Best = 2.200000e+03	FES = 45100	$\mu_F = 0.754$	μ_{CR}
= 0.214 Diversity = 1.6052e-07 $k_t = [1, 1, 1, 1]$			

Generation 460: Best = 2.200000e+03	FES = 46100	$\mu_F = 0.742$	μ_{CR}
= 0.224 Diversity = 7.1991e-08	$k_t = [4, 2, 3, 4]$		
Generation 470: Best = 2.200000e+03	FES = 47100	$\mu_F = 0.728$	μ_{CR}
= 0.226 Diversity = 3.1375e-08	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.200000e+03	FES = 48100	$\mu_F = 0.738$	μ_{CR}
= 0.228 Diversity = 1.4549e-08	$k_t = [3, 3, 2, 4]$		
Generation 490: Best = 2.200000e+03	FES = 49100	$\mu_F = 0.731$	μ_{CR}
= 0.227 Diversity = 7.1858e-09	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 2.200000e+03	FES = 50100	$\mu_F = 0.733$	μ_{CR}
= 0.233 Diversity = 2.9634e-09	$k_t = [3, 4, 2, 3]$		
Generation 510: Best = 2.200000e+03	FES = 51100	$\mu_F = 0.727$	μ_{CR}
= 0.232 Diversity = 1.3551e-09	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.200000e+03	FES = 52100	$\mu_F = 0.733$	μ_{CR}
= 0.235 Diversity = 6.4953e-10	$k_t = [3, 3, 2, 2]$		
Generation 530: Best = 2.200000e+03	FES = 53100	$\mu_F = 0.735$	μ_{CR}
= 0.231 Diversity = 3.3963e-10	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.200000e+03	FES = 54100	$\mu_F = 0.734$	μ_{CR}
= 0.231 Diversity = 1.4315e-10	$k_t = [4, 4, 4, 3]$		
Generation 550: Best = 2.200000e+03	FES = 55100	$\mu_F = 0.725$	μ_{CR}
= 0.238 Diversity = 6.5646e-11	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.200000e+03	FES = 56100	$\mu_F = 0.726$	μ_{CR}
= 0.233 Diversity = 3.1582e-11	$k_t = [2, 4, 4, 2]$		
Generation 570: Best = 2.200000e+03	FES = 57100	$\mu_F = 0.732$	μ_{CR}
= 0.231 Diversity = 1.7167e-11	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.200000e+03	FES = 58100	$\mu_F = 0.739$	μ_{CR}
= 0.233 Diversity = 9.2425e-12	$k_t = [2, 4, 2, 2]$		
Generation 590: Best = 2.200000e+03	FES = 59100	$\mu_F = 0.738$	μ_{CR}
= 0.228 Diversity = 5.6966e-12	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.200000e+03	FES = 60100	$\mu_F = 0.744$	μ_{CR}
= 0.223 Diversity = 3.3839e-12	$k_t = [2, 3, 3, 3]$		
Generation 610: Best = 2.200000e+03	FES = 61100	$\mu_F = 0.756$	μ_{CR}
= 0.218 Diversity = 2.6406e-12	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.200000e+03	FES = 62100	$\mu_F = 0.765$	μ_{CR}
= 0.211 Diversity = 2.0340e-12	$k_t = [2, 4, 4, 3]$		
Generation 630: Best = 2.200000e+03	FES = 63100	$\mu_F = 0.775$	μ_{CR}
= 0.210 Diversity = 1.6309e-12	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.200000e+03	FES = 64100	$\mu_F = 0.783$	μ_{CR}
= 0.205 Diversity = 1.2842e-12	$k_t = [2, 2, 3, 4]$		
Generation 650: Best = 2.200000e+03	FES = 65100	$\mu_F = 0.791$	μ_{CR}
= 0.201 Diversity = 1.3487e-12	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.200000e+03	FES = 66100	$\mu_F = 0.797$	μ_{CR}
= 0.201 Diversity = 1.1832e-12	$k_t = [3, 4, 2, 3]$		
Generation 670: Best = 2.200000e+03	FES = 67100	$\mu_F = 0.804$	μ_{CR}
= 0.207 Diversity = 1.1902e-12	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.200000e+03	FES = 68100	$\mu_F = 0.814$	μ_{CR}
= 0.205 Diversity = 9.9118e-13	$k_t = [2, 2, 2, 4]$		
Generation 690: Best = 2.200000e+03	FES = 69100	$\mu_F = 0.824$	μ_{CR}
= 0.207 Diversity = 1.0480e-12	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.200000e+03	FES = 70100	$\mu_F = 0.839$	μ_{CR}
= 0.202 Diversity = 7.7917e-13	$k_t = [3, 3, 3, 4]$		

Generation 710: Best = 2.200000e+03	FES = 71100	$\mu_F = 0.856$	μ_{CR}
= 0.204 Diversity = 7.8605e-13	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.200000e+03	FES = 72100	$\mu_F = 0.857$	μ_{CR}
= 0.195 Diversity = 7.2633e-13	$k_t = [3, 3, 2, 2]$		
Generation 730: Best = 2.200000e+03	FES = 73100	$\mu_F = 0.848$	μ_{CR}
= 0.193 Diversity = 7.0867e-13	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.200000e+03	FES = 74100	$\mu_F = 0.843$	μ_{CR}
= 0.189 Diversity = 4.3249e-13	$k_t = [3, 2, 2, 4]$		
Generation 750: Best = 2.200000e+03	FES = 75100	$\mu_F = 0.836$	μ_{CR}
= 0.188 Diversity = 4.2431e-13	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.200000e+03	FES = 76100	$\mu_F = 0.835$	μ_{CR}
= 0.185 Diversity = 4.1381e-13	$k_t = [3, 3, 4, 4]$		
Generation 770: Best = 2.200000e+03	FES = 77100	$\mu_F = 0.821$	μ_{CR}
= 0.185 Diversity = 4.1420e-13	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 2.200000e+03	FES = 78100	$\mu_F = 0.813$	μ_{CR}
= 0.179 Diversity = 4.0817e-13	$k_t = [4, 4, 4, 2]$		
Generation 790: Best = 2.200000e+03	FES = 79100	$\mu_F = 0.807$	μ_{CR}
= 0.178 Diversity = 3.9626e-13	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 2.200000e+03	FES = 80100	$\mu_F = 0.800$	μ_{CR}
= 0.169 Diversity = 3.9511e-13	$k_t = [4, 2, 3, 4]$		
Generation 810: Best = 2.200000e+03	FES = 81100	$\mu_F = 0.809$	μ_{CR}
= 0.173 Diversity = 4.1064e-13	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 2.200000e+03	FES = 82100	$\mu_F = 0.804$	μ_{CR}
= 0.170 Diversity = 4.2069e-13	$k_t = [2, 3, 2, 4]$		
Generation 830: Best = 2.200000e+03	FES = 83100	$\mu_F = 0.794$	μ_{CR}
= 0.169 Diversity = 4.2641e-13	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 2.200000e+03	FES = 84100	$\mu_F = 0.801$	μ_{CR}
= 0.173 Diversity = 4.2760e-13	$k_t = [4, 3, 4, 3]$		
Generation 850: Best = 2.200000e+03	FES = 85100	$\mu_F = 0.803$	μ_{CR}
= 0.173 Diversity = 4.2542e-13	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 2.200000e+03	FES = 86100	$\mu_F = 0.807$	μ_{CR}
= 0.177 Diversity = 4.0996e-13	$k_t = [3, 4, 2, 4]$		
Generation 870: Best = 2.200000e+03	FES = 87100	$\mu_F = 0.811$	μ_{CR}
= 0.171 Diversity = 3.9419e-13	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 2.200000e+03	FES = 88100	$\mu_F = 0.813$	μ_{CR}
= 0.176 Diversity = 3.7715e-13	$k_t = [3, 2, 4, 3]$		
Generation 890: Best = 2.200000e+03	FES = 89100	$\mu_F = 0.807$	μ_{CR}
= 0.172 Diversity = 4.0330e-13	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 2.200000e+03	FES = 90100	$\mu_F = 0.795$	μ_{CR}
= 0.170 Diversity = 4.0122e-13	$k_t = [2, 3, 4, 2]$		
Generation 910: Best = 2.200000e+03	FES = 91100	$\mu_F = 0.792$	μ_{CR}
= 0.169 Diversity = 4.3877e-13	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 2.200000e+03	FES = 92100	$\mu_F = 0.788$	μ_{CR}
= 0.166 Diversity = 4.3041e-13	$k_t = [3, 2, 3, 3]$		
Generation 930: Best = 2.200000e+03	FES = 93100	$\mu_F = 0.782$	μ_{CR}
= 0.169 Diversity = 4.2473e-13	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 2.200000e+03	FES = 94100	$\mu_F = 0.789$	μ_{CR}
= 0.167 Diversity = 3.7008e-13	$k_t = [2, 3, 3, 4]$		
Generation 950: Best = 2.200000e+03	FES = 95100	$\mu_F = 0.793$	μ_{CR}
= 0.166 Diversity = 3.9951e-13	$k_t = [1, 1, 1, 1]$		

```
Generation 960: Best = 2.200000e+03 FES = 96100  $\mu_F$  = 0.801  $\mu_{CR}$ 
= 0.162 Diversity = 4.2887e-13  $k_t$  = [4, 4, 2, 2]
Generation 970: Best = 2.200000e+03 FES = 97100  $\mu_F$  = 0.802  $\mu_{CR}$ 
= 0.164 Diversity = 4.4082e-13  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 2.200000e+03 FES = 98100  $\mu_F$  = 0.807  $\mu_{CR}$ 
= 0.165 Diversity = 4.2910e-13  $k_t$  = [2, 4, 3, 2]
Generation 990: Best = 2.200000e+03 FES = 99100  $\mu_F$  = 0.799  $\mu_{CR}$ 
= 0.169 Diversity = 4.1039e-13  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 2.200000e+03

Total Function Evals = 100000

Total Successful Updates = 45833

```
=====
=====
```

Results:

Best Fitness : 2.200000e+03

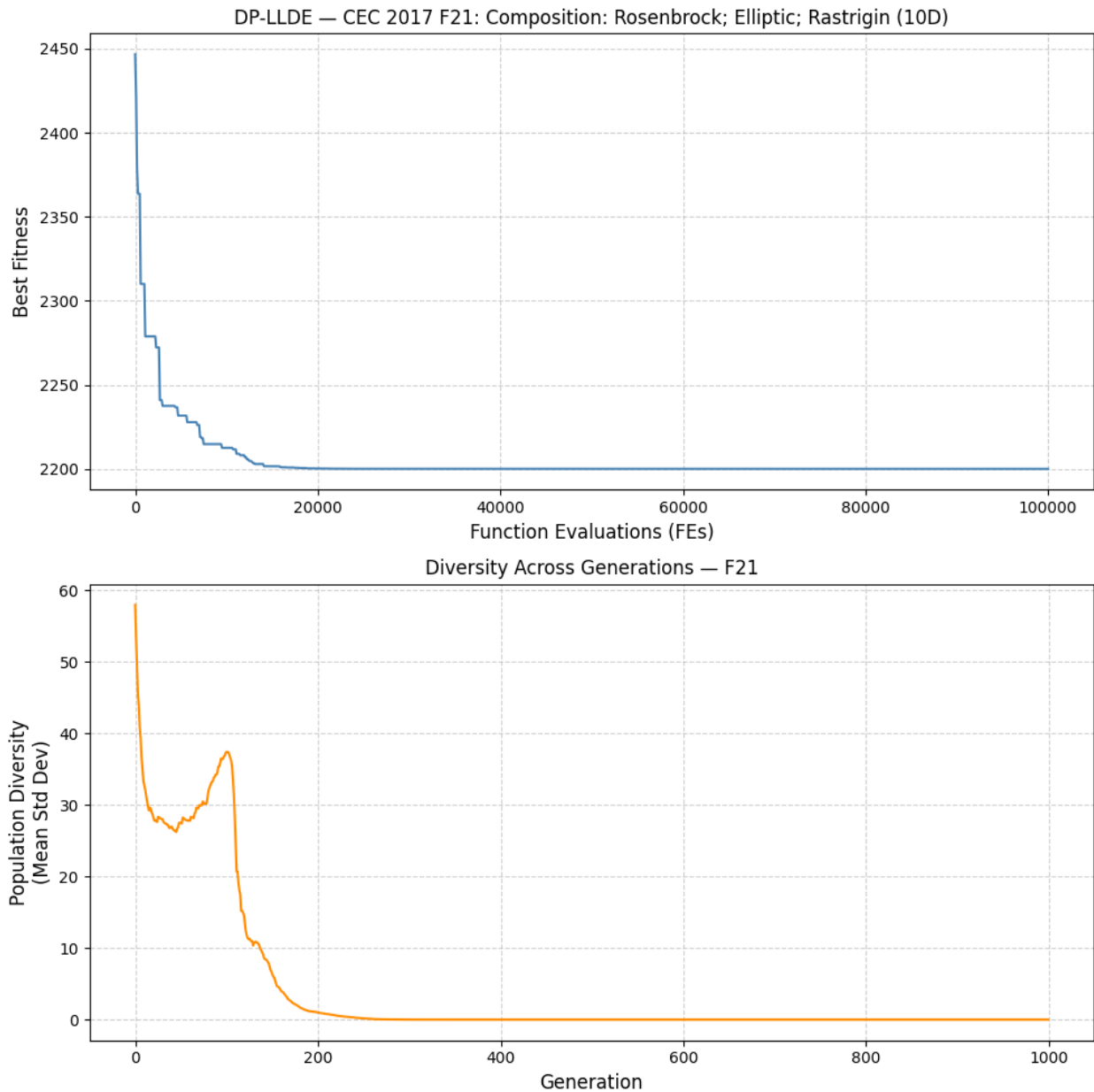
Global Optimum : 2.100000e+03

Error (|best-optimum|) : 1.000000e+02

Best Solution (5 dim) : [37.15806064 -63.08043142 -40.97860684 -
71.06301346 -23.17762719]

D^(0) (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F21.png

```
# =====
# F22 – Composition Function 2 (Rastrigin; Griewank; Schwefel)
# =====
run_and_plot('F22', make_wrapper(composition.f22), 2200.0,
'Composition: Rastrigin; Griewank; Schwefel')
```

```
=====
=====
DP-LLDE on CEC 2017 F22 – Single Run
=====
```

=====

Function : Composition: Rastrigin; Griewank; Schwefel (10D)

Bounds : [-100, 100]

Global optimum : 2200.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 3.528003e+03

Generation 10: Best = 2.399917e+03 FES = 1100 $\mu_F = 0.578$ $\mu_{CR} = 0.353$ Diversity = 1.9860e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 2.319026e+03 FES = 2100 $\mu_F = 0.619$ $\mu_{CR} = 0.356$ Diversity = 5.2010e+00 $k_t = [3, 4, 2, 3]$

Generation 30: Best = 2.311166e+03 FES = 3100 $\mu_F = 0.656$ $\mu_{CR} = 0.358$ Diversity = 1.5548e+00 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 2.305764e+03 FES = 4100 $\mu_F = 0.666$ $\mu_{CR} = 0.372$ Diversity = 8.6435e-01 $k_t = [3, 3, 2, 4]$

Generation 50: Best = 2.301901e+03 FES = 5100 $\mu_F = 0.650$ $\mu_{CR} = 0.348$ Diversity = 7.6769e-01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 2.301901e+03 FES = 6100 $\mu_F = 0.642$ $\mu_{CR} = 0.297$ Diversity = 7.2439e-01 $k_t = [2, 2, 4, 4]$

Generation 70: Best = 2.301901e+03 FES = 7100 $\mu_F = 0.632$ $\mu_{CR} = 0.283$ Diversity = 6.5863e-01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.301706e+03 FES = 8100 $\mu_F = 0.638$ $\mu_{CR} = 0.239$ Diversity = 5.4645e-01 $k_t = [4, 4, 2, 4]$

Generation 90: Best = 2.300829e+03 FES = 9100 $\mu_F = 0.620$ $\mu_{CR} = 0.207$ Diversity = 5.0793e-01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.300726e+03 FES = 10100 $\mu_F = 0.635$ $\mu_{CR} = 0.187$ Diversity = 4.6142e-01 $k_t = [2, 3, 4, 2]$

Generation 110: Best = 2.300548e+03 FES = 11100 $\mu_F = 0.651$ $\mu_{CR} = 0.166$ Diversity = 4.2247e-01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.300401e+03 FES = 12100 $\mu_F = 0.689$ $\mu_{CR} = 0.150$ Diversity = 3.7924e-01 $k_t = [2, 3, 4, 3]$

Generation 130: Best = 2.300205e+03 FES = 13100 $\mu_F = 0.690$ $\mu_{CR} = 0.155$ Diversity = 3.4162e-01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.300157e+03 FES = 14100 $\mu_F = 0.696$ $\mu_{CR} = 0.154$ Diversity = 2.8357e-01 $k_t = [3, 4, 3, 2]$

Generation 150: Best = 2.300107e+03 FES = 15100 $\mu_F = 0.692$ $\mu_{CR} = 0.161$ Diversity = 2.6350e-01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.300062e+03 FES = 16100 $\mu_F = 0.708$ $\mu_{CR} = 0.146$ Diversity = 2.4621e-01 $k_t = [2, 2, 2, 2]$

Generation 170: Best = 2.300043e+03 FES = 17100 $\mu_F = 0.704$ $\mu_{CR} = 0.143$ Diversity = 2.1392e-01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 2.300025e+03 FES = 18100 $\mu_F = 0.720$ $\mu_{CR} = 0.153$ Diversity = 1.5385e-01 $k_t = [3, 3, 2, 4]$

Generation 190: Best = 2.300021e+03 FES = 19100 $\mu_F = 0.712$ $\mu_{CR} = 0.157$ Diversity = 1.4902e-01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 2.300013e+03 FES = 20100 $\mu_F = 0.716$ $\mu_{CR} = 0.161$ Diversity = 1.4502e-01 $k_t = [2, 2, 2, 2]$

Generation 210: Best = 2.300008e+03 FES = 21100 $\mu_F = 0.722$ $\mu_{CR} = 0.172$ Diversity = 1.4226e-01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 2.300006e+03 FES = 22100 $\mu_F = 0.716$ μ_{CR}

= 0.175	Diversity = 6.5422e-02	k_t = [3, 3, 2, 3]	μ_F = 0.713	μ_{CR}
Generation 230:	Best = 2.300003e+03	FES = 23100		
= 0.179	Diversity = 2.1214e-02	k_t = [1, 1, 1, 1]	μ_F = 0.728	μ_{CR}
Generation 240:	Best = 2.300002e+03	FES = 24100		
= 0.177	Diversity = 6.5230e-03	k_t = [4, 4, 3, 3]	μ_F = 0.731	μ_{CR}
Generation 250:	Best = 2.300001e+03	FES = 25100		
= 0.183	Diversity = 3.0582e-03	k_t = [1, 1, 1, 1]	μ_F = 0.728	μ_{CR}
Generation 260:	Best = 2.300001e+03	FES = 26100		
= 0.183	Diversity = 1.3730e-03	k_t = [4, 2, 3, 4]	μ_F = 0.736	μ_{CR}
Generation 270:	Best = 2.300000e+03	FES = 27100		
= 0.189	Diversity = 7.8914e-04	k_t = [1, 1, 1, 1]	μ_F = 0.748	μ_{CR}
Generation 280:	Best = 2.300000e+03	FES = 28100		
= 0.200	Diversity = 4.1158e-04	k_t = [3, 4, 4, 2]	μ_F = 0.757	μ_{CR}
Generation 290:	Best = 2.300000e+03	FES = 29100		
= 0.198	Diversity = 2.4895e-04	k_t = [1, 1, 1, 1]	μ_F = 0.759	μ_{CR}
Generation 300:	Best = 2.300000e+03	FES = 30100		
= 0.192	Diversity = 1.4239e-04	k_t = [3, 3, 2, 2]	μ_F = 0.762	μ_{CR}
Generation 310:	Best = 2.300000e+03	FES = 31100		
= 0.206	Diversity = 9.2146e-05	k_t = [1, 1, 1, 1]	μ_F = 0.754	μ_{CR}
Generation 320:	Best = 2.300000e+03	FES = 32100		
= 0.206	Diversity = 4.1664e-05	k_t = [4, 2, 4, 4]	μ_F = 0.768	μ_{CR}
Generation 330:	Best = 2.300000e+03	FES = 33100		
= 0.202	Diversity = 1.9917e-05	k_t = [1, 1, 1, 1]	μ_F = 0.759	μ_{CR}
Generation 340:	Best = 2.300000e+03	FES = 34100		
= 0.210	Diversity = 9.6378e-06	k_t = [4, 3, 4, 2]	μ_F = 0.752	μ_{CR}
Generation 350:	Best = 2.300000e+03	FES = 35100		
= 0.203	Diversity = 4.5112e-06	k_t = [1, 1, 1, 1]	μ_F = 0.752	μ_{CR}
Generation 360:	Best = 2.300000e+03	FES = 36100		
= 0.209	Diversity = 2.0744e-06	k_t = [3, 4, 3, 3]	μ_F = 0.744	μ_{CR}
Generation 370:	Best = 2.300000e+03	FES = 37100		
= 0.208	Diversity = 1.0223e-06	k_t = [1, 1, 1, 1]	μ_F = 0.739	μ_{CR}
Generation 380:	Best = 2.300000e+03	FES = 38100		
= 0.208	Diversity = 4.7578e-07	k_t = [4, 4, 4, 3]	μ_F = 0.735	μ_{CR}
Generation 390:	Best = 2.300000e+03	FES = 39100		
= 0.205	Diversity = 2.5610e-07	k_t = [1, 1, 1, 1]	μ_F = 0.724	μ_{CR}
Generation 400:	Best = 2.300000e+03	FES = 40100		
= 0.210	Diversity = 1.2996e-07	k_t = [4, 3, 2, 3]	μ_F = 0.723	μ_{CR}
Generation 410:	Best = 2.300000e+03	FES = 41100		
= 0.212	Diversity = 6.4694e-08	k_t = [1, 1, 1, 1]	μ_F = 0.738	μ_{CR}
Generation 420:	Best = 2.300000e+03	FES = 42100		
= 0.213	Diversity = 2.8935e-08	k_t = [4, 3, 3, 4]	μ_F = 0.736	μ_{CR}
Generation 430:	Best = 2.300000e+03	FES = 43100		
= 0.220	Diversity = 1.4584e-08	k_t = [1, 1, 1, 1]	μ_F = 0.740	μ_{CR}
Generation 440:	Best = 2.300000e+03	FES = 44100		
= 0.223	Diversity = 6.5871e-09	k_t = [2, 2, 4, 4]	μ_F = 0.737	μ_{CR}
Generation 450:	Best = 2.300000e+03	FES = 45100		
= 0.228	Diversity = 3.2549e-09	k_t = [1, 1, 1, 1]	μ_F = 0.738	μ_{CR}
Generation 460:	Best = 2.300000e+03	FES = 46100		
= 0.221	Diversity = 1.6595e-09	k_t = [2, 3, 2, 3]		

Generation 470: Best = 2.300000e+03	FES = 47100	$\mu_F = 0.727$	μ_{CR}
= 0.222 Diversity = 8.2957e-10	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.300000e+03	FES = 48100	$\mu_F = 0.724$	μ_{CR}
= 0.226 Diversity = 4.1442e-10	$k_t = [3, 4, 3, 2]$		
Generation 490: Best = 2.300000e+03	FES = 49100	$\mu_F = 0.721$	μ_{CR}
= 0.228 Diversity = 1.8567e-10	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 2.300000e+03	FES = 50100	$\mu_F = 0.730$	μ_{CR}
= 0.225 Diversity = 8.5367e-11	$k_t = [2, 4, 2, 3]$		
Generation 510: Best = 2.300000e+03	FES = 51100	$\mu_F = 0.732$	μ_{CR}
= 0.221 Diversity = 4.5192e-11	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.300000e+03	FES = 52100	$\mu_F = 0.731$	μ_{CR}
= 0.214 Diversity = 2.2968e-11	$k_t = [2, 2, 2, 4]$		
Generation 530: Best = 2.300000e+03	FES = 53100	$\mu_F = 0.741$	μ_{CR}
= 0.214 Diversity = 1.4142e-11	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.300000e+03	FES = 54100	$\mu_F = 0.739$	μ_{CR}
= 0.209 Diversity = 7.1994e-12	$k_t = [3, 2, 2, 3]$		
Generation 550: Best = 2.300000e+03	FES = 55100	$\mu_F = 0.749$	μ_{CR}
= 0.202 Diversity = 4.0589e-12	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.300000e+03	FES = 56100	$\mu_F = 0.753$	μ_{CR}
= 0.200 Diversity = 2.0661e-12	$k_t = [4, 4, 4, 3]$		
Generation 570: Best = 2.300000e+03	FES = 57100	$\mu_F = 0.757$	μ_{CR}
= 0.195 Diversity = 1.4277e-12	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.300000e+03	FES = 58100	$\mu_F = 0.768$	μ_{CR}
= 0.196 Diversity = 9.2336e-13	$k_t = [3, 2, 4, 4]$		
Generation 590: Best = 2.300000e+03	FES = 59100	$\mu_F = 0.781$	μ_{CR}
= 0.194 Diversity = 7.6870e-13	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.300000e+03	FES = 60100	$\mu_F = 0.777$	μ_{CR}
= 0.196 Diversity = 6.5740e-13	$k_t = [4, 2, 2, 3]$		
Generation 610: Best = 2.300000e+03	FES = 61100	$\mu_F = 0.781$	μ_{CR}
= 0.186 Diversity = 6.5891e-13	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.300000e+03	FES = 62100	$\mu_F = 0.785$	μ_{CR}
= 0.186 Diversity = 6.3003e-13	$k_t = [2, 3, 4, 2]$		
Generation 630: Best = 2.300000e+03	FES = 63100	$\mu_F = 0.782$	μ_{CR}
= 0.191 Diversity = 6.7373e-13	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.300000e+03	FES = 64100	$\mu_F = 0.782$	μ_{CR}
= 0.184 Diversity = 5.9086e-13	$k_t = [4, 4, 2, 3]$		
Generation 650: Best = 2.300000e+03	FES = 65100	$\mu_F = 0.789$	μ_{CR}
= 0.180 Diversity = 5.9063e-13	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.300000e+03	FES = 66100	$\mu_F = 0.794$	μ_{CR}
= 0.184 Diversity = 5.5228e-13	$k_t = [3, 4, 3, 2]$		
Generation 670: Best = 2.300000e+03	FES = 67100	$\mu_F = 0.809$	μ_{CR}
= 0.192 Diversity = 5.3809e-13	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.300000e+03	FES = 68100	$\mu_F = 0.816$	μ_{CR}
= 0.189 Diversity = 4.5441e-13	$k_t = [3, 2, 3, 4]$		
Generation 690: Best = 2.300000e+03	FES = 69100	$\mu_F = 0.827$	μ_{CR}
= 0.193 Diversity = 4.9874e-13	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.300000e+03	FES = 70100	$\mu_F = 0.838$	μ_{CR}
= 0.196 Diversity = 4.3648e-13	$k_t = [3, 4, 3, 2]$		
Generation 710: Best = 2.300000e+03	FES = 71100	$\mu_F = 0.846$	μ_{CR}

= 0.196	Diversity = 4.2033e-13	k_t = [1, 1, 1, 1]	μ_F = 0.846	μ_{CR}
Generation 720:	Best = 2.300000e+03	FES = 72100		
= 0.194	Diversity = 3.4367e-13	k_t = [2, 4, 3, 3]	μ_F = 0.851	μ_{CR}
Generation 730:	Best = 2.300000e+03	FES = 73100		
= 0.195	Diversity = 3.6519e-13	k_t = [1, 1, 1, 1]	μ_F = 0.851	μ_{CR}
Generation 740:	Best = 2.300000e+03	FES = 74100		
= 0.197	Diversity = 2.5892e-13	k_t = [2, 3, 3, 4]	μ_F = 0.834	μ_{CR}
Generation 750:	Best = 2.300000e+03	FES = 75100		
= 0.200	Diversity = 2.2598e-13	k_t = [1, 1, 1, 1]	μ_F = 0.825	μ_{CR}
Generation 760:	Best = 2.300000e+03	FES = 76100		
= 0.200	Diversity = 2.1046e-13	k_t = [3, 3, 4, 4]	μ_F = 0.813	μ_{CR}
Generation 770:	Best = 2.300000e+03	FES = 77100		
= 0.201	Diversity = 2.1430e-13	k_t = [1, 1, 1, 1]	μ_F = 0.808	μ_{CR}
Generation 780:	Best = 2.300000e+03	FES = 78100		
= 0.197	Diversity = 2.0131e-13	k_t = [3, 4, 4, 2]	μ_F = 0.800	μ_{CR}
Generation 790:	Best = 2.300000e+03	FES = 79100		
= 0.191	Diversity = 1.9909e-13	k_t = [1, 1, 1, 1]	μ_F = 0.797	μ_{CR}
Generation 800:	Best = 2.300000e+03	FES = 80100		
= 0.184	Diversity = 1.9916e-13	k_t = [3, 2, 3, 2]	μ_F = 0.794	μ_{CR}
Generation 810:	Best = 2.300000e+03	FES = 81100		
= 0.182	Diversity = 2.0338e-13	k_t = [1, 1, 1, 1]	μ_F = 0.792	μ_{CR}
Generation 820:	Best = 2.300000e+03	FES = 82100		
= 0.179	Diversity = 1.8459e-13	k_t = [4, 4, 3, 4]	μ_F = 0.787	μ_{CR}
Generation 830:	Best = 2.300000e+03	FES = 83100		
= 0.183	Diversity = 1.8617e-13	k_t = [1, 1, 1, 1]	μ_F = 0.779	μ_{CR}
Generation 840:	Best = 2.300000e+03	FES = 84100		
= 0.171	Diversity = 1.8747e-13	k_t = [3, 2, 4, 2]	μ_F = 0.782	μ_{CR}
Generation 850:	Best = 2.300000e+03	FES = 85100		
= 0.171	Diversity = 1.8656e-13	k_t = [1, 1, 1, 1]	μ_F = 0.794	μ_{CR}
Generation 860:	Best = 2.300000e+03	FES = 86100		
= 0.169	Diversity = 1.8581e-13	k_t = [3, 2, 2, 3]	μ_F = 0.802	μ_{CR}
Generation 870:	Best = 2.300000e+03	FES = 87100		
= 0.156	Diversity = 1.8331e-13	k_t = [1, 1, 1, 1]	μ_F = 0.811	μ_{CR}
Generation 880:	Best = 2.300000e+03	FES = 88100		
= 0.151	Diversity = 1.6656e-13	k_t = [4, 3, 2, 3]	μ_F = 0.813	μ_{CR}
Generation 890:	Best = 2.300000e+03	FES = 89100		
= 0.148	Diversity = 1.6540e-13	k_t = [1, 1, 1, 1]	μ_F = 0.811	μ_{CR}
Generation 900:	Best = 2.300000e+03	FES = 90100		
= 0.150	Diversity = 1.5214e-13	k_t = [4, 2, 4, 4]	μ_F = 0.826	μ_{CR}
Generation 910:	Best = 2.300000e+03	FES = 91100		
= 0.153	Diversity = 1.6110e-13	k_t = [1, 1, 1, 1]	μ_F = 0.824	μ_{CR}
Generation 920:	Best = 2.300000e+03	FES = 92100		
= 0.151	Diversity = 1.5513e-13	k_t = [4, 2, 4, 2]	μ_F = 0.829	μ_{CR}
Generation 930:	Best = 2.300000e+03	FES = 93100		
= 0.156	Diversity = 1.5590e-13	k_t = [1, 1, 1, 1]	μ_F = 0.824	μ_{CR}
Generation 940:	Best = 2.300000e+03	FES = 94100		
= 0.163	Diversity = 1.5136e-13	k_t = [4, 4, 2, 4]	μ_F = 0.822	μ_{CR}
Generation 950:	Best = 2.300000e+03	FES = 95100		
= 0.162	Diversity = 1.4699e-13	k_t = [1, 1, 1, 1]		

```
Generation 960: Best = 2.300000e+03 FES = 96100  $\mu_F$  = 0.818  $\mu_{CR}$ 
= 0.158 Diversity = 1.3807e-13  $k_t$  = [4, 4, 2, 2]
Generation 970: Best = 2.300000e+03 FES = 97100  $\mu_F$  = 0.822  $\mu_{CR}$ 
= 0.162 Diversity = 1.4929e-13  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 2.300000e+03 FES = 98100  $\mu_F$  = 0.824  $\mu_{CR}$ 
= 0.165 Diversity = 1.6248e-13  $k_t$  = [2, 4, 3, 2]
Generation 990: Best = 2.300000e+03 FES = 99100  $\mu_F$  = 0.821  $\mu_{CR}$ 
= 0.172 Diversity = 1.7523e-13  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 2.300000e+03

Total Function Evals = 100000

Total Successful Updates = 47484

=====

Results:

Best Fitness : 2.300000e+03

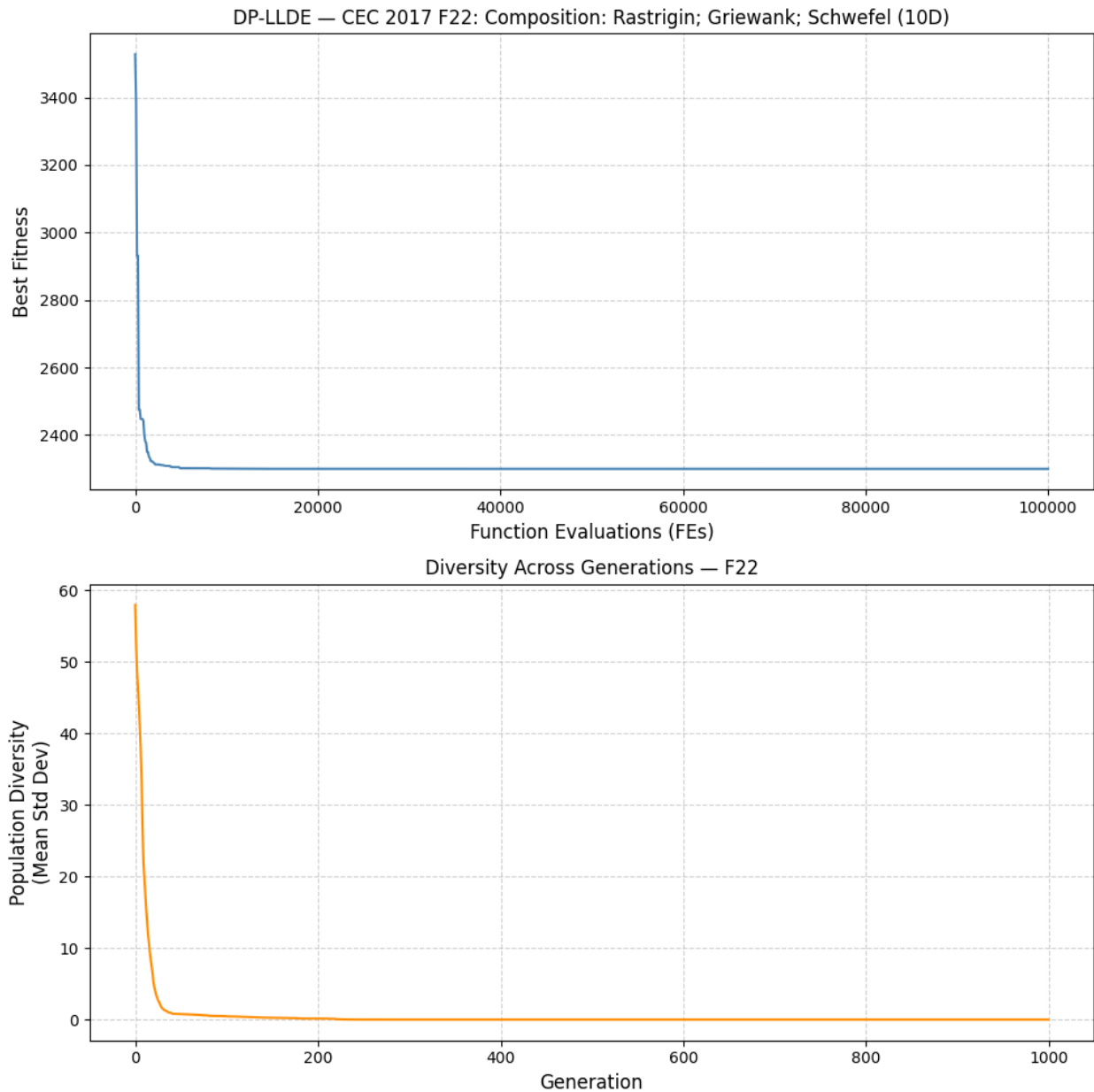
Global Optimum : 2.200000e+03

Error (|best-optimum|) : 1.000000e+02

Best Solution (5 dim) : [-52.41266063 11.16831479 23.05456506
35.08347775 30.48935237]

D^0 (init. diversity): 5.7928e+01

=====



Graph saved as convergence_F22.png

```
# =====
# F23 – Composition Function 3 (Rosenbrock; Ackley; Schwefel;
# Rastrigin)
# =====
run_and_plot('F23', make_wrapper(composition.f23), 2300.0,
'Composition: Rosenbrock; Ackley; Schwefel; Rastrigin')
```

DP-LLDE on CEC 2017 F23 – Single Run

```

=====
Function      : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin
(10D)
Bounds       : [-100, 100]
Global optimum : 2300.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.773921e+03
Generation 10: Best = 2.678874e+03 FES = 1100  $\mu_F$  = 0.593  $\mu_{CR}$ 
= 0.351 Diversity = 2.8216e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 2.651353e+03 FES = 2100  $\mu_F$  = 0.631  $\mu_{CR}$ 
= 0.344 Diversity = 2.0842e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 2.643389e+03 FES = 3100  $\mu_F$  = 0.659  $\mu_{CR}$ 
= 0.316 Diversity = 1.9091e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 2.640096e+03 FES = 4100  $\mu_F$  = 0.697  $\mu_{CR}$ 
= 0.292 Diversity = 1.7992e+01  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 2.640096e+03 FES = 5100  $\mu_F$  = 0.723  $\mu_{CR}$ 
= 0.265 Diversity = 1.7341e+01  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 2.630197e+03 FES = 6100  $\mu_F$  = 0.734  $\mu_{CR}$ 
= 0.211 Diversity = 1.7130e+01  $k_t$  = [3, 3, 4, 3]
Generation 70: Best = 2.626706e+03 FES = 7100  $\mu_F$  = 0.735  $\mu_{CR}$ 
= 0.204 Diversity = 1.7251e+01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 2.626706e+03 FES = 8100  $\mu_F$  = 0.766  $\mu_{CR}$ 
= 0.160 Diversity = 1.6941e+01  $k_t$  = [2, 2, 3, 2]
Generation 90: Best = 2.626706e+03 FES = 9100  $\mu_F$  = 0.753  $\mu_{CR}$ 
= 0.142 Diversity = 1.6935e+01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 2.623789e+03 FES = 10100  $\mu_F$  = 0.780  $\mu_{CR}$ 
= 0.155 Diversity = 1.7123e+01  $k_t$  = [4, 2, 4, 3]
Generation 110: Best = 2.623610e+03 FES = 11100  $\mu_F$  = 0.801  $\mu_{CR}$ 
= 0.139 Diversity = 1.6937e+01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 2.623529e+03 FES = 12100  $\mu_F$  = 0.799  $\mu_{CR}$ 
= 0.116 Diversity = 1.6680e+01  $k_t$  = [3, 3, 2, 4]
Generation 130: Best = 2.623529e+03 FES = 13100  $\mu_F$  = 0.788  $\mu_{CR}$ 
= 0.095 Diversity = 1.6735e+01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 2.621301e+03 FES = 14100  $\mu_F$  = 0.663  $\mu_{CR}$ 
= 0.095 Diversity = 1.6329e+01  $k_t$  = [4, 4, 2, 4]
Generation 150: Best = 2.620842e+03 FES = 15100  $\mu_F$  = 0.707  $\mu_{CR}$ 
= 0.087 Diversity = 1.6285e+01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 2.619933e+03 FES = 16100  $\mu_F$  = 0.762  $\mu_{CR}$ 
= 0.082 Diversity = 1.5785e+01  $k_t$  = [3, 2, 3, 4]
Generation 170: Best = 2.619933e+03 FES = 17100  $\mu_F$  = 0.754  $\mu_{CR}$ 
= 0.082 Diversity = 1.5709e+01  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 2.612647e+03 FES = 18100  $\mu_F$  = 0.752  $\mu_{CR}$ 
= 0.082 Diversity = 1.5460e+01  $k_t$  = [3, 4, 3, 4]
Generation 190: Best = 2.612647e+03 FES = 19100  $\mu_F$  = 0.737  $\mu_{CR}$ 
= 0.087 Diversity = 1.5417e+01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 2.612647e+03 FES = 20100  $\mu_F$  = 0.711  $\mu_{CR}$ 
= 0.086 Diversity = 1.5207e+01  $k_t$  = [2, 2, 4, 3]
Generation 210: Best = 2.612633e+03 FES = 21100  $\mu_F$  = 0.674  $\mu_{CR}$ 

```

= 0.093	Diversity = 1.5071e+01	k_t = [1, 1, 1, 1]	μ_F = 0.679	μ_{CR}
Generation 220:	Best = 2.612633e+03	FES = 22100		
= 0.089	Diversity = 1.4909e+01	k_t = [4, 2, 3, 4]	μ_F = 0.650	μ_{CR}
Generation 230:	Best = 2.612527e+03	FES = 23100		
= 0.111	Diversity = 1.4892e+01	k_t = [1, 1, 1, 1]	μ_F = 0.732	μ_{CR}
Generation 240:	Best = 2.612527e+03	FES = 24100		
= 0.118	Diversity = 1.4861e+01	k_t = [2, 4, 3, 2]	μ_F = 0.723	μ_{CR}
Generation 250:	Best = 2.610559e+03	FES = 25100		
= 0.105	Diversity = 1.4846e+01	k_t = [1, 1, 1, 1]	μ_F = 0.766	μ_{CR}
Generation 260:	Best = 2.610559e+03	FES = 26100		
= 0.094	Diversity = 1.4622e+01	k_t = [3, 2, 4, 3]	μ_F = 0.686	μ_{CR}
Generation 270:	Best = 2.610559e+03	FES = 27100		
= 0.107	Diversity = 1.4613e+01	k_t = [1, 1, 1, 1]	μ_F = 0.647	μ_{CR}
Generation 280:	Best = 2.608331e+03	FES = 28100		
= 0.108	Diversity = 1.4439e+01	k_t = [4, 2, 4, 2]	μ_F = 0.645	μ_{CR}
Generation 290:	Best = 2.608271e+03	FES = 29100		
= 0.117	Diversity = 1.4466e+01	k_t = [1, 1, 1, 1]	μ_F = 0.581	μ_{CR}
Generation 300:	Best = 2.608271e+03	FES = 30100		
= 0.095	Diversity = 1.4447e+01	k_t = [3, 2, 4, 4]	μ_F = 0.645	μ_{CR}
Generation 310:	Best = 2.608271e+03	FES = 31100		
= 0.088	Diversity = 1.4382e+01	k_t = [1, 1, 1, 1]	μ_F = 0.687	μ_{CR}
Generation 320:	Best = 2.608271e+03	FES = 32100		
= 0.071	Diversity = 1.4088e+01	k_t = [2, 4, 2, 4]	μ_F = 0.605	μ_{CR}
Generation 330:	Best = 2.608271e+03	FES = 33100		
= 0.067	Diversity = 1.4042e+01	k_t = [1, 1, 1, 1]	μ_F = 0.541	μ_{CR}
Generation 340:	Best = 2.608091e+03	FES = 34100		
= 0.069	Diversity = 1.3936e+01	k_t = [3, 3, 3, 4]	μ_F = 0.516	μ_{CR}
Generation 350:	Best = 2.608091e+03	FES = 35100		
= 0.087	Diversity = 1.3913e+01	k_t = [1, 1, 1, 1]	μ_F = 0.498	μ_{CR}
Generation 360:	Best = 2.608091e+03	FES = 36100		
= 0.096	Diversity = 1.3892e+01	k_t = [2, 4, 2, 3]	μ_F = 0.478	μ_{CR}
Generation 370:	Best = 2.607843e+03	FES = 37100		
= 0.116	Diversity = 1.3871e+01	k_t = [1, 1, 1, 1]	μ_F = 0.398	μ_{CR}
Generation 380:	Best = 2.607824e+03	FES = 38100		
= 0.105	Diversity = 1.3864e+01	k_t = [4, 4, 3, 4]	μ_F = 0.391	μ_{CR}
Generation 390:	Best = 2.607824e+03	FES = 39100		
= 0.107	Diversity = 1.3761e+01	k_t = [1, 1, 1, 1]	μ_F = 0.357	μ_{CR}
Generation 400:	Best = 2.607824e+03	FES = 40100		
= 0.096	Diversity = 1.3644e+01	k_t = [4, 4, 4, 4]	μ_F = 0.375	μ_{CR}
Generation 410:	Best = 2.607824e+03	FES = 41100		
= 0.116	Diversity = 1.3585e+01	k_t = [1, 1, 1, 1]	μ_F = 0.428	μ_{CR}
Generation 420:	Best = 2.607824e+03	FES = 42100		
= 0.103	Diversity = 1.3535e+01	k_t = [4, 4, 2, 3]	μ_F = 0.458	μ_{CR}
Generation 430:	Best = 2.607824e+03	FES = 43100		
= 0.090	Diversity = 1.3484e+01	k_t = [1, 1, 1, 1]	μ_F = 0.485	μ_{CR}
Generation 440:	Best = 2.607824e+03	FES = 44100		
= 0.109	Diversity = 1.3387e+01	k_t = [4, 2, 3, 2]	μ_F = 0.527	μ_{CR}
Generation 450:	Best = 2.607824e+03	FES = 45100		
= 0.089	Diversity = 1.3385e+01	k_t = [1, 1, 1, 1]		

Generation 460: Best = 2.607824e+03	FES = 46100	$\mu_F = 0.477$	μ_{CR}
= 0.082 Diversity = 1.3429e+01	$k_t = [4, 2, 2, 4]$		
Generation 470: Best = 2.607791e+03	FES = 47100	$\mu_F = 0.388$	μ_{CR}
= 0.092 Diversity = 1.3426e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.607791e+03	FES = 48100	$\mu_F = 0.387$	μ_{CR}
= 0.086 Diversity = 1.3430e+01	$k_t = [3, 4, 2, 3]$		
Generation 490: Best = 2.607407e+03	FES = 49100	$\mu_F = 0.368$	μ_{CR}
= 0.077 Diversity = 1.3412e+01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 2.607402e+03	FES = 50100	$\mu_F = 0.404$	μ_{CR}
= 0.079 Diversity = 1.3401e+01	$k_t = [2, 3, 3, 3]$		
Generation 510: Best = 2.607402e+03	FES = 51100	$\mu_F = 0.366$	μ_{CR}
= 0.087 Diversity = 1.3397e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.607402e+03	FES = 52100	$\mu_F = 0.388$	μ_{CR}
= 0.068 Diversity = 1.3402e+01	$k_t = [4, 2, 2, 3]$		
Generation 530: Best = 2.607386e+03	FES = 53100	$\mu_F = 0.329$	μ_{CR}
= 0.096 Diversity = 1.3396e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.607385e+03	FES = 54100	$\mu_F = 0.328$	μ_{CR}
= 0.108 Diversity = 1.3370e+01	$k_t = [4, 4, 2, 2]$		
Generation 550: Best = 2.607242e+03	FES = 55100	$\mu_F = 0.290$	μ_{CR}
= 0.088 Diversity = 1.3376e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.607242e+03	FES = 56100	$\mu_F = 0.319$	μ_{CR}
= 0.094 Diversity = 1.3361e+01	$k_t = [2, 3, 4, 3]$		
Generation 570: Best = 2.606909e+03	FES = 57100	$\mu_F = 0.380$	μ_{CR}
= 0.087 Diversity = 1.3362e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.606894e+03	FES = 58100	$\mu_F = 0.379$	μ_{CR}
= 0.106 Diversity = 1.3281e+01	$k_t = [3, 3, 4, 4]$		
Generation 590: Best = 2.606785e+03	FES = 59100	$\mu_F = 0.429$	μ_{CR}
= 0.106 Diversity = 1.3188e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.606785e+03	FES = 60100	$\mu_F = 0.461$	μ_{CR}
= 0.141 Diversity = 1.3145e+01	$k_t = [2, 2, 2, 4]$		
Generation 610: Best = 2.606785e+03	FES = 61100	$\mu_F = 0.463$	μ_{CR}
= 0.125 Diversity = 1.3161e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.606574e+03	FES = 62100	$\mu_F = 0.424$	μ_{CR}
= 0.124 Diversity = 1.3038e+01	$k_t = [2, 4, 4, 2]$		
Generation 630: Best = 2.606551e+03	FES = 63100	$\mu_F = 0.484$	μ_{CR}
= 0.112 Diversity = 1.3065e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.606551e+03	FES = 64100	$\mu_F = 0.394$	μ_{CR}
= 0.100 Diversity = 1.3075e+01	$k_t = [4, 2, 4, 2]$		
Generation 650: Best = 2.606500e+03	FES = 65100	$\mu_F = 0.309$	μ_{CR}
= 0.104 Diversity = 1.3069e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.606500e+03	FES = 66100	$\mu_F = 0.252$	μ_{CR}
= 0.110 Diversity = 1.3066e+01	$k_t = [2, 4, 2, 2]$		
Generation 670: Best = 2.606500e+03	FES = 67100	$\mu_F = 0.225$	μ_{CR}
= 0.109 Diversity = 1.3059e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.606500e+03	FES = 68100	$\mu_F = 0.272$	μ_{CR}
= 0.103 Diversity = 1.2909e+01	$k_t = [3, 3, 3, 4]$		
Generation 690: Best = 2.605639e+03	FES = 69100	$\mu_F = 0.257$	μ_{CR}
= 0.116 Diversity = 1.2889e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.605639e+03	FES = 70100	$\mu_F = 0.265$	μ_{CR}

= 0.106	Diversity = 1.2886e+01	k_t = [2, 3, 3, 3]	μ_F = 0.334	μ_{CR}
Generation 710:	Best = 2.605639e+03	FES = 71100		
= 0.105	Diversity = 1.2887e+01	k_t = [1, 1, 1, 1]	μ_F = 0.260	μ_{CR}
Generation 720:	Best = 2.605639e+03	FES = 72100		
= 0.113	Diversity = 1.2886e+01	k_t = [2, 3, 4, 3]	μ_F = 0.309	μ_{CR}
Generation 730:	Best = 2.605461e+03	FES = 73100		
= 0.108	Diversity = 1.2886e+01	k_t = [1, 1, 1, 1]	μ_F = 0.302	μ_{CR}
Generation 740:	Best = 2.605461e+03	FES = 74100		
= 0.092	Diversity = 1.2877e+01	k_t = [3, 4, 3, 3]	μ_F = 0.319	μ_{CR}
Generation 750:	Best = 2.605311e+03	FES = 75100		
= 0.077	Diversity = 1.2869e+01	k_t = [1, 1, 1, 1]	μ_F = 0.260	μ_{CR}
Generation 760:	Best = 2.605005e+03	FES = 76100		
= 0.091	Diversity = 1.2869e+01	k_t = [4, 2, 4, 3]	μ_F = 0.184	μ_{CR}
Generation 770:	Best = 2.605005e+03	FES = 77100		
= 0.088	Diversity = 1.2851e+01	k_t = [1, 1, 1, 1]	μ_F = 0.269	μ_{CR}
Generation 780:	Best = 2.604995e+03	FES = 78100		
= 0.086	Diversity = 1.2839e+01	k_t = [2, 3, 3, 3]	μ_F = 0.235	μ_{CR}
Generation 790:	Best = 2.604980e+03	FES = 79100		
= 0.092	Diversity = 1.2837e+01	k_t = [1, 1, 1, 1]	μ_F = 0.175	μ_{CR}
Generation 800:	Best = 2.604954e+03	FES = 80100		
= 0.105	Diversity = 1.2830e+01	k_t = [2, 2, 4, 2]	μ_F = 0.210	μ_{CR}
Generation 810:	Best = 2.604947e+03	FES = 81100		
= 0.090	Diversity = 1.2809e+01	k_t = [1, 1, 1, 1]	μ_F = 0.208	μ_{CR}
Generation 820:	Best = 2.604787e+03	FES = 82100		
= 0.090	Diversity = 1.2664e+01	k_t = [4, 4, 4, 4]	μ_F = 0.260	μ_{CR}
Generation 830:	Best = 2.604787e+03	FES = 83100		
= 0.110	Diversity = 1.2664e+01	k_t = [1, 1, 1, 1]	μ_F = 0.233	μ_{CR}
Generation 840:	Best = 2.604660e+03	FES = 84100		
= 0.107	Diversity = 1.2675e+01	k_t = [4, 3, 4, 3]	μ_F = 0.197	μ_{CR}
Generation 850:	Best = 2.604660e+03	FES = 85100		
= 0.097	Diversity = 1.2673e+01	k_t = [1, 1, 1, 1]	μ_F = 0.145	μ_{CR}
Generation 860:	Best = 2.604660e+03	FES = 86100		
= 0.101	Diversity = 1.2661e+01	k_t = [3, 2, 3, 2]	μ_F = 0.123	μ_{CR}
Generation 870:	Best = 2.604660e+03	FES = 87100		
= 0.102	Diversity = 1.2661e+01	k_t = [1, 1, 1, 1]	μ_F = 0.119	μ_{CR}
Generation 880:	Best = 2.604660e+03	FES = 88100		
= 0.110	Diversity = 1.2649e+01	k_t = [3, 2, 4, 2]	μ_F = 0.130	μ_{CR}
Generation 890:	Best = 2.604609e+03	FES = 89100		
= 0.107	Diversity = 1.2643e+01	k_t = [1, 1, 1, 1]	μ_F = 0.179	μ_{CR}
Generation 900:	Best = 2.604609e+03	FES = 90100		
= 0.096	Diversity = 1.2599e+01	k_t = [3, 2, 2, 4]	μ_F = 0.221	μ_{CR}
Generation 910:	Best = 2.604609e+03	FES = 91100		
= 0.104	Diversity = 1.2600e+01	k_t = [1, 1, 1, 1]	μ_F = 0.224	μ_{CR}
Generation 920:	Best = 2.604609e+03	FES = 92100		
= 0.109	Diversity = 1.2488e+01	k_t = [4, 2, 3, 3]	μ_F = 0.199	μ_{CR}
Generation 930:	Best = 2.604596e+03	FES = 93100		
= 0.097	Diversity = 1.2489e+01	k_t = [1, 1, 1, 1]	μ_F = 0.254	μ_{CR}
Generation 940:	Best = 2.604481e+03	FES = 94100		
= 0.117	Diversity = 1.2267e+01	k_t = [3, 3, 3, 4]		


```
Generation 950: Best = 2.604450e+03 FES = 95100  $\mu_F$  = 0.264  $\mu_{CR}$ 
= 0.092 Diversity = 1.2267e+01  $k_t$  = [1, 1, 1, 1]
Generation 960: Best = 2.604450e+03 FES = 96100  $\mu_F$  = 0.381  $\mu_{CR}$ 
= 0.081 Diversity = 1.2065e+01  $k_t$  = [2, 2, 2, 4]
Generation 970: Best = 2.604450e+03 FES = 97100  $\mu_F$  = 0.447  $\mu_{CR}$ 
= 0.090 Diversity = 1.2064e+01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 2.604450e+03 FES = 98100  $\mu_F$  = 0.371  $\mu_{CR}$ 
= 0.078 Diversity = 1.2061e+01  $k_t$  = [3, 2, 4, 2]
Generation 990: Best = 2.604392e+03 FES = 99100  $\mu_F$  = 0.313  $\mu_{CR}$ 
= 0.072 Diversity = 1.2062e+01  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 2.604392e+03

Total Function Evals = 100000

Total Successful Updates = 4019

```
=====
=====
```

Results:

Best Fitness : 2.604392e+03

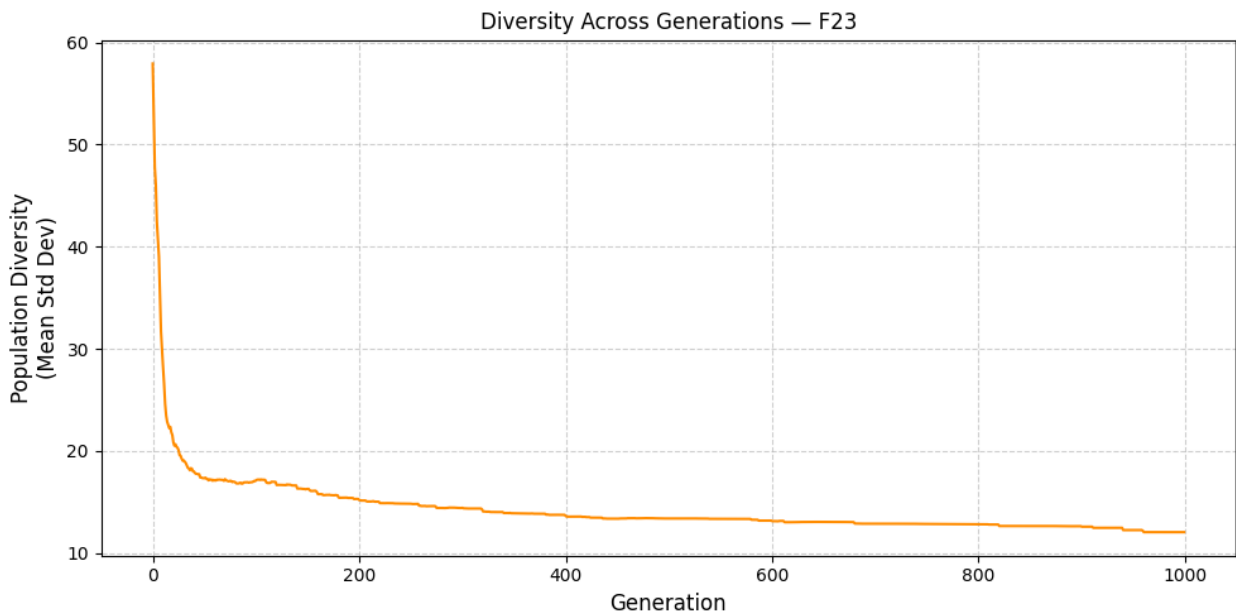
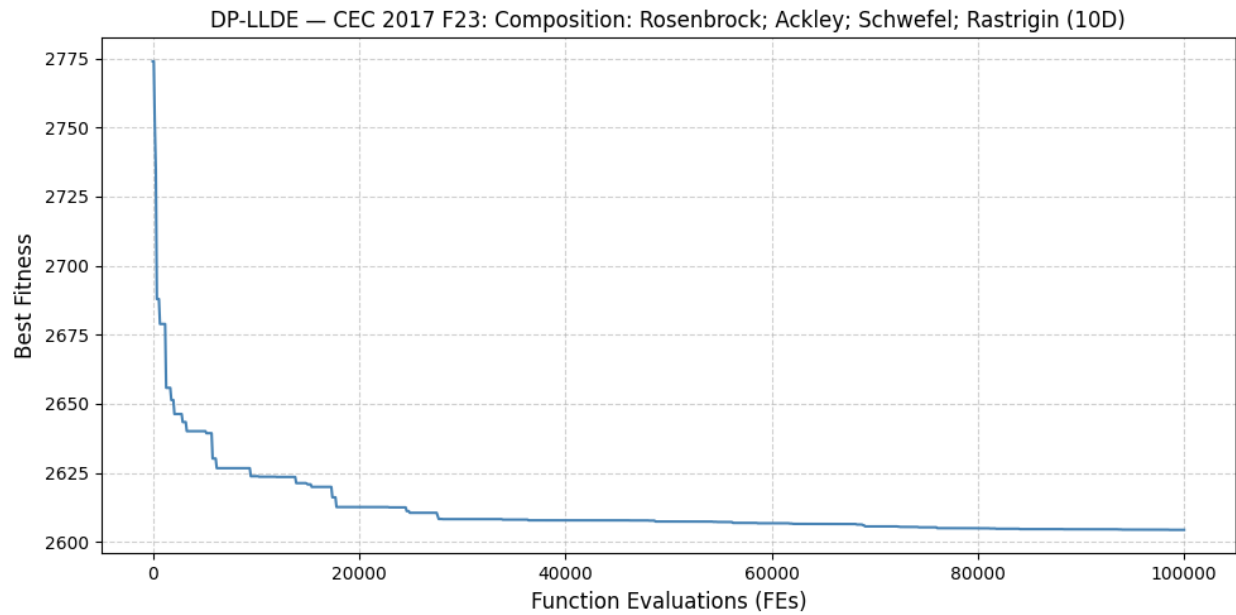
Global Optimum : 2.300000e+03

Error (|best-optimum|) : 3.043923e+02

Best Solution (5 dim) : [-93.24397048 -69.6742498 -62.04986359
70.62559985 -61.6711932]

D^0 (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F23.png

```
# =====
# F24 – Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# =====
run_and_plot('F24', make_wrapper(composition.f24), 2400.0,
'Composition: Ackley; Elliptic; Griewank; Rastrigin')
```

DP-LLDE on CEC 2017 F24 – Single Run

=====

Function : Composition: Ackley; Elliptic; Griewank; Rastrigin
(10D)

Bounds : [-100, 100]

Global optimum : 2400.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.997998e+03

Generation 10: Best = 2.812275e+03 FES = 1100 $\mu_F = 0.591$ μ_{CR}
= 0.349 Diversity = 2.4795e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 2.778736e+03 FES = 2100 $\mu_F = 0.638$ μ_{CR}
= 0.340 Diversity = 1.8120e+01 $k_t = [3, 4, 2, 3]$

Generation 30: Best = 2.776240e+03 FES = 3100 $\mu_F = 0.678$ μ_{CR}
= 0.330 Diversity = 1.5938e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 2.764085e+03 FES = 4100 $\mu_F = 0.715$ μ_{CR}
= 0.287 Diversity = 1.4678e+01 $k_t = [3, 3, 2, 4]$

Generation 50: Best = 2.756150e+03 FES = 5100 $\mu_F = 0.733$ μ_{CR}
= 0.244 Diversity = 1.4519e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 2.754124e+03 FES = 6100 $\mu_F = 0.753$ μ_{CR}
= 0.237 Diversity = 1.4212e+01 $k_t = [3, 3, 2, 4]$

Generation 70: Best = 2.750449e+03 FES = 7100 $\mu_F = 0.672$ μ_{CR}
= 0.183 Diversity = 1.4118e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.745708e+03 FES = 8100 $\mu_F = 0.654$ μ_{CR}
= 0.145 Diversity = 1.3618e+01 $k_t = [3, 2, 3, 3]$

Generation 90: Best = 2.743445e+03 FES = 9100 $\mu_F = 0.622$ μ_{CR}
= 0.114 Diversity = 1.3227e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.743445e+03 FES = 10100 $\mu_F = 0.627$ μ_{CR}
= 0.095 Diversity = 1.3046e+01 $k_t = [4, 4, 4, 4]$

Generation 110: Best = 2.741573e+03 FES = 11100 $\mu_F = 0.640$ μ_{CR}
= 0.099 Diversity = 1.2808e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.741573e+03 FES = 12100 $\mu_F = 0.626$ μ_{CR}
= 0.098 Diversity = 1.2753e+01 $k_t = [2, 2, 4, 3]$

Generation 130: Best = 2.741573e+03 FES = 13100 $\mu_F = 0.566$ μ_{CR}
= 0.112 Diversity = 1.2612e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.739785e+03 FES = 14100 $\mu_F = 0.588$ μ_{CR}
= 0.110 Diversity = 1.2283e+01 $k_t = [4, 2, 4, 4]$

Generation 150: Best = 2.739688e+03 FES = 15100 $\mu_F = 0.644$ μ_{CR}
= 0.096 Diversity = 1.2414e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.736893e+03 FES = 16100 $\mu_F = 0.685$ μ_{CR}
= 0.093 Diversity = 1.2107e+01 $k_t = [3, 4, 3, 3]$

Generation 170: Best = 2.736893e+03 FES = 17100 $\mu_F = 0.708$ μ_{CR}
= 0.091 Diversity = 1.2148e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 2.736800e+03 FES = 18100 $\mu_F = 0.725$ μ_{CR}
= 0.094 Diversity = 1.2096e+01 $k_t = [4, 4, 4, 3]$

Generation 190: Best = 2.736600e+03 FES = 19100 $\mu_F = 0.737$ μ_{CR}
= 0.090 Diversity = 1.2017e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 2.734160e+03 FES = 20100 $\mu_F = 0.721$ μ_{CR}
= 0.086 Diversity = 1.2015e+01 $k_t = [3, 3, 3, 2]$

Generation 210: Best = 2.734160e+03 FES = 21100 $\mu_F = 0.704$ μ_{CR}
= 0.095 Diversity = 1.1815e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 2.734147e+03	FES = 22100	$\mu_F = 0.651$	μ_{CR}
= 0.098 Diversity = 1.1796e+01	$k_t = [3, 2, 3, 3]$		
Generation 230: Best = 2.734147e+03	FES = 23100	$\mu_F = 0.653$	μ_{CR}
= 0.098 Diversity = 1.1601e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 2.734147e+03	FES = 24100	$\mu_F = 0.610$	μ_{CR}
= 0.111 Diversity = 1.1590e+01	$k_t = [2, 3, 2, 2]$		
Generation 250: Best = 2.733510e+03	FES = 25100	$\mu_F = 0.511$	μ_{CR}
= 0.089 Diversity = 1.1556e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.732543e+03	FES = 26100	$\mu_F = 0.505$	μ_{CR}
= 0.076 Diversity = 1.1509e+01	$k_t = [3, 2, 4, 3]$		
Generation 270: Best = 2.732177e+03	FES = 27100	$\mu_F = 0.533$	μ_{CR}
= 0.083 Diversity = 1.1484e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 2.732177e+03	FES = 28100	$\mu_F = 0.509$	μ_{CR}
= 0.091 Diversity = 1.1427e+01	$k_t = [3, 3, 3, 3]$		
Generation 290: Best = 2.732081e+03	FES = 29100	$\mu_F = 0.485$	μ_{CR}
= 0.081 Diversity = 1.1447e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 2.731698e+03	FES = 30100	$\mu_F = 0.494$	μ_{CR}
= 0.098 Diversity = 1.1329e+01	$k_t = [2, 4, 3, 2]$		
Generation 310: Best = 2.731321e+03	FES = 31100	$\mu_F = 0.515$	μ_{CR}
= 0.097 Diversity = 1.1301e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 2.729816e+03	FES = 32100	$\mu_F = 0.571$	μ_{CR}
= 0.116 Diversity = 1.1303e+01	$k_t = [4, 3, 2, 4]$		
Generation 330: Best = 2.729799e+03	FES = 33100	$\mu_F = 0.611$	μ_{CR}
= 0.102 Diversity = 1.1310e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.729548e+03	FES = 34100	$\mu_F = 0.617$	μ_{CR}
= 0.114 Diversity = 1.1193e+01	$k_t = [4, 2, 4, 4]$		
Generation 350: Best = 2.729548e+03	FES = 35100	$\mu_F = 0.730$	μ_{CR}
= 0.118 Diversity = 1.1202e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 2.729548e+03	FES = 36100	$\mu_F = 0.656$	μ_{CR}
= 0.107 Diversity = 1.1152e+01	$k_t = [3, 3, 2, 3]$		
Generation 370: Best = 2.729548e+03	FES = 37100	$\mu_F = 0.512$	μ_{CR}
= 0.114 Diversity = 1.1076e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 2.729548e+03	FES = 38100	$\mu_F = 0.540$	μ_{CR}
= 0.116 Diversity = 1.0976e+01	$k_t = [2, 3, 3, 4]$		
Generation 390: Best = 2.729548e+03	FES = 39100	$\mu_F = 0.434$	μ_{CR}
= 0.106 Diversity = 1.0958e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 2.729326e+03	FES = 40100	$\mu_F = 0.424$	μ_{CR}
= 0.096 Diversity = 1.0726e+01	$k_t = [4, 4, 4, 4]$		
Generation 410: Best = 2.729280e+03	FES = 41100	$\mu_F = 0.385$	μ_{CR}
= 0.108 Diversity = 1.0714e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 2.729113e+03	FES = 42100	$\mu_F = 0.425$	μ_{CR}
= 0.098 Diversity = 1.0649e+01	$k_t = [4, 2, 3, 2]$		
Generation 430: Best = 2.728797e+03	FES = 43100	$\mu_F = 0.514$	μ_{CR}
= 0.108 Diversity = 1.0708e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 2.728796e+03	FES = 44100	$\mu_F = 0.474$	μ_{CR}
= 0.110 Diversity = 1.0695e+01	$k_t = [4, 2, 3, 2]$		
Generation 450: Best = 2.728796e+03	FES = 45100	$\mu_F = 0.371$	μ_{CR}
= 0.111 Diversity = 1.0686e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 2.728788e+03	FES = 46100	$\mu_F = 0.368$	μ_{CR}

= 0.114	Diversity = 1.0702e+01	k_t = [4, 4, 3, 4]	μ_F = 0.391	μ_{CR}
Generation 470:	Best = 2.728788e+03	FES = 47100		
= 0.091	Diversity = 1.0511e+01	k_t = [1, 1, 1, 1]	μ_F = 0.384	μ_{CR}
Generation 480:	Best = 2.728738e+03	FES = 48100		
= 0.087	Diversity = 1.0518e+01	k_t = [3, 4, 2, 3]	μ_F = 0.369	μ_{CR}
Generation 490:	Best = 2.728738e+03	FES = 49100		
= 0.112	Diversity = 1.0505e+01	k_t = [1, 1, 1, 1]	μ_F = 0.383	μ_{CR}
Generation 500:	Best = 2.728738e+03	FES = 50100		
= 0.117	Diversity = 1.0507e+01	k_t = [4, 3, 2, 3]	μ_F = 0.309	μ_{CR}
Generation 510:	Best = 2.728716e+03	FES = 51100		
= 0.097	Diversity = 1.0519e+01	k_t = [1, 1, 1, 1]	μ_F = 0.332	μ_{CR}
Generation 520:	Best = 2.728716e+03	FES = 52100		
= 0.118	Diversity = 1.0529e+01	k_t = [3, 3, 3, 2]	μ_F = 0.372	μ_{CR}
Generation 530:	Best = 2.728716e+03	FES = 53100		
= 0.117	Diversity = 1.0530e+01	k_t = [1, 1, 1, 1]	μ_F = 0.460	μ_{CR}
Generation 540:	Best = 2.728716e+03	FES = 54100		
= 0.090	Diversity = 1.0485e+01	k_t = [3, 3, 3, 4]	μ_F = 0.483	μ_{CR}
Generation 550:	Best = 2.728695e+03	FES = 55100		
= 0.078	Diversity = 1.0459e+01	k_t = [1, 1, 1, 1]	μ_F = 0.420	μ_{CR}
Generation 560:	Best = 2.728691e+03	FES = 56100		
= 0.084	Diversity = 1.0462e+01	k_t = [3, 2, 4, 4]	μ_F = 0.416	μ_{CR}
Generation 570:	Best = 2.728691e+03	FES = 57100		
= 0.085	Diversity = 1.0463e+01	k_t = [1, 1, 1, 1]	μ_F = 0.436	μ_{CR}
Generation 580:	Best = 2.728674e+03	FES = 58100		
= 0.086	Diversity = 1.0466e+01	k_t = [2, 4, 2, 3]	μ_F = 0.436	μ_{CR}
Generation 590:	Best = 2.728674e+03	FES = 59100		
= 0.091	Diversity = 1.0470e+01	k_t = [1, 1, 1, 1]	μ_F = 0.546	μ_{CR}
Generation 600:	Best = 2.728674e+03	FES = 60100		
= 0.101	Diversity = 1.0417e+01	k_t = [4, 3, 3, 2]	μ_F = 0.516	μ_{CR}
Generation 610:	Best = 2.728674e+03	FES = 61100		
= 0.100	Diversity = 1.0364e+01	k_t = [1, 1, 1, 1]	μ_F = 0.477	μ_{CR}
Generation 620:	Best = 2.728669e+03	FES = 62100		
= 0.084	Diversity = 1.0136e+01	k_t = [4, 3, 4, 4]	μ_F = 0.475	μ_{CR}
Generation 630:	Best = 2.728669e+03	FES = 63100		
= 0.091	Diversity = 1.0138e+01	k_t = [1, 1, 1, 1]	μ_F = 0.446	μ_{CR}
Generation 640:	Best = 2.728669e+03	FES = 64100		
= 0.078	Diversity = 1.0137e+01	k_t = [2, 4, 3, 3]	μ_F = 0.607	μ_{CR}
Generation 650:	Best = 2.728638e+03	FES = 65100		
= 0.080	Diversity = 9.8778e+00	k_t = [1, 1, 1, 1]	μ_F = 0.562	μ_{CR}
Generation 660:	Best = 2.728629e+03	FES = 66100		
= 0.096	Diversity = 9.7930e+00	k_t = [4, 2, 4, 4]	μ_F = 0.577	μ_{CR}
Generation 670:	Best = 2.728629e+03	FES = 67100		
= 0.080	Diversity = 9.6556e+00	k_t = [1, 1, 1, 1]	μ_F = 0.553	μ_{CR}
Generation 680:	Best = 2.728625e+03	FES = 68100		
= 0.068	Diversity = 9.6593e+00	k_t = [4, 4, 3, 3]	μ_F = 0.622	μ_{CR}
Generation 690:	Best = 2.728625e+03	FES = 69100		
= 0.100	Diversity = 9.4899e+00	k_t = [1, 1, 1, 1]	μ_F = 0.632	μ_{CR}
Generation 700:	Best = 2.728625e+03	FES = 70100		
= 0.082	Diversity = 9.3495e+00	k_t = [2, 3, 3, 4]		

Generation 710: Best = 2.728625e+03	FES = 71100	$\mu_F = 0.574$	μ_{CR}
= 0.130 Diversity = 9.3100e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.728625e+03	FES = 72100	$\mu_F = 0.570$	μ_{CR}
= 0.123 Diversity = 9.3080e+00	$k_t = [2, 3, 3, 2]$		
Generation 730: Best = 2.728625e+03	FES = 73100	$\mu_F = 0.465$	μ_{CR}
= 0.113 Diversity = 9.3045e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.728625e+03	FES = 74100	$\mu_F = 0.403$	μ_{CR}
= 0.128 Diversity = 9.2140e+00	$k_t = [2, 4, 2, 2]$		
Generation 750: Best = 2.728614e+03	FES = 75100	$\mu_F = 0.348$	μ_{CR}
= 0.108 Diversity = 9.2172e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.728614e+03	FES = 76100	$\mu_F = 0.283$	μ_{CR}
= 0.104 Diversity = 9.0820e+00	$k_t = [2, 3, 2, 2]$		
Generation 770: Best = 2.728614e+03	FES = 77100	$\mu_F = 0.268$	μ_{CR}
= 0.099 Diversity = 9.0851e+00	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 2.728614e+03	FES = 78100	$\mu_F = 0.300$	μ_{CR}
= 0.110 Diversity = 8.9936e+00	$k_t = [2, 4, 4, 4]$		
Generation 790: Best = 2.728614e+03	FES = 79100	$\mu_F = 0.247$	μ_{CR}
= 0.076 Diversity = 8.9788e+00	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 2.728614e+03	FES = 80100	$\mu_F = 0.279$	μ_{CR}
= 0.074 Diversity = 8.9807e+00	$k_t = [4, 2, 2, 2]$		
Generation 810: Best = 2.728614e+03	FES = 81100	$\mu_F = 0.269$	μ_{CR}
= 0.095 Diversity = 8.9933e+00	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 2.728614e+03	FES = 82100	$\mu_F = 0.294$	μ_{CR}
= 0.090 Diversity = 8.9904e+00	$k_t = [2, 4, 3, 4]$		
Generation 830: Best = 2.728614e+03	FES = 83100	$\mu_F = 0.361$	μ_{CR}
= 0.080 Diversity = 8.9595e+00	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 2.728614e+03	FES = 84100	$\mu_F = 0.346$	μ_{CR}
= 0.080 Diversity = 8.8702e+00	$k_t = [4, 4, 4, 4]$		
Generation 850: Best = 2.728614e+03	FES = 85100	$\mu_F = 0.397$	μ_{CR}
= 0.082 Diversity = 8.8682e+00	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 2.728612e+03	FES = 86100	$\mu_F = 0.442$	μ_{CR}
= 0.088 Diversity = 8.8204e+00	$k_t = [3, 2, 4, 4]$		
Generation 870: Best = 2.728612e+03	FES = 87100	$\mu_F = 0.469$	μ_{CR}
= 0.100 Diversity = 8.7466e+00	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 2.728611e+03	FES = 88100	$\mu_F = 0.368$	μ_{CR}
= 0.087 Diversity = 8.7467e+00	$k_t = [4, 2, 2, 2]$		
Generation 890: Best = 2.728611e+03	FES = 89100	$\mu_F = 0.407$	μ_{CR}
= 0.078 Diversity = 8.6679e+00	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 2.728610e+03	FES = 90100	$\mu_F = 0.347$	μ_{CR}
= 0.076 Diversity = 8.6582e+00	$k_t = [4, 4, 4, 4]$		
Generation 910: Best = 2.728610e+03	FES = 91100	$\mu_F = 0.297$	μ_{CR}
= 0.074 Diversity = 8.6609e+00	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 2.728610e+03	FES = 92100	$\mu_F = 0.203$	μ_{CR}
= 0.088 Diversity = 8.6588e+00	$k_t = [2, 3, 4, 2]$		
Generation 930: Best = 2.728607e+03	FES = 93100	$\mu_F = 0.191$	μ_{CR}
= 0.095 Diversity = 8.6559e+00	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 2.728601e+03	FES = 94100	$\mu_F = 0.219$	μ_{CR}
= 0.094 Diversity = 8.6192e+00	$k_t = [4, 3, 2, 4]$		
Generation 950: Best = 2.728601e+03	FES = 95100	$\mu_F = 0.290$	μ_{CR}

```
= 0.090 Diversity = 8.6195e+00 k_t = [1, 1, 1, 1]
Generation 960: Best = 2.728601e+03 FES = 96100  $\mu_F$  = 0.267  $\mu_{CR}$ 
= 0.143 Diversity = 8.4746e+00 k_t = [3, 4, 3, 4]
Generation 970: Best = 2.728599e+03 FES = 97100  $\mu_F$  = 0.230  $\mu_{CR}$ 
= 0.125 Diversity = 8.4757e+00 k_t = [1, 1, 1, 1]
Generation 980: Best = 2.728599e+03 FES = 98100  $\mu_F$  = 0.213  $\mu_{CR}$ 
= 0.141 Diversity = 8.4182e+00 k_t = [3, 3, 4, 4]
Generation 990: Best = 2.728599e+03 FES = 99100  $\mu_F$  = 0.321  $\mu_{CR}$ 
= 0.101 Diversity = 8.3512e+00 k_t = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 2.728598e+03

Total Function Evals = 100000

Total Successful Updates = 4254

```
=====
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```

Results:

Best Fitness : 2.728598e+03

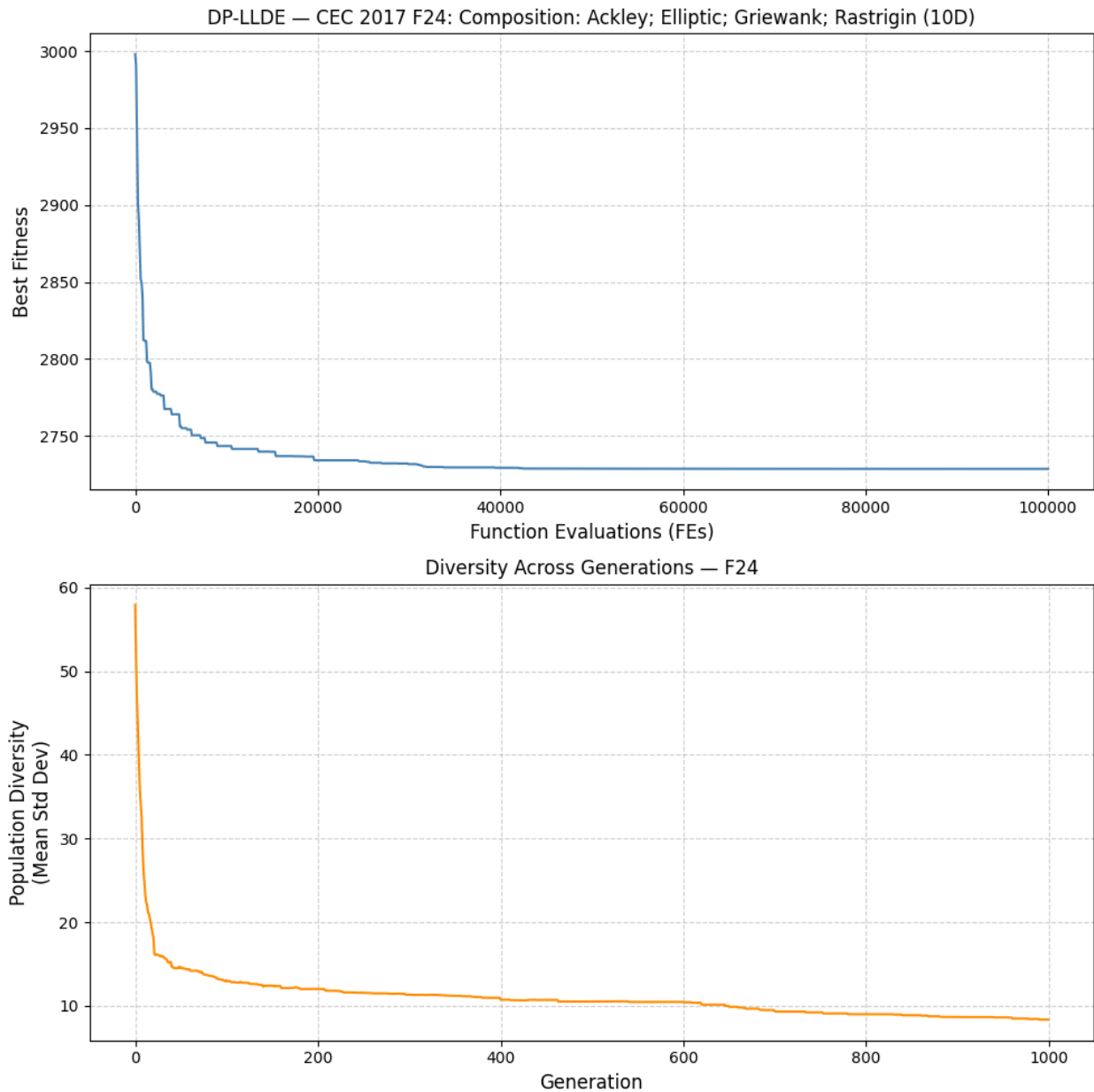
Global Optimum : 2.400000e+03

Error (|best-optimum|) : 3.285984e+02

Best Solution (5 dim) : [51.20677958 64.50697141 -65.85268521 -
75.65964795 65.16368371]

D^(0) (init. diversity): 5.7928e+01

```
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```



Graph saved as convergence_F24.png

```
# =====
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus;
# Rosenbrock)
# =====
run_and_plot('F25', make_wrapper(composition.f25), 2500.0,
'Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock')
```

DP-LLDE on CEC 2017 F25 — Single Run


```

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=====
Function      : Composition: Rastrigin; HappyCat; Ackley; Discus;
Rosenbrock (10D)
Bounds       : [-100, 100]
Global optimum : 2500.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 4.081798e+03
Generation 10: Best = 2.977029e+03 FES = 1100  $\mu_F$  = 0.572  $\mu_{CR}$ 
= 0.351 Diversity = 2.0240e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 2.948651e+03 FES = 2100  $\mu_F$  = 0.619  $\mu_{CR}$ 
= 0.356 Diversity = 6.4753e+00  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 2.945764e+03 FES = 3100  $\mu_F$  = 0.641  $\mu_{CR}$ 
= 0.364 Diversity = 2.6812e+00  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 2.944735e+03 FES = 4100  $\mu_F$  = 0.660  $\mu_{CR}$ 
= 0.378 Diversity = 1.5435e+00  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 2.944272e+03 FES = 5100  $\mu_F$  = 0.662  $\mu_{CR}$ 
= 0.368 Diversity = 1.0547e+00  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 2.943808e+03 FES = 6100  $\mu_F$  = 0.662  $\mu_{CR}$ 
= 0.346 Diversity = 8.6036e-01  $k_t$  = [2, 2, 4, 4]
Generation 70: Best = 2.943808e+03 FES = 7100  $\mu_F$  = 0.648  $\mu_{CR}$ 
= 0.350 Diversity = 8.1885e-01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 2.943808e+03 FES = 8100  $\mu_F$  = 0.613  $\mu_{CR}$ 
= 0.332 Diversity = 7.7794e-01  $k_t$  = [3, 3, 2, 4]
Generation 90: Best = 2.943808e+03 FES = 9100  $\mu_F$  = 0.649  $\mu_{CR}$ 
= 0.317 Diversity = 7.3036e-01  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 2.943808e+03 FES = 10100  $\mu_F$  = 0.652  $\mu_{CR}$ 
= 0.299 Diversity = 7.0095e-01  $k_t$  = [3, 3, 3, 4]
Generation 110: Best = 2.943808e+03 FES = 11100  $\mu_F$  = 0.628  $\mu_{CR}$ 
= 0.265 Diversity = 6.6677e-01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 2.943798e+03 FES = 12100  $\mu_F$  = 0.611  $\mu_{CR}$ 
= 0.233 Diversity = 6.4382e-01  $k_t$  = [2, 2, 3, 3]
Generation 130: Best = 2.943724e+03 FES = 13100  $\mu_F$  = 0.614  $\mu_{CR}$ 
= 0.222 Diversity = 6.4820e-01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 2.943642e+03 FES = 14100  $\mu_F$  = 0.663  $\mu_{CR}$ 
= 0.197 Diversity = 6.3132e-01  $k_t$  = [2, 3, 2, 3]
Generation 150: Best = 2.943642e+03 FES = 15100  $\mu_F$  = 0.611  $\mu_{CR}$ 
= 0.170 Diversity = 6.2523e-01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 2.943642e+03 FES = 16100  $\mu_F$  = 0.616  $\mu_{CR}$ 
= 0.142 Diversity = 6.2597e-01  $k_t$  = [3, 3, 2, 4]
Generation 170: Best = 2.943642e+03 FES = 17100  $\mu_F$  = 0.604  $\mu_{CR}$ 
= 0.119 Diversity = 6.2085e-01  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 2.943610e+03 FES = 18100  $\mu_F$  = 0.569  $\mu_{CR}$ 
= 0.118 Diversity = 6.1099e-01  $k_t$  = [3, 3, 3, 3]
Generation 190: Best = 2.943566e+03 FES = 19100  $\mu_F$  = 0.579  $\mu_{CR}$ 
= 0.100 Diversity = 6.0603e-01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 2.943552e+03 FES = 20100  $\mu_F$  = 0.612  $\mu_{CR}$ 
= 0.086 Diversity = 6.0829e-01  $k_t$  = [3, 2, 2, 3]
Generation 210: Best = 2.943539e+03 FES = 21100  $\mu_F$  = 0.631  $\mu_{CR}$ 

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= 0.091	Diversity = 6.0938e-01	k_t = [1, 1, 1, 1]	μ_F = 0.670	μ_{CR}
Generation 220:	Best = 2.943539e+03	FES = 22100		
= 0.101	Diversity = 5.9209e-01	k_t = [2, 4, 4, 4]	μ_F = 0.651	μ_{CR}
Generation 230:	Best = 2.943539e+03	FES = 23100		
= 0.094	Diversity = 5.8822e-01	k_t = [1, 1, 1, 1]	μ_F = 0.690	μ_{CR}
Generation 240:	Best = 2.943539e+03	FES = 24100		
= 0.082	Diversity = 5.8873e-01	k_t = [3, 4, 3, 2]	μ_F = 0.648	μ_{CR}
Generation 250:	Best = 2.943535e+03	FES = 25100		
= 0.091	Diversity = 5.8598e-01	k_t = [1, 1, 1, 1]	μ_F = 0.664	μ_{CR}
Generation 260:	Best = 2.943535e+03	FES = 26100		
= 0.103	Diversity = 5.8284e-01	k_t = [2, 3, 4, 3]	μ_F = 0.554	μ_{CR}
Generation 270:	Best = 2.943535e+03	FES = 27100		
= 0.095	Diversity = 5.8592e-01	k_t = [1, 1, 1, 1]	μ_F = 0.529	μ_{CR}
Generation 280:	Best = 2.943523e+03	FES = 28100		
= 0.085	Diversity = 5.8591e-01	k_t = [2, 2, 3, 3]	μ_F = 0.579	μ_{CR}
Generation 290:	Best = 2.943520e+03	FES = 29100		
= 0.069	Diversity = 5.8421e-01	k_t = [1, 1, 1, 1]	μ_F = 0.561	μ_{CR}
Generation 300:	Best = 2.943491e+03	FES = 30100		
= 0.098	Diversity = 5.8161e-01	k_t = [2, 2, 4, 2]	μ_F = 0.541	μ_{CR}
Generation 310:	Best = 2.943491e+03	FES = 31100		
= 0.102	Diversity = 5.8187e-01	k_t = [1, 1, 1, 1]	μ_F = 0.467	μ_{CR}
Generation 320:	Best = 2.943491e+03	FES = 32100		
= 0.098	Diversity = 5.8218e-01	k_t = [4, 2, 2, 3]	μ_F = 0.463	μ_{CR}
Generation 330:	Best = 2.943491e+03	FES = 33100		
= 0.109	Diversity = 5.8230e-01	k_t = [1, 1, 1, 1]	μ_F = 0.462	μ_{CR}
Generation 340:	Best = 2.943475e+03	FES = 34100		
= 0.112	Diversity = 5.8260e-01	k_t = [3, 4, 2, 3]	μ_F = 0.481	μ_{CR}
Generation 350:	Best = 2.943474e+03	FES = 35100		
= 0.094	Diversity = 5.8340e-01	k_t = [1, 1, 1, 1]	μ_F = 0.433	μ_{CR}
Generation 360:	Best = 2.943470e+03	FES = 36100		
= 0.097	Diversity = 5.8823e-01	k_t = [4, 2, 4, 4]	μ_F = 0.436	μ_{CR}
Generation 370:	Best = 2.943470e+03	FES = 37100		
= 0.088	Diversity = 5.8494e-01	k_t = [1, 1, 1, 1]	μ_F = 0.418	μ_{CR}
Generation 380:	Best = 2.943469e+03	FES = 38100		
= 0.091	Diversity = 5.8481e-01	k_t = [4, 2, 3, 2]	μ_F = 0.460	μ_{CR}
Generation 390:	Best = 2.943469e+03	FES = 39100		
= 0.085	Diversity = 5.7926e-01	k_t = [1, 1, 1, 1]	μ_F = 0.470	μ_{CR}
Generation 400:	Best = 2.943469e+03	FES = 40100		
= 0.087	Diversity = 5.7764e-01	k_t = [4, 3, 2, 2]	μ_F = 0.425	μ_{CR}
Generation 410:	Best = 2.943469e+03	FES = 41100		
= 0.091	Diversity = 5.7821e-01	k_t = [1, 1, 1, 1]	μ_F = 0.493	μ_{CR}
Generation 420:	Best = 2.943469e+03	FES = 42100		
= 0.085	Diversity = 5.7961e-01	k_t = [4, 4, 2, 4]	μ_F = 0.392	μ_{CR}
Generation 430:	Best = 2.943469e+03	FES = 43100		
= 0.076	Diversity = 5.7973e-01	k_t = [1, 1, 1, 1]	μ_F = 0.366	μ_{CR}
Generation 440:	Best = 2.943469e+03	FES = 44100		
= 0.066	Diversity = 5.7127e-01	k_t = [3, 2, 3, 4]	μ_F = 0.377	μ_{CR}
Generation 450:	Best = 2.943469e+03	FES = 45100		
= 0.078	Diversity = 5.7103e-01	k_t = [1, 1, 1, 1]		

Generation 460: Best = 2.943469e+03	FES = 46100	$\mu_F = 0.382$	μ_{CR}
= 0.074 Diversity = 5.7248e-01	$k_t = [2, 3, 3, 4]$		
Generation 470: Best = 2.943469e+03	FES = 47100	$\mu_F = 0.389$	μ_{CR}
= 0.082 Diversity = 5.6979e-01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.943469e+03	FES = 48100	$\mu_F = 0.356$	μ_{CR}
= 0.082 Diversity = 5.6985e-01	$k_t = [2, 2, 2, 2]$		
Generation 490: Best = 2.943469e+03	FES = 49100	$\mu_F = 0.298$	μ_{CR}
= 0.085 Diversity = 5.6928e-01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 2.943468e+03	FES = 50100	$\mu_F = 0.275$	μ_{CR}
= 0.098 Diversity = 5.6869e-01	$k_t = [4, 2, 3, 4]$		
Generation 510: Best = 2.943468e+03	FES = 51100	$\mu_F = 0.318$	μ_{CR}
= 0.084 Diversity = 5.6854e-01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.943468e+03	FES = 52100	$\mu_F = 0.380$	μ_{CR}
= 0.086 Diversity = 5.6855e-01	$k_t = [3, 4, 2, 4]$		
Generation 530: Best = 2.943468e+03	FES = 53100	$\mu_F = 0.363$	μ_{CR}
= 0.079 Diversity = 5.6813e-01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.943467e+03	FES = 54100	$\mu_F = 0.334$	μ_{CR}
= 0.077 Diversity = 5.6803e-01	$k_t = [2, 2, 4, 3]$		
Generation 550: Best = 2.943467e+03	FES = 55100	$\mu_F = 0.318$	μ_{CR}
= 0.079 Diversity = 5.6795e-01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.943467e+03	FES = 56100	$\mu_F = 0.284$	μ_{CR}
= 0.104 Diversity = 5.6684e-01	$k_t = [2, 3, 3, 4]$		
Generation 570: Best = 2.943467e+03	FES = 57100	$\mu_F = 0.245$	μ_{CR}
= 0.111 Diversity = 5.6707e-01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.943466e+03	FES = 58100	$\mu_F = 0.235$	μ_{CR}
= 0.099 Diversity = 5.6644e-01	$k_t = [4, 4, 2, 3]$		
Generation 590: Best = 2.943465e+03	FES = 59100	$\mu_F = 0.217$	μ_{CR}
= 0.108 Diversity = 5.6646e-01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.943465e+03	FES = 60100	$\mu_F = 0.269$	μ_{CR}
= 0.090 Diversity = 5.6505e-01	$k_t = [3, 3, 3, 4]$		
Generation 610: Best = 2.943465e+03	FES = 61100	$\mu_F = 0.405$	μ_{CR}
= 0.083 Diversity = 5.6536e-01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.943462e+03	FES = 62100	$\mu_F = 0.411$	μ_{CR}
= 0.086 Diversity = 5.6537e-01	$k_t = [4, 4, 2, 2]$		
Generation 630: Best = 2.943462e+03	FES = 63100	$\mu_F = 0.433$	μ_{CR}
= 0.078 Diversity = 5.6522e-01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.943462e+03	FES = 64100	$\mu_F = 0.451$	μ_{CR}
= 0.095 Diversity = 5.6248e-01	$k_t = [3, 4, 3, 4]$		
Generation 650: Best = 2.943461e+03	FES = 65100	$\mu_F = 0.347$	μ_{CR}
= 0.074 Diversity = 5.6069e-01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.943459e+03	FES = 66100	$\mu_F = 0.384$	μ_{CR}
= 0.121 Diversity = 5.6077e-01	$k_t = [4, 2, 4, 2]$		
Generation 670: Best = 2.943459e+03	FES = 67100	$\mu_F = 0.355$	μ_{CR}
= 0.107 Diversity = 5.6076e-01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.943459e+03	FES = 68100	$\mu_F = 0.317$	μ_{CR}
= 0.108 Diversity = 5.6067e-01	$k_t = [2, 4, 2, 2]$		
Generation 690: Best = 2.943459e+03	FES = 69100	$\mu_F = 0.251$	μ_{CR}
= 0.103 Diversity = 5.6053e-01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.943459e+03	FES = 70100	$\mu_F = 0.203$	μ_{CR}

= 0.116	Diversity = 5.6086e-01	k_t = [4, 3, 4, 3]		
Generation 710:	Best = 2.943458e+03	FES = 71100	$\mu_F = 0.213$	μ_{CR}
= 0.088	Diversity = 5.6112e-01	k_t = [1, 1, 1, 1]		
Generation 720:	Best = 2.943458e+03	FES = 72100	$\mu_F = 0.248$	μ_{CR}
= 0.101	Diversity = 5.5985e-01	k_t = [4, 4, 2, 4]		
Generation 730:	Best = 2.943458e+03	FES = 73100	$\mu_F = 0.286$	μ_{CR}
= 0.098	Diversity = 5.6017e-01	k_t = [1, 1, 1, 1]		
Generation 740:	Best = 2.943458e+03	FES = 74100	$\mu_F = 0.261$	μ_{CR}
= 0.093	Diversity = 5.6021e-01	k_t = [3, 4, 3, 3]		
Generation 750:	Best = 2.943458e+03	FES = 75100	$\mu_F = 0.320$	μ_{CR}
= 0.113	Diversity = 5.5985e-01	k_t = [1, 1, 1, 1]		
Generation 760:	Best = 2.943458e+03	FES = 76100	$\mu_F = 0.429$	μ_{CR}
= 0.118	Diversity = 5.5980e-01	k_t = [2, 2, 2, 2]		
Generation 770:	Best = 2.943458e+03	FES = 77100	$\mu_F = 0.399$	μ_{CR}
= 0.099	Diversity = 5.5978e-01	k_t = [1, 1, 1, 1]		
Generation 780:	Best = 2.943458e+03	FES = 78100	$\mu_F = 0.412$	μ_{CR}
= 0.104	Diversity = 5.5434e-01	k_t = [2, 4, 2, 4]		
Generation 790:	Best = 2.943458e+03	FES = 79100	$\mu_F = 0.419$	μ_{CR}
= 0.095	Diversity = 5.5431e-01	k_t = [1, 1, 1, 1]		
Generation 800:	Best = 2.943458e+03	FES = 80100	$\mu_F = 0.396$	μ_{CR}
= 0.126	Diversity = 5.5202e-01	k_t = [2, 4, 3, 2]		
Generation 810:	Best = 2.943458e+03	FES = 81100	$\mu_F = 0.328$	μ_{CR}
= 0.085	Diversity = 5.5192e-01	k_t = [1, 1, 1, 1]		
Generation 820:	Best = 2.943458e+03	FES = 82100	$\mu_F = 0.444$	μ_{CR}
= 0.102	Diversity = 5.4861e-01	k_t = [2, 3, 3, 4]		
Generation 830:	Best = 2.943458e+03	FES = 83100	$\mu_F = 0.548$	μ_{CR}
= 0.097	Diversity = 5.4853e-01	k_t = [1, 1, 1, 1]		
Generation 840:	Best = 2.943458e+03	FES = 84100	$\mu_F = 0.550$	μ_{CR}
= 0.107	Diversity = 5.4539e-01	k_t = [3, 4, 2, 4]		
Generation 850:	Best = 2.943458e+03	FES = 85100	$\mu_F = 0.439$	μ_{CR}
= 0.083	Diversity = 5.4549e-01	k_t = [1, 1, 1, 1]		
Generation 860:	Best = 2.943458e+03	FES = 86100	$\mu_F = 0.470$	μ_{CR}
= 0.085	Diversity = 5.4504e-01	k_t = [2, 2, 3, 4]		
Generation 870:	Best = 2.943458e+03	FES = 87100	$\mu_F = 0.430$	μ_{CR}
= 0.085	Diversity = 5.4386e-01	k_t = [1, 1, 1, 1]		
Generation 880:	Best = 2.943458e+03	FES = 88100	$\mu_F = 0.405$	μ_{CR}
= 0.094	Diversity = 5.4384e-01	k_t = [3, 2, 2, 3]		
Generation 890:	Best = 2.943458e+03	FES = 89100	$\mu_F = 0.287$	μ_{CR}
= 0.107	Diversity = 5.4373e-01	k_t = [1, 1, 1, 1]		
Generation 900:	Best = 2.943458e+03	FES = 90100	$\mu_F = 0.225$	μ_{CR}
= 0.124	Diversity = 5.4354e-01	k_t = [2, 4, 2, 3]		
Generation 910:	Best = 2.943458e+03	FES = 91100	$\mu_F = 0.239$	μ_{CR}
= 0.102	Diversity = 5.4344e-01	k_t = [1, 1, 1, 1]		
Generation 920:	Best = 2.943458e+03	FES = 92100	$\mu_F = 0.228$	μ_{CR}
= 0.081	Diversity = 5.4342e-01	k_t = [2, 3, 4, 3]		
Generation 930:	Best = 2.943458e+03	FES = 93100	$\mu_F = 0.220$	μ_{CR}
= 0.077	Diversity = 5.4272e-01	k_t = [1, 1, 1, 1]		
Generation 940:	Best = 2.943458e+03	FES = 94100	$\mu_F = 0.173$	μ_{CR}
= 0.080	Diversity = 5.4269e-01	k_t = [2, 3, 4, 3]		

```
Generation 950: Best = 2.943458e+03 FES = 95100  $\mu_F$  = 0.181  $\mu_{CR}$ 
= 0.092 Diversity = 5.4272e-01  $k_t$  = [1, 1, 1, 1]
Generation 960: Best = 2.943458e+03 FES = 96100  $\mu_F$  = 0.204  $\mu_{CR}$ 
= 0.117 Diversity = 5.4258e-01  $k_t$  = [3, 4, 4, 3]
Generation 970: Best = 2.943458e+03 FES = 97100  $\mu_F$  = 0.175  $\mu_{CR}$ 
= 0.100 Diversity = 5.4254e-01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 2.943458e+03 FES = 98100  $\mu_F$  = 0.149  $\mu_{CR}$ 
= 0.107 Diversity = 5.4219e-01  $k_t$  = [3, 4, 4, 2]
Generation 990: Best = 2.943458e+03 FES = 99100  $\mu_F$  = 0.190  $\mu_{CR}$ 
= 0.137 Diversity = 5.4231e-01  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 2.943458e+03

Total Function Evals = 100000

Total Successful Updates = 5615

```
=====
=====
```

Results:

Best Fitness : 2.943458e+03

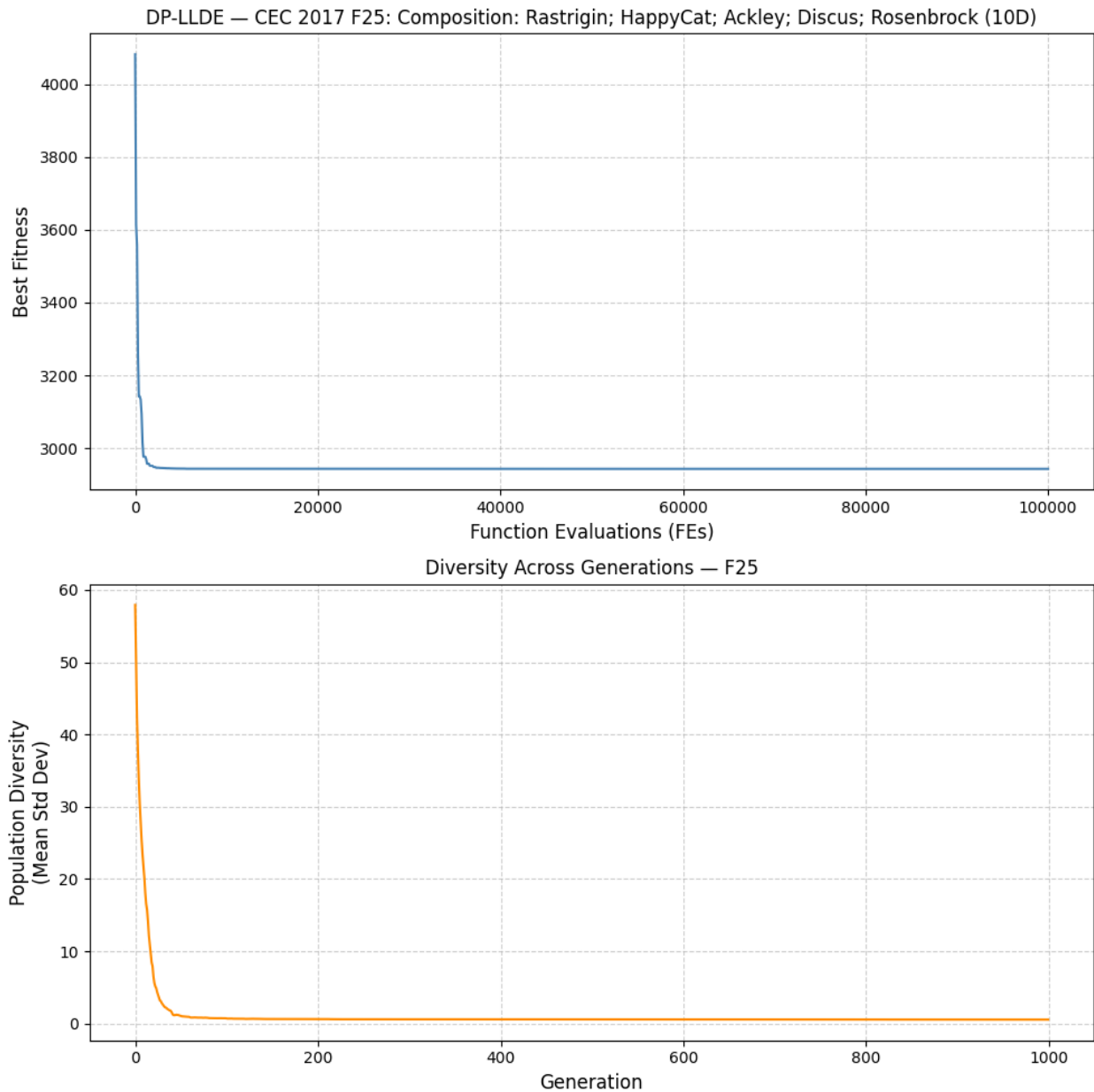
Global Optimum : 2.500000e+03

Error (|best-optimum|) : 4.434576e+02

Best Solution (5 dim) : [13.97500788 72.46300517 -46.7690107 -
17.80435741 45.82463612]

D^(0) (init. diversity): 5.7928e+01

```
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```



Graph saved as convergence_F25.png

```
# =====
# F26 – Composition Function 6 (Schaffer; Schwefel; Griewank;
# Rastrigin)
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0,
'Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin')
```

DP-LLDE on CEC 2017 F26 – Single Run

```

=====
Function      : Composition: Schaffer; Schwefel; Griewank;
Rosenbrock; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 2600.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 4.061684e+03
Generation 10: Best = 3.397413e+03 FES = 1100  $\mu_F$  = 0.571  $\mu_{CR}$ 
= 0.350 Diversity = 3.2254e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 3.081676e+03 FES = 2100  $\mu_F$  = 0.631  $\mu_{CR}$ 
= 0.352 Diversity = 1.0918e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 2.967309e+03 FES = 3100  $\mu_F$  = 0.671  $\mu_{CR}$ 
= 0.354 Diversity = 6.8081e+00  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 2.914891e+03 FES = 4100  $\mu_F$  = 0.700  $\mu_{CR}$ 
= 0.359 Diversity = 3.8712e+00  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 2.908524e+03 FES = 5100  $\mu_F$  = 0.722  $\mu_{CR}$ 
= 0.356 Diversity = 1.5815e+00  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 2.903552e+03 FES = 6100  $\mu_F$  = 0.716  $\mu_{CR}$ 
= 0.355 Diversity = 7.3544e-01  $k_t$  = [2, 2, 4, 4]
Generation 70: Best = 2.901291e+03 FES = 7100  $\mu_F$  = 0.709  $\mu_{CR}$ 
= 0.354 Diversity = 3.8407e-01  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 2.900568e+03 FES = 8100  $\mu_F$  = 0.707  $\mu_{CR}$ 
= 0.350 Diversity = 1.7555e-01  $k_t$  = [3, 3, 2, 4]
Generation 90: Best = 2.900253e+03 FES = 9100  $\mu_F$  = 0.712  $\mu_{CR}$ 
= 0.356 Diversity = 7.7956e-02  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 2.900098e+03 FES = 10100  $\mu_F$  = 0.713  $\mu_{CR}$ 
= 0.350 Diversity = 2.9880e-02  $k_t$  = [2, 3, 4, 4]
Generation 110: Best = 2.900042e+03 FES = 11100  $\mu_F$  = 0.714  $\mu_{CR}$ 
= 0.347 Diversity = 1.3347e-02  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 2.900018e+03 FES = 12100  $\mu_F$  = 0.727  $\mu_{CR}$ 
= 0.348 Diversity = 6.0660e-03  $k_t$  = [2, 2, 3, 2]
Generation 130: Best = 2.900011e+03 FES = 13100  $\mu_F$  = 0.718  $\mu_{CR}$ 
= 0.348 Diversity = 3.1363e-03  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 2.900005e+03 FES = 14100  $\mu_F$  = 0.717  $\mu_{CR}$ 
= 0.349 Diversity = 1.3797e-03  $k_t$  = [3, 2, 2, 4]
Generation 150: Best = 2.900003e+03 FES = 15100  $\mu_F$  = 0.711  $\mu_{CR}$ 
= 0.343 Diversity = 6.9791e-04  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 2.900001e+03 FES = 16100  $\mu_F$  = 0.692  $\mu_{CR}$ 
= 0.336 Diversity = 3.2014e-04  $k_t$  = [2, 2, 2, 3]
Generation 170: Best = 2.900001e+03 FES = 17100  $\mu_F$  = 0.688  $\mu_{CR}$ 
= 0.328 Diversity = 1.6185e-04  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 2.900000e+03 FES = 18100  $\mu_F$  = 0.693  $\mu_{CR}$ 
= 0.326 Diversity = 6.6044e-05  $k_t$  = [2, 2, 4, 3]
Generation 190: Best = 2.900000e+03 FES = 19100  $\mu_F$  = 0.698  $\mu_{CR}$ 
= 0.321 Diversity = 3.0540e-05  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 2.900000e+03 FES = 20100  $\mu_F$  = 0.705  $\mu_{CR}$ 
= 0.317 Diversity = 1.1676e-05  $k_t$  = [3, 2, 3, 4]
Generation 210: Best = 2.900000e+03 FES = 21100  $\mu_F$  = 0.717  $\mu_{CR}$ 

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= 0.312	Diversity = 5.3027e-06	k_t = [1, 1, 1, 1]	μ_F = 0.720	μ_{CR}
Generation 220:	Best = 2.900000e+03	FES = 22100		
= 0.311	Diversity = 2.2228e-06	k_t = [2, 2, 4, 2]	μ_F = 0.708	μ_{CR}
Generation 230:	Best = 2.900000e+03	FES = 23100		
= 0.316	Diversity = 1.1204e-06	k_t = [1, 1, 1, 1]	μ_F = 0.713	μ_{CR}
Generation 240:	Best = 2.900000e+03	FES = 24100		
= 0.311	Diversity = 4.8829e-07	k_t = [3, 3, 3, 3]	μ_F = 0.712	μ_{CR}
Generation 250:	Best = 2.900000e+03	FES = 25100		
= 0.301	Diversity = 2.4142e-07	k_t = [1, 1, 1, 1]	μ_F = 0.711	μ_{CR}
Generation 260:	Best = 2.900000e+03	FES = 26100		
= 0.300	Diversity = 1.0395e-07	k_t = [4, 3, 4, 4]	μ_F = 0.710	μ_{CR}
Generation 270:	Best = 2.900000e+03	FES = 27100		
= 0.299	Diversity = 4.5350e-08	k_t = [1, 1, 1, 1]	μ_F = 0.708	μ_{CR}
Generation 280:	Best = 2.900000e+03	FES = 28100		
= 0.299	Diversity = 1.9368e-08	k_t = [4, 2, 2, 3]	μ_F = 0.704	μ_{CR}
Generation 290:	Best = 2.900000e+03	FES = 29100		
= 0.298	Diversity = 9.3569e-09	k_t = [1, 1, 1, 1]	μ_F = 0.703	μ_{CR}
Generation 300:	Best = 2.900000e+03	FES = 30100		
= 0.304	Diversity = 4.1086e-09	k_t = [4, 2, 3, 2]	μ_F = 0.703	μ_{CR}
Generation 310:	Best = 2.900000e+03	FES = 31100		
= 0.299	Diversity = 1.9897e-09	k_t = [1, 1, 1, 1]	μ_F = 0.709	μ_{CR}
Generation 320:	Best = 2.900000e+03	FES = 32100		
= 0.297	Diversity = 8.2462e-10	k_t = [3, 3, 4, 4]	μ_F = 0.721	μ_{CR}
Generation 330:	Best = 2.900000e+03	FES = 33100		
= 0.286	Diversity = 4.0563e-10	k_t = [1, 1, 1, 1]	μ_F = 0.720	μ_{CR}
Generation 340:	Best = 2.900000e+03	FES = 34100		
= 0.288	Diversity = 1.6525e-10	k_t = [2, 2, 4, 4]	μ_F = 0.721	μ_{CR}
Generation 350:	Best = 2.900000e+03	FES = 35100		
= 0.293	Diversity = 7.2129e-11	k_t = [1, 1, 1, 1]	μ_F = 0.721	μ_{CR}
Generation 360:	Best = 2.900000e+03	FES = 36100		
= 0.293	Diversity = 3.2841e-11	k_t = [3, 3, 4, 2]	μ_F = 0.714	μ_{CR}
Generation 370:	Best = 2.900000e+03	FES = 37100		
= 0.289	Diversity = 1.6176e-11	k_t = [1, 1, 1, 1]	μ_F = 0.716	μ_{CR}
Generation 380:	Best = 2.900000e+03	FES = 38100		
= 0.280	Diversity = 6.9181e-12	k_t = [3, 2, 3, 3]	μ_F = 0.718	μ_{CR}
Generation 390:	Best = 2.900000e+03	FES = 39100		
= 0.278	Diversity = 3.4484e-12	k_t = [1, 1, 1, 1]	μ_F = 0.719	μ_{CR}
Generation 400:	Best = 2.900000e+03	FES = 40100		
= 0.279	Diversity = 1.6374e-12	k_t = [4, 2, 4, 2]	μ_F = 0.716	μ_{CR}
Generation 410:	Best = 2.900000e+03	FES = 41100		
= 0.280	Diversity = 7.7208e-13	k_t = [1, 1, 1, 1]	μ_F = 0.715	μ_{CR}
Generation 420:	Best = 2.900000e+03	FES = 42100		
= 0.272	Diversity = 3.7657e-13	k_t = [3, 4, 2, 3]	μ_F = 0.728	μ_{CR}
Generation 430:	Best = 2.900000e+03	FES = 43100		
= 0.266	Diversity = 2.2284e-13	k_t = [1, 1, 1, 1]	μ_F = 0.746	μ_{CR}
Generation 440:	Best = 2.900000e+03	FES = 44100		
= 0.259	Diversity = 1.3666e-13	k_t = [2, 3, 4, 4]	μ_F = 0.757	μ_{CR}
Generation 450:	Best = 2.900000e+03	FES = 45100		
= 0.259	Diversity = 1.1808e-13	k_t = [1, 1, 1, 1]		

Generation 460: Best = 2.900000e+03	FES = 46100	$\mu_F = 0.770$	μ_{CR}
= 0.245 Diversity = 9.5782e-14	$k_t = [3, 2, 2, 2]$		
Generation 470: Best = 2.900000e+03	FES = 47100	$\mu_F = 0.774$	μ_{CR}
= 0.236 Diversity = 9.5200e-14	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.900000e+03	FES = 48100	$\mu_F = 0.784$	μ_{CR}
= 0.233 Diversity = 8.5550e-14	$k_t = [4, 4, 3, 3]$		
Generation 490: Best = 2.900000e+03	FES = 49100	$\mu_F = 0.795$	μ_{CR}
= 0.230 Diversity = 8.3884e-14	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 2.900000e+03	FES = 50100	$\mu_F = 0.805$	μ_{CR}
= 0.223 Diversity = 7.5482e-14	$k_t = [2, 3, 3, 4]$		
Generation 510: Best = 2.900000e+03	FES = 51100	$\mu_F = 0.808$	μ_{CR}
= 0.218 Diversity = 7.4670e-14	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.900000e+03	FES = 52100	$\mu_F = 0.818$	μ_{CR}
= 0.214 Diversity = 7.0938e-14	$k_t = [3, 4, 3, 3]$		
Generation 530: Best = 2.900000e+03	FES = 53100	$\mu_F = 0.809$	μ_{CR}
= 0.209 Diversity = 7.1552e-14	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.900000e+03	FES = 54100	$\mu_F = 0.799$	μ_{CR}
= 0.201 Diversity = 6.9052e-14	$k_t = [4, 4, 3, 2]$		
Generation 550: Best = 2.900000e+03	FES = 55100	$\mu_F = 0.792$	μ_{CR}
= 0.193 Diversity = 6.9174e-14	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.900000e+03	FES = 56100	$\mu_F = 0.788$	μ_{CR}
= 0.183 Diversity = 7.2356e-14	$k_t = [4, 4, 2, 2]$		
Generation 570: Best = 2.900000e+03	FES = 57100	$\mu_F = 0.789$	μ_{CR}
= 0.184 Diversity = 6.8405e-14	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.900000e+03	FES = 58100	$\mu_F = 0.794$	μ_{CR}
= 0.183 Diversity = 7.0740e-14	$k_t = [4, 4, 4, 3]$		
Generation 590: Best = 2.900000e+03	FES = 59100	$\mu_F = 0.787$	μ_{CR}
= 0.180 Diversity = 7.6147e-14	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.900000e+03	FES = 60100	$\mu_F = 0.797$	μ_{CR}
= 0.181 Diversity = 7.1989e-14	$k_t = [4, 2, 2, 3]$		
Generation 610: Best = 2.900000e+03	FES = 61100	$\mu_F = 0.801$	μ_{CR}
= 0.178 Diversity = 7.0132e-14	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.900000e+03	FES = 62100	$\mu_F = 0.800$	μ_{CR}
= 0.179 Diversity = 6.8961e-14	$k_t = [2, 4, 4, 2]$		
Generation 630: Best = 2.900000e+03	FES = 63100	$\mu_F = 0.807$	μ_{CR}
= 0.178 Diversity = 6.7372e-14	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.900000e+03	FES = 64100	$\mu_F = 0.809$	μ_{CR}
= 0.176 Diversity = 6.7174e-14	$k_t = [2, 4, 4, 3]$		
Generation 650: Best = 2.900000e+03	FES = 65100	$\mu_F = 0.810$	μ_{CR}
= 0.177 Diversity = 6.8734e-14	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.900000e+03	FES = 66100	$\mu_F = 0.804$	μ_{CR}
= 0.174 Diversity = 6.4480e-14	$k_t = [3, 3, 2, 3]$		
Generation 670: Best = 2.900000e+03	FES = 67100	$\mu_F = 0.806$	μ_{CR}
= 0.167 Diversity = 6.3682e-14	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.900000e+03	FES = 68100	$\mu_F = 0.798$	μ_{CR}
= 0.165 Diversity = 6.1674e-14	$k_t = [3, 4, 4, 3]$		
Generation 690: Best = 2.900000e+03	FES = 69100	$\mu_F = 0.807$	μ_{CR}
= 0.171 Diversity = 6.2364e-14	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.900000e+03	FES = 70100	$\mu_F = 0.804$	μ_{CR}

= 0.173	Diversity = 6.4207e-14	k_t = [2, 2, 4, 2]	μ_F = 0.808	μ_{CR}
Generation 710:	Best = 2.900000e+03	FES = 71100		
= 0.168	Diversity = 6.4977e-14	k_t = [1, 1, 1, 1]	μ_F = 0.806	μ_{CR}
Generation 720:	Best = 2.900000e+03	FES = 72100		
= 0.170	Diversity = 6.7064e-14	k_t = [3, 2, 4, 2]	μ_F = 0.815	μ_{CR}
Generation 730:	Best = 2.900000e+03	FES = 73100		
= 0.165	Diversity = 6.9337e-14	k_t = [1, 1, 1, 1]	μ_F = 0.816	μ_{CR}
Generation 740:	Best = 2.900000e+03	FES = 74100		
= 0.164	Diversity = 6.5810e-14	k_t = [4, 4, 3, 4]	μ_F = 0.817	μ_{CR}
Generation 750:	Best = 2.900000e+03	FES = 75100		
= 0.165	Diversity = 6.8435e-14	k_t = [1, 1, 1, 1]	μ_F = 0.814	μ_{CR}
Generation 760:	Best = 2.900000e+03	FES = 76100		
= 0.166	Diversity = 6.7689e-14	k_t = [4, 2, 2, 3]	μ_F = 0.819	μ_{CR}
Generation 770:	Best = 2.900000e+03	FES = 77100		
= 0.168	Diversity = 6.9083e-14	k_t = [1, 1, 1, 1]	μ_F = 0.822	μ_{CR}
Generation 780:	Best = 2.900000e+03	FES = 78100		
= 0.170	Diversity = 7.0427e-14	k_t = [3, 2, 4, 2]	μ_F = 0.815	μ_{CR}
Generation 790:	Best = 2.900000e+03	FES = 79100		
= 0.172	Diversity = 6.7434e-14	k_t = [1, 1, 1, 1]	μ_F = 0.805	μ_{CR}
Generation 800:	Best = 2.900000e+03	FES = 80100		
= 0.170	Diversity = 6.9242e-14	k_t = [4, 2, 4, 3]	μ_F = 0.803	μ_{CR}
Generation 810:	Best = 2.900000e+03	FES = 81100		
= 0.174	Diversity = 6.7482e-14	k_t = [1, 1, 1, 1]	μ_F = 0.808	μ_{CR}
Generation 820:	Best = 2.900000e+03	FES = 82100		
= 0.176	Diversity = 6.9417e-14	k_t = [2, 4, 4, 2]	μ_F = 0.819	μ_{CR}
Generation 830:	Best = 2.900000e+03	FES = 83100		
= 0.171	Diversity = 6.8383e-14	k_t = [1, 1, 1, 1]	μ_F = 0.817	μ_{CR}
Generation 840:	Best = 2.900000e+03	FES = 84100		
= 0.174	Diversity = 6.8828e-14	k_t = [3, 2, 4, 3]	μ_F = 0.818	μ_{CR}
Generation 850:	Best = 2.900000e+03	FES = 85100		
= 0.176	Diversity = 7.0604e-14	k_t = [1, 1, 1, 1]	μ_F = 0.809	μ_{CR}
Generation 860:	Best = 2.900000e+03	FES = 86100		
= 0.177	Diversity = 6.5741e-14	k_t = [4, 3, 4, 3]	μ_F = 0.810	μ_{CR}
Generation 870:	Best = 2.900000e+03	FES = 87100		
= 0.171	Diversity = 6.5813e-14	k_t = [1, 1, 1, 1]	μ_F = 0.809	μ_{CR}
Generation 880:	Best = 2.900000e+03	FES = 88100		
= 0.170	Diversity = 6.7745e-14	k_t = [4, 4, 3, 2]	μ_F = 0.800	μ_{CR}
Generation 890:	Best = 2.900000e+03	FES = 89100		
= 0.169	Diversity = 6.2850e-14	k_t = [1, 1, 1, 1]	μ_F = 0.797	μ_{CR}
Generation 900:	Best = 2.900000e+03	FES = 90100		
= 0.168	Diversity = 6.7609e-14	k_t = [4, 4, 3, 3]	μ_F = 0.793	μ_{CR}
Generation 910:	Best = 2.900000e+03	FES = 91100		
= 0.166	Diversity = 6.8929e-14	k_t = [1, 1, 1, 1]	μ_F = 0.792	μ_{CR}
Generation 920:	Best = 2.900000e+03	FES = 92100		
= 0.170	Diversity = 7.3754e-14	k_t = [2, 4, 3, 3]	μ_F = 0.784	μ_{CR}
Generation 930:	Best = 2.900000e+03	FES = 93100		
= 0.172	Diversity = 6.8918e-14	k_t = [1, 1, 1, 1]	μ_F = 0.783	μ_{CR}
Generation 940:	Best = 2.900000e+03	FES = 94100		
= 0.170	Diversity = 7.0494e-14	k_t = [3, 2, 2, 4]		

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Generation 950: Best = 2.900000e+03 FES = 95100  $\mu_F$  = 0.784  $\mu_{CR}$ 
= 0.178 Diversity = 6.3408e-14  $k_t$  = [1, 1, 1, 1]
Generation 960: Best = 2.900000e+03 FES = 96100  $\mu_F$  = 0.789  $\mu_{CR}$ 
= 0.176 Diversity = 6.5905e-14  $k_t$  = [2, 2, 4, 2]
Generation 970: Best = 2.900000e+03 FES = 97100  $\mu_F$  = 0.785  $\mu_{CR}$ 
= 0.172 Diversity = 6.4220e-14  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 2.900000e+03 FES = 98100  $\mu_F$  = 0.783  $\mu_{CR}$ 
= 0.171 Diversity = 5.8062e-14  $k_t$  = [2, 4, 4, 3]
Generation 990: Best = 2.900000e+03 FES = 99100  $\mu_F$  = 0.791  $\mu_{CR}$ 
= 0.174 Diversity = 6.3882e-14  $k_t$  = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 2.900000e+03

Total Function Evals = 100000

Total Successful Updates = 54061

```

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Results:

Best Fitness : 2.900000e+03

Global Optimum : 2.600000e+03

Error (|best-optimum|) : 3.000000e+02

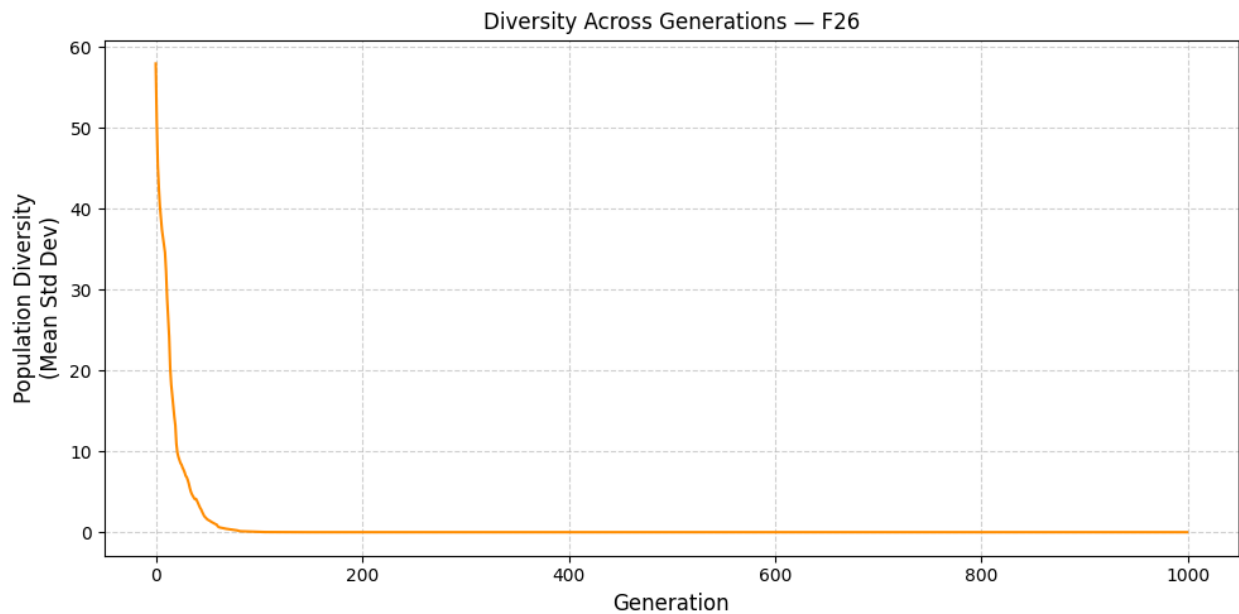
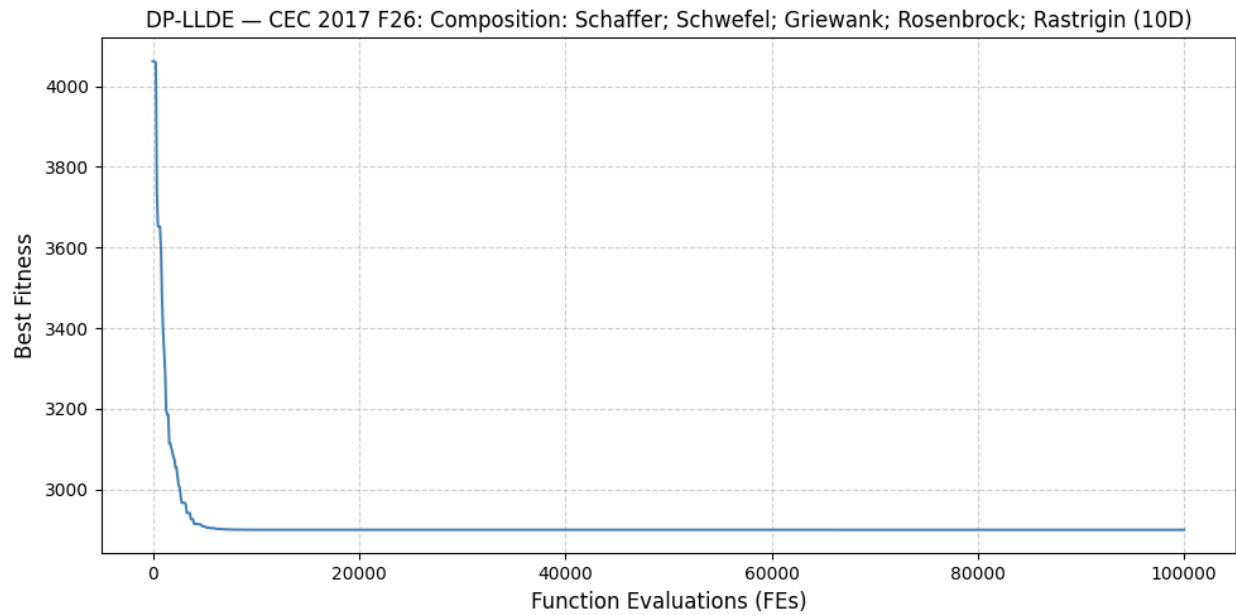
Best Solution (5 dim) : [24.746414 -25.74629564 74.04615394 -
62.35027147 35.80221832]

$D^0(0)$ (init. diversity): 5.7928e+01

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Graph saved as convergence_F26.png

```
# =====  
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent  
# Cigar; Elliptic; Schaffer)  
# =====  
run_and_plot('F27', make_wrapper(composition.f27), 2700.0,  
'Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;  
Schaffer')
```

```
=====
```

DP-LLDE on CEC 2017 F27 – Single Run

```

=====
Function      : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar;
Elliptic; Schaffer (10D)
Bounds       : [-100, 100]
Global optimum : 2700.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 3.284356e+03
Generation 10: Best = 3.124749e+03 FES = 1100  $\mu_F$  = 0.588  $\mu_{CR}$ 
= 0.349 Diversity = 4.3346e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 3.102940e+03 FES = 2100  $\mu_F$  = 0.646  $\mu_{CR}$ 
= 0.354 Diversity = 1.5493e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 3.096732e+03 FES = 3100  $\mu_F$  = 0.685  $\mu_{CR}$ 
= 0.346 Diversity = 9.3682e+00  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 3.095745e+03 FES = 4100  $\mu_F$  = 0.727  $\mu_{CR}$ 
= 0.345 Diversity = 7.4525e+00  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 3.094239e+03 FES = 5100  $\mu_F$  = 0.726  $\mu_{CR}$ 
= 0.328 Diversity = 7.1983e+00  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 3.092968e+03 FES = 6100  $\mu_F$  = 0.751  $\mu_{CR}$ 
= 0.291 Diversity = 6.7787e+00  $k_t$  = [2, 4, 2, 2]
Generation 70: Best = 3.091909e+03 FES = 7100  $\mu_F$  = 0.772  $\mu_{CR}$ 
= 0.270 Diversity = 6.3731e+00  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 3.091909e+03 FES = 8100  $\mu_F$  = 0.804  $\mu_{CR}$ 
= 0.246 Diversity = 5.7879e+00  $k_t$  = [3, 2, 3, 2]
Generation 90: Best = 3.091316e+03 FES = 9100  $\mu_F$  = 0.811  $\mu_{CR}$ 
= 0.227 Diversity = 4.7832e+00  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 3.090671e+03 FES = 10100  $\mu_F$  = 0.803  $\mu_{CR}$ 
= 0.216 Diversity = 3.7260e+00  $k_t$  = [2, 3, 4, 4]
Generation 110: Best = 3.090305e+03 FES = 11100  $\mu_F$  = 0.797  $\mu_{CR}$ 
= 0.193 Diversity = 3.3569e+00  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 3.090088e+03 FES = 12100  $\mu_F$  = 0.806  $\mu_{CR}$ 
= 0.167 Diversity = 3.2518e+00  $k_t$  = [2, 4, 3, 2]
Generation 130: Best = 3.089989e+03 FES = 13100  $\mu_F$  = 0.795  $\mu_{CR}$ 
= 0.157 Diversity = 3.1655e+00  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 3.089989e+03 FES = 14100  $\mu_F$  = 0.819  $\mu_{CR}$ 
= 0.149 Diversity = 2.7985e+00  $k_t$  = [2, 4, 4, 2]
Generation 150: Best = 3.089875e+03 FES = 15100  $\mu_F$  = 0.810  $\mu_{CR}$ 
= 0.138 Diversity = 2.4612e+00  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 3.089751e+03 FES = 16100  $\mu_F$  = 0.797  $\mu_{CR}$ 
= 0.135 Diversity = 1.6482e+00  $k_t$  = [4, 2, 4, 4]
Generation 170: Best = 3.089687e+03 FES = 17100  $\mu_F$  = 0.790  $\mu_{CR}$ 
= 0.136 Diversity = 1.0026e+00  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 3.089613e+03 FES = 18100  $\mu_F$  = 0.789  $\mu_{CR}$ 
= 0.132 Diversity = 5.3358e-01  $k_t$  = [4, 4, 2, 4]
Generation 190: Best = 3.089564e+03 FES = 19100  $\mu_F$  = 0.791  $\mu_{CR}$ 
= 0.139 Diversity = 4.6010e-01  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 3.089537e+03 FES = 20100  $\mu_F$  = 0.793  $\mu_{CR}$ 
= 0.137 Diversity = 2.8907e-01  $k_t$  = [3, 4, 2, 4]

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Generation 210: Best = 3.089524e+03	FES = 21100	$\mu_F = 0.787$	μ_{CR}
= 0.138 Diversity = 2.0178e-01	$k_t = [1, 1, 1, 1]$		
Generation 220: Best = 3.089522e+03	FES = 22100	$\mu_F = 0.787$	μ_{CR}
= 0.142 Diversity = 1.2421e-01	$k_t = [2, 4, 3, 4]$		
Generation 230: Best = 3.089520e+03	FES = 23100	$\mu_F = 0.771$	μ_{CR}
= 0.146 Diversity = 8.3175e-02	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 3.089519e+03	FES = 24100	$\mu_F = 0.758$	μ_{CR}
= 0.145 Diversity = 5.8062e-02	$k_t = [4, 2, 4, 2]$		
Generation 250: Best = 3.089518e+03	FES = 25100	$\mu_F = 0.763$	μ_{CR}
= 0.139 Diversity = 3.9768e-02	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 3.089518e+03	FES = 26100	$\mu_F = 0.775$	μ_{CR}
= 0.149 Diversity = 2.6603e-02	$k_t = [3, 4, 3, 3]$		
Generation 270: Best = 3.089518e+03	FES = 27100	$\mu_F = 0.779$	μ_{CR}
= 0.151 Diversity = 1.7675e-02	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 3.089518e+03	FES = 28100	$\mu_F = 0.797$	μ_{CR}
= 0.150 Diversity = 1.3370e-02	$k_t = [4, 2, 3, 2]$		
Generation 290: Best = 3.089518e+03	FES = 29100	$\mu_F = 0.795$	μ_{CR}
= 0.136 Diversity = 9.1958e-03	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 3.089518e+03	FES = 30100	$\mu_F = 0.795$	μ_{CR}
= 0.132 Diversity = 6.1186e-03	$k_t = [3, 3, 3, 3]$		
Generation 310: Best = 3.089518e+03	FES = 31100	$\mu_F = 0.786$	μ_{CR}
= 0.147 Diversity = 4.4982e-03	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 3.089518e+03	FES = 32100	$\mu_F = 0.784$	μ_{CR}
= 0.152 Diversity = 2.9838e-03	$k_t = [3, 2, 3, 3]$		
Generation 330: Best = 3.089518e+03	FES = 33100	$\mu_F = 0.782$	μ_{CR}
= 0.157 Diversity = 1.8987e-03	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 3.089518e+03	FES = 34100	$\mu_F = 0.763$	μ_{CR}
= 0.159 Diversity = 1.1388e-03	$k_t = [4, 2, 3, 4]$		
Generation 350: Best = 3.089518e+03	FES = 35100	$\mu_F = 0.753$	μ_{CR}
= 0.166 Diversity = 6.9535e-04	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 3.089518e+03	FES = 36100	$\mu_F = 0.758$	μ_{CR}
= 0.163 Diversity = 4.5364e-04	$k_t = [3, 4, 3, 4]$		
Generation 370: Best = 3.089518e+03	FES = 37100	$\mu_F = 0.746$	μ_{CR}
= 0.160 Diversity = 3.2248e-04	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 3.089518e+03	FES = 38100	$\mu_F = 0.749$	μ_{CR}
= 0.155 Diversity = 2.2879e-04	$k_t = [4, 2, 3, 3]$		
Generation 390: Best = 3.089518e+03	FES = 39100	$\mu_F = 0.753$	μ_{CR}
= 0.147 Diversity = 1.7186e-04	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 3.089518e+03	FES = 40100	$\mu_F = 0.748$	μ_{CR}
= 0.156 Diversity = 1.1169e-04	$k_t = [2, 2, 4, 4]$		
Generation 410: Best = 3.089518e+03	FES = 41100	$\mu_F = 0.761$	μ_{CR}
= 0.163 Diversity = 6.1785e-05	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 3.089518e+03	FES = 42100	$\mu_F = 0.752$	μ_{CR}
= 0.157 Diversity = 4.1467e-05	$k_t = [4, 2, 2, 2]$		
Generation 430: Best = 3.089518e+03	FES = 43100	$\mu_F = 0.755$	μ_{CR}
= 0.148 Diversity = 2.7296e-05	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 3.089518e+03	FES = 44100	$\mu_F = 0.772$	μ_{CR}
= 0.150 Diversity = 1.7541e-05	$k_t = [3, 3, 2, 4]$		
Generation 450: Best = 3.089518e+03	FES = 45100	$\mu_F = 0.762$	μ_{CR}

= 0.153	Diversity = 1.1354e-05	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 3.089518e+03	FES = 46100	$\mu_F = 0.759$	μ_{CR}
= 0.152	Diversity = 7.5619e-06	k_t = [3, 3, 4, 2]		
Generation 470:	Best = 3.089518e+03	FES = 47100	$\mu_F = 0.770$	μ_{CR}
= 0.148	Diversity = 5.1672e-06	k_t = [1, 1, 1, 1]		
Generation 480:	Best = 3.089518e+03	FES = 48100	$\mu_F = 0.779$	μ_{CR}
= 0.149	Diversity = 3.3086e-06	k_t = [4, 4, 2, 3]		
Generation 490:	Best = 3.089518e+03	FES = 49100	$\mu_F = 0.774$	μ_{CR}
= 0.147	Diversity = 2.6332e-06	k_t = [1, 1, 1, 1]		
Generation 500:	Best = 3.089518e+03	FES = 50100	$\mu_F = 0.759$	μ_{CR}
= 0.150	Diversity = 1.6337e-06	k_t = [3, 4, 2, 4]		
Generation 510:	Best = 3.089518e+03	FES = 51100	$\mu_F = 0.771$	μ_{CR}
= 0.145	Diversity = 1.2861e-06	k_t = [1, 1, 1, 1]		
Generation 520:	Best = 3.089518e+03	FES = 52100	$\mu_F = 0.768$	μ_{CR}
= 0.144	Diversity = 1.0448e-06	k_t = [3, 4, 3, 2]		
Generation 530:	Best = 3.089518e+03	FES = 53100	$\mu_F = 0.756$	μ_{CR}
= 0.144	Diversity = 9.0346e-07	k_t = [1, 1, 1, 1]		
Generation 540:	Best = 3.089518e+03	FES = 54100	$\mu_F = 0.767$	μ_{CR}
= 0.143	Diversity = 7.4375e-07	k_t = [4, 3, 4, 2]		
Generation 550:	Best = 3.089518e+03	FES = 55100	$\mu_F = 0.765$	μ_{CR}
= 0.152	Diversity = 6.8158e-07	k_t = [1, 1, 1, 1]		
Generation 560:	Best = 3.089518e+03	FES = 56100	$\mu_F = 0.772$	μ_{CR}
= 0.151	Diversity = 6.5343e-07	k_t = [2, 4, 2, 4]		
Generation 570:	Best = 3.089518e+03	FES = 57100	$\mu_F = 0.787$	μ_{CR}
= 0.149	Diversity = 6.6004e-07	k_t = [1, 1, 1, 1]		
Generation 580:	Best = 3.089518e+03	FES = 58100	$\mu_F = 0.792$	μ_{CR}
= 0.149	Diversity = 4.6205e-07	k_t = [3, 2, 2, 3]		
Generation 590:	Best = 3.089518e+03	FES = 59100	$\mu_F = 0.778$	μ_{CR}
= 0.148	Diversity = 4.4458e-07	k_t = [1, 1, 1, 1]		
Generation 600:	Best = 3.089518e+03	FES = 60100	$\mu_F = 0.789$	μ_{CR}
= 0.146	Diversity = 4.2452e-07	k_t = [2, 4, 4, 4]		
Generation 610:	Best = 3.089518e+03	FES = 61100	$\mu_F = 0.789$	μ_{CR}
= 0.158	Diversity = 4.2119e-07	k_t = [1, 1, 1, 1]		
Generation 620:	Best = 3.089518e+03	FES = 62100	$\mu_F = 0.796$	μ_{CR}
= 0.146	Diversity = 4.2187e-07	k_t = [4, 3, 3, 2]		
Generation 630:	Best = 3.089518e+03	FES = 63100	$\mu_F = 0.810$	μ_{CR}
= 0.149	Diversity = 4.2851e-07	k_t = [1, 1, 1, 1]		
Generation 640:	Best = 3.089518e+03	FES = 64100	$\mu_F = 0.802$	μ_{CR}
= 0.141	Diversity = 4.1899e-07	k_t = [3, 2, 2, 4]		
Generation 650:	Best = 3.089518e+03	FES = 65100	$\mu_F = 0.806$	μ_{CR}
= 0.137	Diversity = 4.2846e-07	k_t = [1, 1, 1, 1]		
Generation 660:	Best = 3.089518e+03	FES = 66100	$\mu_F = 0.795$	μ_{CR}
= 0.135	Diversity = 4.1446e-07	k_t = [4, 3, 4, 4]		
Generation 670:	Best = 3.089518e+03	FES = 67100	$\mu_F = 0.808$	μ_{CR}
= 0.135	Diversity = 4.3682e-07	k_t = [1, 1, 1, 1]		
Generation 680:	Best = 3.089518e+03	FES = 68100	$\mu_F = 0.808$	μ_{CR}
= 0.139	Diversity = 4.2717e-07	k_t = [3, 4, 4, 2]		
Generation 690:	Best = 3.089518e+03	FES = 69100	$\mu_F = 0.799$	μ_{CR}
= 0.145	Diversity = 4.3048e-07	k_t = [1, 1, 1, 1]		

Generation 700: Best = 3.089518e+03	FES = 70100	$\mu_F = 0.793$	μ_{CR}
= 0.142 Diversity = 3.9638e-07 $k_t = [2, 4, 2, 3]$			
Generation 710: Best = 3.089518e+03	FES = 71100	$\mu_F = 0.791$	μ_{CR}
= 0.134 Diversity = 3.9083e-07 $k_t = [1, 1, 1, 1]$			
Generation 720: Best = 3.089518e+03	FES = 72100	$\mu_F = 0.791$	μ_{CR}
= 0.142 Diversity = 3.9097e-07 $k_t = [4, 3, 3, 3]$			
Generation 730: Best = 3.089518e+03	FES = 73100	$\mu_F = 0.788$	μ_{CR}
= 0.138 Diversity = 3.8403e-07 $k_t = [1, 1, 1, 1]$			
Generation 740: Best = 3.089518e+03	FES = 74100	$\mu_F = 0.800$	μ_{CR}
= 0.130 Diversity = 3.7917e-07 $k_t = [4, 2, 4, 2]$			
Generation 750: Best = 3.089518e+03	FES = 75100	$\mu_F = 0.793$	μ_{CR}
= 0.131 Diversity = 4.0715e-07 $k_t = [1, 1, 1, 1]$			
Generation 760: Best = 3.089518e+03	FES = 76100	$\mu_F = 0.795$	μ_{CR}
= 0.132 Diversity = 3.7002e-07 $k_t = [3, 3, 3, 4]$			
Generation 770: Best = 3.089518e+03	FES = 77100	$\mu_F = 0.782$	μ_{CR}
= 0.128 Diversity = 4.0938e-07 $k_t = [1, 1, 1, 1]$			
Generation 780: Best = 3.089518e+03	FES = 78100	$\mu_F = 0.783$	μ_{CR}
= 0.130 Diversity = 3.9088e-07 $k_t = [3, 3, 3, 2]$			
Generation 790: Best = 3.089518e+03	FES = 79100	$\mu_F = 0.787$	μ_{CR}
= 0.130 Diversity = 3.8448e-07 $k_t = [1, 1, 1, 1]$			
Generation 800: Best = 3.089518e+03	FES = 80100	$\mu_F = 0.785$	μ_{CR}
= 0.126 Diversity = 3.6573e-07 $k_t = [3, 4, 2, 3]$			
Generation 810: Best = 3.089518e+03	FES = 81100	$\mu_F = 0.787$	μ_{CR}
= 0.126 Diversity = 3.7797e-07 $k_t = [1, 1, 1, 1]$			
Generation 820: Best = 3.089518e+03	FES = 82100	$\mu_F = 0.788$	μ_{CR}
= 0.128 Diversity = 3.9055e-07 $k_t = [4, 4, 3, 2]$			
Generation 830: Best = 3.089518e+03	FES = 83100	$\mu_F = 0.789$	μ_{CR}
= 0.133 Diversity = 3.8563e-07 $k_t = [1, 1, 1, 1]$			
Generation 840: Best = 3.089518e+03	FES = 84100	$\mu_F = 0.761$	μ_{CR}
= 0.135 Diversity = 3.6050e-07 $k_t = [3, 3, 2, 4]$			
Generation 850: Best = 3.089518e+03	FES = 85100	$\mu_F = 0.768$	μ_{CR}
= 0.140 Diversity = 3.8189e-07 $k_t = [1, 1, 1, 1]$			
Generation 860: Best = 3.089518e+03	FES = 86100	$\mu_F = 0.750$	μ_{CR}
= 0.135 Diversity = 3.6643e-07 $k_t = [3, 3, 4, 2]$			
Generation 870: Best = 3.089518e+03	FES = 87100	$\mu_F = 0.748$	μ_{CR}
= 0.131 Diversity = 3.6797e-07 $k_t = [1, 1, 1, 1]$			
Generation 880: Best = 3.089518e+03	FES = 88100	$\mu_F = 0.757$	μ_{CR}
= 0.132 Diversity = 3.5749e-07 $k_t = [2, 4, 2, 4]$			
Generation 890: Best = 3.089518e+03	FES = 89100	$\mu_F = 0.766$	μ_{CR}
= 0.134 Diversity = 3.7063e-07 $k_t = [1, 1, 1, 1]$			
Generation 900: Best = 3.089518e+03	FES = 90100	$\mu_F = 0.764$	μ_{CR}
= 0.134 Diversity = 3.7797e-07 $k_t = [2, 2, 4, 2]$			
Generation 910: Best = 3.089518e+03	FES = 91100	$\mu_F = 0.783$	μ_{CR}
= 0.137 Diversity = 3.7982e-07 $k_t = [1, 1, 1, 1]$			
Generation 920: Best = 3.089518e+03	FES = 92100	$\mu_F = 0.777$	μ_{CR}
= 0.140 Diversity = 3.4720e-07 $k_t = [3, 3, 4, 4]$			
Generation 930: Best = 3.089518e+03	FES = 93100	$\mu_F = 0.780$	μ_{CR}
= 0.145 Diversity = 3.7887e-07 $k_t = [1, 1, 1, 1]$			
Generation 940: Best = 3.089518e+03	FES = 94100	$\mu_F = 0.767$	μ_{CR}


```

= 0.150 Diversity = 3.8392e-07 k_t = [3, 4, 3, 3]
Generation 950: Best = 3.089518e+03 FES = 95100  $\mu_F$  = 0.760  $\mu_{CR}$ 
= 0.152 Diversity = 3.7749e-07 k_t = [1, 1, 1, 1]
Generation 960: Best = 3.089518e+03 FES = 96100  $\mu_F$  = 0.750  $\mu_{CR}$ 
= 0.150 Diversity = 3.6960e-07 k_t = [2, 3, 4, 2]
Generation 970: Best = 3.089518e+03 FES = 97100  $\mu_F$  = 0.748  $\mu_{CR}$ 
= 0.139 Diversity = 3.7789e-07 k_t = [1, 1, 1, 1]
Generation 980: Best = 3.089518e+03 FES = 98100  $\mu_F$  = 0.746  $\mu_{CR}$ 
= 0.138 Diversity = 3.6668e-07 k_t = [3, 3, 3, 4]
Generation 990: Best = 3.089518e+03 FES = 99100  $\mu_F$  = 0.738  $\mu_{CR}$ 
= 0.137 Diversity = 3.5766e-07 k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 3.089518e+03

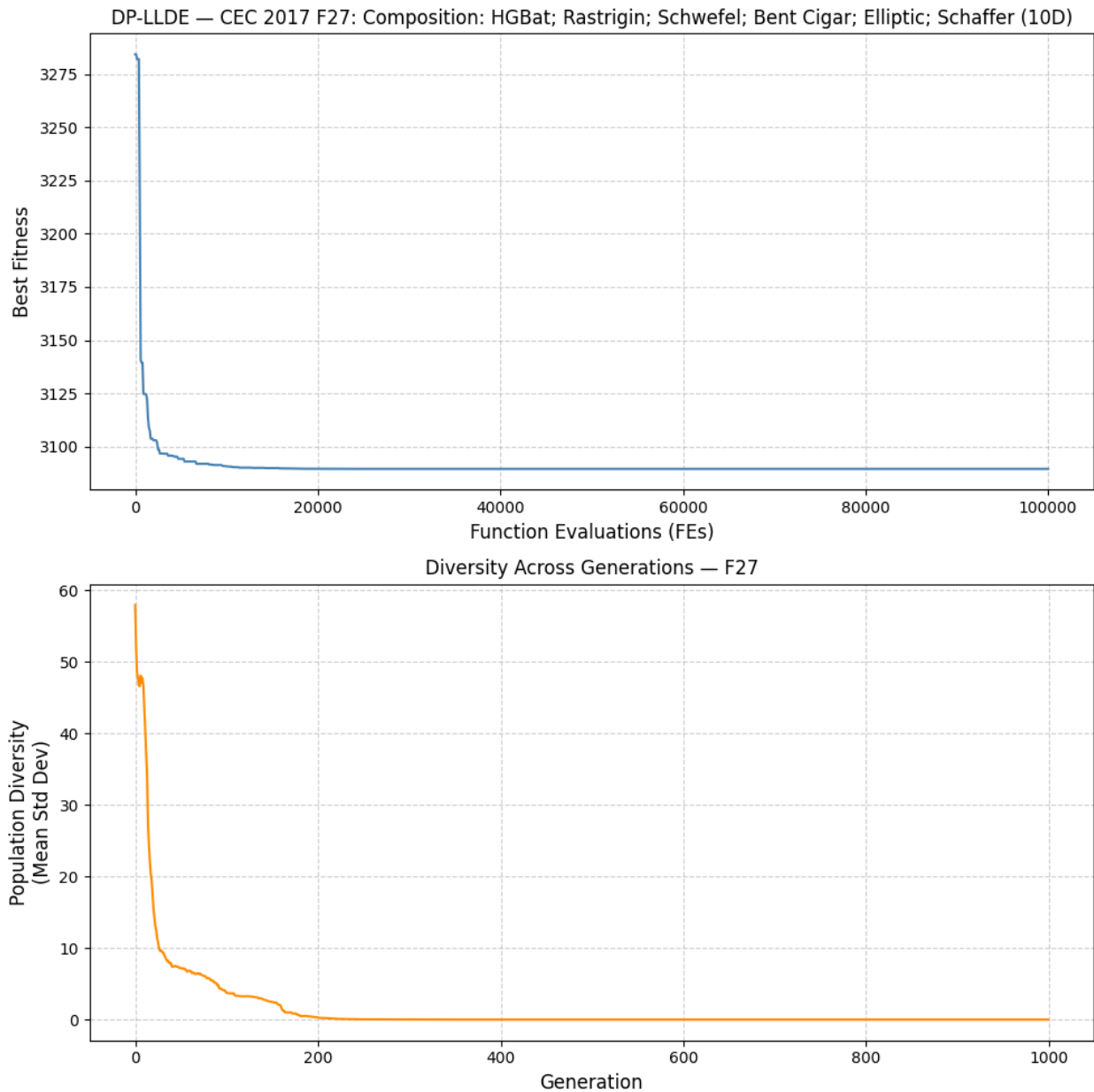
Total Function Evals = 100000

Total Successful Updates = 29753

```

=====
=====
Results:
  Best Fitness      : 3.089518e+03
  Global Optimum    : 2.700000e+03
  Error (|best-optimum|) : 3.895179e+02
  Best Solution (5 dim) : [-80.73873614 -85.37641845 -93.47564993 -
88.07468589 100.      ]
  D^(0) (init. diversity): 5.7928e+01
=====
=====

```



Graph saved as convergence_F27.png

```
# =====
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock;
# HappyCat; Schaffer)
# =====
run_and_plot('F28', make_wrapper(composition.f28), 2800.0,
'Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat;
Schaffer')
```

```
=====
=====
```

DP-LLDE on CEC 2017 F28 – Single Run

```

=====
Function      : Composition: Ackley; Griewank; Discus; Rosenbrock;
HappyCat; Schaffer (10D)
Bounds       : [-100, 100]
Global optimum : 2800.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 4.188047e+03
Generation 10: Best = 3.520386e+03 FES = 1100  $\mu_F$  = 0.568  $\mu_{CR}$ 
= 0.351 Diversity = 2.8889e+01  $k_t$  = [1, 1, 1, 1]
Generation 20: Best = 3.270856e+03 FES = 2100  $\mu_F$  = 0.616  $\mu_{CR}$ 
= 0.348 Diversity = 2.8435e+01  $k_t$  = [3, 4, 2, 3]
Generation 30: Best = 3.189122e+03 FES = 3100  $\mu_F$  = 0.663  $\mu_{CR}$ 
= 0.351 Diversity = 1.3269e+01  $k_t$  = [1, 1, 1, 1]
Generation 40: Best = 3.175829e+03 FES = 4100  $\mu_F$  = 0.693  $\mu_{CR}$ 
= 0.361 Diversity = 5.3224e+00  $k_t$  = [3, 3, 2, 4]
Generation 50: Best = 3.169058e+03 FES = 5100  $\mu_F$  = 0.710  $\mu_{CR}$ 
= 0.366 Diversity = 2.8406e+00  $k_t$  = [1, 1, 1, 1]
Generation 60: Best = 3.166441e+03 FES = 6100  $\mu_F$  = 0.717  $\mu_{CR}$ 
= 0.350 Diversity = 2.3325e+00  $k_t$  = [2, 2, 4, 4]
Generation 70: Best = 3.159266e+03 FES = 7100  $\mu_F$  = 0.736  $\mu_{CR}$ 
= 0.368 Diversity = 2.7937e+00  $k_t$  = [1, 1, 1, 1]
Generation 80: Best = 3.138698e+03 FES = 8100  $\mu_F$  = 0.759  $\mu_{CR}$ 
= 0.377 Diversity = 2.0992e+00  $k_t$  = [3, 3, 2, 4]
Generation 90: Best = 3.128115e+03 FES = 9100  $\mu_F$  = 0.772  $\mu_{CR}$ 
= 0.378 Diversity = 1.5060e+00  $k_t$  = [1, 1, 1, 1]
Generation 100: Best = 3.115054e+03 FES = 10100  $\mu_F$  = 0.772  $\mu_{CR}$ 
= 0.372 Diversity = 9.5107e-01  $k_t$  = [2, 3, 4, 4]
Generation 110: Best = 3.108285e+03 FES = 11100  $\mu_F$  = 0.767  $\mu_{CR}$ 
= 0.371 Diversity = 7.8663e-01  $k_t$  = [1, 1, 1, 1]
Generation 120: Best = 3.104466e+03 FES = 12100  $\mu_F$  = 0.768  $\mu_{CR}$ 
= 0.372 Diversity = 5.6743e-01  $k_t$  = [2, 2, 2, 2]
Generation 130: Best = 3.103246e+03 FES = 13100  $\mu_F$  = 0.754  $\mu_{CR}$ 
= 0.360 Diversity = 3.8837e-01  $k_t$  = [1, 1, 1, 1]
Generation 140: Best = 3.101047e+03 FES = 14100  $\mu_F$  = 0.739  $\mu_{CR}$ 
= 0.360 Diversity = 2.1793e-01  $k_t$  = [3, 2, 2, 4]
Generation 150: Best = 3.100596e+03 FES = 15100  $\mu_F$  = 0.733  $\mu_{CR}$ 
= 0.360 Diversity = 1.2402e-01  $k_t$  = [1, 1, 1, 1]
Generation 160: Best = 3.100331e+03 FES = 16100  $\mu_F$  = 0.718  $\mu_{CR}$ 
= 0.356 Diversity = 5.4541e-02  $k_t$  = [2, 2, 2, 3]
Generation 170: Best = 3.100113e+03 FES = 17100  $\mu_F$  = 0.722  $\mu_{CR}$ 
= 0.344 Diversity = 2.9189e-02  $k_t$  = [1, 1, 1, 1]
Generation 180: Best = 3.100074e+03 FES = 18100  $\mu_F$  = 0.723  $\mu_{CR}$ 
= 0.337 Diversity = 1.2632e-02  $k_t$  = [3, 2, 3, 4]
Generation 190: Best = 3.100038e+03 FES = 19100  $\mu_F$  = 0.729  $\mu_{CR}$ 
= 0.337 Diversity = 6.5492e-03  $k_t$  = [1, 1, 1, 1]
Generation 200: Best = 3.100016e+03 FES = 20100  $\mu_F$  = 0.711  $\mu_{CR}$ 
= 0.335 Diversity = 2.8843e-03  $k_t$  = [2, 2, 2, 3]

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Generation 210: Best = 3.100006e+03	FES = 21100	$\mu_F = 0.715$	μ_{CR}
= 0.336 Diversity = 1.4477e-03 $k_t = [1, 1, 1, 1]$			
Generation 220: Best = 3.100003e+03	FES = 22100	$\mu_F = 0.715$	μ_{CR}
= 0.335 Diversity = 6.6930e-04 $k_t = [4, 4, 3, 2]$			
Generation 230: Best = 3.100001e+03	FES = 23100	$\mu_F = 0.706$	μ_{CR}
= 0.335 Diversity = 2.7485e-04 $k_t = [1, 1, 1, 1]$			
Generation 240: Best = 3.100001e+03	FES = 24100	$\mu_F = 0.715$	μ_{CR}
= 0.331 Diversity = 1.1094e-04 $k_t = [3, 3, 3, 3]$			
Generation 250: Best = 3.100000e+03	FES = 25100	$\mu_F = 0.723$	μ_{CR}
= 0.328 Diversity = 5.5899e-05 $k_t = [1, 1, 1, 1]$			
Generation 260: Best = 3.100000e+03	FES = 26100	$\mu_F = 0.721$	μ_{CR}
= 0.326 Diversity = 2.2162e-05 $k_t = [4, 3, 4, 4]$			
Generation 270: Best = 3.100000e+03	FES = 27100	$\mu_F = 0.720$	μ_{CR}
= 0.319 Diversity = 9.3383e-06 $k_t = [1, 1, 1, 1]$			
Generation 280: Best = 3.100000e+03	FES = 28100	$\mu_F = 0.719$	μ_{CR}
= 0.311 Diversity = 4.2201e-06 $k_t = [4, 3, 3, 2]$			
Generation 290: Best = 3.100000e+03	FES = 29100	$\mu_F = 0.720$	μ_{CR}
= 0.316 Diversity = 1.8309e-06 $k_t = [1, 1, 1, 1]$			
Generation 300: Best = 3.100000e+03	FES = 30100	$\mu_F = 0.719$	μ_{CR}
= 0.319 Diversity = 8.0870e-07 $k_t = [4, 2, 3, 2]$			
Generation 310: Best = 3.100000e+03	FES = 31100	$\mu_F = 0.719$	μ_{CR}
= 0.316 Diversity = 3.9023e-07 $k_t = [1, 1, 1, 1]$			
Generation 320: Best = 3.100000e+03	FES = 32100	$\mu_F = 0.739$	μ_{CR}
= 0.312 Diversity = 1.3561e-07 $k_t = [4, 2, 3, 4]$			
Generation 330: Best = 3.100000e+03	FES = 33100	$\mu_F = 0.738$	μ_{CR}
= 0.306 Diversity = 6.2869e-08 $k_t = [1, 1, 1, 1]$			
Generation 340: Best = 3.100000e+03	FES = 34100	$\mu_F = 0.740$	μ_{CR}
= 0.307 Diversity = 3.3351e-08 $k_t = [4, 4, 4, 2]$			
Generation 350: Best = 3.100000e+03	FES = 35100	$\mu_F = 0.729$	μ_{CR}
= 0.310 Diversity = 1.6021e-08 $k_t = [1, 1, 1, 1]$			
Generation 360: Best = 3.100000e+03	FES = 36100	$\mu_F = 0.726$	μ_{CR}
= 0.311 Diversity = 7.3220e-09 $k_t = [3, 3, 4, 2]$			
Generation 370: Best = 3.100000e+03	FES = 37100	$\mu_F = 0.722$	μ_{CR}
= 0.318 Diversity = 3.5127e-09 $k_t = [1, 1, 1, 1]$			
Generation 380: Best = 3.100000e+03	FES = 38100	$\mu_F = 0.723$	μ_{CR}
= 0.316 Diversity = 1.3993e-09 $k_t = [3, 2, 3, 3]$			
Generation 390: Best = 3.100000e+03	FES = 39100	$\mu_F = 0.711$	μ_{CR}
= 0.319 Diversity = 6.2929e-10 $k_t = [1, 1, 1, 1]$			
Generation 400: Best = 3.100000e+03	FES = 40100	$\mu_F = 0.702$	μ_{CR}
= 0.320 Diversity = 2.6829e-10 $k_t = [4, 4, 3, 2]$			
Generation 410: Best = 3.100000e+03	FES = 41100	$\mu_F = 0.710$	μ_{CR}
= 0.319 Diversity = 1.2421e-10 $k_t = [1, 1, 1, 1]$			
Generation 420: Best = 3.100000e+03	FES = 42100	$\mu_F = 0.711$	μ_{CR}
= 0.318 Diversity = 5.8236e-11 $k_t = [2, 3, 3, 2]$			
Generation 430: Best = 3.100000e+03	FES = 43100	$\mu_F = 0.710$	μ_{CR}
= 0.324 Diversity = 2.7562e-11 $k_t = [1, 1, 1, 1]$			
Generation 440: Best = 3.100000e+03	FES = 44100	$\mu_F = 0.710$	μ_{CR}
= 0.318 Diversity = 1.1004e-11 $k_t = [2, 4, 2, 4]$			
Generation 450: Best = 3.100000e+03	FES = 45100	$\mu_F = 0.716$	μ_{CR}

= 0.319	Diversity = 5.2832e-12	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 3.100000e+03	FES = 46100	$\mu_F = 0.717$	μ_{CR}
= 0.319	Diversity = 2.0869e-12	k_t = [2, 4, 4, 4]		
Generation 470:	Best = 3.100000e+03	FES = 47100	$\mu_F = 0.705$	μ_{CR}
= 0.313	Diversity = 9.2200e-13	k_t = [1, 1, 1, 1]		
Generation 480:	Best = 3.100000e+03	FES = 48100	$\mu_F = 0.715$	μ_{CR}
= 0.315	Diversity = 4.3852e-13	k_t = [2, 3, 2, 4]		
Generation 490:	Best = 3.100000e+03	FES = 49100	$\mu_F = 0.714$	μ_{CR}
= 0.313	Diversity = 2.2635e-13	k_t = [1, 1, 1, 1]		
Generation 500:	Best = 3.100000e+03	FES = 50100	$\mu_F = 0.723$	μ_{CR}
= 0.314	Diversity = 1.1305e-13	k_t = [3, 4, 4, 4]		
Generation 510:	Best = 3.100000e+03	FES = 51100	$\mu_F = 0.751$	μ_{CR}
= 0.305	Diversity = 9.8855e-14	k_t = [1, 1, 1, 1]		
Generation 520:	Best = 3.100000e+03	FES = 52100	$\mu_F = 0.767$	μ_{CR}
= 0.298	Diversity = 7.6643e-14	k_t = [4, 4, 3, 4]		
Generation 530:	Best = 3.100000e+03	FES = 53100	$\mu_F = 0.787$	μ_{CR}
= 0.297	Diversity = 6.6354e-14	k_t = [1, 1, 1, 1]		
Generation 540:	Best = 3.100000e+03	FES = 54100	$\mu_F = 0.796$	μ_{CR}
= 0.294	Diversity = 6.2624e-14	k_t = [2, 4, 3, 2]		
Generation 550:	Best = 3.100000e+03	FES = 55100	$\mu_F = 0.794$	μ_{CR}
= 0.293	Diversity = 5.8301e-14	k_t = [1, 1, 1, 1]		
Generation 560:	Best = 3.100000e+03	FES = 56100	$\mu_F = 0.800$	μ_{CR}
= 0.287	Diversity = 5.8955e-14	k_t = [2, 3, 4, 3]		
Generation 570:	Best = 3.100000e+03	FES = 57100	$\mu_F = 0.807$	μ_{CR}
= 0.277	Diversity = 5.5723e-14	k_t = [1, 1, 1, 1]		
Generation 580:	Best = 3.100000e+03	FES = 58100	$\mu_F = 0.816$	μ_{CR}
= 0.277	Diversity = 5.5385e-14	k_t = [3, 4, 2, 3]		
Generation 590:	Best = 3.100000e+03	FES = 59100	$\mu_F = 0.811$	μ_{CR}
= 0.269	Diversity = 5.3565e-14	k_t = [1, 1, 1, 1]		
Generation 600:	Best = 3.100000e+03	FES = 60100	$\mu_F = 0.812$	μ_{CR}
= 0.264	Diversity = 5.2853e-14	k_t = [2, 3, 2, 2]		
Generation 610:	Best = 3.100000e+03	FES = 61100	$\mu_F = 0.816$	μ_{CR}
= 0.246	Diversity = 5.2553e-14	k_t = [1, 1, 1, 1]		
Generation 620:	Best = 3.100000e+03	FES = 62100	$\mu_F = 0.816$	μ_{CR}
= 0.234	Diversity = 5.7190e-14	k_t = [2, 2, 4, 3]		
Generation 630:	Best = 3.100000e+03	FES = 63100	$\mu_F = 0.800$	μ_{CR}
= 0.223	Diversity = 5.7291e-14	k_t = [1, 1, 1, 1]		
Generation 640:	Best = 3.100000e+03	FES = 64100	$\mu_F = 0.791$	μ_{CR}
= 0.215	Diversity = 5.1328e-14	k_t = [3, 2, 4, 3]		
Generation 650:	Best = 3.100000e+03	FES = 65100	$\mu_F = 0.793$	μ_{CR}
= 0.207	Diversity = 5.5162e-14	k_t = [1, 1, 1, 1]		
Generation 660:	Best = 3.100000e+03	FES = 66100	$\mu_F = 0.789$	μ_{CR}
= 0.203	Diversity = 5.5442e-14	k_t = [4, 3, 4, 3]		
Generation 670:	Best = 3.100000e+03	FES = 67100	$\mu_F = 0.785$	μ_{CR}
= 0.201	Diversity = 5.6648e-14	k_t = [1, 1, 1, 1]		
Generation 680:	Best = 3.100000e+03	FES = 68100	$\mu_F = 0.796$	μ_{CR}
= 0.196	Diversity = 5.3522e-14	k_t = [3, 4, 4, 3]		
Generation 690:	Best = 3.100000e+03	FES = 69100	$\mu_F = 0.799$	μ_{CR}
= 0.201	Diversity = 5.6182e-14	k_t = [1, 1, 1, 1]		

Generation 700: Best = 3.100000e+03	FES = 70100	$\mu_F = 0.812$	μ_{CR}
= 0.199 Diversity = 5.6727e-14	$k_t = [4, 3, 4, 4]$		
Generation 710: Best = 3.100000e+03	FES = 71100	$\mu_F = 0.806$	μ_{CR}
= 0.198 Diversity = 5.1321e-14	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 3.100000e+03	FES = 72100	$\mu_F = 0.808$	μ_{CR}
= 0.199 Diversity = 5.5326e-14	$k_t = [4, 3, 2, 3]$		
Generation 730: Best = 3.100000e+03	FES = 73100	$\mu_F = 0.813$	μ_{CR}
= 0.200 Diversity = 6.0536e-14	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 3.100000e+03	FES = 74100	$\mu_F = 0.819$	μ_{CR}
= 0.197 Diversity = 5.4754e-14	$k_t = [3, 3, 4, 3]$		
Generation 750: Best = 3.100000e+03	FES = 75100	$\mu_F = 0.821$	μ_{CR}
= 0.196 Diversity = 5.1980e-14	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 3.100000e+03	FES = 76100	$\mu_F = 0.831$	μ_{CR}
= 0.188 Diversity = 5.1702e-14	$k_t = [4, 4, 3, 2]$		
Generation 770: Best = 3.100000e+03	FES = 77100	$\mu_F = 0.827$	μ_{CR}
= 0.186 Diversity = 5.0851e-14	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 3.100000e+03	FES = 78100	$\mu_F = 0.819$	μ_{CR}
= 0.183 Diversity = 4.9044e-14	$k_t = [2, 2, 3, 2]$		
Generation 790: Best = 3.100000e+03	FES = 79100	$\mu_F = 0.808$	μ_{CR}
= 0.179 Diversity = 5.1791e-14	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 3.100000e+03	FES = 80100	$\mu_F = 0.802$	μ_{CR}
= 0.170 Diversity = 5.2395e-14	$k_t = [3, 4, 4, 3]$		
Generation 810: Best = 3.100000e+03	FES = 81100	$\mu_F = 0.804$	μ_{CR}
= 0.169 Diversity = 5.1683e-14	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 3.100000e+03	FES = 82100	$\mu_F = 0.799$	μ_{CR}
= 0.162 Diversity = 5.3657e-14	$k_t = [2, 2, 3, 3]$		
Generation 830: Best = 3.100000e+03	FES = 83100	$\mu_F = 0.800$	μ_{CR}
= 0.161 Diversity = 5.1373e-14	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 3.100000e+03	FES = 84100	$\mu_F = 0.806$	μ_{CR}
= 0.162 Diversity = 5.5936e-14	$k_t = [3, 2, 4, 3]$		
Generation 850: Best = 3.100000e+03	FES = 85100	$\mu_F = 0.808$	μ_{CR}
= 0.165 Diversity = 5.4390e-14	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 3.100000e+03	FES = 86100	$\mu_F = 0.802$	μ_{CR}
= 0.164 Diversity = 5.2046e-14	$k_t = [2, 4, 4, 4]$		
Generation 870: Best = 3.100000e+03	FES = 87100	$\mu_F = 0.808$	μ_{CR}
= 0.156 Diversity = 5.2607e-14	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 3.100000e+03	FES = 88100	$\mu_F = 0.806$	μ_{CR}
= 0.158 Diversity = 5.1257e-14	$k_t = [3, 2, 4, 3]$		
Generation 890: Best = 3.100000e+03	FES = 89100	$\mu_F = 0.817$	μ_{CR}
= 0.156 Diversity = 5.0857e-14	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 3.100000e+03	FES = 90100	$\mu_F = 0.814$	μ_{CR}
= 0.157 Diversity = 5.2233e-14	$k_t = [3, 2, 2, 2]$		
Generation 910: Best = 3.100000e+03	FES = 91100	$\mu_F = 0.820$	μ_{CR}
= 0.160 Diversity = 5.0588e-14	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 3.100000e+03	FES = 92100	$\mu_F = 0.828$	μ_{CR}
= 0.158 Diversity = 5.1024e-14	$k_t = [4, 2, 3, 4]$		
Generation 930: Best = 3.100000e+03	FES = 93100	$\mu_F = 0.828$	μ_{CR}
= 0.166 Diversity = 5.0797e-14	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 3.100000e+03	FES = 94100	$\mu_F = 0.819$	μ_{CR}

```

= 0.172 Diversity = 5.0935e-14 k_t = [3, 2, 2, 4]
Generation 950: Best = 3.100000e+03 FES = 95100  $\mu_F$  = 0.821  $\mu_{CR}$ 
= 0.171 Diversity = 5.3942e-14 k_t = [1, 1, 1, 1]
Generation 960: Best = 3.100000e+03 FES = 96100  $\mu_F$  = 0.821  $\mu_{CR}$ 
= 0.169 Diversity = 5.3530e-14 k_t = [4, 4, 4, 2]
Generation 970: Best = 3.100000e+03 FES = 97100  $\mu_F$  = 0.813  $\mu_{CR}$ 
= 0.170 Diversity = 5.4316e-14 k_t = [1, 1, 1, 1]
Generation 980: Best = 3.100000e+03 FES = 98100  $\mu_F$  = 0.814  $\mu_{CR}$ 
= 0.171 Diversity = 4.8738e-14 k_t = [3, 4, 2, 4]
Generation 990: Best = 3.100000e+03 FES = 99100  $\mu_F$  = 0.800  $\mu_{CR}$ 
= 0.168 Diversity = 5.0438e-14 k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 3.100000e+03

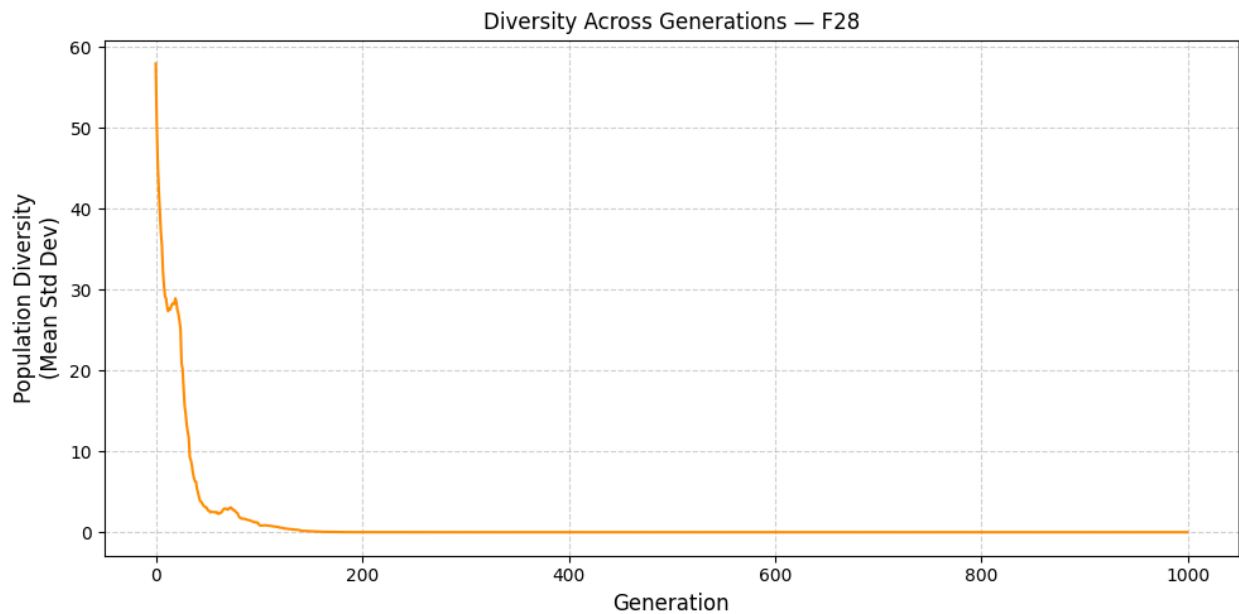
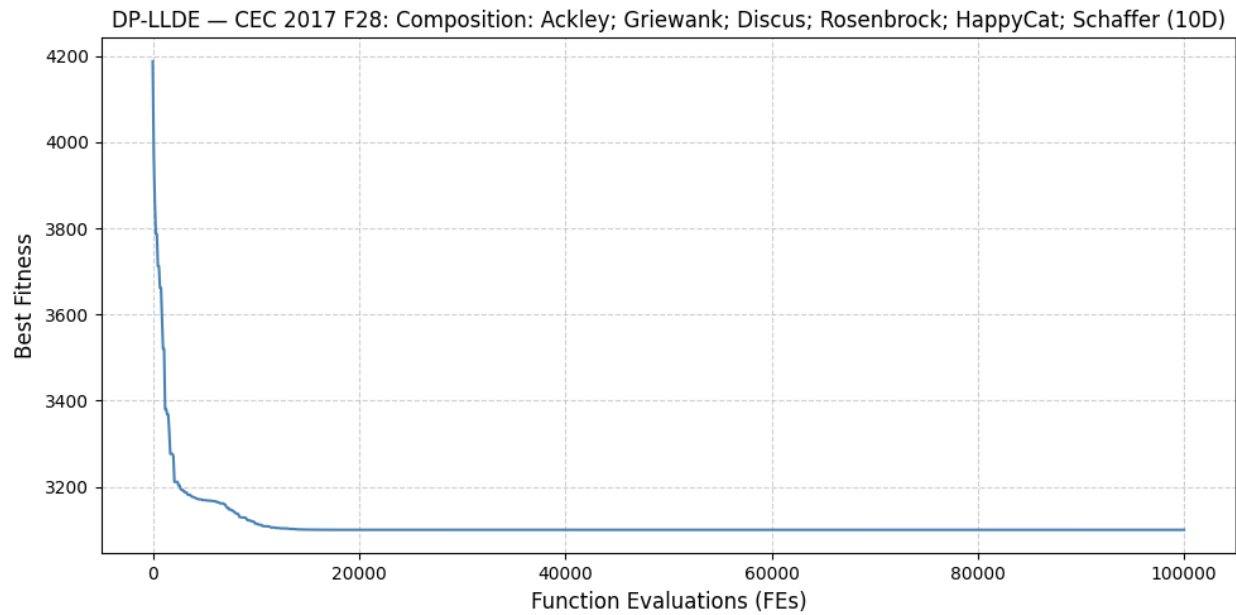
Total Function Evals = 100000

Total Successful Updates = 49656

```

=====
=====
Results:
  Best Fitness      : 3.100000e+03
  Global Optimum    : 2.800000e+03
  Error (|best-optimum|) : 3.000000e+02
  Best Solution (5 dim) : [-26.37193452 -34.00844064 -49.92961782
47.6166024 -66.54538272]
  D^(0) (init. diversity): 5.7928e+01
=====
=====

```



Graph saved as convergence_F28.png

```
# =====
# F29 – Composition Function 9 (F15; F16; F17)
# =====
run_and_plot('F29', make_wrapper(composition.f29), 2900.0,
'Composition: F15; F16; F17')
```

```
=====
=====
DP-LLDE on CEC 2017 F29 – Single Run
=====
```


=====

Function : Composition: F15; F16; F17 (10D)

Bounds : [-100, 100]

Global optimum : 2900.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = $5.7928e+00$

Generation 0: Best Fitness = $3.805371e+03$

Generation 10: Best = $3.498177e+03$ FES = 1100 $\mu_F = 0.580$ μ_{CR}

= 0.346 Diversity = $3.8537e+01$ $k_t = [1, 1, 1, 1]$

Generation 20: Best = $3.328327e+03$ FES = 2100 $\mu_F = 0.644$ μ_{CR}

= 0.336 Diversity = $3.8258e+01$ $k_t = [2, 4, 2, 2]$

Generation 30: Best = $3.328327e+03$ FES = 3100 $\mu_F = 0.700$ μ_{CR}

= 0.319 Diversity = $3.7333e+01$ $k_t = [1, 1, 1, 1]$

Generation 40: Best = $3.310138e+03$ FES = 4100 $\mu_F = 0.734$ μ_{CR}

= 0.327 Diversity = $3.5926e+01$ $k_t = [3, 4, 2, 3]$

Generation 50: Best = $3.205632e+03$ FES = 5100 $\mu_F = 0.741$ μ_{CR}

= 0.311 Diversity = $3.7670e+01$ $k_t = [1, 1, 1, 1]$

Generation 60: Best = $3.205632e+03$ FES = 6100 $\mu_F = 0.773$ μ_{CR}

= 0.268 Diversity = $3.5958e+01$ $k_t = [2, 4, 4, 4]$

Generation 70: Best = $3.205632e+03$ FES = 7100 $\mu_F = 0.779$ μ_{CR}

= 0.231 Diversity = $3.4211e+01$ $k_t = [1, 1, 1, 1]$

Generation 80: Best = $3.205632e+03$ FES = 8100 $\mu_F = 0.789$ μ_{CR}

= 0.200 Diversity = $3.1865e+01$ $k_t = [2, 4, 3, 4]$

Generation 90: Best = $3.194979e+03$ FES = 9100 $\mu_F = 0.793$ μ_{CR}

= 0.210 Diversity = $3.1989e+01$ $k_t = [1, 1, 1, 1]$

Generation 100: Best = $3.194281e+03$ FES = 10100 $\mu_F = 0.793$ μ_{CR}

= 0.188 Diversity = $3.1164e+01$ $k_t = [3, 2, 3, 4]$

Generation 110: Best = $3.194281e+03$ FES = 11100 $\mu_F = 0.756$ μ_{CR}

= 0.141 Diversity = $3.1675e+01$ $k_t = [1, 1, 1, 1]$

Generation 120: Best = $3.192129e+03$ FES = 12100 $\mu_F = 0.789$ μ_{CR}

= 0.144 Diversity = $3.1003e+01$ $k_t = [3, 2, 2, 2]$

Generation 130: Best = $3.190572e+03$ FES = 13100 $\mu_F = 0.788$ μ_{CR}

= 0.117 Diversity = $3.0996e+01$ $k_t = [1, 1, 1, 1]$

Generation 140: Best = $3.172244e+03$ FES = 14100 $\mu_F = 0.730$ μ_{CR}

= 0.112 Diversity = $3.0465e+01$ $k_t = [2, 2, 2, 3]$

Generation 150: Best = $3.172244e+03$ FES = 15100 $\mu_F = 0.725$ μ_{CR}

= 0.093 Diversity = $2.9885e+01$ $k_t = [1, 1, 1, 1]$

Generation 160: Best = $3.172244e+03$ FES = 16100 $\mu_F = 0.754$ μ_{CR}

= 0.086 Diversity = $2.8199e+01$ $k_t = [3, 3, 3, 4]$

Generation 170: Best = $3.164164e+03$ FES = 17100 $\mu_F = 0.773$ μ_{CR}

= 0.115 Diversity = $2.7784e+01$ $k_t = [1, 1, 1, 1]$

Generation 180: Best = $3.164164e+03$ FES = 18100 $\mu_F = 0.775$ μ_{CR}

= 0.089 Diversity = $2.7601e+01$ $k_t = [2, 3, 3, 2]$

Generation 190: Best = $3.160498e+03$ FES = 19100 $\mu_F = 0.763$ μ_{CR}

= 0.102 Diversity = $2.7064e+01$ $k_t = [1, 1, 1, 1]$

Generation 200: Best = $3.160498e+03$ FES = 20100 $\mu_F = 0.745$ μ_{CR}

= 0.114 Diversity = $2.6329e+01$ $k_t = [4, 3, 3, 4]$

Generation 210: Best = $3.160498e+03$ FES = 21100 $\mu_F = 0.658$ μ_{CR}

= 0.089 Diversity = $2.5939e+01$ $k_t = [1, 1, 1, 1]$

Generation 220: Best = $3.160498e+03$ FES = 22100 $\mu_F = 0.658$ μ_{CR}

= 0.091	Diversity = 2.5162e+01	k_t = [3, 2, 4, 4]		
Generation 230:	Best = 3.160498e+03	FES = 23100	$\mu_F = 0.572$	μ_{CR}
= 0.097	Diversity = 2.4516e+01	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 3.160498e+03	FES = 24100	$\mu_F = 0.476$	μ_{CR}
= 0.104	Diversity = 2.4413e+01	k_t = [3, 2, 4, 2]		
Generation 250:	Best = 3.160129e+03	FES = 25100	$\mu_F = 0.480$	μ_{CR}
= 0.105	Diversity = 2.4178e+01	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 3.160129e+03	FES = 26100	$\mu_F = 0.476$	μ_{CR}
= 0.112	Diversity = 2.3707e+01	k_t = [4, 3, 4, 4]		
Generation 270:	Best = 3.160129e+03	FES = 27100	$\mu_F = 0.526$	μ_{CR}
= 0.122	Diversity = 2.3556e+01	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 3.160129e+03	FES = 28100	$\mu_F = 0.564$	μ_{CR}
= 0.098	Diversity = 2.3309e+01	k_t = [4, 2, 4, 3]		
Generation 290:	Best = 3.160129e+03	FES = 29100	$\mu_F = 0.499$	μ_{CR}
= 0.113	Diversity = 2.3107e+01	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 3.160129e+03	FES = 30100	$\mu_F = 0.544$	μ_{CR}
= 0.098	Diversity = 2.3074e+01	k_t = [4, 4, 4, 3]		
Generation 310:	Best = 3.160129e+03	FES = 31100	$\mu_F = 0.444$	μ_{CR}
= 0.086	Diversity = 2.2685e+01	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 3.160129e+03	FES = 32100	$\mu_F = 0.514$	μ_{CR}
= 0.090	Diversity = 2.2578e+01	k_t = [2, 2, 2, 4]		
Generation 330:	Best = 3.160129e+03	FES = 33100	$\mu_F = 0.540$	μ_{CR}
= 0.079	Diversity = 2.2574e+01	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 3.160129e+03	FES = 34100	$\mu_F = 0.500$	μ_{CR}
= 0.085	Diversity = 2.2506e+01	k_t = [4, 2, 3, 3]		
Generation 350:	Best = 3.160129e+03	FES = 35100	$\mu_F = 0.398$	μ_{CR}
= 0.085	Diversity = 2.2308e+01	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 3.160129e+03	FES = 36100	$\mu_F = 0.399$	μ_{CR}
= 0.070	Diversity = 2.1997e+01	k_t = [3, 3, 2, 2]		
Generation 370:	Best = 3.160129e+03	FES = 37100	$\mu_F = 0.446$	μ_{CR}
= 0.102	Diversity = 2.1900e+01	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 3.160129e+03	FES = 38100	$\mu_F = 0.369$	μ_{CR}
= 0.083	Diversity = 2.1780e+01	k_t = [3, 2, 4, 3]		
Generation 390:	Best = 3.160129e+03	FES = 39100	$\mu_F = 0.434$	μ_{CR}
= 0.104	Diversity = 2.1767e+01	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 3.152862e+03	FES = 40100	$\mu_F = 0.387$	μ_{CR}
= 0.092	Diversity = 2.1683e+01	k_t = [2, 4, 2, 3]		
Generation 410:	Best = 3.152862e+03	FES = 41100	$\mu_F = 0.425$	μ_{CR}
= 0.090	Diversity = 2.1532e+01	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 3.152862e+03	FES = 42100	$\mu_F = 0.508$	μ_{CR}
= 0.088	Diversity = 2.1510e+01	k_t = [3, 4, 4, 2]		
Generation 430:	Best = 3.152862e+03	FES = 43100	$\mu_F = 0.477$	μ_{CR}
= 0.075	Diversity = 2.1506e+01	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 3.152862e+03	FES = 44100	$\mu_F = 0.414$	μ_{CR}
= 0.080	Diversity = 2.1583e+01	k_t = [3, 4, 3, 2]		
Generation 450:	Best = 3.151539e+03	FES = 45100	$\mu_F = 0.433$	μ_{CR}
= 0.104	Diversity = 2.1441e+01	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 3.151539e+03	FES = 46100	$\mu_F = 0.303$	μ_{CR}
= 0.113	Diversity = 2.1088e+01	k_t = [3, 4, 3, 4]		

Generation 470: Best = 3.151539e+03	FES = 47100	$\mu_F = 0.333$	μ_{CR}
= 0.101 Diversity = 2.1088e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 3.149144e+03	FES = 48100	$\mu_F = 0.311$	μ_{CR}
= 0.089 Diversity = 2.1069e+01	$k_t = [2, 4, 3, 3]$		
Generation 490: Best = 3.149144e+03	FES = 49100	$\mu_F = 0.377$	μ_{CR}
= 0.094 Diversity = 2.0974e+01	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 3.149144e+03	FES = 50100	$\mu_F = 0.280$	μ_{CR}
= 0.091 Diversity = 2.0897e+01	$k_t = [3, 3, 4, 4]$		
Generation 510: Best = 3.149144e+03	FES = 51100	$\mu_F = 0.245$	μ_{CR}
= 0.084 Diversity = 2.0867e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 3.149144e+03	FES = 52100	$\mu_F = 0.372$	μ_{CR}
= 0.088 Diversity = 2.0927e+01	$k_t = [2, 4, 2, 3]$		
Generation 530: Best = 3.149144e+03	FES = 53100	$\mu_F = 0.301$	μ_{CR}
= 0.123 Diversity = 2.0759e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 3.149144e+03	FES = 54100	$\mu_F = 0.405$	μ_{CR}
= 0.128 Diversity = 2.0850e+01	$k_t = [3, 3, 3, 4]$		
Generation 550: Best = 3.147149e+03	FES = 55100	$\mu_F = 0.343$	μ_{CR}
= 0.123 Diversity = 2.0810e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 3.146891e+03	FES = 56100	$\mu_F = 0.359$	μ_{CR}
= 0.103 Diversity = 2.0668e+01	$k_t = [2, 3, 3, 4]$		
Generation 570: Best = 3.146395e+03	FES = 57100	$\mu_F = 0.367$	μ_{CR}
= 0.072 Diversity = 2.0792e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 3.141993e+03	FES = 58100	$\mu_F = 0.424$	μ_{CR}
= 0.084 Diversity = 2.0438e+01	$k_t = [4, 3, 3, 4]$		
Generation 590: Best = 3.141993e+03	FES = 59100	$\mu_F = 0.356$	μ_{CR}
= 0.099 Diversity = 2.0437e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 3.141993e+03	FES = 60100	$\mu_F = 0.431$	μ_{CR}
= 0.076 Diversity = 2.0385e+01	$k_t = [4, 3, 2, 4]$		
Generation 610: Best = 3.141993e+03	FES = 61100	$\mu_F = 0.434$	μ_{CR}
= 0.065 Diversity = 2.0344e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 3.141993e+03	FES = 62100	$\mu_F = 0.320$	μ_{CR}
= 0.130 Diversity = 2.0314e+01	$k_t = [2, 2, 4, 3]$		
Generation 630: Best = 3.141993e+03	FES = 63100	$\mu_F = 0.345$	μ_{CR}
= 0.124 Diversity = 2.0422e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 3.141993e+03	FES = 64100	$\mu_F = 0.449$	μ_{CR}
= 0.118 Diversity = 2.0262e+01	$k_t = [2, 4, 2, 4]$		
Generation 650: Best = 3.141993e+03	FES = 65100	$\mu_F = 0.379$	μ_{CR}
= 0.119 Diversity = 2.0401e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 3.141993e+03	FES = 66100	$\mu_F = 0.396$	μ_{CR}
= 0.118 Diversity = 2.0401e+01	$k_t = [3, 2, 3, 4]$		
Generation 670: Best = 3.141993e+03	FES = 67100	$\mu_F = 0.438$	μ_{CR}
= 0.084 Diversity = 2.0377e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 3.141993e+03	FES = 68100	$\mu_F = 0.381$	μ_{CR}
= 0.086 Diversity = 2.0371e+01	$k_t = [2, 2, 4, 2]$		
Generation 690: Best = 3.141993e+03	FES = 69100	$\mu_F = 0.416$	μ_{CR}
= 0.084 Diversity = 2.0292e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 3.141993e+03	FES = 70100	$\mu_F = 0.324$	μ_{CR}
= 0.081 Diversity = 2.0290e+01	$k_t = [2, 4, 2, 4]$		
Generation 710: Best = 3.141760e+03	FES = 71100	$\mu_F = 0.374$	μ_{CR}

= 0.078	Diversity = 2.0346e+01	k_t = [1, 1, 1, 1]	μ_F = 0.379	μ_{CR}
Generation 720:	Best = 3.141760e+03	FES = 72100		
= 0.080	Diversity = 2.0349e+01	k_t = [3, 2, 3, 2]	μ_F = 0.338	μ_{CR}
Generation 730:	Best = 3.141760e+03	FES = 73100		
= 0.074	Diversity = 2.0348e+01	k_t = [1, 1, 1, 1]	μ_F = 0.304	μ_{CR}
Generation 740:	Best = 3.141760e+03	FES = 74100		
= 0.070	Diversity = 2.0284e+01	k_t = [2, 4, 3, 4]	μ_F = 0.236	μ_{CR}
Generation 750:	Best = 3.141760e+03	FES = 75100		
= 0.062	Diversity = 2.0284e+01	k_t = [1, 1, 1, 1]	μ_F = 0.356	μ_{CR}
Generation 760:	Best = 3.139373e+03	FES = 76100		
= 0.104	Diversity = 2.0555e+01	k_t = [2, 3, 4, 4]	μ_F = 0.392	μ_{CR}
Generation 770:	Best = 3.139373e+03	FES = 77100		
= 0.089	Diversity = 2.0668e+01	k_t = [1, 1, 1, 1]	μ_F = 0.469	μ_{CR}
Generation 780:	Best = 3.139373e+03	FES = 78100		
= 0.075	Diversity = 2.0664e+01	k_t = [3, 3, 3, 3]	μ_F = 0.489	μ_{CR}
Generation 790:	Best = 3.139373e+03	FES = 79100		
= 0.079	Diversity = 2.0857e+01	k_t = [1, 1, 1, 1]	μ_F = 0.553	μ_{CR}
Generation 800:	Best = 3.139373e+03	FES = 80100		
= 0.065	Diversity = 2.1044e+01	k_t = [2, 2, 3, 3]	μ_F = 0.689	μ_{CR}
Generation 810:	Best = 3.139298e+03	FES = 81100		
= 0.058	Diversity = 2.1209e+01	k_t = [1, 1, 1, 1]	μ_F = 0.656	μ_{CR}
Generation 820:	Best = 3.139298e+03	FES = 82100		
= 0.074	Diversity = 2.1050e+01	k_t = [4, 2, 2, 2]	μ_F = 0.593	μ_{CR}
Generation 830:	Best = 3.139298e+03	FES = 83100		
= 0.078	Diversity = 2.1050e+01	k_t = [1, 1, 1, 1]	μ_F = 0.596	μ_{CR}
Generation 840:	Best = 3.139298e+03	FES = 84100		
= 0.086	Diversity = 2.1026e+01	k_t = [2, 2, 3, 3]	μ_F = 0.623	μ_{CR}
Generation 850:	Best = 3.139298e+03	FES = 85100		
= 0.086	Diversity = 2.1020e+01	k_t = [1, 1, 1, 1]	μ_F = 0.679	μ_{CR}
Generation 860:	Best = 3.139298e+03	FES = 86100		
= 0.092	Diversity = 2.1040e+01	k_t = [3, 3, 2, 3]	μ_F = 0.680	μ_{CR}
Generation 870:	Best = 3.139298e+03	FES = 87100		
= 0.075	Diversity = 2.0922e+01	k_t = [1, 1, 1, 1]	μ_F = 0.685	μ_{CR}
Generation 880:	Best = 3.139298e+03	FES = 88100		
= 0.099	Diversity = 2.0871e+01	k_t = [4, 2, 4, 3]	μ_F = 0.702	μ_{CR}
Generation 890:	Best = 3.139298e+03	FES = 89100		
= 0.086	Diversity = 2.0871e+01	k_t = [1, 1, 1, 1]	μ_F = 0.590	μ_{CR}
Generation 900:	Best = 3.139298e+03	FES = 90100		
= 0.090	Diversity = 2.0783e+01	k_t = [4, 2, 3, 2]	μ_F = 0.688	μ_{CR}
Generation 910:	Best = 3.139298e+03	FES = 91100		
= 0.080	Diversity = 2.0971e+01	k_t = [1, 1, 1, 1]	μ_F = 0.648	μ_{CR}
Generation 920:	Best = 3.139298e+03	FES = 92100		
= 0.078	Diversity = 2.0968e+01	k_t = [3, 4, 3, 2]	μ_F = 0.672	μ_{CR}
Generation 930:	Best = 3.139298e+03	FES = 93100		
= 0.081	Diversity = 2.0789e+01	k_t = [1, 1, 1, 1]	μ_F = 0.576	μ_{CR}
Generation 940:	Best = 3.139298e+03	FES = 94100		
= 0.061	Diversity = 2.0794e+01	k_t = [2, 2, 4, 3]	μ_F = 0.543	μ_{CR}
Generation 950:	Best = 3.139298e+03	FES = 95100		
= 0.080	Diversity = 2.0758e+01	k_t = [1, 1, 1, 1]		

```
Generation 960: Best = 3.139298e+03 FES = 96100  $\mu_F$  = 0.457  $\mu_{CR}$ 
= 0.090 Diversity = 2.0766e+01  $k_t$  = [3, 2, 3, 3]
Generation 970: Best = 3.139298e+03 FES = 97100  $\mu_F$  = 0.384  $\mu_{CR}$ 
= 0.091 Diversity = 2.0731e+01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 3.139298e+03 FES = 98100  $\mu_F$  = 0.368  $\mu_{CR}$ 
= 0.111 Diversity = 2.0657e+01  $k_t$  = [3, 3, 2, 3]
Generation 990: Best = 3.139298e+03 FES = 99100  $\mu_F$  = 0.265  $\mu_{CR}$ 
= 0.114 Diversity = 2.0644e+01  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 3.139298e+03

Total Function Evals = 100000

Total Successful Updates = 2565

```
=====
=====
```

Results:

Best Fitness : 3.139298e+03

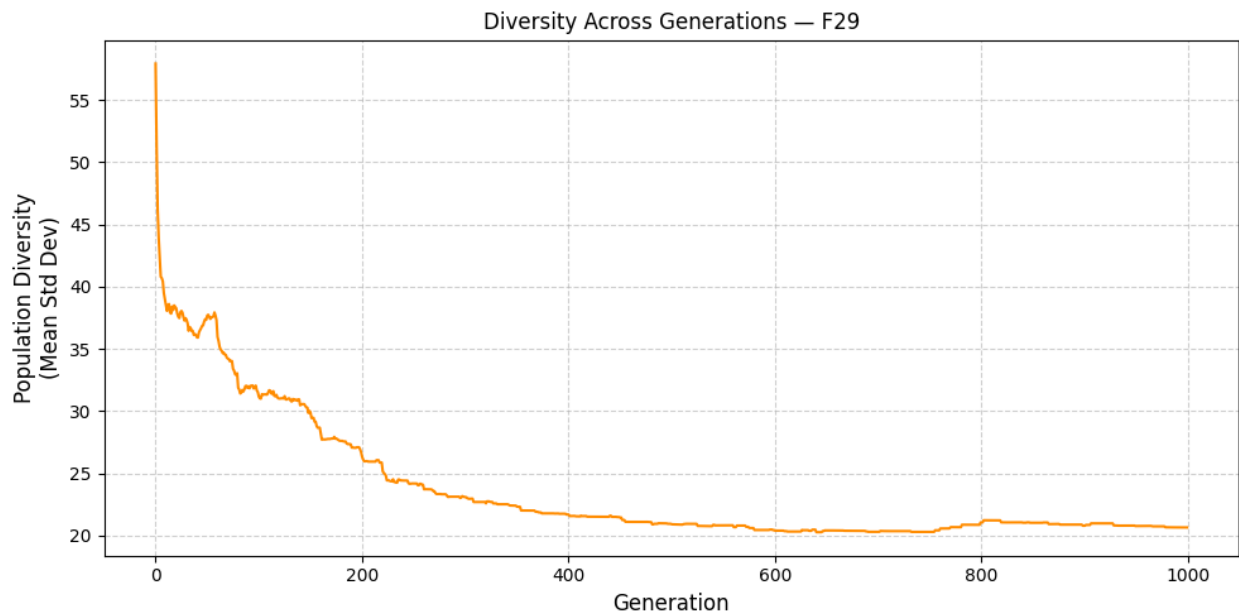
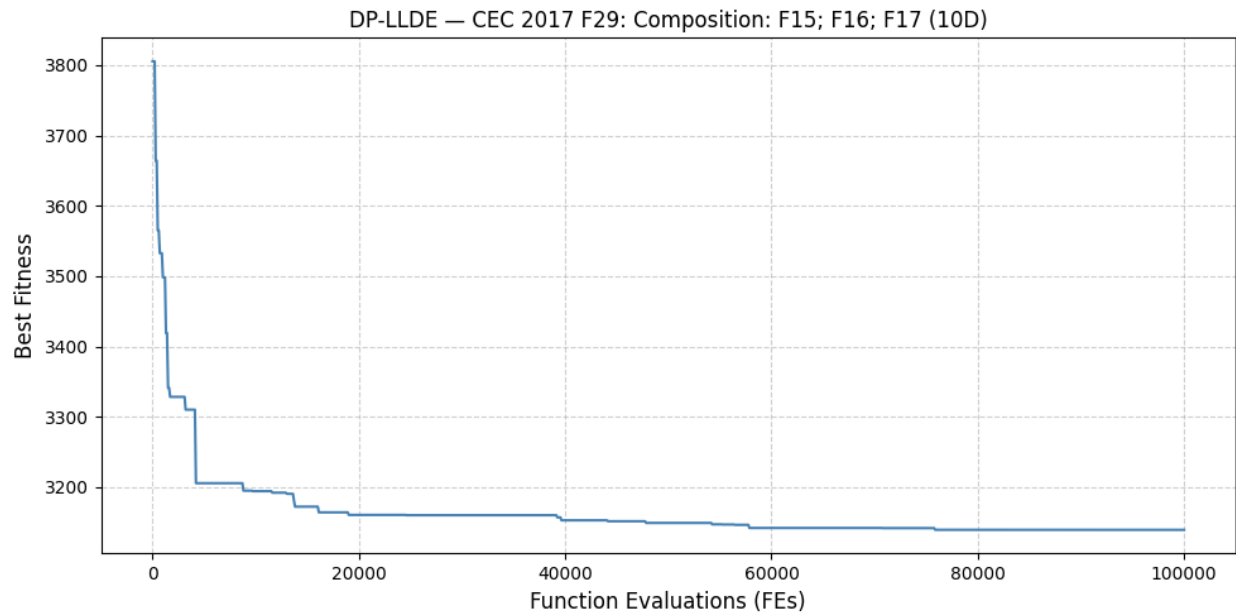
Global Optimum : 2.900000e+03

Error (|best-optimum|) : 2.392977e+02

Best Solution (5 dim) : [-37.36551691 -72.40833529 -68.95234747
81.79330718 -8.75329707]

D^(0) (init. diversity): 5.7928e+01

```
=====
=====
```



Graph saved as convergence_F29.png

```
# =====
# F30 — Composition Function 10 (F15; F18; F19)
# =====
run_and_plot('F30', make_wrapper(composition.f30), 3000.0,
'Composition: F15; F18; F19')
```

```
=====
DP-LLDE on CEC 2017 F30 — Single Run
=====
```

=====

Function : Composition: F15; F18; F19 (10D)

Bounds : [-100, 100]

Global optimum : 3000.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = $5.7928e+00$

Generation 0: Best Fitness = $1.165390e+08$

Generation 10:	Best = $2.119589e+07$	FES = 1100	$\mu_F = 0.573$	μ_{CR}
= 0.336 Diversity = $4.4369e+01$ $k_t = [1, 1, 1, 1]$				
Generation 20:	Best = $1.160791e+07$	FES = 2100	$\mu_F = 0.630$	μ_{CR}
= 0.324 Diversity = $4.1194e+01$ $k_t = [3, 4, 2, 3]$				
Generation 30:	Best = $1.949409e+06$	FES = 3100	$\mu_F = 0.658$	μ_{CR}
= 0.276 Diversity = $3.9813e+01$ $k_t = [1, 1, 1, 1]$				
Generation 40:	Best = $1.214605e+06$	FES = 4100	$\mu_F = 0.694$	μ_{CR}
= 0.230 Diversity = $3.8664e+01$ $k_t = [3, 2, 4, 2]$				
Generation 50:	Best = $9.126511e+05$	FES = 5100	$\mu_F = 0.724$	μ_{CR}
= 0.217 Diversity = $3.6127e+01$ $k_t = [1, 1, 1, 1]$				
Generation 60:	Best = $4.350937e+05$	FES = 6100	$\mu_F = 0.764$	μ_{CR}
= 0.189 Diversity = $3.2302e+01$ $k_t = [2, 2, 4, 2]$				
Generation 70:	Best = $8.186981e+04$	FES = 7100	$\mu_F = 0.795$	μ_{CR}
= 0.134 Diversity = $2.9039e+01$ $k_t = [1, 1, 1, 1]$				
Generation 80:	Best = $4.356181e+04$	FES = 8100	$\mu_F = 0.787$	μ_{CR}
= 0.126 Diversity = $2.4085e+01$ $k_t = [3, 2, 3, 2]$				
Generation 90:	Best = $2.184460e+04$	FES = 9100	$\mu_F = 0.769$	μ_{CR}
= 0.119 Diversity = $2.0247e+01$ $k_t = [1, 1, 1, 1]$				
Generation 100:	Best = $1.439775e+04$	FES = 10100	$\mu_F = 0.761$	μ_{CR}
= 0.098 Diversity = $1.7724e+01$ $k_t = [2, 3, 2, 2]$				
Generation 110:	Best = $9.324865e+03$	FES = 11100	$\mu_F = 0.763$	μ_{CR}
= 0.073 Diversity = $1.6715e+01$ $k_t = [1, 1, 1, 1]$				
Generation 120:	Best = $5.561263e+03$	FES = 12100	$\mu_F = 0.756$	μ_{CR}
= 0.074 Diversity = $1.5552e+01$ $k_t = [3, 3, 2, 4]$				
Generation 130:	Best = $5.561263e+03$	FES = 13100	$\mu_F = 0.771$	μ_{CR}
= 0.088 Diversity = $1.3845e+01$ $k_t = [1, 1, 1, 1]$				
Generation 140:	Best = $5.242196e+03$	FES = 14100	$\mu_F = 0.773$	μ_{CR}
= 0.098 Diversity = $1.2589e+01$ $k_t = [4, 3, 2, 4]$				
Generation 150:	Best = $4.651010e+03$	FES = 15100	$\mu_F = 0.790$	μ_{CR}
= 0.097 Diversity = $1.3317e+01$ $k_t = [1, 1, 1, 1]$				
Generation 160:	Best = $4.246292e+03$	FES = 16100	$\mu_F = 0.793$	μ_{CR}
= 0.098 Diversity = $1.0440e+01$ $k_t = [2, 2, 3, 2]$				
Generation 170:	Best = $4.167556e+03$	FES = 17100	$\mu_F = 0.781$	μ_{CR}
= 0.088 Diversity = $8.0775e+00$ $k_t = [1, 1, 1, 1]$				
Generation 180:	Best = $3.995914e+03$	FES = 18100	$\mu_F = 0.774$	μ_{CR}
= 0.094 Diversity = $7.6305e+00$ $k_t = [2, 3, 2, 3]$				
Generation 190:	Best = $3.862832e+03$	FES = 19100	$\mu_F = 0.800$	μ_{CR}
= 0.094 Diversity = $8.3146e+00$ $k_t = [1, 1, 1, 1]$				
Generation 200:	Best = $3.841186e+03$	FES = 20100	$\mu_F = 0.795$	μ_{CR}
= 0.108 Diversity = $7.7582e+00$ $k_t = [2, 2, 2, 4]$				
Generation 210:	Best = $3.813483e+03$	FES = 21100	$\mu_F = 0.810$	μ_{CR}
= 0.100 Diversity = $6.3707e+00$ $k_t = [1, 1, 1, 1]$				
Generation 220:	Best = $3.780969e+03$	FES = 22100	$\mu_F = 0.827$	μ_{CR}

= 0.108	Diversity = 6.5304e+00	k_t = [4, 4, 3, 2]		
Generation 230:	Best = 3.753384e+03	FES = 23100	$\mu_F = 0.841$	μ_{CR}
= 0.102	Diversity = 7.3583e+00	k_t = [1, 1, 1, 1]		
Generation 240:	Best = 3.734465e+03	FES = 24100	$\mu_F = 0.844$	μ_{CR}
= 0.102	Diversity = 7.4829e+00	k_t = [3, 2, 3, 4]		
Generation 250:	Best = 3.711887e+03	FES = 25100	$\mu_F = 0.849$	μ_{CR}
= 0.103	Diversity = 7.6609e+00	k_t = [1, 1, 1, 1]		
Generation 260:	Best = 3.688291e+03	FES = 26100	$\mu_F = 0.842$	μ_{CR}
= 0.103	Diversity = 8.3976e+00	k_t = [2, 2, 2, 4]		
Generation 270:	Best = 3.674078e+03	FES = 27100	$\mu_F = 0.849$	μ_{CR}
= 0.109	Diversity = 7.7312e+00	k_t = [1, 1, 1, 1]		
Generation 280:	Best = 3.652823e+03	FES = 28100	$\mu_F = 0.827$	μ_{CR}
= 0.109	Diversity = 7.7892e+00	k_t = [3, 2, 2, 4]		
Generation 290:	Best = 3.652823e+03	FES = 29100	$\mu_F = 0.813$	μ_{CR}
= 0.109	Diversity = 7.7789e+00	k_t = [1, 1, 1, 1]		
Generation 300:	Best = 3.652823e+03	FES = 30100	$\mu_F = 0.830$	μ_{CR}
= 0.114	Diversity = 6.2777e+00	k_t = [2, 3, 3, 4]		
Generation 310:	Best = 3.640027e+03	FES = 31100	$\mu_F = 0.829$	μ_{CR}
= 0.114	Diversity = 6.0404e+00	k_t = [1, 1, 1, 1]		
Generation 320:	Best = 3.635538e+03	FES = 32100	$\mu_F = 0.835$	μ_{CR}
= 0.127	Diversity = 4.9130e+00	k_t = [2, 2, 3, 3]		
Generation 330:	Best = 3.632434e+03	FES = 33100	$\mu_F = 0.828$	μ_{CR}
= 0.127	Diversity = 3.1104e+00	k_t = [1, 1, 1, 1]		
Generation 340:	Best = 3.615940e+03	FES = 34100	$\mu_F = 0.818$	μ_{CR}
= 0.117	Diversity = 3.4360e+00	k_t = [3, 3, 4, 2]		
Generation 350:	Best = 3.608243e+03	FES = 35100	$\mu_F = 0.810$	μ_{CR}
= 0.106	Diversity = 3.4577e+00	k_t = [1, 1, 1, 1]		
Generation 360:	Best = 3.604021e+03	FES = 36100	$\mu_F = 0.829$	μ_{CR}
= 0.089	Diversity = 3.6811e+00	k_t = [3, 3, 3, 2]		
Generation 370:	Best = 3.596243e+03	FES = 37100	$\mu_F = 0.835$	μ_{CR}
= 0.081	Diversity = 3.7138e+00	k_t = [1, 1, 1, 1]		
Generation 380:	Best = 3.584388e+03	FES = 38100	$\mu_F = 0.855$	μ_{CR}
= 0.083	Diversity = 3.8217e+00	k_t = [2, 4, 4, 3]		
Generation 390:	Best = 3.575143e+03	FES = 39100	$\mu_F = 0.843$	μ_{CR}
= 0.071	Diversity = 4.0967e+00	k_t = [1, 1, 1, 1]		
Generation 400:	Best = 3.570615e+03	FES = 40100	$\mu_F = 0.841$	μ_{CR}
= 0.094	Diversity = 4.4237e+00	k_t = [4, 2, 4, 3]		
Generation 410:	Best = 3.562067e+03	FES = 41100	$\mu_F = 0.840$	μ_{CR}
= 0.085	Diversity = 4.2245e+00	k_t = [1, 1, 1, 1]		
Generation 420:	Best = 3.554627e+03	FES = 42100	$\mu_F = 0.832$	μ_{CR}
= 0.078	Diversity = 3.2562e+00	k_t = [2, 3, 4, 3]		
Generation 430:	Best = 3.538040e+03	FES = 43100	$\mu_F = 0.826$	μ_{CR}
= 0.083	Diversity = 2.6409e+00	k_t = [1, 1, 1, 1]		
Generation 440:	Best = 3.528555e+03	FES = 44100	$\mu_F = 0.835$	μ_{CR}
= 0.083	Diversity = 2.2313e+00	k_t = [3, 4, 2, 4]		
Generation 450:	Best = 3.525722e+03	FES = 45100	$\mu_F = 0.828$	μ_{CR}
= 0.083	Diversity = 2.0410e+00	k_t = [1, 1, 1, 1]		
Generation 460:	Best = 3.519175e+03	FES = 46100	$\mu_F = 0.825$	μ_{CR}
= 0.088	Diversity = 1.9620e+00	k_t = [4, 3, 3, 2]		

Generation 470: Best = 3.510402e+03	FES = 47100	$\mu_F = 0.802$	μ_{CR}
= 0.083 Diversity = 1.8961e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 3.508771e+03	FES = 48100	$\mu_F = 0.777$	μ_{CR}
= 0.079 Diversity = 1.6818e+00	$k_t = [2, 4, 3, 4]$		
Generation 490: Best = 3.508771e+03	FES = 49100	$\mu_F = 0.764$	μ_{CR}
= 0.076 Diversity = 1.5262e+00	$k_t = [1, 1, 1, 1]$		
Generation 500: Best = 3.508092e+03	FES = 50100	$\mu_F = 0.745$	μ_{CR}
= 0.082 Diversity = 1.2878e+00	$k_t = [4, 3, 3, 3]$		
Generation 510: Best = 3.506224e+03	FES = 51100	$\mu_F = 0.747$	μ_{CR}
= 0.083 Diversity = 1.2589e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 3.504668e+03	FES = 52100	$\mu_F = 0.752$	μ_{CR}
= 0.085 Diversity = 1.2219e+00	$k_t = [3, 4, 3, 4]$		
Generation 530: Best = 3.503575e+03	FES = 53100	$\mu_F = 0.719$	μ_{CR}
= 0.085 Diversity = 1.1634e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 3.503575e+03	FES = 54100	$\mu_F = 0.704$	μ_{CR}
= 0.082 Diversity = 1.1103e+00	$k_t = [2, 3, 4, 3]$		
Generation 550: Best = 3.501955e+03	FES = 55100	$\mu_F = 0.683$	μ_{CR}
= 0.091 Diversity = 1.1080e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 3.501879e+03	FES = 56100	$\mu_F = 0.683$	μ_{CR}
= 0.095 Diversity = 1.0521e+00	$k_t = [4, 4, 2, 4]$		
Generation 570: Best = 3.500480e+03	FES = 57100	$\mu_F = 0.744$	μ_{CR}
= 0.093 Diversity = 1.0118e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 3.499988e+03	FES = 58100	$\mu_F = 0.752$	μ_{CR}
= 0.086 Diversity = 1.0848e+00	$k_t = [3, 4, 2, 3]$		
Generation 590: Best = 3.499988e+03	FES = 59100	$\mu_F = 0.747$	μ_{CR}
= 0.079 Diversity = 1.0832e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 3.498990e+03	FES = 60100	$\mu_F = 0.755$	μ_{CR}
= 0.077 Diversity = 1.0346e+00	$k_t = [4, 3, 4, 4]$		
Generation 610: Best = 3.494801e+03	FES = 61100	$\mu_F = 0.751$	μ_{CR}
= 0.087 Diversity = 9.6720e-01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 3.494801e+03	FES = 62100	$\mu_F = 0.736$	μ_{CR}
= 0.091 Diversity = 9.4672e-01	$k_t = [2, 4, 3, 4]$		
Generation 630: Best = 3.494801e+03	FES = 63100	$\mu_F = 0.744$	μ_{CR}
= 0.090 Diversity = 8.9753e-01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 3.494801e+03	FES = 64100	$\mu_F = 0.740$	μ_{CR}
= 0.091 Diversity = 9.2512e-01	$k_t = [4, 4, 2, 2]$		
Generation 650: Best = 3.494801e+03	FES = 65100	$\mu_F = 0.748$	μ_{CR}
= 0.091 Diversity = 9.4908e-01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 3.494512e+03	FES = 66100	$\mu_F = 0.754$	μ_{CR}
= 0.105 Diversity = 8.8447e-01	$k_t = [2, 4, 4, 2]$		
Generation 670: Best = 3.494128e+03	FES = 67100	$\mu_F = 0.730$	μ_{CR}
= 0.106 Diversity = 8.5498e-01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 3.493808e+03	FES = 68100	$\mu_F = 0.726$	μ_{CR}
= 0.114 Diversity = 8.1060e-01	$k_t = [3, 3, 3, 3]$		
Generation 690: Best = 3.484521e+03	FES = 69100	$\mu_F = 0.756$	μ_{CR}
= 0.111 Diversity = 7.9638e-01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 3.484521e+03	FES = 70100	$\mu_F = 0.736$	μ_{CR}
= 0.113 Diversity = 7.3889e-01	$k_t = [3, 3, 4, 4]$		
Generation 710: Best = 3.484521e+03	FES = 71100	$\mu_F = 0.715$	μ_{CR}

= 0.109	Diversity = 7.2309e-01	k_t = [1, 1, 1, 1]		
Generation 720:	Best = 3.484521e+03	FES = 72100	μ_F = 0.718	μ_{CR}
= 0.108	Diversity = 7.6804e-01	k_t = [3, 2, 2, 4]		
Generation 730:	Best = 3.484521e+03	FES = 73100	μ_F = 0.745	μ_{CR}
= 0.110	Diversity = 7.9000e-01	k_t = [1, 1, 1, 1]		
Generation 740:	Best = 3.484372e+03	FES = 74100	μ_F = 0.736	μ_{CR}
= 0.105	Diversity = 7.5341e-01	k_t = [4, 2, 3, 3]		
Generation 750:	Best = 3.483547e+03	FES = 75100	μ_F = 0.717	μ_{CR}
= 0.105	Diversity = 7.4545e-01	k_t = [1, 1, 1, 1]		
Generation 760:	Best = 3.483547e+03	FES = 76100	μ_F = 0.701	μ_{CR}
= 0.102	Diversity = 6.7700e-01	k_t = [3, 3, 2, 2]		
Generation 770:	Best = 3.483279e+03	FES = 77100	μ_F = 0.644	μ_{CR}
= 0.095	Diversity = 6.3746e-01	k_t = [1, 1, 1, 1]		
Generation 780:	Best = 3.483279e+03	FES = 78100	μ_F = 0.604	μ_{CR}
= 0.078	Diversity = 6.1557e-01	k_t = [3, 4, 4, 2]		
Generation 790:	Best = 3.483279e+03	FES = 79100	μ_F = 0.537	μ_{CR}
= 0.077	Diversity = 5.9959e-01	k_t = [1, 1, 1, 1]		
Generation 800:	Best = 3.482476e+03	FES = 80100	μ_F = 0.557	μ_{CR}
= 0.078	Diversity = 6.0716e-01	k_t = [4, 2, 4, 4]		
Generation 810:	Best = 3.482476e+03	FES = 81100	μ_F = 0.620	μ_{CR}
= 0.069	Diversity = 5.9900e-01	k_t = [1, 1, 1, 1]		
Generation 820:	Best = 3.481864e+03	FES = 82100	μ_F = 0.618	μ_{CR}
= 0.076	Diversity = 6.4664e-01	k_t = [4, 4, 4, 3]		
Generation 830:	Best = 3.481671e+03	FES = 83100	μ_F = 0.598	μ_{CR}
= 0.073	Diversity = 5.7139e-01	k_t = [1, 1, 1, 1]		
Generation 840:	Best = 3.481127e+03	FES = 84100	μ_F = 0.609	μ_{CR}
= 0.073	Diversity = 5.0521e-01	k_t = [2, 3, 2, 4]		
Generation 850:	Best = 3.481127e+03	FES = 85100	μ_F = 0.587	μ_{CR}
= 0.071	Diversity = 5.0100e-01	k_t = [1, 1, 1, 1]		
Generation 860:	Best = 3.481056e+03	FES = 86100	μ_F = 0.614	μ_{CR}
= 0.064	Diversity = 5.0985e-01	k_t = [4, 2, 4, 4]		
Generation 870:	Best = 3.481056e+03	FES = 87100	μ_F = 0.654	μ_{CR}
= 0.069	Diversity = 5.3902e-01	k_t = [1, 1, 1, 1]		
Generation 880:	Best = 3.480765e+03	FES = 88100	μ_F = 0.645	μ_{CR}
= 0.072	Diversity = 5.1059e-01	k_t = [3, 4, 4, 2]		
Generation 890:	Best = 3.480262e+03	FES = 89100	μ_F = 0.640	μ_{CR}
= 0.072	Diversity = 5.1732e-01	k_t = [1, 1, 1, 1]		
Generation 900:	Best = 3.480007e+03	FES = 90100	μ_F = 0.677	μ_{CR}
= 0.068	Diversity = 4.5910e-01	k_t = [4, 2, 2, 4]		
Generation 910:	Best = 3.480007e+03	FES = 91100	μ_F = 0.652	μ_{CR}
= 0.072	Diversity = 4.5047e-01	k_t = [1, 1, 1, 1]		
Generation 920:	Best = 3.480007e+03	FES = 92100	μ_F = 0.621	μ_{CR}
= 0.062	Diversity = 4.2962e-01	k_t = [2, 4, 2, 3]		
Generation 930:	Best = 3.480007e+03	FES = 93100	μ_F = 0.589	μ_{CR}
= 0.058	Diversity = 3.8644e-01	k_t = [1, 1, 1, 1]		
Generation 940:	Best = 3.480007e+03	FES = 94100	μ_F = 0.708	μ_{CR}
= 0.055	Diversity = 3.7879e-01	k_t = [4, 2, 3, 4]		
Generation 950:	Best = 3.480007e+03	FES = 95100	μ_F = 0.657	μ_{CR}
= 0.053	Diversity = 3.7736e-01	k_t = [1, 1, 1, 1]		

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Generation 960: Best = 3.480007e+03 FES = 96100  $\mu_F$  = 0.580  $\mu_{CR}$ 
= 0.056 Diversity = 3.8378e-01  $k_t$  = [4, 4, 2, 3]
Generation 970: Best = 3.479662e+03 FES = 97100  $\mu_F$  = 0.650  $\mu_{CR}$ 
= 0.050 Diversity = 3.8065e-01  $k_t$  = [1, 1, 1, 1]
Generation 980: Best = 3.479662e+03 FES = 98100  $\mu_F$  = 0.673  $\mu_{CR}$ 
= 0.049 Diversity = 3.7811e-01  $k_t$  = [4, 3, 4, 3]
Generation 990: Best = 3.479662e+03 FES = 99100  $\mu_F$  = 0.598  $\mu_{CR}$ 
= 0.049 Diversity = 3.7475e-01  $k_t$  = [1, 1, 1, 1]
```

Optimization complete.

Best Fitness = 3.479662e+03

Total Function Evals = 100000

Total Successful Updates = 9668

=====

Results:

Best Fitness : 3.479662e+03

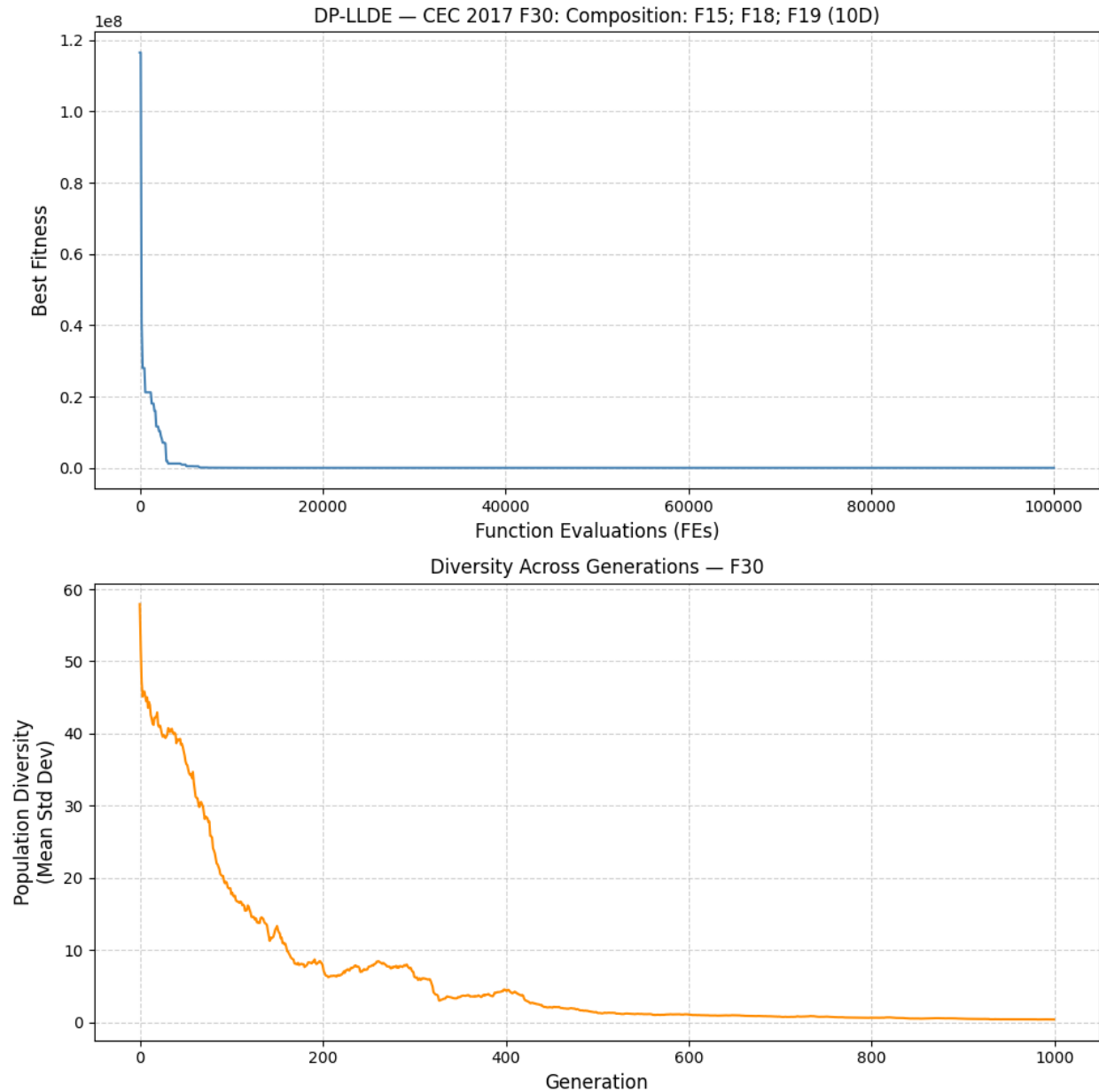
Global Optimum : 3.000000e+03

Error (|best-optimum|) : 4.796619e+02

Best Solution (5 dim) : [-52.63224752 71.71359887 97.31099067
99.48902325 -89.40368893]

D^0 (init. diversity): 5.7928e+01

=====



Graph saved as convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
print('=' * 80)
print('SUMMARY — DP-LLDE on CEC 2017 (10D), Single Run, Seed=42')
```

```

print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |best-opt|":>18}')
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{"fn":<10} {"N/A (deprecated)":>18}')
    {r["f_global"]:>18.2f} {"N/A":>18}')
    else:
        print(f'{"fn":<10} {"r["best_fit"]:>18.6e} {"r["f_global"]:>18.2f} {"r["error"]:>18.6e}')
print('=' * 80)

```

=====

SUMMARY – DP-LLDE on CEC 2017 (10D), Single Run, Seed=42

=====

Function	Best Fitness	Global Optimum	Error best-opt
F1	1.000000e+02	100.00	1.421085e-14
F2	2.000001e+02	200.00	8.558419e-05
F3	3.000007e+02	300.00	6.879505e-04
F4	4.015519e+02	400.00	1.551874e+00
F5	5.022566e+02	500.00	2.256606e+00
F6	6.000000e+02	600.00	6.101186e-08
F7	7.120259e+02	700.00	1.202587e+01
F8	8.030705e+02	800.00	3.070500e+00
F9	9.000000e+02	900.00	0.000000e+00
F10	1.041861e+03	1000.00	4.186088e+01
F11	1.100000e+03	1100.00	8.412826e-11
F12	1.218469e+03	1200.00	1.846903e+01
F13	1.300005e+03	1300.00	5.376836e-03
F14	1.400000e+03	1400.00	1.726725e-05
F15	1.500002e+03	1500.00	2.265975e-03
F16	1.600541e+03	1600.00	5.408353e-01
F17	1.700438e+03	1700.00	4.377342e-01
F18	1.800998e+03	1800.00	9.980688e-01
F19	1.900002e+03	1900.00	2.205140e-03
F20	2.000000e+03	2000.00	2.547147e-05
F21	2.200000e+03	2100.00	1.000000e+02
F22	2.300000e+03	2200.00	1.000000e+02
F23	2.604392e+03	2300.00	3.043923e+02
F24	2.728598e+03	2400.00	3.285984e+02
F25	2.943458e+03	2500.00	4.434576e+02
F26	2.900000e+03	2600.00	3.000000e+02
F27	3.089518e+03	2700.00	3.895179e+02

F28	3.100000e+03	2800.00	3.000000e+02
F29	3.139298e+03	2900.00	2.392977e+02
F30	3.479662e+03	3000.00	4.796619e+02
=====			
=====			